

Lectures on General Relativity

“To understand the universe we must understand how space and time bend”

Saturday 19th September, 2026



LECTURES ON GENERAL RELATIVITY

Miguel Jafert Serrano Mantilla

Professor: Fabio Duván Lora Clavijo

*Universidad Industrial de Santander
Sciences faculty
Physics school
Group of optics and signals treatment*

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Abstract

These notes build the theory of general relativity from the ground up. After tracing how our understanding of gravity evolved from the ancient Greeks to Einstein's own revolution, the book develops the differential geometry needed to speak of curved space-time: manifolds, charts, tangent vectors and the metric tensor. With that language in hand, it turns to the theory itself, deriving the geodesic equation, the Riemann curvature tensor and the Einstein field equations that relate the curvature of space-time to the matter and energy within it. The final chapter linearises these field equations around a flat background to obtain the wave equation for gravitational waves, and studies their propagation and polarisation properties.

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Chapter 1

The History of General Relativity

Every revolution in physics begins as a quiet correction to the certainties of the age before it.

“If I have seen further it is by standing on the shoulders of Giants.”

– Isaac Newton

Before we bend space and curve time we must first understand why anyone would ever have thought to try. **The history of general relativity is not the history of a lone genius working in isolation, it is the history of nearly twenty five centuries of argument about a single, deceptively simple question:** why do bodies move the way they do? Every civilisation that has looked up at the night sky or dropped a stone from a height has had to answer this question in its own way, and the theory we are about to build in this book is, in a very real sense, the latest instalment of that argument rather than its final word. In this chapter we are going to walk through that argument from the Greeks to the physics of today, and along the way we will meet the geometers whose abstract mathematics quietly waited, sometimes for decades, for a physicist to notice that it was exactly what nature needed.

1.1 Motion in the Ancient World

1.1.1 The Aristotelian Cosmos

For the better part of two thousand years the Western understanding of motion belonged to Aristotle (384 to 322 B.C.), whose *Physics* divided all motion into two families. *Natural motion* was the motion a body underwent when left alone, and it depended entirely on what the body was made of: earth and water naturally fell towards the centre of the cosmos, while air and fire naturally rose away from it, and the heavenly bodies, made of a fifth incorruptible element he called *aether*, naturally moved in perfect eternal circles. *Violent motion*, by contrast, was any motion imposed on a body against its nature, and Aristotle's central postulate about it is one every student of mechanics eventually collides with: **a body moves only while something is in contact with it and pushing it**, so that the moment the pushing stops the motion must stop too (Aristotle, 1984). This is why a thrown spear troubled him so much. If the hand that threw it is no longer touching it, what keeps it moving through the air after it leaves the fingers? Aristotle's answer, the doctrine of *antiperistasis*, was that the air parted by the spear's tip rushes around to its rear and pushes it onwards, a mechanism that is elegant on paper and hopeless the moment one actually thinks about throwing something, and it would take the better part of two millennia and the combined efforts of Alexandria, Baghdad and Paris to finally dismantle it.

Underneath this physics of pushes and pulls lay a much grander architecture, the geocentric cosmos, in which a stationary Earth sat at the centre of a finite universe and everything else, the Moon, the Sun, the planets and the fixed stars, was carried around it on nested, concentric, crystalline spheres. It is worth pausing on how seriously this was meant: for Aristotle the centre of the cosmos was not merely a convenient coordinate origin, it was the natural resting place towards which heavy matter was drawn, so that geometry and physics were one and the same fact.

1.1.2 Ptolemy and the Mathematics of the Heavens

The trouble with perfect circles around a stationary Earth is that the planets simply refuse to move on them, drifting faster and slower across the year and occasionally, most embarrassingly

of all, reversing direction in the sky entirely, a phenomenon we now call retrograde motion and understand as nothing more than the view from a moving Earth overtaking a slower outer planet. Claudius Ptolemy, working in Alexandria around 150 A.D., inherited several centuries of Greek attempts to rescue the circles (Apollonius and Hipparchus among them) and assembled them into the *Almagest*, the most complete and mathematically serious astronomical text the ancient world produced (Ptolemy, 1998). Ptolemy kept the philosophical commitment to circular motion but smuggled in three devices to make the mathematics fit the sky: the *epicycle*, a small circle whose centre rides upon a larger circle called the *deferent*; the *eccentric*, which quietly moves the Earth slightly off the true centre of the deferent; and, most cunning of all, the *equant*, an empty point in space from which the planet's motion *appears* uniform even though it is uniform nowhere else. The equant in particular is a fascinating piece of intellectual dishonesty dressed up as rigour, because it abandons the very principle of uniform circular motion it was invented to save, and it is precisely this point, as we shall see in a moment, that the astronomers of Islam would spend centuries trying to repair.

1.2 The Arabic Custodians of Motion

When the intellectual centre of gravity of the ancient world shifted from Alexandria towards Baghdad, Damascus and later Maragha and Samarkand, it carried Aristotle and Ptolemy with it, but it did not simply preserve them, it interrogated them. The crucial bridge figure is not, strictly speaking, an Arab at all: John Philoponus, a Christian philosopher working in sixth century Alexandria, wrote a commentary on Aristotle's *Physics* in which he attacked antiperistasis directly, proposing instead that the thrower impresses onto the projectile itself a finite, internal quantity of motive power, an *incorporeal kinetic force*, which then dissipates gradually as the projectile travels, explaining why a spear eventually falls without needing the air to conspire behind it. Philoponus's Greek text was largely forgotten in Byzantium but survived and was studied intensely by scholars writing in Arabic, and it is through that transmission, not through the Latin West, that his idea re-entered the mainstream of physics.

The Persian polymath Ibn Sina, known in Latin as Avicenna (980 to 1037), sharpened Philoponus's idea considerably in his encyclopaedic *Kitab al-Shifa* (The Book of Healing). He argued that the impressed force, which he called *mayl* (inclination), is a genuine, quantifiable property of the moving body itself rather than of the medium around it, and, in a passage that reads almost eerily like a draft of Newton's first law, he claimed that in a void this *mayl* would never be consumed and the body would continue moving forever, since nothing in a true vacuum could exist to wear it down (Rashed, 1996). This is not yet inertia in the modern sense, because Ibn Sina still thought of *mayl* as a cause that must be continuously present to sustain motion rather than as the mere absence of a cause to stop it, but the intuition that unresisted motion is self-perpetuating had, for the first time, been stated clearly and defended on physical grounds rather than dismissed as an impossible thought experiment about an empty universe Aristotle himself had declared could not exist.

Astronomy saw an even more consequential development. Frustrated by the equant's abandonment of true uniform circular motion, the mathematicians of the Maragha observatory in thirteenth century

Persia, led by Nasir al-Din al-Tusi, invented a purely geometrical device, now called the *Tusi couple*, a small circle rolling inside a larger one, whose combined motion can reproduce a straight line oscillation using only uniformly rotating circles, with no equant required. A century later in Damascus, Ibn al-Shatir used exactly this device to rebuild the entire Ptolemaic planetary system without a single equant, producing models that were mathematically superior to Ptolemy's own (Saliba, 1987). What makes this genuinely remarkable, and worth pausing on for a student of physics rather than only of history, is that when Nicolaus Copernicus published *De Revolutionibus Orbium Coelestium* in 1543, the geometrical diagrams he used to eliminate the equant from his own, now heliocentric, system are mathematically identical to Ibn al-Shatir's, down to the labelling of points in some manuscripts. Historians still debate exactly how the Maragha school's mathematics reached Copernicus, whether through Byzantine Greek intermediaries in Italy or some now lost channel, but the technical debt is not seriously disputed, and it is a useful reminder that the Copernican revolution did not spring from nowhere: it stood, quite literally, on Islamic geometrical shoulders.

1.3 The Latin West Breaks with Aristotle

1.3.1 Buridan's Impetus

The idea of an impressed motive force crossed into Latin Europe through translations of Arabic and Greek texts flooding into the universities of the thirteenth century, and it found its sharpest medieval expression in Jean Buridan, rector of the University of Paris in the 1340s. Buridan proposed that the mover impresses upon the projectile a genuine, permanent quantity he called *impetus*, proportional to the amount of matter in the body and to its speed (a description that should already sound suspiciously familiar to anyone who has met the word *momentum*), and that this impetus would in principle persist forever were it not continuously eroded by air resistance and by the body's own natural tendency to fall (Clagett, 1959). Buridan went further than Ibn Sina in one crucial respect: he applied impetus theory to the heavens themselves, suggesting that God, at the Creation, might simply have impressed upon the celestial spheres an impetus that meets no resistance in the frictionless heavens and therefore never decays, a proposal that quietly detaches the eternal circular motion of the planets from any need for a continuously acting mover, Aristotle's prime mover included. His student Nicole Oresme pushed the mathematics further still, inventing a graphical method of plotting a body's speed against time as a geometrical figure and showing that the area under such a figure gives the distance travelled, a genuinely startling anticipation, two and a half centuries early, of the mean speed theorem that Galileo would later use to characterise uniformly accelerated fall.

1.3.2 Copernicus and Kepler

Copernicus's 1543 relocation of the Sun to the centre of the cosmos was, in itself, primarily a mathematical and aesthetic argument (he found the resulting system simpler and more harmonious), and it would take Johannes Kepler, working with the extraordinarily precise naked eye observations

that Tycho Brahe had spent a lifetime accumulating, to turn heliocentrism into a genuinely predictive physical theory. Between 1609 and 1619 Kepler announced three laws that any general relativity student should keep close at hand, because they are the observational bedrock that Newton, and much later Einstein, would both have to reproduce: planets move on ellipses with the Sun at one focus, they sweep out equal areas in equal times, and the square of their orbital period is proportional to the cube of their orbit's semi-major axis. Kepler arrived at the ellipse only after years of fighting his own preference for circles, famously calling his discovery a *cartload of dung* in a fit of frustration with the messy, imperfect shape nature had apparently chosen over the geometrical perfection he had hoped for, and it is a nice historical irony that this same ellipse would reappear, itself only an approximation, when Einstein finally explained in 1915 why Mercury's own elliptical orbit slowly, stubbornly, refuses to close.

1.3.3 Galileo and the First Principle of Relativity

It is with Galileo Galilei that impetus theory finally becomes something closer to modern inertia. In his *Two New Sciences* of 1638, written under house arrest and smuggled out of Italy for publication in the Protestant Netherlands, Galileo established through careful (and, for the tools available to him, remarkably precise) experiments with balls rolling down inclined planes that a freely falling body accelerates uniformly, covering distances in the ratio of the odd numbers one, three, five and so on in successive equal intervals of time, and he reasoned that a body moving on a perfectly smooth horizontal surface, meeting no resistance and no incline either to speed it up or slow it down, would continue moving forever at constant speed (Galilei, 1974). This is, for all practical purposes, Newton's first law stated half a century early.

Just as important for our story, though, is an idea Galileo had already set out six years earlier, in 1632, in the *Dialogue Concerning the Two Chief World Systems*. There he asks the reader to imagine themselves below decks on a smoothly sailing ship, with no portholes, watching fish swim in a bowl, butterflies fly and drops of water fall from a bottle, and he argues that not one of these phenomena would betray to the observer whether the ship is standing still in harbour or sailing at constant velocity across a perfectly calm sea (Galilei, 1953). **No mechanical experiment performed entirely inside a closed cabin can distinguish uniform motion from rest**, and this is the first clean statement of what we now call the *principle of relativity*, the very principle that, generalised first to electromagnetism by Einstein in 1905 and then to accelerated frames by the equivalence principle we shall meet in chapter 3, becomes the seed from which the entire theory of general relativity eventually grows. Tradition, almost certainly apocryphal but too delicious to leave out, has Galileo muttering *E pur si muove* ("and yet it moves") under his breath after being forced by the Roman Inquisition in 1633 to formally recant the Copernican system his own physics had done so much to defend.

1.4 Newton and the Absolute Stage

Isaac Newton's *Philosophiae Naturalis Principia Mathematica*, published in 1687, is the book that finally welded Galileo's terrestrial kinematics to Kepler's celestial geometry into a single mechanics. Newton's three laws of motion, and above all his law of universal gravitation, showed that the very same force that pulls an apple to the ground also holds the Moon in its orbit and drives the planets along Kepler's ellipses, a genuinely staggering unification for its time (Newton, 1999). But Newton needed a stage on which this mechanics could play out, and he built one deliberately: *absolute space*, an unmoving, featureless, infinite container existing independently of anything inside it, and *absolute time*, flowing uniformly and identically for every observer everywhere in the universe. He even offered an experimental argument for taking absolute space seriously, the famous rotating bucket: spin a bucket of water and its surface climbs the sides into a concave shape, and since this happens whether or not the bucket is rotating relative to any other nearby matter, Newton argued that the water must be rotating relative to something, namely space itself.

What is often forgotten by students who only encounter Newton through the tidy inverse square law is how deeply uncomfortable Newton himself was with one consequence of his own theory: gravity, as he had written it down, acts instantaneously at a distance across empty space, with no mediating mechanism whatsoever, and this troubled him far more than it troubles most introductory textbooks today. In a 1693 letter to the theologian Richard Bentley he wrote, with striking force, that

“that gravity should be innate, inherent and essential to matter, so that one body may act upon another at a distance through a vacuum, without the mediation of any thing else, by and through which their action and force may be conveyed from one to another, is to me so great an absurdity that I believe no man who has in philosophical matters a competent faculty of thinking can ever fall into it.” (Newton, 1958)

Newton had no candidate mechanism to offer in its place, and so, with characteristic caution, he famously declined to speculate publicly, writing in the General Scholium he added to later editions of the *Principia* that *hypotheses non fingo*, “I feign no hypotheses” (Newton, 1999). It would take more than two centuries, and an entirely new branch of geometry that did not yet exist in Newton's lifetime, before anyone could offer a mechanism at all: general relativity's answer, that gravity is not a force transmitted through space but the curvature of space-time itself, is in a very real sense the belated reply to Newton's own unease.

1.5 The Geometers Who Prepared the Ground

1.5.1 Gauss and the Curvature Hidden Inside a Surface

For gravity to become geometry, geometry first had to learn how to describe curvature without appealing to anything outside itself, and this is exactly what Carl Friedrich Gauss achieved in 1827

in his *Disquisitiones Generales circa Superficies Curvas* (General Investigations of Curved Surfaces) (Gauss, 1828). Gauss showed that the curvature of a surface, a quantity that seems at first to depend entirely on how that surface bends through the three dimensional space surrounding it, can in fact be computed using only measurements made *within* the surface itself, lengths and angles as measured by an ant living on it, with no reference whatsoever to the ambient space. He was so struck by this fact that he named it his *Theorema Egregium*, the “remarkable theorem”, and remarkable it is: it is the very first proof that curvature is an *intrinsic* property of a space, not something a space merely borrows from a higher dimensional stage on which it happens to sit. This is precisely the property that will let us, in chapter 2, speak of the curvature of space-time without ever having to ask what space-time is supposedly curving *into*.

It is often repeated, and worth mentioning with the appropriate degree of caution, that Gauss was not only a mathematician but also, in his day job, the director of a two decade long geodetic survey of the Kingdom of Hanover, and that he measured the angles of an enormous triangle formed by the peaks of the Brocken, the Hohenhagen and the Inselsberg, partly to test whether the angles of a very large physical triangle really do sum to exactly 180 degrees, as flat Euclidean geometry demands, or whether the physical space we live in might, just possibly, be very slightly curved. Whether Gauss genuinely intended this as a test of the geometry of physical space, or whether the story has been embellished by later admirers eager to cast him as a prophet of general relativity, remains disputed among historians, and no deviation from 180 degrees was in any case ever detected, the curvature of space-time being far too gentle at the scale of a mountain range for eighteenth century theodolites to have any hope of catching it. What is beyond dispute is that Gauss had, at the very least, begun to entertain the question of whether Euclidean geometry is a truth of pure reason or an empirical fact about our particular universe, and that question is the hinge on which this entire chapter, and indeed this entire book, turns.

1.5.2 Riemann’s Prophecy

Gauss’s intrinsic geometry described curved surfaces, objects with only two dimensions. The decisive generalisation, to spaces of any number of dimensions whatsoever, was the achievement of Gauss’s own student, Bernhard Riemann, and it arrived in a single lecture. In 1854, required by University of Göttingen custom to deliver a probationary lecture (a *Habilitationsvortrag*) before the assembled faculty in order to earn the right to teach, the shy and sickly Riemann chose, on Gauss’s own suggestion, the topic “Über die Hypothesen, welche der Geometrie zu Grunde liegen” (On the Hypotheses which lie at the Bases of Geometry). Gauss himself, by then an old man near the end of his life and famously difficult to impress, sat in the audience, and by every surviving account he was profoundly moved by what he heard.

In that lecture Riemann introduced what we would now call an n -dimensional manifold, a space that need only look locally like ordinary flat Euclidean space, and he showed how to equip such a space with a quantity, the *metric*, encoding the distance between any two infinitesimally close points, from which a notion of curvature at every point could then be built by pure calculation, generalising Gauss’s Theorema Egregium to any dimension and to spaces nobody could ever hope to visualise by

embedding them in something larger. This is the mathematical object we will meet formally as $g_{\mu\nu}$ in chapter 2, and without it there simply is no general relativity to write down. But Riemann did something else that evening, something far stranger for a lecture ostensibly about pure mathematics, something translated from the German by the English mathematician William Kingdon Clifford two decades later and read ever since by physicists with a peculiar chill of recognition: he wondered aloud, in the lecture's closing paragraphs, whether the geometry of physical space might not be fixed and given after all, but might instead be determined by the matter distributed within it.

“we must seek the ground of its metric relations outside it, in binding forces which act upon it” (Riemann, 1873)

is how Clifford rendered Riemann's own conjecture, and Riemann explicitly named this a question for physics, not mathematics, to eventually answer, adding that the answer could only come from experience and observation of the actual physical world. Riemann died of tuberculosis in 1866, at only thirty nine years old, on the shores of Lake Maggiore in Italy, sixty years before anyone would show that he had, in that single lecture, more or less written down the mathematical vessel into which Albert Einstein would one day pour a physical theory of gravity.

1.6 Einstein's Long Road to General Relativity

Albert Einstein's 1905 theory of special relativity had already overthrown Newton's absolute time, welding space and time together into a single four dimensional structure and showing that simultaneity itself depends on the observer's motion. But special relativity, as its name honestly admits, is special: it applies only to unaccelerated, inertial observers, and it could offer no account whatsoever of gravity, since Newtonian gravity's instantaneous action at a distance is flatly incompatible with special relativity's insistence that no signal, gravitational or otherwise, can outrun the speed of light.

The idea that would eventually resolve this came to Einstein in 1907, while he was still working at the Swiss patent office in Bern, and he later called it, in a 1922 lecture in Kyoto, *der glücklichste Gedanke meines Lebens*, “the happiest thought of my life”: a person falling freely from a rooftop feels no gravity whatsoever, their own weight vanishing entirely for as long as the fall lasts. From this simple, almost childlike observation Einstein extracted his *equivalence principle*, the proposal that being at rest in a gravitational field is locally indistinguishable from accelerating through empty space with no gravity at all, a principle that, as we shall see explicitly in chapter 3, is nothing less than Galileo's shipboard relativity extended from uniform motion to acceleration itself, and it is the true conceptual seed of everything that follows in this book.

Turning that seed into a working theory took Einstein eight more gruelling years. He understood by around 1912, while teaching in Prague, that a theory built on the equivalence principle would need a gravitational field capable of bending the paths of light rays and of clocks running at different rates in different places, and that flat Minkowski space-time (the space-time of special relativity) simply could not accommodate this. What he needed was some kind of curved geometry, and he had no idea such a thing already existed in full generality, sitting in the mathematics literature

since 1854, until his old university classmate from Zurich, the mathematician Marcel Grossmann, came to his rescue. According to Einstein's own later recollection, and as reported by his biographer Abraham Pais, Einstein turned to Grossmann in something close to desperation with the words "*Grossmann, du musst mir helfen, sonst werd' ich verrückt!*" ("Grossmann, you must help me or I shall go crazy!") (Pais, 1982). Grossmann answered by introducing Einstein to precisely Riemann's geometry, together with the *absolute differential calculus* that the Italian mathematicians Gregorio Ricci-Curbastro and Tullio Levi-Civita had spent the intervening decades building on top of it, the very tensor calculus this book will spend the whole of chapter 2 constructing from scratch.

In 1913 Einstein and Grossmann published together a first attempt, the so called *Entwurf* (draft) theory, which used Riemann's curved metric but, out of a mixture of genuine physical worry and outright mathematical error, stopped short of demanding that the field equations hold in every coordinate system (Einstein and Grossmann, 1913). The *Entwurf* theory predicted the wrong value for the century old, already precisely measured anomaly in the precession of Mercury's perihelion, and Einstein knew it. The final months, in Berlin during the autumn of 1915, were by his own account among the most intense of his life: in a sequence of four papers delivered to the Prussian Academy of Sciences on four successive Thursdays in November, he corrected his errors in full public view, week by week, until on 25 November 1915 he presented the field equations essentially in the form we use today (Einstein, 1915), and applied them almost immediately to recover Mercury's precession exactly, a moment he later described as having given him heart palpitations severe enough to worry he might be physically unwell.

1.7 Riemann, Hilbert and the Question of Theft

It is sometimes claimed, in the looser corners of popular science writing, that Einstein "stole" general relativity from Riemann, and it is worth being precise about why this particular accusation does not survive contact with the historical record, even though a genuine and fascinating priority dispute does sit at the heart of this story, just with a different antagonist. Riemann died in 1866, forty nine years before Einstein wrote down a single field equation, and he never worked on gravity, never proposed a field equation, and never claimed to have solved the problem this chapter has been tracing since Aristotle: Riemann supplied a mathematical *language*, the geometry of curved manifolds, not a physical theory of gravitation, and Einstein always said so plainly and gratefully, both in his 1913 paper with Grossmann and later in his own *Autobiographical Notes* (Einstein, 1949). Crediting Riemann with the mathematics he actually built, generously and by name, is simply accurate history, not theft.

The real, and rather more dramatic, priority dispute concerns not Riemann but David Hilbert, the towering Göttingen mathematician who was, quite independently, hunting for the same field equations in the very same autumn of 1915. Einstein had visited Göttingen that summer and lectured on his still unfinished theory, and Hilbert, characteristically supremely confident (he is reported to have remarked, only half in jest, that "*every boy in the streets of Göttingen understands more about four dimensional geometry than Einstein*"), so thoroughly had David Hilbert's own city absorbed the mathematics Einstein was still struggling to apply), set to work on the problem from a different

direction entirely, a variational principle built from the very Einstein-Hilbert action we shall derive in chapter 3. Hilbert submitted a paper to the Göttingen Academy titled “Die Grundlagen der Physik” on 20 November 1915, five days before Einstein’s own final, decisive paper of 25 November (Hilbert, 1915), and for decades afterwards a persistent rumour circulated that Einstein had somehow seen and silently borrowed from Hilbert’s unpublished proofs, or conversely that Hilbert had quietly lifted the final covariant equations from Einstein’s own Göttingen lecture.

The matter was only properly settled in 1997, when the historians of science Leo Corry, Jürgen Renn and John Stachel tracked down the actual surviving galley proofs of Hilbert’s paper, printer’s proofs dated December 1915, sitting quietly in the Göttingen archive (Corry et al., 1997). Those proofs, it turned out, did *not* yet contain the fully generally covariant field equations that appeared in Hilbert’s paper when it was finally published, months later, in 1916: Hilbert had revised his own manuscript after that publication date, almost certainly after seeing Einstein’s completed November papers, to bring it into line with Einstein’s result. **Einstein reached the correct, generally covariant field equations first, and independently**, while Hilbert arrived at essentially the same equations by a beautiful and genuinely independent variational route, close behind him and very possibly influenced, in the final polish of his own paper, by Einstein’s success. Hilbert himself, to his considerable credit, never once publicly disputed Einstein’s priority and always referred to the result as “Einstein’s equations”, and the two men remained on entirely cordial, mutually respectful terms for the rest of their lives. If there is a lesson here for a student just beginning this subject, it is that the discovery of general relativity was not a single flash of individual genius but the convergence of an entire community, Gauss’s intrinsic curvature, Riemann’s manifolds, Ricci and Levi-Civita’s calculus, Grossmann’s tutoring, and Hilbert’s own parallel and generous rivalry, all funnelled through Einstein’s singular physical insight that gravity simply *is* the shape of space-time.

1.8 Confirmation: From the Trenches to the Eclipse

Extraordinary claims of this kind demand to be tested, and their testing produced two of the most romantic episodes in the whole history of physics. The first came almost immediately. Karl Schwarzschild, a distinguished German astrophysicist who had volunteered for military service and was at the time calculating artillery trajectories on the Russian front, read Einstein’s November 1915 papers and, within a matter of weeks, working between his military duties, found the first exact, non approximate solution to Einstein’s notoriously difficult field equations, the solution describing the curved space-time around a single spherical, non rotating mass, which we will study closely in chapter 4 and which today bears his name (Schwarzschild, 1916). Schwarzschild mailed his solution to Einstein directly from the front; Einstein, visibly delighted, presented it to the Prussian Academy on Schwarzschild’s behalf in January 1916. Schwarzschild had, tragically, already contracted a rare and painful autoimmune skin disease at the front, and he died only a few months later, in May 1916, never learning that his solution would one day predict the existence of black holes.

The second episode made Einstein a global celebrity almost overnight. General relativity predicted that starlight passing close to the Sun should be deflected by the Sun’s gravity by twice the amount Newtonian gravity alone would predict, an effect only observable during a total solar eclipse, when the

Sun’s own overwhelming brightness is briefly blotted out and stars that ordinarily hide directly behind it become visible. On 29 May 1919, two expeditions organised chiefly by the British astronomers Arthur Eddington and Frank Dyson, one to the island of Príncipe off the coast of West Africa and one to Sobral in northern Brazil, photographed the stars around the eclipsed Sun and found, after months of careful measurement, a deflection matching Einstein’s relativistic prediction and decisively ruling out the smaller Newtonian value (Dyson et al., 1920). When the results were announced at a joint meeting of the Royal Society and the Royal Astronomical Society in London that November, the news spread with astonishing speed: the *New York Times* greeted it on 10 November 1919 with the headline “Lights All Askew in the Heavens”, and within weeks Einstein, until then a name known chiefly to physicists, had become the most famous scientist on Earth.

1.9 Beyond General Relativity

The theory did not stop working once it was confirmed, and it has continued, ever since, to reveal consequences even its own author found hard to accept. In 1917 Einstein had inserted an extra term, the *cosmological constant*, into his own field equations purely to force them to describe a static, unchanging universe, since a static cosmos was simply what everyone, Einstein included, assumed the real universe to be. The Russian mathematician Alexander Friedmann in 1922 (Friedmann, 1922) and, independently, the Belgian priest and astronomer Georges Lemaître in 1927 (Lemaître, 1927), showed that Einstein’s own unmodified equations naturally describe an expanding, evolving universe instead, a result Einstein initially, and famously, resisted. It took Edwin Hubble’s 1929 observation that distant galaxies are indeed systematically receding from us, and at a rate proportional to their distance, exactly as Friedmann and Lemaître had predicted, to settle the matter (Hubble, 1929), and Einstein is widely (if perhaps not entirely reliably) reported to have called his own cosmological constant the biggest blunder of his career.

General relativity’s most extreme prediction took even longer to be taken seriously. In 1939 J. Robert Oppenheimer and his student Hartland Snyder showed mathematically that a sufficiently massive star, having exhausted its nuclear fuel, must collapse without limit under its own gravity, its surface eventually crossing a point of no return beyond which not even light can climb back out (Oppenheimer and Snyder, 1939), but the paper was largely ignored for nearly three decades, overshadowed by the Second World War and by a lingering, widely shared conviction among physicists, Einstein among them, that nature would surely find some way to forbid so pathological an outcome. It was the physicist John Archibald Wheeler who, in a 1967 lecture, popularised a short, vivid name for these collapsed objects, *black holes*, a term that stuck instantly where more cumbersome earlier phrases such as “gravitationally completely collapsed object” never had (Wheeler and Ford, 1998), and it was Wheeler, too, who gave us the aphorism already quoted on this chapter’s own title page: spacetime tells matter how to move, and matter tells spacetime how to curve.

Einstein’s 1916 prediction of gravitational waves, tiny ripples of curvature radiating outward from accelerating masses, waited even longer for its confirmation, precisely one century, until on 14 September 2015 the two detectors of the Laser Interferometer Gravitational Wave Observatory (LIGO) simultaneously registered the unmistakable signature of two black holes, thirty six and

twenty nine times the mass of the Sun, spiralling into one another and merging over a billion years ago, a discovery announced to the world in February 2016 and one we shall return to properly in chapter 5 (Abbott, B. P. et al., 2016).

And there the story remains genuinely, actively unfinished. General relativity describes gravity magnificently at the scale of planets, stars and galaxies, yet it still refuses to speak the same language as quantum mechanics, the theory that governs the very small, and no one yet possesses a fully working theory of *quantum gravity* capable of describing, for instance, what truly happens at the very centre of a black hole or in the first unimaginably brief instant of the universe's own existence. String theory, loop quantum gravity and a handful of other, still more speculative approaches are today's competing attempts to write the next chapter, each borrowing, in one way or another, from the very same geometrical inheritance that runs through this one: Gauss's intrinsic curvature, Riemann's manifolds, and Einstein's insight that geometry itself can be physics. It is worth carrying this long procession of names in mind as we turn, in chapter 2, to build the mathematics properly from first principles, because every symbol we are about to write down, the metric, the connection, the curvature tensor, was earned by someone, often across a lifetime and occasionally at real personal cost, and general relativity is best understood not as a single flash of individual genius but as the latest, and very probably not the last, chapter in humanity's long and stubborn argument about why things fall.

Chapter 2

Differential Geometry

*Before space-time could bend, mathematics first
had to learn how a surface curves upon itself.*

“Geometry is not true, it is advantageous.”

– Henri Poincaré

2.1 A brief history of geometry

Differential geometry is the language in which general relativity is written, and like every language it was not designed in advance for the use we are about to make of it, it grew slowly, over more than four thousand years, out of problems that had nothing to do with gravity. Before we start defining manifolds and tangent spaces it is worth spending a few pages on where these ideas came from, because **almost every object we are going to construct in this chapter was invented by someone who had no idea that physics would one day need it**, and knowing that history makes the definitions feel far less arbitrary than they otherwise might.

2.1.1 Geometry as the measurement of the earth

The word itself confesses the origin: *geometry* comes from the Greek *γεωμετρία*, built from *γῆ* (earth) and *μέτρον* (measure), so that the name of the discipline literally means *earth measurement*. Herodotus, writing in the fifth century B.C., tells us in the second book of his *Histories* that geometry was invented in Egypt for an entirely practical reason: the Nile flooded every year, washing away the boundary markers of the fields along its banks, and since the pharaoh taxed each landholder in proportion to the area he held, somebody had to go out after every flood and measure the land again (Herodotus, 2003). The men who did this were the *harpedonaptai*, the rope stretchers, and they worked with knotted cords that let them lay out right angles in the field by the simple trick of forming a triangle with sides in the ratio three, four and five.

The Babylonians were, if anything, further ahead. The clay tablet catalogued as Plimpton 322, dated to roughly 1800 B.C., lists what we would now call Pythagorean triples, several centuries before Pythagoras was born, and the Egyptian Rhind papyrus of about 1650 B.C. contains worked recipes for the areas of triangles and circles together with a remarkably good approximation to π . What none of these documents contain, however, is a single proof. **Geometry began as an empirical and practical science, a catalogue of recipes that worked**, with no axioms, no deductive structure and no claim whatsoever to necessary truth, and it would take the Greeks to change that.

2.1.2 Euclid and the tyranny of the fifth postulate

Around 300 B.C., in the Alexandria we already visited in chapter 1, Euclid gathered essentially everything the Greek world knew about geometry into the thirteen books of the *Elements*, and in doing so he invented something far more valuable than any individual theorem: the axiomatic method itself (Euclid, 1908). Euclid begins with definitions, five common notions and five postulates, and from that small seed he deduces four hundred and sixty five propositions, each one following from what came before with a logical necessity that had never been demanded of mathematics before. For the next two thousand years the *Elements* was not merely a textbook of geometry, it was the model of what secure knowledge was supposed to look like, and Newton's *Principia*, as we saw, is written deliberately in its style.

There was, however, a splinter in this perfect structure, and everybody could feel it. Euclid's first four postulates are short, obvious and almost embarrassingly modest: a straight line can be drawn between any two points, a finite line can be extended indefinitely, a circle can be drawn with any centre and radius, and all right angles are equal. The fifth is nothing like them. In Euclid's own formulation it states that if a straight line falling on two straight lines makes the interior angles on the same side less than two right angles, then those two lines, extended far enough, will meet on that side, a statement so long, so conditional and so manifestly *unobvious* that it reads like a theorem somebody failed to prove and quietly promoted. Euclid himself seems to have been uneasy about it, because he avoids using it for as long as he possibly can, delaying its first appearance until Proposition 29. By the fifth century A.D. Proclus was already arguing openly that the fifth postulate ought to be struck from the list of assumptions and derived from the other four, and **the attempt to do exactly that became the longest running unsolved problem in the history of mathematics.**

2.1.3 The long assault on the parallel postulate

The assault was genuinely international and it lasted the better part of two millennia. Working in the same Islamic mathematical tradition we met in chapter 1, Ibn al-Haytham attacked the problem in the eleventh century using a quadrilateral with three right angles, the Persian poet and mathematician Omar Khayyam (1048 to 1131) refined the approach considerably, and Nasir al-Din al-Tusi (1201 to 1274), whose Tusi couple we have already encountered, produced a treatment careful enough that it eventually reached Europe and was discussed by John Wallis at Oxford in the seventeenth century (Kline, 1972).

The decisive attempt, and the most poignant, came from the Italian Jesuit Girolamo Saccheri, who published in 1733, the year of his death, a book with the triumphant title *Euclides ab omni naevo vindicatus*, Euclid freed of every flaw (Saccheri, 1733). Saccheri's strategy was *reductio ad absurdum*: he assumed the fifth postulate to be false, and then set out to derive a contradiction. He worked with enormous patience and skill, deducing theorem after theorem from the negated postulate, and what he found was that the consequences were strange (triangles whose angles sum to less than two right angles, no similar figures of different sizes, an absolute unit of length built into space itself) but that they were also, maddeningly, perfectly consistent with one another. No contradiction ever came. At the end, unable to produce the absurdity his whole book was built to produce, Saccheri simply declared that the resulting propositions were *repugnant to the nature of the straight line* and announced the job done. **Saccheri had derived, theorem by theorem, a substantial part of hyperbolic geometry, and then refused to believe what he was looking at.** Johann Heinrich Lambert went further still a few decades later, and came even closer to accepting the conclusion, but he too stopped short of publishing it as a discovery rather than a failure.

2.1.4 The birth of non-Euclidean geometry

The break, when it finally came, came three times at once. Gauss had convinced himself privately, somewhere in the 1810s and 1820s, that a perfectly consistent geometry could be built in which the parallel postulate fails, and he never published a word of it. His reason, given in a letter to Bessel of January 1829, is one of the more human documents in the history of mathematics: he feared, he wrote, *das Geschrei der Bötier*, the clamour of the Boeotians, the Boeotians being the Greek byword for provincial stupidity, and he had no appetite for the controversy that publication would bring down on him (Kline, 1972).

Nikolai Ivanovich Lobachevsky, working at the University of Kazan far out on the Russian frontier, had no such caution and published his account of what he called imaginary geometry in 1829 (Lobachevsky, 1829). Independently, in Hungary, the young army officer János Bolyai reached the same conclusions and published them in 1832 as a twenty four page appendix to a mathematics textbook written by his father Farkas (Bolyai, 1832). Bolyai had written to his father about the discovery nine years earlier, in November 1823, with a line that every student of geometry should know: *out of nothing I have created a strange new universe*. Farkas, who had himself wasted years on the parallel postulate and who happened to be an old friend of Gauss, proudly forwarded the appendix to Göttingen for an opinion. Gauss's reply was, by any reasonable standard, a disaster: he explained that he could not praise the work, because to praise it would be to praise himself, since he had held the same views for thirty or thirty five years already. János Bolyai, twenty nine years old and convinced he had been robbed, never published mathematics again.

Set aside the human wreckage and the mathematical content is this: **the parallel postulate is independent of the other four, so Euclidean geometry is not the only possible geometry, it is merely one of several**, and therefore the question of which geometry describes the physical space we actually inhabit stops being a matter of pure logic and becomes an empirical question, to be settled by measurement. That single shift in status is what eventually made general relativity thinkable at all.

2.1.5 Gauss, Riemann and the turn inwards

The second decisive step was Gauss's, and we have already met it in chapter 1, though it is worth restating here from the mathematician's side rather than the physicist's. In the *Disquisitiones generales circa superficies curvas* of 1827 Gauss proved his *Theorema Egregium*, showing that the curvature of a surface can be computed entirely from measurements made within the surface itself (Gauss, 1828). The consequence is philosophical as much as technical: a two dimensional creature confined to a curved surface, with no access whatsoever to a third dimension, could still determine that its world is curved, simply by measuring the angles of triangles or the circumferences of circles and comparing them with what flat geometry predicts. **Curvature stopped being something a space is seen to have from outside, and became something a space possesses in itself.**

Bernhard Riemann's 1854 habilitation lecture, delivered before Gauss himself, then took this intrinsic point of view and released it from the restriction to two dimensional surfaces sitting in

ordinary space (Riemann, 1873). Riemann introduced what we would now call an n dimensional manifold, a space that need only look like $\mathbb{R}^n$ in the neighbourhood of each of its points, and he showed how to put a *metric* on such a space, a rule assigning a length to every infinitesimal displacement, from which curvature could then be computed at every point by pure calculation. Every single object in the remainder of this chapter, the manifold, the chart, the tangent space, the metric and ultimately the curvature tensor, descends directly from that one lecture, and Riemann's closing suggestion that the metric of physical space might be determined by the matter within it is the prophecy we discussed at length in chapter 1.

2.1.6 From absolute differential calculus to the modern theory

Riemann had the ideas but not yet a usable calculus, and supplying one occupied the next half century. Elwin Bruno Christoffel worked out in 1869 how the derivatives of the metric combine into the symbols that now bear his name and that we shall meet in chapter 3 (Christoffel, 1869), and Gregorio Ricci-Curbastro, together with his student Tullio Levi-Civita, spent the 1880s and 1890s systematising the whole subject into what they published in 1900 under the name *absolute differential calculus* (Ricci-Curbastro and Levi-Civita, 1900). Their central concern was exactly the one that matters to a physicist: how to write equations whose form does not depend on the coordinate system used to express them, which is to say, how to do calculus with tensors. This is the paper, and this the machinery, that Marcel Grossmann put into Einstein's hands in 1912. Levi-Civita then added the missing geometrical idea in 1917, the notion of *parallel transport*, which finally explained what it means to compare vectors at different points of a curved space (Levi-Civita, 1917).

Alongside this ran a second, more structural current. Felix Klein's Erlangen programme of 1872 proposed that a geometry is best understood not through its objects but through its symmetries, as the study of those properties left invariant by a group of transformations (Klein, 1872), a point of view that runs straight into modern physics and that we shall use implicitly every time we demand that a physical law be invariant under Lorentz transformations. Élie Cartan rebuilt the subject again in the 1920s with his moving frames and his exterior calculus of differential forms, Hermann Weyl wrote the first genuinely geometrical textbook treatment of relativity in *Raum, Zeit, Materie* in 1918 (Weyl, 1918), and it was only in 1936 that Hassler Whitney finally proved the theorems that let us treat an abstract manifold as an object in its own right rather than as a surface embedded in some larger Euclidean space (Whitney, 1936). That date is worth pausing on, because it is twenty one years *after* the field equations were published: **Einstein built general relativity out of mathematics that was still being invented around him**, and the clean, coordinate free formulation that we now teach to students was largely assembled afterwards, in response to the theory rather than in preparation for it.

2.1.7 Why differential geometry is the language of relativity

It remains to say plainly why a physicist should care about any of this, and the answer is that every structural feature of general relativity is already a demand for differential geometry. The equivalence

principle, which we shall state carefully in chapter 3, says that a freely falling observer detects no gravitational field in a sufficiently small neighbourhood of themselves, which is to say that *space-time looks flat locally even when it is curved globally*, and this is precisely and exactly the defining property of a manifold, a space that is locally $\mathbb{R}^n$ without being globally anything of the sort. The principle of general covariance, that the laws of physics cannot depend on the coordinates chosen to write them down, forces us to work with objects defined independently of any chart, which is to say with tensors, and it is the reason we shall spend so much of this chapter on how components transform. Gauss's theorem that curvature is intrinsic is what allows us to speak of the curvature of space-time at all without embarrassing ourselves by having to ask what higher dimensional space it is supposedly curving into, since there is none and none is needed. Tidal forces, the one part of Newtonian gravity that genuinely cannot be transformed away by falling freely, turn out to be nothing other than the curvature tensor acting on a family of neighbouring geodesics, as we shall prove when we reach the equation of geodesic deviation. And the metric tensor, Riemann's rule for measuring infinitesimal distances, does double duty in relativity as both the ruler that measures space-time intervals and the gravitational potential itself, so that **in general relativity geometry and gravitation are not analogous to one another, they are the same thing described in two vocabularies**. What follows in this chapter is the construction of that vocabulary, from the ground up.

2.2 Differentiable manifolds

This is one of the most attractive theories in physics and mathematics, because it allowed physicists to arrive at one of the most important theories of modern physics, general relativity, a geometrical theory of space and time bound together into a single entity called *space-time*, with all the implications for our reality that this carries, since the theory allowed us to see beyond our planet, to dream of travelling beyond this earth and to describe genuinely strange systems, galaxies or universes. Here we are going to study briefly the foundations of differential geometry and of differentiable manifolds.

If we wanted to make a completely rigorous study of differential geometry we should begin with set theory, then topology, then manifolds, and only then arrive at differentiable manifolds. We shall take a shorter road, but it is worth knowing which stations we are skipping.

2.2.1 A dictionary

Before we start it is worth confronting a difficulty which has nothing to do with the mathematics itself and everything to do with the fact that mathematicians and physicists have spent a century describing the very same objects with different words. A student arriving here from a physics degree already knows most of these objects perfectly well and has been computing with them since the second year, so what makes the first encounter with differential geometry so uncomfortable is usually not that the content is new but that the names are. Because of that, and before writing down a single definition, we are going to lay out a small dictionary in table 2.1, which the reader should feel entirely free to come back to as often as necessary, since almost everything appearing in it will only be properly defined in the pages that follow.

Geometrical object	Components	What a physicist calls it, and where it appears
Smooth function, $\binom{0}{0}$	f	Scalar field: a temperature, the Newtonian potential φ
Tangent vector, $\binom{1}{0}$	X^μ	Vector, or contravariant vector: the four velocity u^μ
Covector, $\binom{0}{1}$	ω_μ	Covariant vector, or one-form: the gradient $\partial_\mu f$, the four momentum p_μ
$\binom{1}{1}$ tensor	A^μ_ν	A linear operator, in practice a matrix: the Kronecker delta δ^μ_ν
Symmetric $\binom{0}{2}$ tensor	$g_{\mu\nu}$	The metric: it measures intervals and doubles as the gravitational potential
Antisymmetric $\binom{0}{2}$ tensor	$F_{\mu\nu}$	The Faraday tensor: the electric and magnetic fields packed into one object
Symmetric $\binom{2}{0}$ tensor	$T^{\mu\nu}$	The energy momentum tensor: the source term in Einstein's equations
$\binom{1}{3}$ tensor	$R^\rho_{\sigma\mu\nu}$	The Riemann tensor: the curvature, which is to say the tidal field

Table 2.1: A dictionary between the coordinate free language of differential geometry, the index notation of the working relativist, and the objects a physicist has already met somewhere else.

The same translation problem happens with the vocabulary itself, so it is worth listing the most important pairs before we meet them one by one. When a mathematician says *manifold* a physicist should hear *the space, or the space-time, in which everything happens*; when the mathematician says *chart* the physicist should hear *coordinate system*, or more physically *reference frame*; an *atlas* is nothing more sinister than the complete collection of coordinate patches needed to cover everything without leaving holes; *smooth* means differentiable as many times as we happen to need; the *tangent space* $T_p(M)$ is the space of all the velocities a particle could possibly have as it goes through that event; and a *diffeomorphism* is, for almost all of our purposes, a change of coordinates. **None of the geometry in this chapter is harder than the physics the reader already knows, it is simply written in a language built to survive without coordinates**, and that last property is exactly why general relativity cannot do without it. Readers who want the same material developed at greater length and with a physicist's priorities throughout will find it in Schutz (1980) and in Nakahara (2003), while Wald (1984) is the standard place to look when we want the mathematician's version of a statement we have treated lightly here.

2.2.2 Manifolds, charts and atlases

Definition 1 *An n -dimensional differentiable manifold is a set M together with a collection of subsets $\mathcal{O}_\alpha$ such that:*

1. $\bigcup_\alpha \mathcal{O}_\alpha = M$. This means that the union of all the subsets covers the manifold M , so that every point of M lies in at least one of them.

2. For every α there exists a bijective map ϕ_α onto an open subset $U_\alpha \subset \mathbb{R}^n$,

$$\begin{aligned} \phi_\alpha : \mathcal{O}_\alpha &\longrightarrow U_\alpha \subset \mathbb{R}^n \\ p &\longmapsto \phi_\alpha(p) = (x_\alpha^1(p), \dots, x_\alpha^n(p)). \end{aligned} \quad (2.1)$$

3. If $\mathcal{O}_\alpha$ and $\mathcal{O}_\beta$ intersect, so that $\mathcal{O}_\alpha \cap \mathcal{O}_\beta \neq \emptyset$, then the composition $\phi_\beta \circ \phi_\alpha^{-1}$ maps

$$\begin{aligned} \phi_\beta \circ \phi_\alpha^{-1} : \phi_\alpha(\mathcal{O}_\alpha \cap \mathcal{O}_\beta) &\subset \mathbb{R}^n \longrightarrow \phi_\beta(\mathcal{O}_\alpha \cap \mathcal{O}_\beta) \subset \mathbb{R}^n \\ x &\longmapsto (\phi_\beta \circ \phi_\alpha^{-1})(x), \end{aligned} \quad (2.2)$$

and this map, which is a map between open subsets of $\mathbb{R}^n$ and therefore an ordinary object of multivariable calculus, must be smooth, that is, infinitely differentiable.

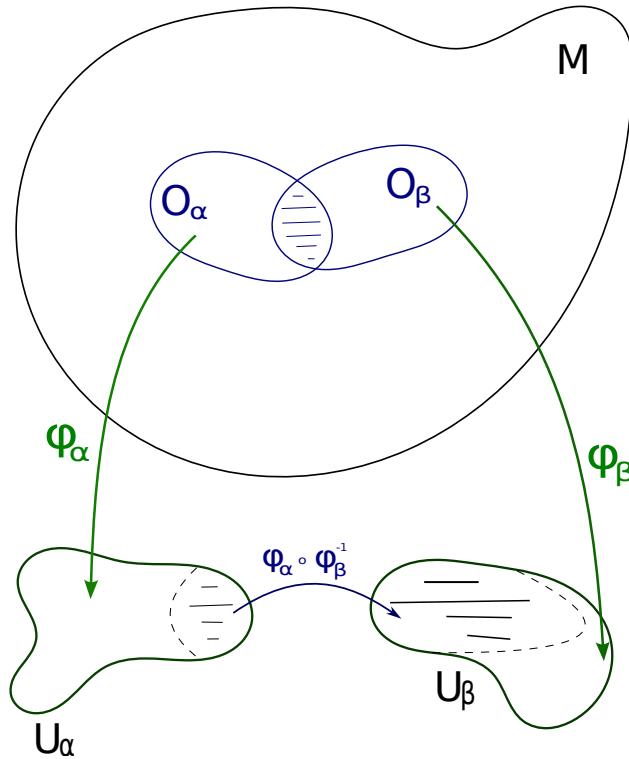


Figure 2.1: Schematic representation of the third postulate: in a differentiable manifold we can always pass from one coordinate chart to another.

Here we must make some comments and observations:

- The maps ϕ_α are usually called coordinate charts or coordinate systems, and we will later see that this is directly related to the strong equivalence principle.

- The collection $\{\phi_\alpha\}$ is called an *atlas*.
- A point p represented in a coordinate chart ϕ_α is written as $\phi_\alpha(p) = (x_\alpha^1(p), \dots, x_\alpha^n(p))$, and the components x_α^i are called the coordinates of p in that chart.

We will often use the word *smooth*, which refers to the differentiability of the geometrical object: we say something is smooth if it is of class C^∞ , meaning that it is infinitely differentiable. Such objects are sometimes hard to come by in practice, especially in physics, so one often works instead with objects of class C^k , that is, k times differentiable. Let us see some examples:

1. $\mathbb{R}^n$ is itself a manifold, whose atlas consists of the single identity chart

$$\begin{aligned}\phi : \mathbb{R}^n &\longrightarrow \mathbb{R}^n \\ x &\longmapsto \phi(x) = x,\end{aligned}\tag{2.3}$$

which leaves every n -tuple $x = (x^1, \dots, x^n)$ exactly where it already was, so a single chart is enough here and a second one is never needed. This is a manifold of dimension n .

2. S^1 , also called the unit circle, $S^1 = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 = 1\}$. It is a one dimensional manifold, because we only need one parameter to describe it, and at the same time it is a subset of $\mathbb{R}^2$ given by $(\cos \theta, \sin \theta)$ with $\theta \in \mathbb{R}$. Let us see how to cover this manifold.

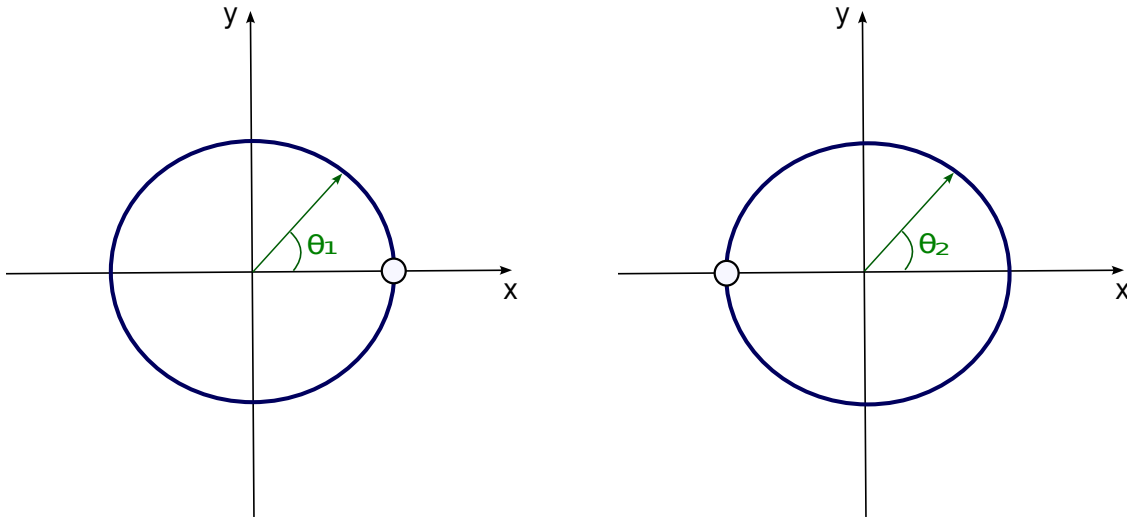


Figure 2.2: On the left there is the first chart ϕ_1 , which covers almost all of the circumference except a point p , and on the right there is the other chart, which covers almost all of the circumference except a point q diametrically opposite to p .

We cannot use a single closed interval, say $0 \leq \theta \leq 2\pi$, because then the map $\phi = \theta$ would fail to be bijective, since $\theta = 0$ and $\theta = 2\pi$ label the very same point of the circle. Because of

this we choose two coordinate charts. Removing the point $p = (1, 0)$ we define the domain $\mathcal{O}_1 = S^1 \setminus \{p\}$ and the chart

$$\begin{aligned}\phi_1 : \mathcal{O}_1 &\longrightarrow (0, 2\pi) \\ s &\longmapsto \phi_1(s) = \theta_1,\end{aligned}\tag{2.4}$$

where θ_1 is the angle of the point $s = (\cos \theta_1, \sin \theta_1)$ measured anticlockwise starting from the removed point p . Removing instead the diametrically opposite point $q = (-1, 0)$ we define $\mathcal{O}_2 = S^1 \setminus \{q\}$ together with

$$\begin{aligned}\phi_2 : \mathcal{O}_2 &\longrightarrow (-\pi, \pi) \\ s &\longmapsto \phi_2(s) = \theta_2,\end{aligned}\tag{2.5}$$

where now θ_2 is measured from the point diametrically opposite to q , so that the cut of this second chart falls where the first one was perfectly well behaved.

Neither chart alone covers the circle, but between them they leave nothing out, so the atlas for this manifold is

$$A = \{\phi_1, \phi_2\}.\tag{2.6}$$

Now, how are θ_1 and θ_2 related on the region where both charts are defined? Notice first that the overlap $\mathcal{O}_1 \cap \mathcal{O}_2$ is the circle with *two* points removed, so it consists of two disjoint arcs, the upper one and the lower one, and the transition function has to be given separately on each of them. From the third postulate we obtain

$$\theta_2 = (\phi_2 \circ \phi_1^{-1})(\theta_1) = \begin{cases} \theta_1, & 0 < \theta_1 < \pi \quad (\text{upper arc}), \\ \theta_1 - 2\pi, & \pi < \theta_1 < 2\pi \quad (\text{lower arc}). \end{cases}\tag{2.7}$$

On each arc separately this is a smooth (indeed linear) function of θ_1 , which is exactly what the third postulate demands, and the geometrical reading of the second branch is simply that *the two charts disagree by one complete turn on the lower arc*, which is the price of having cut the circle open at two different places.

3. S^2 , the sphere of unit radius, $S^2 = \{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 + z^2 = 1\}$. Here the situation is analogous to S^1 , because any point on the sphere has a small enough neighbourhood that maps one to one onto a disc of $\mathbb{R}^2$, see figure 2.3.

This also illustrates that such a map certainly preserves neither lengths nor angles. Consider the spherical coordinates we always use in our calculus courses, taking the polar angle $\vartheta = x$ and the azimuthal angle $\varphi = y$, so that the sphere appears to be mapped onto the rectangle

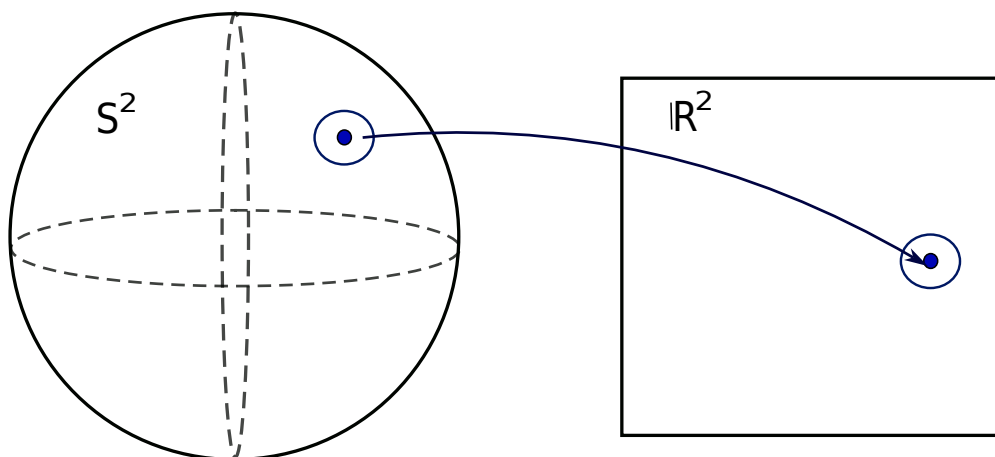


Figure 2.3: Each point on the sphere has a small neighbourhood which maps onto a disc in $\mathbb{R}^2$.

$$0 \leq x \leq \pi, \quad 0 \leq y \leq 2\pi.$$

The map breaks down at the pole, $\vartheta = 0$. The whole pole, which is a single point of the manifold, is smeared out along the entire line $x = 0$, $0 \leq y \leq 2\pi$, see figure 2.4, so at the pole there is no bijection and therefore no chart at all.

A second problem arises at the points whose azimuthal coordinate is $\varphi = 0$, since they are sent to both $y = 0$ and $y = 2\pi$ at once, so again the correspondence fails to be one to one.

Both problems disappear if instead of a closed rectangle we take the open one, $0 < x < \pi$ and $0 < y < 2\pi$, and then use a second chart to cover what the first one lost, choosing for instance a chart whose own excluded line is the meridian $\varphi = \pi$, so that its azimuthal coordinate runs from $\varphi = \pi/2$ to $\varphi = 3\pi/2$.

There is another, more elegant way to choose the charts for S^2 , called the stereographic projection of the manifold onto $\mathbb{R}^2$.

The sphere is tangent to the plane, and from the point N diametrically opposite to the point of tangency we draw a straight line which cuts the sphere at P and the plane $\mathbb{R}^2$ at Q . This defines the map: P is sent to Q , so that the coordinates of Q in $\mathbb{R}^2$ serve as the coordinates of P in S^2 . It is one to one everywhere except at N itself, since as P approaches N the image point Q runs off to infinity, and to cover N we simply choose a second stereographic chart projecting from the opposite pole.

Definition 2 *Two atlases are compatible if their union is again an atlas. The union of all mutually compatible atlases is called the **complete atlas**, and it is characterised by the property that it is not contained in any larger atlas.*

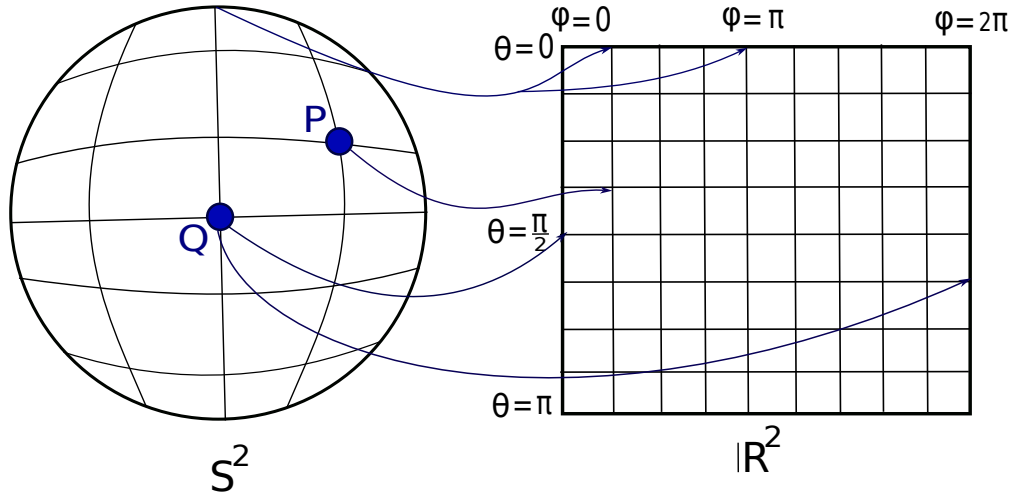


Figure 2.4: Representation of the traditional map from S^2 to $\mathbb{R}^2$. We can see that there are multivalued points which will not allow us to cover the manifold S^2 with a single chart.

2.2.3 Smooth functions

If $\phi : \mathcal{O} \rightarrow U$ is a coordinate chart and $f : M \rightarrow \mathbb{R}$ is a function on the manifold, then the composition $f \circ \phi^{-1}$ is a map from $U \subset \mathbb{R}^n$ to $\mathbb{R}$, and this is the object we can actually differentiate, because it is an ordinary function of n real variables. This is the standard manoeuvre of the whole subject: **we never differentiate on the manifold itself, we always push the problem down into $\mathbb{R}^n$ with a chart, do the calculus there, and then check that the answer did not depend on which chart we chose.**

Definition 3 *A function*

$$\begin{aligned} f : M &\longrightarrow \mathbb{R} \\ p &\longmapsto f(p) \end{aligned} \tag{2.8}$$

is smooth if and only if, for every coordinate chart ϕ , the map $F \equiv f \circ \phi^{-1}$ defined by

$$\begin{aligned} F : U \subset \mathbb{R}^n &\longrightarrow \mathbb{R} \\ x &\longmapsto F(x) = f(\phi^{-1}(x)) \end{aligned} \tag{2.9}$$

is a smooth function in the ordinary sense of multivariable calculus.

In general relativity a function $f : M \rightarrow \mathbb{R}$ is often called a *scalar field*. The set of all such smooth functions on M is denoted $C^\infty(M)$, and we shall need it in a moment, because tangent vectors are going to be defined as operators acting on it.

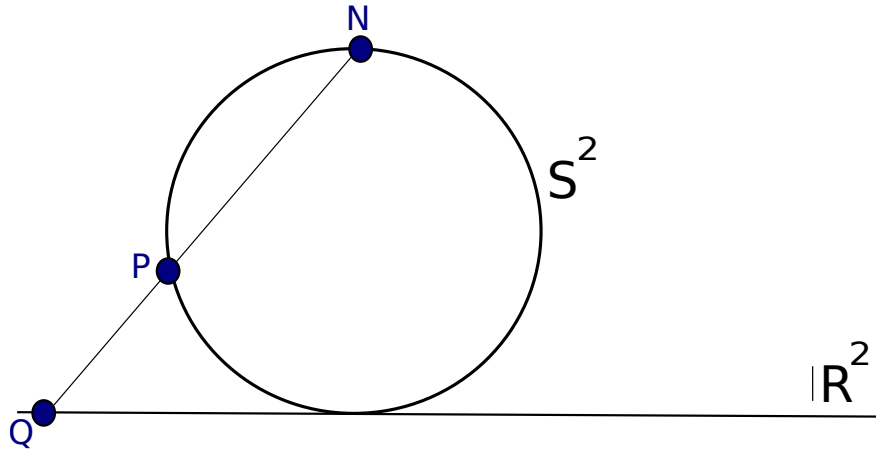


Figure 2.5: Graphical representation used to find the stereographic projection of S^2 onto $\mathbb{R}^2$.

2.2.4 Curves and vectors

Definition 4 A smooth curve on a differentiable manifold M is a smooth map

$$\begin{aligned}\lambda : I &\longrightarrow M \\ t &\longmapsto \lambda(t),\end{aligned}\tag{2.10}$$

where I is an open interval of $\mathbb{R}$.

This means that $\phi_\alpha \circ \lambda$ is a map from I into $\mathbb{R}^n$ for every coordinate chart ϕ_α , and this is what is represented in figure 2.7.

The elements of I can be thought of as the parameters that parameterise the curve. Given

$$f : M \longrightarrow \mathbb{R} \text{ and } \lambda : I \longrightarrow M,\tag{2.11}$$

we can define the composition

$$\begin{aligned}f \circ \lambda : I &\longrightarrow \mathbb{R} \\ t &\longmapsto f(\lambda(t)),\end{aligned}\tag{2.12}$$

which is in fact a parameterised function $f(\lambda(t))$ of a single real variable, so we may perfectly well speak about its derivative with respect to the parameter t . To compute that derivative we must, as always, go down into a chart: writing $F = f \circ \phi^{-1}$ and $x^\mu(t) = x^\mu(\lambda(t))$ for the coordinates of the curve, the chain rule gives

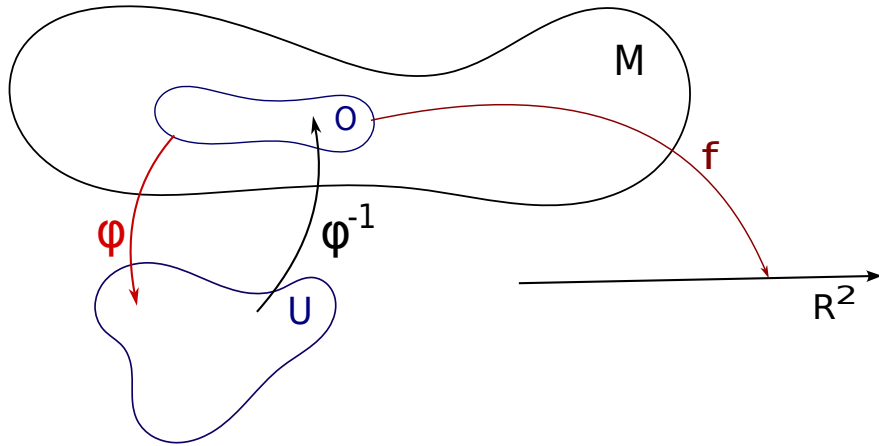


Figure 2.6: Schematic representation of the map f , which takes elements of the manifold and maps them onto the real line $\mathbb{R}$.

$$\frac{d}{dt} [(f \circ \lambda)(t)] = \frac{d}{dt} F(x(t)) = \frac{\partial F}{\partial x^\mu} \frac{dx^\mu}{dt}, \quad (2.13)$$

where a sum over the repeated index μ is understood. Notice that it would make no sense to write $\partial f / \partial \lambda$, since λ takes values on the manifold and there is no notion of dividing by a point.

Definition 5 Let $\lambda : I \rightarrow M$ be a smooth curve with $\lambda(0) = p$, so that the parameter value $t = 0$ picks out the point p on the manifold, see figure 2.7. The vector tangent to λ at p is the linear map

$$X_p : C^\infty(M) \longrightarrow \mathbb{R} \\ f \longmapsto X_p(f) = \left\{ \frac{d}{dt} [f(\lambda(t))] \right\}_{t=0}. \quad (2.14)$$

It is worth pausing on how strange this definition looks the first time one meets it. **A tangent vector is not an arrow, it is a directional derivative**, an operator that eats a function and returns the rate at which that function changes as we move along the curve. The reason for this apparently perverse choice is that arrows need an ambient space to live in, whereas directional derivatives need nothing but the manifold and its smooth functions, so this is the definition that survives when we throw the embedding space away.

This tangent vector satisfies two really important algebraic properties.

1. The tangent vector is a linear map, so that for all $f, g \in C^\infty(M)$ and all $\alpha \in \mathbb{R}$,

$$X_p(f + g) = X_p(f) + X_p(g), \quad (2.15)$$

$$X_p(\alpha f) = \alpha X_p(f). \quad (2.16)$$

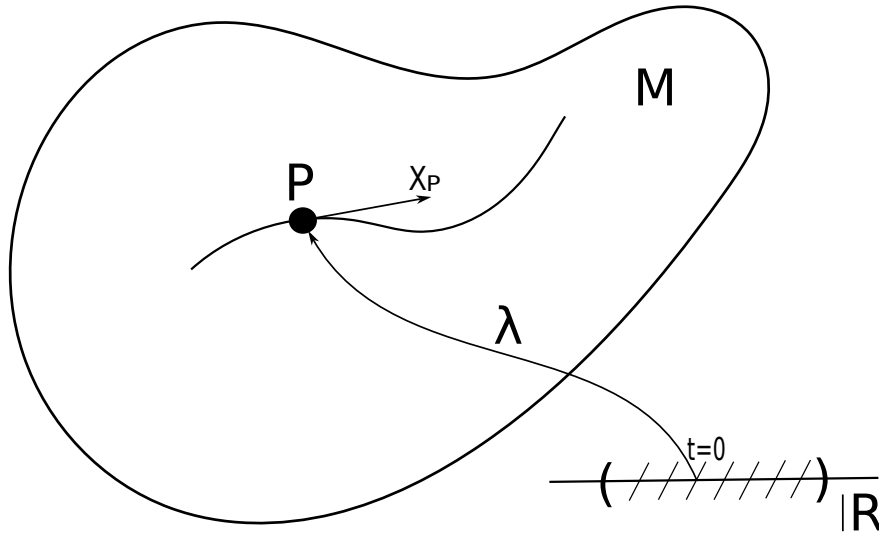


Figure 2.7: The curve λ maps an open interval of the reals into the manifold. On the manifold this map looks like (remember this is only a scheme) the kind of curve we are used to seeing, and at a point of that curve we can define a tangent vector X_p .

2. The tangent vector satisfies Leibniz's rule,

$$X_p(fg) = f(p)X_p(g) + g(p)X_p(f), \quad (2.17)$$

where the functions multiplying the derivatives have to be evaluated at the point p , since $X_p(fg)$ is a number and not a function.

Let us now consider a coordinate chart $\phi = (x^1, \dots, x^n)$ defined in a neighbourhood of p , and the function $F = f \circ \phi^{-1}$. Because ϕ is a bijection we may write

$$f \circ \lambda = f \circ \phi^{-1} \circ \phi \circ \lambda, \quad (2.18)$$

or equivalently

$$f \circ \lambda = F \circ \phi \circ \lambda. \quad (2.19)$$

The tangent vector, using the chain rule exactly as in equation 2.13, is then given by

$$X_p(f) = \left\{ \frac{d}{dt} [f(\lambda(t))] \right\}_{t=0} = \left(\frac{\partial F(x)}{\partial x^\mu} \right)_{\phi(p)} \left(\frac{dx^\mu(\lambda(t))}{dt} \right)_{t=0}, \quad (2.20)$$

which means that in general we can write every tangent vector as a linear combination of some components multiplying some basis elements, and the whole of the next page is devoted to making that statement precise.

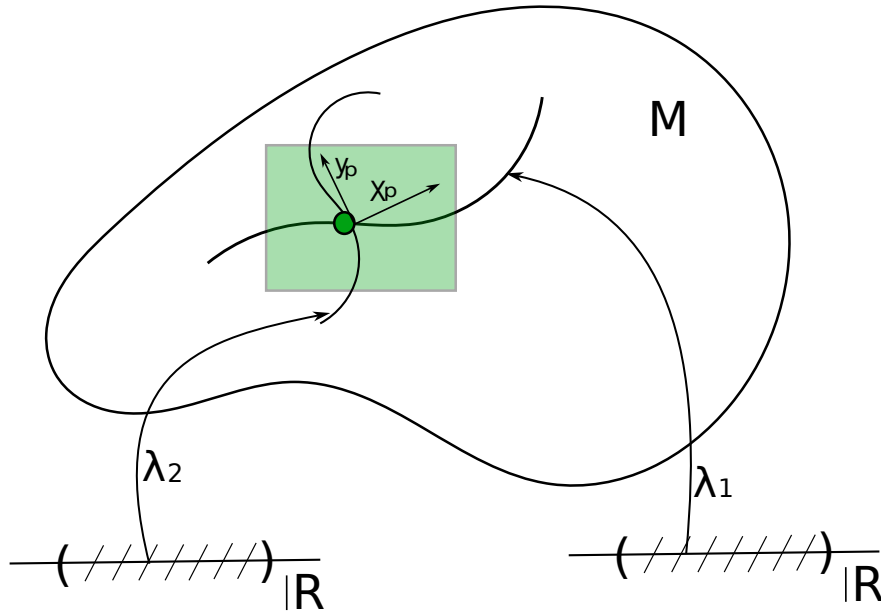


Figure 2.8: Two tangent vectors X_p and Y_p , both tangent at p , living on two different curves (λ_1, λ_2) but belonging to the same vector space $T_p(M)$.

Theorem 1 *The set of all tangent vectors at p forms an n -dimensional vector space, called the tangent space $T_p(M)$.*

Demonstration:

Let λ_1 and λ_2 be two curves such that $\lambda_1(0) = \lambda_2(0) = p$, and let X_p and Y_p be their tangent vectors respectively. We must define multiplication by a constant and addition of two vectors, and to do so we exhibit a curve whose tangent is the required combination. Let $\phi = (x^1, \dots, x^n)$ be a coordinate chart defined in a neighbourhood of p and consider the curve

$$\nu(t) = \phi^{-1} \{ \alpha [\phi(\lambda_1(t)) - \phi(p)] + \beta [\phi(\lambda_2(t)) - \phi(p)] + \phi(p) \}, \quad (2.21)$$

which is well defined for t small enough that the argument stays inside U , and which satisfies $\nu(0) = p$. Let Z_p be the tangent vector to this curve at p , then

$$Z_p(f) = \left(\frac{\partial F(x)}{\partial x^\mu} \right)_{\phi(p)} \left\{ \frac{d}{dt} [\alpha(x^\mu(\lambda_1(t)) - x^\mu(p)) + \beta(x^\mu(\lambda_2(t)) - x^\mu(p)) + x^\mu(p)] \right\}_{t=0}, \quad (2.22)$$

$$Z_p(f) = \left(\frac{\partial F(x)}{\partial x^\mu} \right)_{\phi(p)} \left[\alpha \left(\frac{d}{dt} x^\mu(\lambda_1(t)) \right)_{t=0} + \beta \left(\frac{d}{dt} x^\mu(\lambda_2(t)) \right)_{t=0} \right], \quad (2.23)$$

$$Z_p(f) = \alpha X_p(f) + \beta Y_p(f), \quad (2.24)$$

$$Z_p(f) = (\alpha X_p + \beta Y_p)(f). \quad (2.25)$$

Since this holds for any smooth function f , we conclude that $Z_p = \alpha X_p + \beta Y_p$, and Z_p is by construction tangent to the curve ν at p . Therefore any linear combination of tangent vectors at p is again a tangent vector at p , and the set of tangent vectors at p forms a vector space.

Now we must prove that it is n -dimensional, and to do this we exhibit a basis. Let $1 \leq \mu \leq n$ and consider the curve λ_μ through p defined by

$$\lambda_\mu(t) = \phi^{-1}(x^1(p), \dots, x^{\mu-1}(p), x^\mu(p) + t, x^{\mu+1}(p), \dots, x^n(p)), \quad (2.26)$$

that is, the curve obtained by letting the μ -th coordinate run and holding all the others fixed. The tangent vector to this curve at p is denoted by $\left(\frac{\partial}{\partial x^\mu} \right)_p$, and it acts on functions exactly as the notation suggests,

$$\left(\frac{\partial}{\partial x^\mu} \right)_p (f) = \left(\frac{\partial F}{\partial x^\mu} \right)_{\phi(p)}, \quad (2.27)$$

which is easy to see because we have defined $F = f \circ \phi^{-1}$ and the chart and its inverse cancel, leaving an object evaluated at p rather than at $\phi(p)$.

Let us now check that these n tangent vectors are linearly independent, that is, that

$$\sum_{\mu=1}^n a_\mu \left(\frac{\partial}{\partial x^\mu} \right)_p = 0 \implies a_1 = a_2 = \dots = a_n = 0. \quad (2.28)$$

The equation on the left is an equation between operators, so it means that the left hand side gives zero when applied to *any* smooth function, and the cleanest way to extract the coefficients is therefore to feed it the most convenient functions we have available, namely the n coordinate functions x^ν themselves. Applying the operator to x^ν and using equation 2.27 we obtain

$$0 = \sum_{\mu=1}^n a_\mu \left(\frac{\partial}{\partial x^\mu} \right)_p (x^\nu) = \sum_{\mu=1}^n a_\mu \left(\frac{\partial x^\nu}{\partial x^\mu} \right)_{\phi(p)} = \sum_{\mu=1}^n a_\mu \delta_\mu^\nu = a_\nu,$$

and since ν was arbitrary this gives

$$a_\nu = 0 \quad \forall \nu \in \{1, \dots, n\}, \quad (2.29)$$

which is precisely the linear independence we wanted.

Finally, having proved that they are linearly independent, equation 2.20 together with equation 2.27 lets us write

$$X_p(f) = \left(\frac{dx^\mu(\lambda(t))}{dt} \right)_{t=0} \left(\frac{\partial}{\partial x^\mu} \right)_p (f), \quad (2.30)$$

and since this is true for every f we may drop it and write the operator identity

$$X_p = X^\mu \left(\frac{\partial}{\partial x^\mu} \right)_p, \quad X^\mu \equiv \left(\frac{dx^\mu(\lambda(t))}{dt} \right)_{t=0}. \quad (2.31)$$

So X_p can be written as a linear combination of the n tangent vectors $\left(\frac{\partial}{\partial x^\mu} \right)_p$, which therefore span $T_p(M)$, and being also linearly independent they form a basis. This establishes that the tangent space is n -dimensional, the same dimension as the manifold itself.

The basis $\left\{ \left(\frac{\partial}{\partial x^\mu} \right)_p : \mu = 1, \dots, n \right\}$ depends on the coordinate chart, since we had to choose a chart ϕ defined in a neighbourhood of p in order to build it, and choosing a different chart would give a different basis of the same space $T_p(M)$. A basis constructed in this way is called a *coordinate basis*, and we will see in due course that not every basis one might want to use is of this type.

We often want to represent the same object in one coordinate system or in another, so we must know how to transform both the basis vectors and the components from one representation to the other. Let $\phi = (x^1, \dots, x^n)$ and $\bar{\phi} = (x^{\bar{1}}, \dots, x^{\bar{n}})$ be two coordinate charts, then

$$\left(\frac{\partial}{\partial x^\mu} \right)_p (f) = \left(\frac{\partial}{\partial x^\mu} (f \circ \phi^{-1}) \right)_{\phi(p)} = \left\{ \frac{\partial}{\partial x^\mu} [(f \circ \bar{\phi}^{-1}) \circ (\bar{\phi} \circ \phi^{-1})] \right\}_{\phi(p)}, \quad (2.32)$$

where we have simply inserted the identity $\bar{\phi}^{-1} \circ \bar{\phi}$, and writing

$$\bar{F} = f \circ \bar{\phi}^{-1} \quad : \quad \bar{F} = \bar{F}(\bar{x}), \quad (2.33)$$

the chain rule gives

$$\begin{aligned}
\left(\frac{\partial}{\partial x^\mu}\right)_p(f) &= \left(\frac{\partial}{\partial x^\mu} [\bar{F}(\bar{x}(x))]\right)_{\phi(p)} \\
&= \left(\frac{\partial x^{\bar{\nu}}}{\partial x^\mu}\right)_{\phi(p)} \left(\frac{\partial \bar{F}(\bar{x})}{\partial x^{\bar{\nu}}}\right)_{\bar{\phi}(p)} \\
&= \left(\frac{\partial x^{\bar{\nu}}}{\partial x^\mu}\right)_{\phi(p)} \left(\frac{\partial}{\partial x^{\bar{\nu}}}\right)_p(f).
\end{aligned}$$

In this way we have found the transformation rule for the *basis vectors* when we pass from one coordinate system to another, which is

$$\left(\frac{\partial}{\partial x^\mu}\right)_p = \left(\frac{\partial x^{\bar{\nu}}}{\partial x^\mu}\right)_{\phi(p)} \left(\frac{\partial}{\partial x^{\bar{\nu}}}\right)_p. \quad (2.34)$$

The components must then transform in the opposite way, so that the vector itself, which is a geometrical object and knows nothing about our choice of chart, remains the same,

$$X_p = X^\nu \left(\frac{\partial}{\partial x^\nu}\right)_p = X^\nu \left(\frac{\partial x^{\bar{\mu}}}{\partial x^\nu}\right)_p \left(\frac{\partial}{\partial x^{\bar{\mu}}}\right)_p = X^{\bar{\mu}} \left(\frac{\partial}{\partial x^{\bar{\mu}}}\right)_p, \quad (2.35)$$

from which we can read off the transformation law for the components,

$$X^{\bar{\mu}} = \left(\frac{\partial x^{\bar{\mu}}}{\partial x^\nu}\right) X^\nu. \quad (2.36)$$

Comparing equations 2.34 and 2.36 we see that components and basis vectors transform with mutually inverse Jacobian matrices, and this is not a coincidence, it is exactly what has to happen for the contraction $X^\mu(\partial/\partial x^\mu)_p$ to come out chart independent. Objects that transform like the components in equation 2.36 are called *contravariant*, and we shall meet their *covariant* counterparts in the next subsection.

2.2.5 Covectors and one-forms

Definition 6 Let V be a real vector space. The dual space V^* of V is the vector space of all linear maps from V to the reals, that is, the space whose elements are the maps

$$\begin{aligned}
\tilde{P} : V &\longrightarrow \mathbb{R} \\
X &\longmapsto \tilde{P}(X).
\end{aligned} \quad (2.37)$$

Lemma: if V is n -dimensional then so is V^* . If $\{\mathbf{e}_\mu : \mu = 1, \dots, n\}$ is a basis of V , then V^* has a basis $\{f^\mu : \mu = 1, \dots, n\}$, called the *dual basis*, defined by the condition $f^\mu(\mathbf{e}_\nu) = \delta^\mu_\nu$. If $X = X^\alpha \mathbf{e}_\alpha$ then the dual basis simply reads off the components,

$$f^\alpha(X) = f^\alpha(X^\beta \mathbf{e}_\beta) = X^\beta f^\alpha(\mathbf{e}_\beta) = X^\beta \delta^\alpha_\beta = X^\alpha. \quad (2.38)$$

Since V and V^* have the same dimension they are isomorphic, so an isomorphism between them certainly exists. It is worth being careful here, though, because **that isomorphism is not canonical**: to build it we have to pick a basis of V first, and a different choice of basis gives a different identification, so there is no natural way to say which covector corresponds to a given vector. In general relativity it is precisely the metric tensor that will supply the missing extra structure and let us raise and lower indices, and the fact that this requires a metric, rather than coming for free, is one of the reasons the metric occupies such a central place in the theory. The situation for the double dual is quite different.

Theorem 2 *If V is finite dimensional, then $(V^*)^*$ is naturally isomorphic to V through the map*

$$\begin{aligned} \Psi : V &\longrightarrow (V^*)^* \\ X &\longmapsto \Psi(X), \quad \text{defined by } \Psi(X)(\omega) = \omega(X) \quad \forall \omega \in V^*. \end{aligned} \quad (2.39)$$

Notice that no choice of basis was needed to write that map down, which is exactly what the word *naturally* is doing in the statement of the theorem.

Definition 7 *The dual space of $T_p(M)$ is called the cotangent space at p and is denoted by $T_p^*(M)$.*

Definition 8 *Let $f : M \rightarrow \mathbb{R}$ be a smooth function. The covector $(df)_p$ is defined by*

$$\begin{aligned} (df)_p : T_p(M) &\longrightarrow \mathbb{R} \\ X &\longmapsto (df)_p(X) = X(f), \end{aligned} \quad (2.40)$$

for any $X \in T_p(M)$. It is called the differential of f at p , or in the older physics literature the gradient of f at p . Covectors, also called one-forms, satisfy the following properties.

1. *If $\eta \in V^*$, then its components in the dual basis are recovered by acting on the basis vectors,*

$$\eta(\mathbf{e}_\mu) = \eta_\nu f^\nu(\mathbf{e}_\mu) = \eta_\nu \delta^\nu_\mu = \eta_\mu. \quad (2.41)$$

2. *If $X \in T_p(M)$, then the action of a one-form on a vector is the contraction of their components,*

$$\eta(X) = X^\alpha \eta_\alpha \in \mathbb{R}. \quad (2.42)$$

A one-form is a really special geometrical object, because we use it constantly in physics and yet it is usually late in our studies that we start to think about what it actually is. If vectors are pictured as directed arrows, what does a covector look like? To see this let us consider the interaction between the vector $\vec{A} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ and the one-form $\tilde{q} = [q_1 \quad q_2]$, so that

$$\tilde{q}(\vec{A}) = q_1 + 2q_2. \quad (2.43)$$

If we ask which one-forms return a fixed value c when fed our vector $\vec{A}$, equation 2.43 tells us that the answer is the straight line $q_1 + 2q_2 = c$ in the space of components, and the family of such lines obtained by varying c is a set of parallel lines whose normal has components $[1, 2]^T$. Turning the picture around, this is the standard way of drawing a one-form: **while a vector is a directed arrow, a one-form is a family of parallel hyperplanes, its level surfaces, and the value $\tilde{q}(\vec{A})$ is the number of those surfaces that the arrow $\vec{A}$ pierces.** A one-form with a large norm is a tightly packed stack of surfaces, and a one-form with a small norm is a loosely spaced one, which is why the picture gets the scaling right without any extra work.

What is the most common operation that takes a scalar field and returns something orthogonal to its level surfaces? The gradient, and indeed the gradient is the archetypal example of a one-form, since it represents precisely the stack of equipotential surfaces of the field, which is the reason we denote the one-form at a point p by $(df)_p$. This picture also lets us realise that in electromagnetism the electric field, written in a Coulomb gauge as $E_i = -\partial_i\varphi$, appears most naturally with its index downstairs, that is, as a one-form built as the differential of the potential, while the object we habitually draw as a field of arrows is really doing something slightly different, a point we shall be able to make completely precise once we have the metric at our disposal.

2.2.6 Tensors

A tensor of type $\binom{r}{s}$ at p is a multilinear map

$$T : \underbrace{T_p^*(M) \times \dots \times T_p^*(M)}_{r \text{ copies}} \times \underbrace{T_p(M) \times \dots \times T_p(M)}_{s \text{ copies}} \longrightarrow \mathbb{R} \quad (2.44)$$

$$(\tilde{q}_1, \dots, \tilde{q}_r, \vec{A}_1, \dots, \vec{A}_s) \longmapsto T(\tilde{q}_1, \dots, \tilde{q}_r, \vec{A}_1, \dots, \vec{A}_s),$$

which maps r one-forms and s vectors into the reals, the word *multilinear* meaning that it is linear in each of its arguments separately when all the others are held fixed. The components of the tensor in a given basis are obtained by feeding it the basis elements themselves,

$$T^{\mu_1 \dots \mu_r}_{\nu_1 \dots \nu_s} = T(f^{\mu_1}, \dots, f^{\mu_r}, \mathbf{e}_{\nu_1}, \dots, \mathbf{e}_{\nu_s}), \quad (2.45)$$

and it is a useful sanity check to note that a $\binom{1}{0}$ tensor is just a vector, a $\binom{0}{1}$ tensor is just a one-form, and a $\binom{0}{0}$ tensor is a scalar.

Definition 9 If S is a tensor of type $\binom{p}{q}$ and T is a tensor of type $\binom{r}{s}$, then the tensor product of S and T , denoted $S \otimes T$, is the tensor of type $\binom{p+r}{q+s}$ defined by

$$(S \otimes T)(\omega_1, \dots, \omega_p, \eta_1, \dots, \eta_r, X_1, \dots, X_q, Y_1, \dots, Y_s) = S(\omega_1, \dots, \omega_p, X_1, \dots, X_q) T(\eta_1, \dots, \eta_r, Y_1, \dots, Y_s). \quad (2.46)$$

Up to this point we have defined a tensor as a multilinear machine, which is the honest and coordinate free way of doing it, but it is not the definition the reader will meet in most physics books, where a tensor is instead introduced as *a collection of numbers which transforms in a particular way under a change of coordinates*. The two definitions are completely equivalent and it is worth deriving the second from the first, because the second one is what we actually use whenever we sit down to compute. Everything follows from the way the bases transform: we already know from equation 2.34 how the coordinate basis of $T_p(M)$ behaves, and inverting that relation gives

$$\mathbf{e}_{\bar{\nu}} = \left(\frac{\partial x^{\beta}}{\partial x^{\bar{\nu}}} \right) \mathbf{e}_{\beta}, \quad (2.47)$$

while the dual basis has to transform in the opposite way, since the defining condition $f^{\mu}(\mathbf{e}_{\nu}) = \delta^{\mu}_{\nu}$ must remain true in every chart, and therefore

$$f^{\bar{\mu}} = \left(\frac{\partial x^{\bar{\mu}}}{\partial x^{\alpha}} \right) f^{\alpha}. \quad (2.48)$$

Now we only have to feed the barred bases into the tensor and use multilinearity to pull every Jacobian out of its slot, which gives

$$T^{\bar{\mu}_1 \dots \bar{\mu}_r}_{\bar{\nu}_1 \dots \bar{\nu}_s} = \frac{\partial x^{\bar{\mu}_1}}{\partial x^{\alpha_1}} \dots \frac{\partial x^{\bar{\mu}_r}}{\partial x^{\alpha_r}} \frac{\partial x^{\beta_1}}{\partial x^{\bar{\nu}_1}} \dots \frac{\partial x^{\beta_s}}{\partial x^{\bar{\nu}_s}} T^{\alpha_1 \dots \alpha_r}_{\beta_1 \dots \beta_s}, \quad (2.49)$$

which is the law every physicist eventually learns by heart. The rule it encodes is simple enough to state in one sentence: **each index brings its own Jacobian, the indices upstairs carry $\partial x^{\bar{\mu}}/\partial x^{\alpha}$ and the indices downstairs carry $\partial x^{\beta}/\partial x^{\bar{\nu}}$** , and the whole content of the older definition of a tensor is the demand that equation 2.49 should hold. That definition sounds circular the first time one hears it, since it seems to say that a tensor is whatever transforms like a tensor, but it is perfectly precise and it is enormously useful in practice, because it converts the question *is this object geometrical?* into a mechanical check that can be carried out with nothing more than the chain rule.

There is one operation on tensors which has no analogue for ordinary numbers and which we shall use constantly from chapter 3 onwards, and it consists in setting one upper index equal to one lower index and summing over it.

Definition 10 Let T be a tensor of type $\binom{r}{s}$ with $r \geq 1$ and $s \geq 1$. The contraction of T on its i -th contravariant and j -th covariant slot is the tensor of type $\binom{r-1}{s-1}$ whose components are obtained by setting those two indices equal and summing,

$$(CT)^{\mu_1 \dots \mu_{r-1}}_{\nu_1 \dots \nu_{s-1}} = T^{\mu_1 \dots \alpha \dots \mu_{r-1}}_{\nu_1 \dots \alpha \dots \nu_{s-1}}. \quad (2.50)$$

The reason this deserves a definition of its own, rather than being dismissed as an obvious piece of index bookkeeping, is that it is genuinely surprising that the result should be a tensor at all, since we have summed over a label which only makes sense once a chart has been chosen. That it works is a small miracle of index placement, and the cleanest way to see it is to contract a $\binom{1}{1}$ tensor and watch what the Jacobians do,

$$T^{\bar{\alpha}}_{\bar{\alpha}} = \frac{\partial x^{\bar{\alpha}}}{\partial x^{\mu}} \frac{\partial x^{\nu}}{\partial x^{\bar{\alpha}}} T^{\mu}_{\nu} = \delta^{\nu}_{\mu} T^{\mu}_{\nu} = T^{\mu}_{\mu},$$

where the two Jacobians marked in red have collapsed into a Kronecker delta precisely because one of them came from an upper index and the other from a lower one, so that they are inverse matrices being multiplied together. **Contraction is only legitimate between one upper and one lower index**, and an expression such as $T^{\mu\mu}$, with both indices upstairs, is not chart independent and therefore does not define anything geometrical, at least not until we have a metric available to lower one of them for us. The reader has in fact been contracting tensors for years without calling it that: the trace of a matrix, the divergence of a vector field, the scalar product written as $\omega_{\mu} X^{\mu}$ and the invariant $F^{\mu\nu} F_{\mu\nu}$ of electromagnetism are all contractions, and in chapter 3 the Ricci tensor and the Ricci scalar will be built out of the Riemann tensor by exactly this operation and nothing else.

Any tensor with two or more indices of the same kind can be split into a part which does not care about the order of those indices and a part which changes sign when they are swapped, and because the physically important tensors of relativity are almost always one or the other it pays to fix the notation now. We write the symmetric and antisymmetric parts with round and square brackets respectively,

$$T_{(\mu\nu)} = \frac{1}{2} (T_{\mu\nu} + T_{\nu\mu}), \quad (2.51)$$

$$T_{[\mu\nu]} = \frac{1}{2} (T_{\mu\nu} - T_{\nu\mu}), \quad (2.52)$$

and adding the two of them recovers the original tensor,

$$T_{\mu\nu} = T_{(\mu\nu)} + T_{[\mu\nu]}, \quad (2.53)$$

so that the splitting loses no information at all. For more than two indices the same brackets mean the average over all the permutations of the enclosed indices, with a minus sign for the

odd permutations in the antisymmetric case, so that for three indices we would write $T_{[\mu\nu\rho]} = \frac{1}{3!} (T_{\mu\nu\rho} + T_{\nu\rho\mu} + T_{\rho\mu\nu} - T_{\nu\mu\rho} - T_{\mu\rho\nu} - T_{\rho\nu\mu})$.

Two properties make this splitting worth the trouble. The first is that **being symmetric or antisymmetric is a statement about the tensor and not about the chart**, which we can check directly by transforming a symmetric tensor and swapping the barred indices,

$$T_{\bar{\nu}\bar{\mu}} = \frac{\partial x^\alpha}{\partial x^{\bar{\nu}}} \frac{\partial x^\beta}{\partial x^{\bar{\mu}}} T_{\alpha\beta} = \frac{\partial x^\beta}{\partial x^{\bar{\mu}}} \frac{\partial x^\alpha}{\partial x^{\bar{\nu}}} T_{\beta\alpha} = T_{\bar{\mu}\bar{\nu}},$$

where we used the symmetry $T_{\alpha\beta} = T_{\beta\alpha}$ and then simply renamed the two dummy indices, so a tensor which is symmetric in one coordinate system is symmetric in all of them, and the same argument with one extra minus sign works for antisymmetry. The second property is an identity we shall use again and again without comment, namely that **the full contraction of a symmetric tensor with an antisymmetric one always vanishes**,

$$S^{\mu\nu} A_{\mu\nu} = 0 \quad \text{whenever } S^{\mu\nu} = S^{\nu\mu} \text{ and } A_{\mu\nu} = -A_{\nu\mu}, \quad (2.54)$$

which follows in one line by using the symmetry of S , then the antisymmetry of A , and then relabelling the dummies, since this produces $S^{\mu\nu} A_{\mu\nu} = -S^{\mu\nu} A_{\mu\nu}$ and a quantity equal to minus itself can only be zero.

Finally it is worth counting, because the counting explains a good deal of physics. In n dimensions a symmetric two index tensor has $n(n+1)/2$ independent components and an antisymmetric one has $n(n-1)/2$, so in the four dimensions of space-time we get ten and six respectively. Those two numbers should look familiar: the metric $g_{\mu\nu}$ is symmetric and has exactly ten independent components, which is why Einstein's field equations are ten equations and not sixteen, while the Faraday tensor $F_{\mu\nu}$ is antisymmetric and has exactly six, which is nothing other than the three components of the electric field plus the three of the magnetic field.

There is one last question of a practical kind which the transformation law leaves open and which a working physicist runs into constantly. Suppose we have built some array of numbers out of a physical situation, an array of conductivities measured in a crystal or of field strengths read off an experiment, and we want to know whether it is a tensor. Checking the transformation law directly means computing sixteen Jacobians and comparing sixteen expressions, which is honest work but exhausting, and very often we are in a much better position than that: we already know that the array, once contracted with something arbitrary, produces something we are sure is a tensor. The quotient theorem says that this alone is enough.

Theorem 3 *Let T_ν^μ be an array of numbers defined in every coordinate chart, and suppose that for every vector V^ν the contraction $W^\mu = T_\nu^\mu V^\nu$ transforms as a vector. Then T_ν^μ is a $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ tensor.*

Demonstration:

Write the hypothesis in a second chart. Because W^μ is assumed to be a vector, its barred components are obtained with a single Jacobian, and the same is true of V^ν , so that

$$\bar{T}^{\bar{\mu}}_{\bar{\nu}} \bar{V}^{\bar{\nu}} = \bar{W}^{\bar{\mu}} = \frac{\partial x^{\bar{\mu}}}{\partial x^\alpha} W^\alpha = \frac{\partial x^{\bar{\mu}}}{\partial x^\alpha} T^\alpha_\beta V^\beta,$$

and replacing $\bar{V}^{\bar{\nu}}$ on the far left by its own transformation law $\bar{V}^{\bar{\nu}} = (\partial x^{\bar{\nu}} / \partial x^\beta) V^\beta$ we may bring everything to one side,

$$\left(\frac{\partial x^{\bar{\mu}}}{\partial x^\alpha} T^\alpha_\beta - \bar{T}^{\bar{\mu}}_{\bar{\nu}} \frac{\partial x^{\bar{\nu}}}{\partial x^\beta} \right) V^\beta = 0.$$

Here comes the only step that matters. This expression vanishes not for some cleverly chosen V but for *every* vector whatsoever, and a linear form that annihilates every vector is the zero form, so the bracket itself must vanish identically. Contracting what is left with the inverse Jacobian $\partial x^\beta / \partial x^{\bar{\sigma}}$, and using that the two Jacobians collapse into a Kronecker delta exactly as they did when we proved that contraction is well defined,

$$\frac{\partial x^{\bar{\mu}}}{\partial x^\alpha} \frac{\partial x^\beta}{\partial x^{\bar{\sigma}}} T^\alpha_\beta = \bar{T}^{\bar{\mu}}_{\bar{\nu}} \frac{\partial x^{\bar{\nu}}}{\partial x^\beta} \frac{\partial x^\beta}{\partial x^{\bar{\sigma}}} = \bar{T}^{\bar{\mu}}_{\bar{\nu}} \delta^{\bar{\nu}}_{\bar{\sigma}} = \bar{T}^{\bar{\mu}}_{\bar{\sigma}},$$

which is word for word the transformation law of a $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ tensor, and the theorem is proved. The same argument, with more indices and no new ideas, gives the general version: if the contraction of an array with an arbitrary tensor is a tensor, then the array was a tensor all along.

The physical value of this result is that it converts a tedious verification into a statement about experiments. Consider the Lorentz force law written relativistically, $ma^\mu = qF^\mu_\nu u^\nu$. The four acceleration on the left is a vector, because it is built from a worldline and nothing else, and the four velocity on the right is a vector too, and the law is supposed to hold for a test charge moving in *any* way we please, which is to say for every u^ν . The quotient theorem then tells us at once that the electromagnetic field strength F^μ_ν is a tensor, and we never had to transform it. The same reasoning is what guarantees that the conductivity of a crystal, the moment of inertia of a rigid body and the polarisability of a molecule are all genuine tensors rather than accidental tables of numbers: in each case an experiment feeds the array an arbitrary vector and gets a vector back.

It is just as instructive to notice where the theorem fails to apply, because that is what protects us from using it carelessly. The Christoffel symbols $\Gamma^\lambda_{\mu\nu}$, which we shall meet in chapter 3, look exactly like the components of a $\begin{pmatrix} 1 \\ 2 \end{pmatrix}$ tensor and are not one, and the reason is that the object they contract with in the covariant derivative is not an arbitrary tensor but a derivative, whose own transformation law carries the inhomogeneous term we found at the end of this chapter. The hypothesis of the theorem is genuinely a hypothesis, and **the words for every in its statement are doing all the work.**

2.2.7 Tensor fields

Definition 11 A vector field is a map X which assigns to every point $p \in M$ a vector $X_p \in T_p(M)$. Acting on a smooth function it produces another function,

$$\begin{aligned} X(f) : M &\longrightarrow \mathbb{R} \\ p &\longmapsto X_p(f). \end{aligned} \tag{2.55}$$

These vector fields tell us how a physical quantity may change from one point of the manifold to another. Usually we picture a vector field as a field of arrows, but that picture belongs to the manifold $\mathbb{R}^n$, and this last definition is far more general, so those abstract arrows are better interpreted as rates of change of our physical quantities. The field is smooth if $X(f)$ is a smooth function for every smooth f .

Definition 12 A covector field, or one-form field, is a map ω which assigns to every point $p \in M$ a covector $\omega_p \in T_p^*(M)$, so that acting on a vector field it produces a function,

$$\begin{aligned} \omega(\vec{X}) : M &\longrightarrow \mathbb{R} \\ p &\longmapsto \omega_p(\vec{X}_p). \end{aligned} \tag{2.56}$$

This is analogous to the vector field, but in this case there is no analogue of a space of arrows, and the right mental picture is instead a foliation of the manifold by the level surfaces we described above. In both cases the smoothness of the field depends on the smoothness of its components and of the functions f on which it acts.

Definition 13 A $\binom{r}{s}$ tensor field is a map T which assigns to every point $p \in M$ a $\binom{r}{s}$ tensor T_p at p ,

$$\begin{aligned} T(\eta_1, \dots, \eta_r, \vec{X}_1, \dots, \vec{X}_s) : M &\longrightarrow \mathbb{R} \\ p &\longmapsto T_p\left((\eta_1)_p, \dots, (\eta_r)_p, (\vec{X}_1)_p, \dots, (\vec{X}_s)_p\right). \end{aligned} \tag{2.57}$$

Tensor fields can be thought of as the generalisation of vector and covector fields, and this is the heart of many physical theories, not only general and special relativity: they appear in solid state physics, in particle physics and even in crystal optics. They are so important because, as we will see later, with a tensor we can define the notion of distance between events ¹ and we can also define physical quantities that do not depend on the coordinate chart, quantities such as the Maxwell

¹this is also related to a topology on the manifold M

stress tensor, which tells us how much electromagnetic energy density is stored in our system, the Faraday electromagnetic tensor, which tells us how the electric and magnetic fields are related and encodes the conservation laws of a dynamical system, or the pressure tensor, which tells us about the pressure in every direction of a fluid, even in a magnetohydrodynamical one.

Tensors were the heart of twentieth century physics, but there are now further geometrical objects, such as spinors, which behave in a sense like fractional rank tensors, or the twistors of Penrose, which contain both tensors and spinors.

2.2.8 Commutator

Let us analyse two vector fields $\vec{X}$ and $\vec{Y}$ acting on a smooth function f . Since $\vec{Y}(f)$ is again a smooth function, we may feed it to $\vec{X}$, and it is natural to ask whether the composition $\vec{X}(\vec{Y}(f))$ is itself a vector field. To answer this we must check whether the new operator satisfies linearity and the Leibniz rule. Linearity is immediate, so let us test Leibniz by applying the composition to a product of functions,

$$\begin{aligned}\vec{X}(\vec{Y}(fg)) &= \vec{X}(f\vec{Y}(g) + g\vec{Y}(f)) \\ &= f\vec{X}(\vec{Y}(g)) + g\vec{X}(\vec{Y}(f)) + \vec{X}(f)\vec{Y}(g) + \vec{X}(g)\vec{Y}(f).\end{aligned}$$

The first two terms are exactly what the Leibniz rule would demand, but the last two are extra, so **the composition $\vec{X} \circ \vec{Y}$ is not a vector field**, essentially because applying two derivatives in a row produces second derivatives and a genuine derivation is only allowed first ones. The remedy is to notice that the two offending terms are completely symmetric under the exchange $\vec{X} \leftrightarrow \vec{Y}$, since repeating the computation with the fields swapped gives

$$\vec{Y}(\vec{X}(fg)) = f\vec{Y}(\vec{X}(g)) + g\vec{Y}(\vec{X}(f)) + \vec{Y}(f)\vec{X}(g) + \vec{Y}(g)\vec{X}(f),$$

so that if we subtract one from the other the unwanted terms cancel exactly and we are left with an operator that does obey Leibniz. This motivates defining a new map $\vec{Z}$ by

$$\vec{Z}(f) = \vec{X}(\vec{Y}(f)) - \vec{Y}(\vec{X}(f)), \quad (2.58)$$

which, by the argument we have just given, is a genuine vector field.

Definition 14 *The commutator of two vector fields $\vec{X}$ and $\vec{Y}$ is the vector field defined by*

$$[\vec{X}, \vec{Y}](f) = \vec{X}(\vec{Y}(f)) - \vec{Y}(\vec{X}(f)) \quad \forall f \in C^\infty(M). \quad (2.59)$$

We can expand the definition of the commutator to see how it operates in a given coordinate chart $\phi = (x^1, \dots, x^n)$,

$$\begin{aligned} [\vec{X}, \vec{Y}](f) &= \vec{X} \left(Y^\alpha \frac{\partial F}{\partial x^\alpha} \right) - \vec{Y} \left(X^\beta \frac{\partial F}{\partial x^\beta} \right) \\ &= X^\beta \frac{\partial}{\partial x^\beta} \left(Y^\alpha \frac{\partial F}{\partial x^\alpha} \right) - Y^\alpha \frac{\partial}{\partial x^\alpha} \left(X^\beta \frac{\partial F}{\partial x^\beta} \right) \\ &= X^\beta \frac{\partial Y^\alpha}{\partial x^\beta} \frac{\partial F}{\partial x^\alpha} + \textcolor{red}{X^\beta Y^\alpha \frac{\partial^2 F}{\partial x^\beta \partial x^\alpha}} - Y^\alpha \frac{\partial X^\beta}{\partial x^\alpha} \frac{\partial F}{\partial x^\beta} - \textcolor{red}{Y^\alpha X^\beta \frac{\partial^2 F}{\partial x^\alpha \partial x^\beta}}, \end{aligned}$$

where the two second derivative terms marked in red cancel one another because the mixed partial derivatives of a smooth function commute, which is exactly the cancellation we anticipated above. Relabelling the dummy index in the first term so that both surviving terms carry the same derivative $\partial F / \partial x^\beta$, we obtain

$$[\vec{X}, \vec{Y}](f) = \left(X^\alpha \frac{\partial Y^\beta}{\partial x^\alpha} - Y^\alpha \frac{\partial X^\beta}{\partial x^\alpha} \right) \frac{\partial F}{\partial x^\beta}, \quad (2.60)$$

so that, comparing with the expansion of a vector field in the coordinate basis,

$$[\vec{X}, \vec{Y}](f) = [\vec{X}, \vec{Y}]^\beta \left(\frac{\partial}{\partial x^\beta} \right)_p (f), \quad (2.61)$$

we can finally read off the components of the commutator,

$$[\vec{X}, \vec{Y}]^\beta = X^\alpha \frac{\partial Y^\beta}{\partial x^\alpha} - Y^\alpha \frac{\partial X^\beta}{\partial x^\alpha}. \quad (2.62)$$

The commutator satisfies four important relations:

1. $[\vec{X}, \vec{Y}] = -[\vec{Y}, \vec{X}]$; the commutator is antisymmetric.
2. $[\vec{X}, \vec{Y} + \vec{Z}] = [\vec{X}, \vec{Y}] + [\vec{X}, \vec{Z}]$; the commutator is linear in each argument.
3. $[\vec{X}, f\vec{Y}] = f[\vec{X}, \vec{Y}] + \vec{X}(f)\vec{Y}$; this is how the commutator behaves under multiplication by a smooth function f , and note that it is *not* simply linear over functions, the extra term being the reason the commutator is a derivation rather than a tensor.
4. $[\vec{X}, [\vec{Y}, \vec{Z}]] + [\vec{Y}, [\vec{Z}, \vec{X}]] + [\vec{Z}, [\vec{X}, \vec{Y}]] = 0$; the Jacobi identity.

We will see later that this commutator carries a great deal of physical information, because the Riemann tensor, which tells us about the geometry of the system, can be written in terms of commutators, which makes its independence of any coordinate chart manifest. The second promise we made, that the commutator would tell us whether a given basis is a coordinate basis or not, we can keep immediately.

Theorem 4 *Let $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$ be n linearly independent smooth vector fields defined on some region of a manifold M . There exists a coordinate chart $\phi = (x^1, \dots, x^n)$ on that region such that $\mathbf{e}_\mu = \partial/\partial x^\mu$ for every μ , which is to say the basis is a coordinate basis, if and only if all of the commutators vanish,*

$$[\mathbf{e}_\mu, \mathbf{e}_\nu] = 0 \quad \forall \mu, \nu. \quad (2.63)$$

Demonstration of the easy direction:

Suppose the basis is a coordinate basis, so that $\mathbf{e}_\mu = \partial/\partial x^\mu$ for some chart. Then applying the commutator to any smooth function and going down into that chart, as we have done throughout this section, gives

$$\left[\frac{\partial}{\partial x^\mu}, \frac{\partial}{\partial x^\nu} \right] (f) = \frac{\partial^2 F}{\partial x^\mu \partial x^\nu} - \frac{\partial^2 F}{\partial x^\nu \partial x^\mu} = 0,$$

which vanishes because F is smooth and the mixed partial derivatives of a smooth function of several real variables do not depend on the order in which they are taken. The converse, that vanishing commutators guarantee that coordinates exist, is genuinely harder and is a consequence of the Frobenius theorem, whose proof we shall not reproduce here and which the interested reader can find in Wald (1984) or in Nakahara (2003).

The geometrical content of equation 2.63 is much more intuitive than the algebra suggests, and it is worth carrying the picture around because exactly the same picture returns when we build the Riemann tensor in chapter 3. Take two vector fields $\vec{X}$ and $\vec{Y}$, start at a point p , travel a parameter distance ϵ along the integral curve of $\vec{X}$ and then a parameter distance ϵ along the integral curve of $\vec{Y}$, and note where you land. Now do the same two journeys in the opposite order. **In general you do not arrive at the same point, and the gap between the two endpoints is $\epsilon^2 [\vec{X}, \vec{Y}]$ to lowest order**, so the commutator is precisely the failure of the little parallelogram to close (Misner et al., 1973). A coordinate system is nothing but a grid, and a grid closes by construction, since going three steps east and then two steps north must land you where going two steps north and then three steps east does, and that is the whole reason coordinate bases are exactly the bases with vanishing commutators.

It is worth finishing with an example which usually surprises people, because it shows that a basis every physicist has used since their first year is not a coordinate basis at all. Take the plane in polar coordinates (r, θ) and the orthonormal basis we all draw on such a diagram, which is

$$\hat{\mathbf{e}}_r = \frac{\partial}{\partial r}, \quad \hat{\mathbf{e}}_\theta = \frac{1}{r} \frac{\partial}{\partial \theta}, \quad (2.64)$$

where the factor $1/r$ is there precisely so that $\hat{\mathbf{e}}_\theta$ has unit length rather than length r . Computing the commutator with equation 2.62, or simply acting on a function and being careful with the product rule, we find

$$[\hat{\mathbf{e}}_r, \hat{\mathbf{e}}_\theta](f) = \frac{\partial}{\partial r} \left(\frac{1}{r} \frac{\partial f}{\partial \theta} \right) - \frac{1}{r} \frac{\partial}{\partial \theta} \left(\frac{\partial f}{\partial r} \right) = -\frac{1}{r^2} \frac{\partial f}{\partial \theta} = -\frac{1}{r} \hat{\mathbf{e}}_\theta(f),$$

so that

$$[\hat{\mathbf{e}}_r, \hat{\mathbf{e}}_\theta] = -\frac{1}{r} \hat{\mathbf{e}}_\theta \neq 0. \quad (2.65)$$

The orthonormal polar basis is therefore not a coordinate basis, and there is no coordinate system anywhere whose partial derivatives are $\hat{\mathbf{e}}_r$ and $\hat{\mathbf{e}}_\theta$, which is the honest reason behind all those extra terms that appear when one writes the divergence or the Laplacian in polar coordinates and which students usually memorise without ever being told where they come from. Bases of this kind, with non vanishing commutators, are called *anholonomic* or non coordinate bases, and far from being pathological they are extremely useful in relativity, since an observer carrying a set of orthonormal rods and a clock is carrying exactly such a basis and not a coordinate one.

2.2.9 Non-coordinate bases

The polar example is worth taking seriously rather than treating as a curiosity, because it shows that we have been quietly assuming something we never needed. Nothing in the definition of the tangent space $T_p M$ requires its basis to come from a chart. A basis is simply a set of n linearly independent vectors, and if we choose one such set smoothly at every point of the manifold we obtain what is called a *frame*, or a moving frame in the older literature, which we shall write as

$$\{e_{(a)}\}_{a=1}^n, \quad e_{(a)} = e_{(a)}^\mu \partial_\mu. \quad (2.66)$$

The round brackets on the index are a warning, and they will stay with us for the rest of these notes. The label a says which vector of the frame we are talking about, and it is not a coordinate index: it does not transform when we change chart, because the frame was chosen without reference to any chart in the first place. The Greek index μ in equation 2.66 is a coordinate index and does transform. The n^2 functions $e_{(a)}^\mu$ are therefore a dictionary between the two languages, and since the

frame is by construction linearly independent the matrix they form is invertible at every point, with inverse written $e_{\mu}^{(a)}$, so that

$$e_{(a)}^{\mu} e_{\nu}^{(a)} = \delta_{\nu}^{\mu}, \quad e_{(a)}^{\mu} e_{\mu}^{(b)} = \delta_a^b. \quad (2.67)$$

The second of these identities is more than an algebraic accident. It says that the one forms $e^{(a)} = e_{\mu}^{(a)} dx^{\mu}$ satisfy $e^{(a)}(e_{(b)}) = \delta_b^a$, which is precisely the condition defining the dual basis of $\{e_{(a)}\}$ in the cotangent space. Choosing a frame in $T_p M$ therefore chooses a coframe in $T_p^* M$ automatically, exactly as the coordinate basis $\{\partial_{\mu}\}$ came with the coordinate coframe $\{dx^{\mu}\}$, and the components of any tensor can be referred to the frame instead of to the chart by contracting with the dictionary in the obvious way,

$$T_{ab} = T_{\mu\nu} e_{(a)}^{\mu} e_{(b)}^{\nu}, \quad T_{\mu\nu} = T_{ab} e_{\mu}^{(a)} e_{\nu}^{(b)}. \quad (2.68)$$

The freedom we have just granted ourselves is enormous. At every point we may act on the frame with any invertible matrix $M_a^b(x)$ and obtain another perfectly good frame, so the choice is a whole general linear group's worth of arbitrariness, one independent copy at every point of the manifold. In practice one never uses all of it, because the whole reason for abandoning the coordinate basis is usually to impose some condition the coordinate basis could not satisfy, and imposing that condition cuts the group down to a subgroup. In relativity the condition will be that the frame be orthonormal with respect to the metric, and the subgroup that survives will turn out to be the Lorentz group, but the metric is not part of the structure we have built so far and that discussion belongs to chapter 4.

What we can already say is how much the frame differs from a coordinate basis, and the criterion 2.63 gives the answer without any extra work. Since the frame vectors are vector fields, their commutator is a vector field, and since the frame is a basis that commutator can be expanded in it,

$$[e_{(a)}, e_{(b)}] = C_{ab}^c e_{(c)}, \quad (2.69)$$

which defines the $n^2(n-1)/2$ functions C_{ab}^c , antisymmetric in their two lower indices and called the *structure functions* of the frame. They are computed directly from the dictionary by feeding equation 2.62 into the definition,

$$C_{ab}^c = e_{\mu}^{(c)} \left(e_{(a)}^{\nu} \partial_{\nu} e_{(b)}^{\mu} - e_{(b)}^{\nu} \partial_{\nu} e_{(a)}^{\mu} \right), \quad (2.70)$$

and the polar example we have just done is the smallest possible illustration, with the single non vanishing structure function $C_{r\theta}^{\theta} = -1/r$. Everything said in this subsection is summarised by the statement that a frame is a coordinate basis if and only if all its structure functions vanish, which is the criterion 2.63 written in the new notation.

Two remarks are worth making before we leave the subject, because both of them are sources of confusion later. The first is that the structure functions are not the components of a tensor.

Equation 2.69 involves derivatives of the frame, and the same argument that showed the derivative of a vector field is not a tensor applies here unchanged. The second, and more important, is that they have nothing whatever to do with curvature. The plane is flat, and its orthonormal polar frame has a non vanishing structure function, so a frame can fail to be holonomic in a perfectly flat space simply because we insisted on a condition, in that case unit length, that the chart could not provide. Curvature will be measured in chapter 3 by an entirely different object, and the reader who keeps these two failures apart from the beginning will be spared a good deal of confusion.

The name the physicist will meet for these objects depends on the dimension and on the tradition. In four dimensions a frame is a *tetrad* or a *vierbein*, in n dimensions a *vielbein*, and in Cartan's own language the coframe $e^{(a)}$ is a set of one forms whose exterior derivatives encode 2.69. They are all the same idea, which is that the basis in which we write our components should be chosen to suit the physics rather than the chart.

2.2.10 Integral curves

In ordinary language the terms curve and vector field are used as though they described separate things, but if we look carefully they are not separate at all. If we go back to our elementary calculus courses we remember that in $\mathbb{R}^3$ a curve is given by a vector whose components are parameterised functions, and that is exactly what we have been developing here: a curve is a map from an interval I , parameterised by t , into the manifold, which in the case of $\mathbb{R}^3$ is represented by the familiar $\lambda(t) = [X(t), Y(t), Z(t)]^T$.

It is time to formalise all of this, and the mental picture to keep in mind is the velocity field of a fluid, which (and not only in special relativity) is usually written as a vector field $\vec{U}(\vec{X}) \in \mathbb{R}^3$. The trajectory followed by a particle carried along by that velocity field is fixed by an initial position $\vec{X}_0$ through

$$\frac{d\vec{X}}{dt} = \vec{U}(\vec{X}(t)) ; \vec{X}(0) = \vec{X}_0, \quad (2.71)$$

and the solution of equation 2.71 is what we call the *integral curve of $\vec{U}$ through $\vec{X}_0$* .

Definition 15 Let $\vec{X}$ be a vector field on a manifold M and let $p \in M$. An integral curve of $\vec{X}$ through p is a curve passing through p whose tangent vector at every point coincides with the value of $\vec{X}$ at that point.

Let λ be an integral curve of $\vec{X}$ with $\lambda(0) = p$, and choose a coordinate chart $\phi = (x^1, \dots, x^n)$. Writing the condition that the tangent of the curve equals the field at every point, component by component, gives the system of ordinary differential equations

$$\frac{d}{dt} [x^\alpha(\lambda(t))] = X^\alpha(\lambda(t)), \quad (2.72)$$

subject to the initial condition

$$x^\alpha(\lambda(0)) = x^\alpha(p). \quad (2.73)$$

Since the components X^α are smooth functions of the coordinates, the standard existence and uniqueness theorem for ordinary differential equations guarantees that through every point p there passes exactly one integral curve, at least for a short enough interval of the parameter t . **The initial condition is what selects one curve out of the whole family**, and if we drop it we are left with the family itself, which is precisely the setting in which one speaks of the flow of the vector field, of autonomous and non autonomous dynamical systems, and eventually of chaos.

2.2.11 A worked example: the two sphere

Everything we have built so far has been abstract on purpose, since the whole point of the construction is that it never refers to an ambient space, but abstraction is much easier to trust once one has seen it produce familiar answers in a case where the answers are already known. In this subsection we are going to take the unit sphere S^2 , which we already met as an example of a manifold, and compute absolutely everything explicitly: the coordinate basis, a curve and its tangent vector, a non coordinate basis, the commutators, a change of chart and an integral curve. The reader who works through this once will find the rest of the book considerably less intimidating.

We begin by working on the region covered by the usual spherical chart, $\phi = (\vartheta, \varphi)$ with $0 < \vartheta < \pi$ and $0 < \varphi < 2\pi$, which as we discussed above misses one meridian and both poles, so strictly speaking we are working on one patch of an atlas which needs at least two. The coordinate basis of the tangent space at each point of the patch is

$$\left\{ \left(\frac{\partial}{\partial \vartheta} \right)_p, \left(\frac{\partial}{\partial \varphi} \right)_p \right\}, \quad (2.74)$$

and since there are two of them the tangent space $T_p(S^2)$ is two dimensional, exactly as the general theorem promised, which is already worth noticing: **the tangent space to the sphere is a plane of two dimensions and not of three**, even though we habitually draw the sphere sitting inside $\mathbb{R}^3$, because the manifold itself is two dimensional and the third dimension in the drawing is our own addition and nothing the sphere knows about.

Consider now the circle of constant latitude at $\vartheta = \vartheta_0$, described by the curve

$$\lambda(t) = \phi^{-1}(\vartheta_0, \varphi_0 + t), \quad (2.75)$$

whose coordinates are $\vartheta(t) = \vartheta_0$ and $\varphi(t) = \varphi_0 + t$. Using equation 2.31 its tangent vector has components $X^\vartheta = d\vartheta/dt = 0$ and $X^\varphi = d\varphi/dt = 1$, so that

$$X_p = \left(\frac{\partial}{\partial \varphi} \right)_p, \quad (2.76)$$

which says that the vector field $\partial/\partial\varphi$ is precisely the field whose integral curves are the circles of latitude, travelled once per 2π of parameter. This is the generator of rotations about the polar axis and we will meet it again in a moment.

Physicists almost never use the coordinate basis on a sphere, because $\partial/\partial\varphi$ does not have unit length, its length being $\sin\vartheta$, which shrinks to nothing as we approach the poles. What we use instead is the orthonormal basis

$$\hat{\mathbf{e}}_\vartheta = \frac{\partial}{\partial\vartheta}, \quad \hat{\mathbf{e}}_\varphi = \frac{1}{\sin\vartheta} \frac{\partial}{\partial\varphi}, \quad (2.77)$$

and by the criterion of equation 2.63 we can now ask whether this is a coordinate basis, which amounts to computing one commutator,

$$\begin{aligned} [\hat{\mathbf{e}}_\vartheta, \hat{\mathbf{e}}_\varphi](f) &= \frac{\partial}{\partial\vartheta} \left(\frac{1}{\sin\vartheta} \frac{\partial f}{\partial\varphi} \right) - \frac{1}{\sin\vartheta} \frac{\partial}{\partial\varphi} \left(\frac{\partial f}{\partial\vartheta} \right) \\ &= -\frac{\cos\vartheta}{\sin^2\vartheta} \frac{\partial f}{\partial\varphi} + \frac{1}{\sin\vartheta} \frac{\partial^2 f}{\partial\vartheta\partial\varphi} - \frac{1}{\sin\vartheta} \frac{\partial^2 f}{\partial\varphi\partial\vartheta} \\ &= -\frac{\cos\vartheta}{\sin\vartheta} \left(\frac{1}{\sin\vartheta} \frac{\partial f}{\partial\varphi} \right) = -\cot\vartheta \hat{\mathbf{e}}_\varphi(f), \end{aligned}$$

where the second derivative terms marked in red cancel as always, and therefore

$$[\hat{\mathbf{e}}_\vartheta, \hat{\mathbf{e}}_\varphi] = -\cot\vartheta \hat{\mathbf{e}}_\varphi \neq 0. \quad (2.78)$$

So the orthonormal basis on the sphere, exactly like the orthonormal polar basis on the plane, is anholonomic, and no coordinate system exists whose partial derivatives are $\hat{\mathbf{e}}_\vartheta$ and $\hat{\mathbf{e}}_\varphi$.

The sphere has three obvious symmetries, the rotations about the three Cartesian axes, and each of them is generated by a vector field on S^2 . Written in our chart they are

$$\vec{X}_{(z)} = \frac{\partial}{\partial\varphi}, \quad (2.79)$$

$$\vec{X}_{(x)} = -\sin\varphi \frac{\partial}{\partial\vartheta} - \cos\varphi \cot\vartheta \frac{\partial}{\partial\varphi}, \quad (2.80)$$

$$\vec{X}_{(y)} = \cos\varphi \frac{\partial}{\partial\vartheta} - \sin\varphi \cot\vartheta \frac{\partial}{\partial\varphi}, \quad (2.81)$$

and computing their commutators with equation 2.62, which is a slightly tedious but completely mechanical exercise that the reader should do at least once, gives

$$[\vec{X}_{(x)}, \vec{X}_{(y)}] = -\vec{X}_{(z)}, \quad (2.82)$$

$$[\vec{X}_{(y)}, \vec{X}_{(z)}] = -\vec{X}_{(x)}, \quad (2.83)$$

$$[\vec{X}_{(z)}, \vec{X}_{(x)}] = -\vec{X}_{(y)}. \quad (2.84)$$

This should stop the reader in their tracks, because it is the angular momentum algebra of quantum mechanics arriving unannounced in the middle of a geometry calculation. If we define the operators $\hat{L}_i = -i\hbar\vec{X}_{(i)}$, which is exactly how the angular momentum operators are built in any quantum mechanics course (Sakurai and Napolitano, 2011), then the three relations above become

$$[\hat{L}_x, \hat{L}_y] = i\hbar\hat{L}_z, \quad (2.85)$$

together with its cyclic partners, and **the commutation relations that a physics student first meets as an algebraic curiosity about operators turn out to be a purely geometrical statement about how rotations fail to commute on the surface of a sphere.** The overall sign is a matter of convention, since flipping the sign of all three fields flips the sign of the structure constants, but the content does not depend on that choice.

Let us now check equation 2.34 on a concrete change of coordinates, using the stereographic projection we described earlier. Projecting from the north pole onto the plane tangent at the south pole gives the chart (u, v) with

$$u = 2 \cot\left(\frac{\vartheta}{2}\right) \cos \varphi, \quad v = 2 \cot\left(\frac{\vartheta}{2}\right) \sin \varphi, \quad (2.86)$$

so that the distance from the origin in the plane is $R = \sqrt{u^2 + v^2} = 2 \cot(\vartheta/2)$, which correctly runs off to infinity as we approach the north pole at $\vartheta = 0$ and vanishes at the south pole at $\vartheta = \pi$. Applying the transformation law for basis vectors, which simply means computing the Jacobian of this change and contracting, we obtain

$$\frac{\partial}{\partial \varphi} = \frac{\partial u}{\partial \varphi} \frac{\partial}{\partial u} + \frac{\partial v}{\partial \varphi} \frac{\partial}{\partial v} = -v \frac{\partial}{\partial u} + u \frac{\partial}{\partial v}, \quad (2.87)$$

$$\frac{\partial}{\partial \vartheta} = \frac{\partial u}{\partial \vartheta} \frac{\partial}{\partial u} + \frac{\partial v}{\partial \vartheta} \frac{\partial}{\partial v} = -\frac{4 + R^2}{4R} \left(u \frac{\partial}{\partial u} + v \frac{\partial}{\partial v} \right). \quad (2.88)$$

The first of these is worth reading twice, because $-v \partial_u + u \partial_v$ is exactly the generator of rotations in the plane that we all wrote down in first year mechanics, and it has appeared here automatically:

the field which rotates the sphere about its polar axis is, seen through the stereographic chart, the field which rotates the plane about its origin. The second one is a pure radial field, which is the projection of the meridians, and both expressions are perfectly regular everywhere except at the origin of the plane and at infinity, which are the two points where this single chart is bound to fail.

Finally let us solve equation 2.72 for the field $\vec{X}_{(z)} = \partial/\partial\varphi$, whose components are $X^\vartheta = 0$ and $X^\varphi = 1$, so that the system reads

$$\frac{d\vartheta}{dt} = 0, \quad \frac{d\varphi}{dt} = 1, \quad (2.89)$$

with the immediate solution $\vartheta(t) = \vartheta_0$ and $\varphi(t) = \varphi_0 + t$, which is the circle of latitude we started from, closing the loop of this example. There is, however, one last thing the sphere has to teach us, and it is topological rather than geometrical: the field $\partial/\partial\varphi$ vanishes at both poles, since there is no preferred direction of rotation once you are standing exactly on the axis, and this is not a defect of our particular choice. **Every smooth vector field on S^2 must vanish somewhere**, a result proved by Brouwer in 1912 and known affectionately as the hairy ball theorem, because it says you cannot comb a hairy sphere flat without leaving at least one crown. The plane and the torus suffer from no such obstruction, so this is a genuine statement about the shape of the sphere and not about our charts, and it is a first hint that manifolds carry global information that no amount of local calculus will ever reveal.

2.2.12 Why the partial derivative is not enough

We are about to start differentiating tensor fields, and before we do it is essential to understand why the obvious candidate fails, because the failure is what forces the whole apparatus of the next section and of the next chapter into existence. The obvious candidate is simply to take the components of a vector field and differentiate them one by one, forming the array $\partial V^\nu/\partial x^\mu$, which certainly looks like the components of a $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ tensor since it carries one index upstairs and one downstairs. Let us test it against equation 2.49 and see whether it really transforms as it should.

Starting from the transformation law for the components of a vector, equation 2.36, and differentiating with the chain rule,

$$\frac{\partial V^{\bar{\mu}}}{\partial x^{\bar{\alpha}}} = \frac{\partial x^\beta}{\partial x^{\bar{\alpha}}} \frac{\partial}{\partial x^\beta} \left(\frac{\partial x^{\bar{\mu}}}{\partial x^\nu} V^\nu \right) = \underbrace{\frac{\partial x^\beta}{\partial x^{\bar{\alpha}}} \frac{\partial x^{\bar{\mu}}}{\partial x^\nu} \frac{\partial V^\nu}{\partial x^\beta}}_{\text{what a tensor would do}} + \underbrace{\frac{\partial x^\beta}{\partial x^{\bar{\alpha}}} \frac{\partial^2 x^{\bar{\mu}}}{\partial x^\beta \partial x^\nu}}_{\text{the unwanted term}} V^\nu,$$

we see that the first term is exactly the transformation law a $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ tensor is supposed to obey, one Jacobian upstairs and one downstairs, but that there is a second term which has no business being there, and **because of it the partial derivative of the components of a vector field is not a tensor**. The offending term is built from the second derivatives of the coordinate change, so it

vanishes if and only if those second derivatives vanish, which happens exactly when the change of coordinates is affine, that is, linear plus a constant shift.

That observation explains a great deal about why relativity is taught the way it is. **Lorentz transformations are linear**, so in special relativity the second derivatives are identically zero, the unwanted term never appears, and a physicist can differentiate the components of a four vector with ∂_μ all day long and still obtain perfectly good tensors. General relativity allows arbitrary smooth changes of coordinates, which are certainly not linear, so the same manoeuvre stops working immediately, and this single fact is the reason the next chapter has to spend so much effort building a new derivative from scratch.

There are two honest ways out of the difficulty and we shall use both. The first is to add extra structure to the manifold, an object called a *connection*, whose own transformation law contains an inhomogeneous piece designed precisely to cancel the unwanted term above, producing the *covariant derivative* ∇_μ that we shall construct in chapter 3 and which is the derivative appearing in the geodesic equation and in the Einstein field equations. The second is to give up on differentiating in a single direction and instead use a second vector field to carry the comparison for us, dragging the tensor along the flow of that field and comparing it with itself, which produces the *Lie derivative* and which needs no extra structure whatsoever beyond the manifold itself. That second route is the subject of the section which follows.

It is worth closing with a small consolation, because the situation is not quite as bad as it looks. If we take a one-form instead of a vector and run the same computation, the unwanted term turns out to be

$$\frac{\partial^2 x^\alpha}{\partial x^{\bar{\nu}} \partial x^{\bar{\mu}}} \omega_\alpha, \quad (2.90)$$

which is manifestly symmetric under the exchange of $\bar{\mu}$ and $\bar{\nu}$, since mixed partial derivatives commute, so if we antisymmetrise the two indices using the brackets we introduced above then the whole offending term disappears and

$$\partial_{[\mu} \omega_{\nu]} \text{ is a tensor,} \quad (2.91)$$

even though $\partial_\mu \omega_\nu$ on its own is not. This is not an accident, it is the first appearance of the *exterior derivative*, the one derivative in differential geometry that needs no connection at all, and it is the reason the electromagnetic field tensor $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ is a perfectly good tensor in curved space-time despite being written entirely with ordinary partial derivatives.

2.3 The Lie derivative

There is a truly important concept in general relativity and differential geometry, the derivative, but there is no single derivative, there are many kinds and all of them have their own meaning and

applications: the ordinary derivative, the partial derivative, the gradient, the divergence, the curl, the Jacobian, the covariant derivative, the fractional derivative. Here we are going to develop one special derivative that allows us to pull or push elements from one manifold to another, and to find the notion of a derivative along a curve together with the relation between manifolds which are in general different. These are the pull back and push forward operations, and with them we can develop the *Lie derivative*, a primitive and primordial derivative in the precise sense that it needs no structure on the manifold beyond the manifold itself, no metric and no connection, and of which the derivative of a scalar along a direction and the plain partial derivative are both particular cases. Its relation to the covariant derivative is subtler than that, and rather than the one the reader might guess, so we shall settle it carefully in chapter 3 once both derivatives are on the table.

2.3.1 Pull back and push forward

Let M and N be two different manifolds, let f be a smooth function mapping from N to $\mathbb{R}$, and let ϕ be a bijection mapping from M to N . There then exists an inverse map $\phi^{-1} : N \rightarrow M$ such that $\phi \circ \phi^{-1} = I_N$, where I_N is the identity function on the manifold N , and $\phi^{-1} \circ \phi = I_M$, where I_M is the identity function on the manifold M .

The question we want to answer is the following. If we have a map between two manifolds, and we have some geometrical object living on one of them, can we transport that object to the other one? The answer is yes for some objects and no for others, and finding out which is which is not a formality: the pattern of what can be transported and in which direction is what the whole of this section rests on, and it is also, as we shall see at the end, the precise sense in which general relativity has no preferred coordinates.

The easiest case is a function, and it goes backwards. Suppose $f : N \rightarrow \mathbb{R}$ assigns a number to every point of N , and suppose we stand at a point p of M and want a number. We have one, because ϕ takes us to the point $\phi(p)$ of N and f gives us a number there. Composing the two maps defines a function on M , which we call the *pull back* of f by ϕ and write

$$\begin{aligned} \phi^* f : M &\rightarrow \mathbb{R} \\ p &\mapsto (\phi^* f)(p) = (f \circ \phi)(p) = f(\phi(p)). \end{aligned} \tag{2.92}$$

Notice the direction. The map ϕ goes from M to N but the pull back carries functions from N to M , against the arrow, and that is where the name comes from: f lived on N and we have pulled it back to M . Notice also that we never used ϕ^{-1} , so equation 2.92 makes perfect sense even when ϕ has no inverse at all, which will matter shortly.

Now take a vector, and it goes forwards. Here we must remember what a vector actually is, because the naive picture of an arrow gives no hint of how to move it from one manifold to another. A vector at p , as we defined it, is a derivation: a machine that eats functions defined near p and returns numbers. So to define a vector at $\phi(p)$ on N we need only say what number it returns when fed a function on N , and there is exactly one sensible answer available to us. Given a function f on

N , pull it back to M with 2.92, feed the result to the vector we already have, and hand over the number. That defines the *push forward* of X ,

$$(\phi_* X)(f) \Big|_{\phi(p)} = X(\phi^* f) \Big|_p = X(f \circ \phi) \Big|_p, \quad (2.93)$$

and the reader should appreciate how little choice we had. Equation 2.93 is not a convention, it is the only definition that could possibly be written down with the ingredients available, and this is a recurring experience in differential geometry: once the objects are defined properly the constructions build themselves.

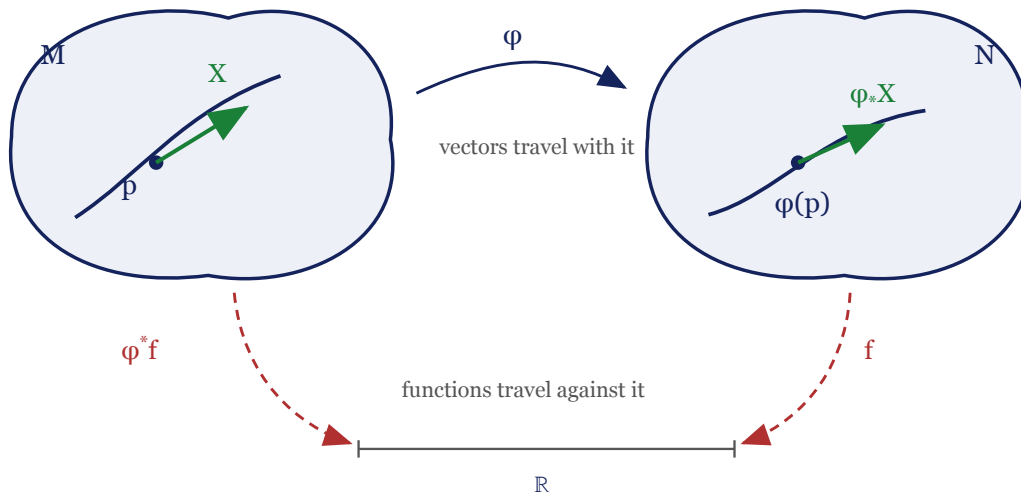


Figure 2.9: The map ϕ carries points of M to points of N . Vectors travel with the arrow, since a curve through p is carried to a curve through $\phi(p)$ and its tangent goes with it, while functions travel against it, since a function on N becomes a function on M by composition. Covectors, being machines that eat vectors, travel against the arrow for the same reason functions do.

To see what 2.93 says in components, put coordinates x^μ on M and y^α on N , so that the map ϕ is described by the n functions $y^\alpha(x^\mu)$. Expanding both sides in the respective coordinate bases and using the chain rule,

$$(\phi_* X)^\alpha \frac{\partial f}{\partial y^\alpha} = X^\mu \frac{\partial (f \circ \phi)}{\partial x^\mu} = X^\mu \frac{\partial y^\alpha}{\partial x^\mu} \frac{\partial f}{\partial y^\alpha},$$

and since this must hold for every function f the coefficients must agree, so that

$$(\phi_* X)^\alpha = \frac{\partial y^\alpha}{\partial x^\mu} X^\mu. \quad (2.94)$$

The push forward is therefore nothing more mysterious than the Jacobian matrix of the map acting on the components, which is a reassuring result: it says that the abstract machinery we have built reduces, when one finally puts coordinates on everything, to the object every physicist already knows from changing variables in an integral. The map ϕ_* is often called the *differential* of ϕ for exactly this reason.

Covectors go backwards, like functions, and the argument is the same one again. A covector at $\phi(p)$ is a machine that eats vectors at $\phi(p)$ and returns numbers, so to build a covector at p we must say what it returns when fed a vector at p , and the only route available is to push that vector forward first,

$$(\phi^* \omega)(X) \Big|_p = \omega(\phi_* X) \Big|_{\phi(p)}, \quad (2.95)$$

whose component version follows immediately from 2.94,

$$(\phi^* \omega)_\mu = \frac{\partial y^\alpha}{\partial x^\mu} \omega_\alpha. \quad (2.96)$$

Let us take stock, because the pattern is worth stating plainly. Vectors push forward and covectors pull back, and neither of the two operations we have defined required ϕ to be invertible. What we cannot do with a general ϕ is the opposite pair: there is no way to pull a vector back or to push a covector forward, and the reason is not a failure of imagination. If ϕ is not injective then two different points of M may be sent to the same point of N , and a vector sitting there would have to be pulled back to both of them at once with no way of deciding how; and if ϕ is not surjective then there are points of N carrying no vector at all, so a pushed forward covector would have nothing to eat over most of the target. Both obstructions disappear at once if ϕ is a bijection with smooth inverse, which is what we called a *diffeomorphism*, and in that case we simply define the missing operations using the inverse map,

$$\phi_* \equiv (\phi^{-1})^*, \quad \phi^* \equiv (\phi^{-1})_*, \quad (2.97)$$

so that for a diffeomorphism everything can be transported in both directions and the distinction between pushing and pulling becomes a matter of bookkeeping rather than of principle. This is why we insisted at the start that ϕ be a bijection.

With both directions available the extension to a general tensor is immediate, since a tensor is built out of vectors and covectors and each slot is transported in whichever direction its own kind of object goes. For a tensor of type $\binom{k}{l}$ the rule is

$$(\phi_* T)^{\alpha_1 \dots \alpha_k}_{\beta_1 \dots \beta_l} = \frac{\partial y^{\alpha_1}}{\partial x^{\mu_1}} \dots \frac{\partial y^{\alpha_k}}{\partial x^{\mu_k}} \frac{\partial x^{\nu_1}}{\partial y^{\beta_1}} \dots \frac{\partial x^{\nu_l}}{\partial y^{\beta_l}} T^{\mu_1 \dots \mu_k}_{\nu_1 \dots \nu_l}, \quad (2.98)$$

one Jacobian for each contravariant index and one inverse Jacobian for each covariant one, and the reader who compares 2.98 with the transformation law 2.49 will notice that the two expressions are identical. That coincidence is the most important thing in this subsection and it deserves to be said out loud rather than left as an observation. A change of coordinates is a passive operation: nothing moves, we merely relabel the same point with different numbers. A diffeomorphism is an active operation: the points genuinely move, and the tensor is genuinely carried from one place to another. The two are conceptually entirely different, and yet they produce exactly the same formula, and therefore no experiment can distinguish them. That is the technical content of the statement that general relativity is diffeomorphism invariant, and it is also, as we saw in chapter 1, precisely the trap Einstein fell into with his hole argument and spent almost three years escaping from.

A small example will fix the ideas. Let $M = \mathbb{R}^2$ with coordinates (θ, z) , let $N = \mathbb{R}^3$ with the usual Cartesian coordinates, and let ϕ be the map that wraps the plane around a cylinder of unit radius,

$$\begin{aligned}\phi: \mathbb{R}^2 &\rightarrow \mathbb{R}^3 \\ (\theta, z) &\mapsto (\cos \theta, \sin \theta, z).\end{aligned}\tag{2.99}$$

This ϕ is not injective, since θ and $\theta + 2\pi$ land on the same point, so it is not a diffeomorphism and we may only push vectors forward and pull functions back, which is exactly what the general theory predicted. Applying 2.94 to the basis vector ∂_θ , whose components are $(1, 0)$, gives

$$\phi_* \partial_\theta = \frac{\partial x}{\partial \theta} \partial_x + \frac{\partial y}{\partial \theta} \partial_y + \frac{\partial z}{\partial \theta} \partial_z = -\sin \theta \partial_x + \cos \theta \partial_y + \mathbf{0} \cdot \partial_z,$$

which is the unit vector tangent to the circle at the corresponding point of the cylinder, as it should be, and pulling back the function $f = x^2 + y^2$ from $\mathbb{R}^3$ gives $\phi^* f = \cos^2 \theta + \sin^2 \theta = 1$, the constant function, which is the analytic way of saying that the cylinder sits at unit distance from the axis everywhere.

2.3.2 The flow of a vector field

Everything we have built so far concerned two different manifolds, but the case we actually need is the one where they are the same. Let X be a vector field on M and recall from the section on integral curves that through every point there passes exactly one curve whose tangent is X . Now do something slightly different from what we did there: instead of following one curve, take *every* point of the manifold at once and slide each of them a parameter distance t along the curve through it. The result is a map from M to itself,

$$\begin{aligned}\sigma_t: M &\rightarrow M \\ p &\mapsto \sigma_t(p) = \text{the point reached from } p \text{ after a parameter } t \text{ along the integral curve of } X,\end{aligned}\tag{2.100}$$

which is called the *flow* generated by X . The name is well chosen, and the physical picture behind it is the honest one: think of X as the velocity field of a fluid filling the manifold, and of σ_t as the operation of letting the fluid run for a time t and asking where each particle has ended up.

The flow is not merely a collection of maps, it is a group, and this is the structural fact that makes the whole construction work. Flowing for no time at all changes nothing, flowing for t and then for s is the same as flowing for $t + s$ because the integral curves do not care where we choose to start counting, and consequently flowing for $-t$ undoes flowing for t ,

$$\sigma_0 = I_M, \quad \sigma_t \circ \sigma_s = \sigma_{t+s}, \quad (\sigma_t)^{-1} = \sigma_{-t}. \quad (2.101)$$

A family of maps satisfying 2.101 is called a *one-parameter group of diffeomorphisms*, and the last of the three identities is the one we care about most, because it says that every σ_t has a smooth inverse and is therefore a diffeomorphism. By the discussion of the previous subsection we may consequently transport any tensor whatsoever along the flow, in either direction, and we did not have to assume it: the invertibility came for free from the group property, which in turn came for free from the uniqueness of integral curves.

It is worth writing down what the flow looks like to first order in t , because that is the only thing we shall ever need. Expanding the integral curve in a Taylor series about $t = 0$ and using the fact that its tangent is X ,

$$\sigma_t^\mu(x) = x^\mu + t X^\mu(x) + \mathcal{O}(t^2), \quad (2.102)$$

so that for infinitesimal t the flow is an infinitesimal displacement in the direction of X , and the vector field can be regarded as the generator of the flow in exactly the sense in which, in quantum mechanics, the momentum operator generates translations. The parallel is not a loose analogy: the commutator 2.62 of two vector fields is the Lie bracket of the corresponding generators, and the algebra $\mathfrak{so}(3)$ we found on the sphere is the same algebra the angular momentum operators obey.

2.3.3 The Lie derivative

We are finally in a position to say what it means to differentiate a tensor field, and to see why the obstruction of the previous section is genuinely overcome rather than swept aside. The difficulty was never the differentiation itself, it was the subtraction. To compute a rate of change at p we must compare the value of the field at p with its value at a nearby point q , and those two values live in different vector spaces, $T_p M$ and $T_q M$, between which no canonical identification exists. Subtracting them is not merely inaccurate, it is meaningless, in the same way that subtracting a vector in one tangent plane of a sphere from a vector in another is meaningless until someone tells us how to compare them.

The flow tells us how. If we take the value of the tensor at the displaced point $\sigma_t(p)$ and drag it back to p with the flow, using $(\sigma_{-t})_* = (\sigma_t)^*$, then both objects sit at the same point, both belong to the same vector space, and the subtraction becomes legitimate. Dividing by t and letting t tend to zero gives a derivative.

Definition 16 Let X be a vector field on M with flow σ_t , and let T be a tensor field of any type. The Lie derivative of T along X is the tensor field defined at each point p by

$$(\mathcal{L}_X T)|_p = \lim_{t \rightarrow 0} \frac{[(\sigma_{-t})_* T]_{\sigma_t(p)} - T|_p}{t} = \frac{d}{dt} [(\sigma_{-t})_* T] \Big|_{t=0}. \quad (2.103)$$

Three features of the definition 2.103 deserve emphasis before we compute anything with it. The first is that it uses nothing but the smooth structure of the manifold. There is no metric in it, no connection, no basis and no coordinates, only the flow of a vector field, which every vector field has. The Lie derivative is therefore available on any manifold whatsoever, and in this precise sense it is the most primitive derivative in differential geometry. The second is that the result is a tensor of the same type as T , because pushing forward preserves type and the limit of tensors of a given type is a tensor of that type, so the objection that sank the partial derivative simply does not arise. The third is that $\mathcal{L}_X T$ at p depends on the field X in a whole neighbourhood of p and not only on its value there, since the flow through nearby points is what does the dragging. We shall see in chapter 3 that this apparently technical remark is exactly what separates the Lie derivative from the covariant derivative.

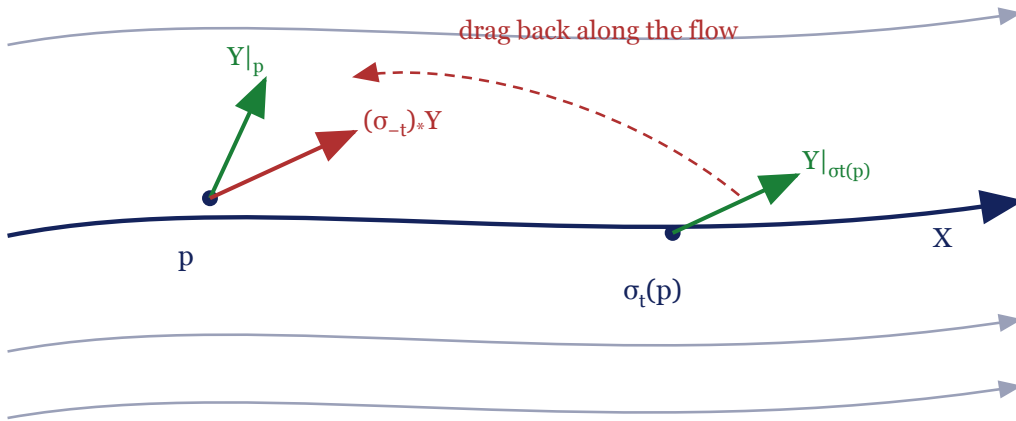


Figure 2.10: The construction behind definition 2.103. The value of the field at p and its value at $\sigma_t(p)$ live in different tangent spaces and cannot be subtracted. Dragging the second one back to p along the flow puts both in $T_p M$, and the difference, divided by t , has a limit as t tends to zero.

Let us now extract the component formulas, starting with the simplest case. For a scalar field there is nothing to drag, since the push forward of a function is just composition, and

$$\mathcal{L}_X f|_p = \lim_{t \rightarrow 0} \frac{f(\sigma_t(p)) - f(p)}{t} = \frac{d}{dt} f(\sigma_t(p)) \Big|_{t=0},$$

which is by definition the derivative of f along the integral curve of X , that is,

$$\mathcal{L}_X f = X(f) = X^\mu \partial_\mu f. \quad (2.104)$$

The case of a vector field is the one that does the real work, and we do it in full. Write everything in a chart and use 2.102. The Jacobian of the map σ_{-t} , which is what 2.94 requires, is obtained by differentiating $\sigma_{-t}^\mu(x) = x^\mu - tX^\mu(x) + \mathcal{O}(t^2)$,

$$\frac{\partial \sigma_{-t}^\mu}{\partial x^\nu} = \delta_\nu^\mu - t \partial_\nu X^\mu + \mathcal{O}(t^2),$$

so that dragging the vector Y from the point $\sigma_t(p)$ back to p gives, to first order,

$$[(\sigma_{-t})_* Y]^\mu \Big|_p = (\delta_\nu^\mu - t \partial_\nu X^\mu) Y^\nu(\sigma_t(p)) + \mathcal{O}(t^2).$$

The components appearing on the right are still evaluated at the displaced point, so we must expand them about p as well, again with 2.102,

$$Y^\nu(\sigma_t(p)) = Y^\nu(p) + t X^\rho \partial_\rho Y^\nu + \mathcal{O}(t^2),$$

and multiplying the two expansions while discarding everything of order t^2 ,

$$\begin{aligned} [(\sigma_{-t})_* Y]^\mu \Big|_p &= (\delta_\nu^\mu - t \partial_\nu X^\mu) (Y^\nu + t X^\rho \partial_\rho Y^\nu) + \mathcal{O}(t^2) \\ &= \textcolor{red}{Y}^\mu + t X^\rho \partial_\rho Y^\mu - t Y^\nu \partial_\nu X^\mu - \textcolor{blue}{t^2 X^\rho \partial_\nu X^\mu \partial_\rho Y^\nu} + \mathcal{O}(t^2), \end{aligned}$$

where the term in blue is of second order and joins the remainder. Substituting this into the definition 2.103, the term in red cancels against $Y^\mu(p)$, which is precisely what had to happen if the limit was to exist at all, and dividing by t leaves

$$(\mathcal{L}_X Y)^\mu = X^\nu \partial_\nu Y^\mu - Y^\nu \partial_\nu X^\mu, \quad (2.105)$$

which the reader will recognise at once. Comparing 2.105 with the component formula 2.62 for the commutator we obtain a result which is one of the small miracles of the subject.

Theorem 5 *The Lie derivative of a vector field Y along a vector field X is their commutator,*

$$\mathcal{L}_X Y = [X, Y]. \quad (2.106)$$

Equation 2.106 closes a circle that has been open since we introduced the commutator, where it appeared as a purely algebraic repair job, the antisymmetric combination invented to cancel the second derivatives that spoiled the Leibniz rule. It now turns out that this artificial looking object was a derivative all along, and the geometrical meaning of the commutator is the one the definition supplies: $[X, Y]$ measures the failure of the field Y to be carried into itself by the flow of X . In particular the criterion 2.63 for a coordinate basis acquires a new reading, since $[\partial_\mu, \partial_\nu] = 0$ now says that each coordinate vector field is dragged into itself by the flow of every other, which is exactly what one would expect of a rigid grid.

The remaining cases follow from 2.104 and 2.105 without any further use of the definition, and the trick is the standard one of applying the Leibniz rule to a scalar. Given a one-form ω and an arbitrary vector field Y , the contraction $\omega(Y) = \omega_\mu Y^\mu$ is a scalar, so on the one hand 2.104 gives

$$\mathcal{L}_X (\omega_\mu Y^\mu) = X^\nu \partial_\nu (\omega_\mu Y^\mu) = X^\nu Y^\mu \partial_\nu \omega_\mu + X^\nu \omega_\mu \partial_\nu Y^\mu,$$

while on the other the Leibniz rule demands

$$\mathcal{L}_X (\omega_\mu Y^\mu) = (\mathcal{L}_X \omega)_\mu Y^\mu + \omega_\mu (\mathcal{L}_X Y)^\mu = (\mathcal{L}_X \omega)_\mu Y^\mu + \omega_\mu X^\nu \partial_\nu Y^\mu - \omega_\mu Y^\nu \partial_\nu X^\mu.$$

Setting the two expressions equal, the terms in red cancel against their counterparts, and after relabelling the dummy indices of the last term so that it carries a free Y^μ we are left with an identity that holds for every Y , whence

$$(\mathcal{L}_X \omega)_\mu = X^\nu \partial_\nu \omega_\mu + \omega_\nu \partial_\mu X^\nu. \quad (2.107)$$

The sign of the second term in 2.107 is opposite to the one in 2.105, and that is not a slip: it is the same reversal we met when vectors pushed forward and covectors pulled back, appearing now at the level of derivatives. Repeating the argument slot by slot gives the general formula, one term with a minus sign for each contravariant index and one with a plus sign for each covariant one,

$$\mathcal{L}_X T^{\mu_1 \dots \mu_k}_{\nu_1 \dots \nu_l} = X^\rho \partial_\rho T^{\mu_1 \dots \mu_k}_{\nu_1 \dots \nu_l} - \sum_{i=1}^k T^{\mu_1 \dots \rho \dots \mu_k}_{\nu_1 \dots \nu_l} \partial_\rho X^{\mu_i} + \sum_{j=1}^l T^{\mu_1 \dots \mu_k}_{\nu_1 \dots \rho \dots \nu_l} \partial_{\nu_j} X^\rho. \quad (2.108)$$

The properties of the operator are now easy to collect, and each of them is either immediate from 2.108 or was built into the definition from the start. The Lie derivative is linear over the real numbers and obeys the Leibniz rule with respect to the tensor product, it maps tensors of type $\binom{k}{l}$ to tensors of the same type, and it commutes with contraction, which is what allowed the argument that produced 2.107. It is linear over the real numbers in the field X as well, but, and this is the property to remember, it is *not* linear over functions in X ,

$$\mathcal{L}_{fX}Y = f \mathcal{L}_X Y - Y(f) X, \quad (2.109)$$

for the reason we gave when the definition was introduced, namely that the dragging depends on the flow in a neighbourhood and not merely on the arrow at the point. Finally, the operators built from different fields do not commute among themselves, and their failure to do so is itself a Lie derivative,

$$[\mathcal{L}_X, \mathcal{L}_Y] = \mathcal{L}_{[X, Y]}, \quad (2.110)$$

which for vector fields is nothing but the Jacobi identity written in a different notation, and which says that the map sending a vector field to its Lie derivative is a homomorphism of Lie algebras. The set of vector fields on a manifold, equipped with the commutator, is an infinite dimensional Lie algebra, and equation 2.110 says that this algebra acts on tensor fields by Lie differentiation. That is the structural statement behind every symmetry argument in the rest of these notes.

The interpretation that matters for physics is the vanishing case. If $\mathcal{L}_X T = 0$ then the tensor T is carried into itself by the flow of X , which is to say that sliding the whole manifold along X leaves T exactly as it was, and T is said to be invariant under the flow. When T happens to be the metric, that condition is the statement that the geometry itself has a symmetry, and the vector field generating it will be called a Killing field. We cannot pursue that here, because we have no metric yet and shall not have one until the next chapter, but it is worth knowing in advance that this is what the whole construction has been for.

Chapter 3

The Theory of General Relativity

*Gravity is not a force reaching across space, it is
the shape of space itself.*

“Spacetime tells matter how to move; matter tells spacetime how to curve.”

– John Archibald Wheeler

3.1 Conventions and notation

Before anything else it is worth fixing, once and for all, the conventions that this chapter and the rest of the book are going to use, because general relativity is a subject in which a good half of the confusion a student suffers comes not from the physics at all but from the plain fact that two books sitting side by side on the same shelf will write the same equation with opposite signs. Nothing that follows in this section is deeper than a choice, and a different choice would be just as legitimate, but once the choice is made we shall keep to it everywhere.

We work throughout with the *mostly plus* signature, so that the Minkowski metric reads $\eta_{\mu\nu} = \text{diag}(-1, 1, 1, 1)$ and the interval of special relativity is $ds^2 = -C^2 dt^2 + dx^2 + dy^2 + dz^2$. With this choice a time-like vector has negative norm, a space-like vector has positive norm, a null vector has zero norm, and the four velocity of any observer satisfies $u^\mu u_\mu = -C^2$, which becomes simply $u^\mu u_\mu = -1$ whenever we set $C = 1$. The opposite convention, mostly minus, is equally respectable and is the one most particle physicists prefer, and the only thing that really matters is never to mix the two in the middle of the same calculation.

Greek indices μ, ν, ρ, σ and so on run over the four space-time coordinates 0, 1, 2, 3, Latin indices taken from the middle of the alphabet, i, j, k , run over the three spatial ones 1, 2, 3, and the first Latin letters a, b, c, d are used as abstract indices, which is to say as labels telling us what kind of object we are handling without committing us to any particular chart. A repeated index, one upstairs and one downstairs, is always summed over without writing the summation sign, which is the Einstein summation convention we have been using since chapter 2, and an index is raised or lowered with the metric, $V^\mu = g^{\mu\nu} V_\nu$, an operation that only makes sense because the metric is non degenerate.

Most of the time we shall work in *geometrised units*, in which $G = C = 1$, because they remove from every equation a forest of constants that carry no physical information and only obscure the geometry. In these units masses, lengths and times are all measured in the same unit, and the conversion back is made by restoring the factors of G and C that dimensional analysis demands, which we shall do explicitly whenever a number is to be compared with an experiment. Table 3.1 collects the remaining pieces of notation.

Notation	Meaning
$\eta_{\mu\nu} = \text{diag}(-1, 1, 1, 1)$	Minkowski metric, mostly plus signature
$g_{\mu\nu}, g$	metric tensor and the determinant of its matrix of components
$\partial_\mu f, f_{,\mu}$	partial derivative with respect to x^μ
$\vec{\nabla}_\mu V^\nu, V^\nu_{;\mu}$	covariant derivative
$\Gamma^\lambda_{\mu\nu}$	Christoffel symbols of the Levi-Civita connection
$T_{(\mu\nu)}, T_{[\mu\nu]}$	symmetric and antisymmetric parts, as defined in chapter 2
$\square = \vec{\nabla}^a \vec{\nabla}_a$	the d'Alembertian, which reduces to $\partial^\alpha \partial_\alpha$ in flat space
$\bar{T}_{\mu\nu}$	the trace reversed version of $T_{\mu\nu}$
$d\Omega^2 = d\vartheta^2 + \sin^2 \vartheta d\varphi^2$	the metric of the unit two sphere

Table 3.1: Notation used throughout this chapter and the two that follow.

One last convention is not mathematical but typographical, and it concerns the long calculations that fill the second half of this chapter. In every derivation carried out step by step we colour in **red** those terms which are about to cancel against one another or to simplify away, and in **blue** those terms or identities which have been deliberately inserted in order to make the next step possible, for instance a factor that is added and subtracted at the same time. The intention is that a reader following a derivation with a pencil should be able to see at a glance where each line came from, since nothing is more discouraging than an algebraic chain in which two consecutive lines differ by a step the author considered too obvious to mention.

3.2 Introduction to general relativity

At the heart of Albert Einstein's general theory of relativity lies a profound shift in our understanding of gravity and space-time. In this section, we'll delve into the qualitative aspects of general relativity and draw comparisons with its predecessor, special relativity.

One of the central tenets of general relativity is the idea that massive objects, like planets and stars, bend the fabric of space-time around them. This bending, often depicted as a curved surface, is what we perceive as gravity. Imagine placing a heavy ball on a stretched rubber sheet. The sheet curves around the ball, and smaller objects nearby naturally roll toward it. Similarly, massive objects create curves in space-time, influencing the motion of other objects in their vicinity.

This insight was a monumental departure from the classical concept of gravity as a force of attraction between masses. Rather than envisioning a force acting at a distance, general relativity describes gravity as a consequence of the geometry of space-time itself.

Special relativity, introduced by Einstein in 1905, focuses on the relationship between space and time, particularly in situations involving relative motion. It famously established the constancy of the speed of light, which led to phenomena like time dilation and length contraction. Special relativity was designed to address situations where velocities approach the speed of light and encompassed the realm of "inertial frames", frames that move at a constant velocity relative to one another.

General relativity extends this framework to non-inertial frames, those undergoing acceleration or influenced by gravitational fields. It unifies gravity with the fabric of space-time, encompassing all motion, including accelerated motion, within its purview. While special relativity deals with flat space-time and relative motion, general relativity introduces the dynamic curvature of space-time itself as a response to mass and energy.

One of the remarkable consequences of general relativity is its application to the cosmological scale. The curvature of space-time influences the trajectory of light, causing gravitational lensing, the bending of light around massive objects. This effect was confirmed during the 1919 solar eclipse observations, where stars near the Sun appeared slightly shifted due to the bending of their light

around the Sun's gravitational field.

General relativity also provides the foundation for understanding black holes, where the gravitational curvature becomes so intense that not even light can escape, leading to the concept of an event horizon. These celestial phenomena have since become crucial tools for studying the universe's most extreme environments and for testing the boundaries of our understanding of physics.

In summary, general relativity profoundly transformed our understanding of gravity by revealing it as the result of the curvature of space-time caused by mass and energy. This departure from the classical Newtonian view revolutionized our comprehension of the cosmos, offering a unified description of gravity and motion on all scales.

While special relativity opened the door to the intertwining of space and time and addressed relative motion in inertial frames, general relativity extended this framework to include accelerated motion and the gravitational influence of massive objects. Together, these theories form the foundation of modern physics, guiding our exploration of the universe's inner workings and shaping our technological advancements.

3.2.1 The incompatibility between the Newtonian gravity with the special relativity

The Newton's universal law of gravity was one of the biggest hits in the history of physics, this is because he could explain almost all the gravitational phenomena in the planet (or any other) and in the solar system, with this theory Edmund Halley could also predict the trajectory of a far asteroid, the well known Halley's comet. The Newton's theory failed with one special planet in the solar system, Mercury, no one could explain the slow precession of the perihelion of this planet using this theory, it was Albert Einstein with his theory of general relativity who could solve this problem and generalize to high gravitational fields and also discovered that his theory of special relativity is incompatible with the Newton's theory.

Lets show that. The SR (special relativity from now on) has preferred observers ¹ this are known as *inertial observers* when we associate then with a set of four coordinates (t, x, y, z) , this becomes an inertial reference frame. The different inertial frames are all related by Lorentz transformations, with this we can enunciate the *principle of special relativity*.

Principle 1 *Principle of special relativity:* *The laws of physics must take the same form in every inertial frame.*

This principle talks about the mathematical forms of the laws of physics this means that no matter how you transform the reference frame, the coordinate system, the mathematical form of the

¹here we must clarify that a preferred observer is not a coordinate chart, an observer is related to world trajectories, therefore the observers are related more to a geometrical objects.

laws must be the same, there is no physical interest in a mathematical formulation from a physical law which has a different form for every inertial frame.

Lets analyze the Newton's laws to see like this the compatibility between this and the SR

$$\nabla^2 \phi = 4\pi G \rho \quad (3.1)$$

where ϕ is the gravitational potential, ρ is the mass density, and G is the gravitational or Cavendish's constant. Due to the Lorentz transformations the spatial and temporal coordinates mix each other it is obvious that this equations won't be longer a Poisson equation under a Lorentz transformation, this implies that *the Newton laws are incompatible with the principle of special relativity*. So now lets analyze the behaviour of the solutions of the equation 3.1, these have the form,

$$\phi(t, \vec{x}) = -G \int \frac{\rho(t, \vec{x}')}{|\vec{x} - \vec{x}'|} d^3 x', \quad (3.2)$$

solutions of this form also violate the principle of SR, because solutions of this form have instantaneous reactions which also violate the principles because *two simultaneous events (spatially separated) in an inertial reference frame, wont be simultaneous en all the other reference frames*.

It is worth asking when special relativity and Newtonian gravity are compatible, we will prove that this compatibility is given under low velocities, when we talk about low velocities we must say respect to what, from now on in the context of the relativity theory usually is respect to the speed of light C , so when $v \ll C$ like in the solar system. All the planets in our solar system travel, in a relatively good approximation, in circular trajectories so we can use the Newton's law and the gravitation law to relate the potential with the speed, the gravitational potential is

$$\phi = -\frac{MG}{|\vec{r}|} \quad (3.3)$$

where M is the sun's mass, and $\vec{r}$ is the positions vector from the sun to the planet, if the movement is perfectly circular there is a centripetal acceleration, thus

$$\frac{GM}{|\vec{r}|^2} = \frac{v^2}{|\vec{r}|} \rightarrow \frac{v^2}{C^2} = \frac{|\phi|}{C^2}. \quad (3.4)$$

Here we can see the direct relation there is between the gravitational potential with the relativistic effects in gravitational fields, intense gravitational fields imply relativistic effects, this also implies, yeh again, *the Newton theory is incompatible with relativity*. When $|\phi|/C^2 \ll 1$ we have that the Newton's law are compatible. There is a certain dimensionless constant that tell us if an object is enough to produce relativistic effects, this is called the compactness c defined by:

$$c = \frac{GM}{C^2 r}. \quad (3.5)$$

This constant for the sun has a values of the order of $\approx 10^{-6}$ and for a neutron star of ≈ 0.2 , thus in the vicinity of a Neutron star the relativistic effects will be really important in contrast with our sun.

3.2.2 Weak equivalence principle

In the Newtonian theory there is a difference between two concepts, the *inertial mass* and the *gravitational mass*, m_I, m_g respectively, the distinctions between these two is really important, because in the common language, they're the same, but we will see that the fact they're numerically the same implies a principle and they are not the same concepts, physically imply two different things.

- The inertial mass is related with the second law of Newton, this mass is an inertia factor which tell us how hard would be to move an object and appears mathematically as

$$\vec{F} = m_I \vec{a}. \quad (3.6)$$

- The gravitational mass is a factor which tells us how an object interacts with a gravitational field it is analogous to charge in the electromagnetism and appears mathematically as

$$\vec{F} = m_g \vec{g}. \quad (3.7)$$

To see how they are related, let's consider a particle falling freely in a gravitational field, thus

$$\vec{a} = \frac{\vec{F}}{m_I} = \frac{m_g}{m_I} \vec{g}. \quad (3.8)$$

It was Galileo, with his so famous experiments in the Pisa tower, who postulate *the relation between the gravitational and inertial mass m_g/m_I are the same for all the materials*. Without any loss of generality, we're saying that in magnitude the gravitational mass and the inertial mass are equal. Later Eötvös proved experimentally that

$$\frac{m_g}{m_I} - 1 \approx \mathcal{O}(10^{-12}). \quad (3.9)$$

This separates gravity from all the other three forces, because all the other forces interact with matter depending on their compositions but this not, the main objective nowadays is to try to violate the relations 3.9 and we can say that *the equality between m_I and m_g for all the*

bodies is a form of the weak equivalence principle.

The relation 3.9, this principle, in fact has a really heavy implication mathematically, because we start from a movement equation like,

$$m_I \ddot{\vec{x}} = m_g \vec{g}[\vec{x}(t), t], \quad (3.10)$$

but using 3.9 we then get

$$\ddot{\vec{x}} = \vec{g}[\vec{x}(t), t]. \quad (3.11)$$

Solutions of this equations are determined uniquely by the initial position and speed of the particle. This implies that **two different particle with the same position and speed will have the same trajectories**. Without this relation this wont be any longer true, there is the strength and beauty of this principle. Lets now formulate this principle in this other, but physically equal, form:

Principle 2 Weak equivalence principle: *The trajectory of a test body falling freely depends only on its initial position and speed and it is independent of its chemical composition.*

We must remember that this is enunciate to test bodies, this kind of test bodies have got auto-gravitational effects negligible respect to the field of the sources and this is not extended, it is almost a particle respect to the field size, this to avoid effects like torques. Lets analyze the mechanical behavior of a reference frame that fulfill the transformations of Galileo's group.

$$t' = t, \quad (3.12)$$

$$\vec{X}' = \vec{X} - \vec{Y}(t), \quad (3.13)$$

$\vec{Y}(t)$ is related with the origin of the new reference frame. And we will set $\ddot{\vec{Y}}(t) = \vec{a}$ so the acceleration of the new reference frame is given by the Galileo's relativity theory by

$$\ddot{\vec{X}}' = \vec{g} - \vec{a} = \vec{g}'. \quad (3.14)$$

Here we can see the acceleration of the new reference frame can be interpreted as a measure of the gravity acceleration from this reference frame, lets analyze some special cases of the equation 3.14.

1. $\vec{g} = 0$ which implies that $\vec{g}' = -\vec{a}$ we can see that *the uniform acceleration is indistinguishable from an uniform gravitational acceleration*.

2. $\vec{g} = \text{constant}$ & $\vec{a} = \vec{g}$. For this kind of situations we can then find an inertial reference frame associated to the free fall, in this reference frame the gravitational effects are zero. This is equivalent to be next to a falling object like a paratrooper.

If we haven't got uniformity, then we will see that *each coordinate chart of the inertial reference frame can be such that this is associated to a geometrical manifold, such manifolds will allow us to define neighborhoods which are locally Minkowskian.*

3.2.3 Experimental evidence for a general relativity theory

Many physics students feel attraction to the theory of general relativity and gravitation because the mathematical structure behind this theory which by its own is really abstract but with this theory it takes a beautiful physical interpretation but many people forget how this theory was born.

Is the experimentation, the experiences we live everyday what allow physicist do physics, because of that there are some experiments which cannot be explained with the special relativity theory of which are the first steps in the search of a general theory of relativity, here we're going to check some of those experiments and analyze their physical information.

3.2.3.1 Eötvös experiment

The Hungarian Baron Loránd Eötvös was a mathematician, engineer and physicist who made an amazing experiment to test the equivalence principle, here we're going to analyze his experiment. Lets suppose a pair of objects to which a force is applied

$$\vec{F}^{(1)} = m_I^{(1)} \vec{a}^{(1)} = m_g^{(1)} \vec{g}, \quad (3.15)$$

$$\vec{F}^{(2)} = m_I^{(2)} \vec{a}^{(2)} = m_g^{(2)} \vec{g}. \quad (3.16)$$

He, in his experiment, postulate that the acceleration of the two objects aren't the same

$$\vec{a}^{(2)} = \frac{m_g^{(2)}}{m_I^{(2)}} \vec{g} \neq \frac{m_g^{(1)}}{m_I^{(1)}} \vec{g} = \vec{a}^{(1)}. \quad (3.17)$$

We're going to have on count the gravitational force

$$\vec{F}_g = -GM_O M_G \frac{\vec{r}}{r^3}, \quad (3.18)$$

and the centrifugal force due to the relative rotational motion between the two forces with angular frequency $\vec{\omega}$, see figure 3.1.

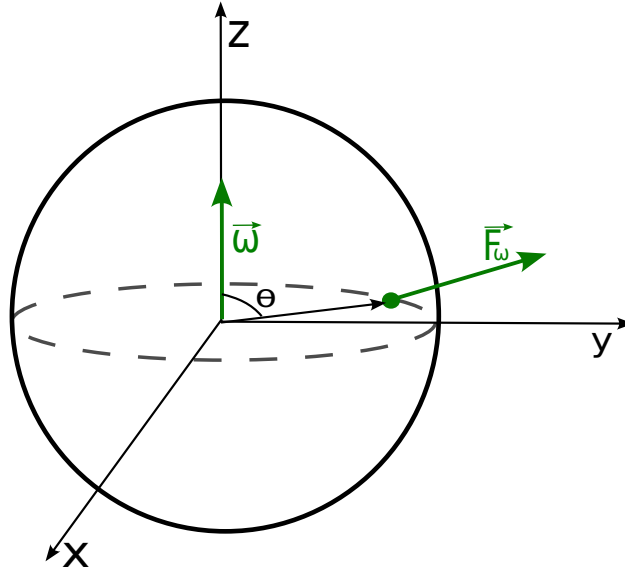


Figure 3.1: Graphical representation of the relative movement between the two test bodies.

$$\vec{F}_\omega = m_I \omega^2 \vec{r}_{axis}, \quad (3.19)$$

$$\vec{r}_{axis} = \vec{r} - \frac{(\vec{\omega} \cdot \vec{r})}{\omega^2} \vec{\omega}, \quad (3.20)$$

$$\vec{F}_\omega = -m_I [\vec{\omega} \times (\vec{\omega} \times \vec{r})]. \quad (3.21)$$

Then the forces balance is

$$\vec{F}^{(i)} = \vec{F}_g^{(i)} + \vec{F}_\omega^{(i)}.$$

And lets define a parameter α which relates the strength relation between the gravitational and centrifugal force as

$$\alpha = \frac{|\vec{F}^{(1)} \times \vec{F}^{(2)}|}{|\vec{F}^{(1)}| |\vec{F}^{(2)}|}, \quad (3.22)$$

if $\alpha = 0$ we'll have that the forces are parallel. In magnitude in the earth the gravitational force is stronger than the centrifugal because

$$\frac{GM_O/|\vec{r}|^2}{\omega^2 |\vec{r}|} = \frac{GM_O}{\omega^2 |\vec{r}|^3} \approx 300.$$

Lets calculate the parameter α so, first we're going to calculate the numerator

$$\begin{aligned}
 \vec{F}^{(1)} \times \vec{F}^{(2)} &= \left(-GM_O m_g^{(1)} \frac{\vec{r}}{r^3} + m_I^{(1)} [\vec{\omega} \times (\vec{\omega} \times \vec{r})] \right) \times \left(-GM_O m_g^{(2)} \frac{\vec{r}}{r^3} + m_I^{(2)} [\vec{\omega} \times (\vec{\omega} \times \vec{r})] \right) \\
 &= - \left(\frac{GM_O m_g^{(1)} m_I^{(2)}}{r^3} \right) (\vec{r} \times \vec{\omega}) \times (\vec{\omega} \times \vec{r}) - \left(\frac{GM_O m_g^{(2)} m_I^{(1)}}{r^3} \right) [\vec{\omega} \times (\vec{\omega} \times \vec{r})] \times \vec{r} \\
 &= - \frac{GM_O \omega^2}{r^3} \left\{ m_g^{(1)} m_I^{(2)} \vec{r} \times \left[\vec{r} - \frac{(\vec{\omega} \cdot \vec{r}) \vec{\omega}}{\omega^2} \right] - m_g^{(2)} m_I^{(1)} \left[\vec{r} - \frac{(\vec{\omega} \cdot \vec{r}) \vec{\omega}}{\omega^2} \right] \times \vec{r} \right\} \\
 &= - \frac{GM_O}{r^3} (\vec{r} \times \vec{\omega}) (\vec{\omega} \cdot \vec{r}) [m_g^{(1)} m_I^{(2)} - m_g^{(2)} m_I^{(1)}], \\
 |\vec{F}^{(1)} \times \vec{F}^{(2)}| &= \frac{GM_O}{r^3} |\vec{r} \times \vec{\omega}| (\vec{\omega} \cdot \vec{r}) [m_g^{(1)} m_I^{(2)} - m_g^{(2)} m_I^{(1)}]. \tag{3.23}
 \end{aligned}$$

Now, for the denominator we must calculate each norm, because both have the same form we'll do for one and only change their respective indices

$$\begin{aligned}
 |\vec{F}^{(i)}| &= \left\{ \frac{(GM_O m_g^{(i)})^2}{r^4} + (m_I^{(i)} \omega^2)^2 \left[\vec{r} - \frac{(\vec{\omega} \cdot \vec{r}) \vec{\omega}}{\omega^2} \right] \cdot \left[\vec{r} - \frac{(\vec{\omega} \cdot \vec{r}) \vec{\omega}}{\omega^2} \right] \right. \\
 &\quad \left. - 2GM_O m_g^{(i)} m_I^{(i)} \frac{\omega^2}{r^3} \vec{r} \cdot \left[\vec{r} - \frac{(\vec{\omega} \cdot \vec{r}) \vec{\omega}}{\omega^2} \right] \right\}^{1/2} \\
 &= \left\{ \frac{(GM_O m_g^{(i)})^2}{r^4} + (m_I^{(i)} \omega^2)^2 \left[r^2 - \frac{2(\vec{\omega} \cdot \vec{r})^2}{\omega^2} + \frac{(\vec{\omega} \cdot \vec{r})^2}{\omega^2} \right] \right. \\
 &\quad \left. - 2GM_O m_g^{(i)} m_I^{(i)} \frac{\omega^2}{r^3} \left[r^2 - \frac{(\vec{\omega} \cdot \vec{r})^2}{\omega^2} \right] \right\}^{1/2} \\
 &= \frac{GM_O m_g^{(i)}}{r^2} \left\{ 1 + \left[\frac{m_I^{(i)} \omega^2}{GM_O m_g^{(i)}} \right] \left[r^2 - \frac{(\vec{\omega} \cdot \vec{r})^2}{\omega^2} \right] \left[\frac{m_I^{(i)} r^4 \omega^2}{GM_O m_g^{(i)}} - 2r \right] \right\}^{1/2}.
 \end{aligned}$$

Taking a first order approximation, or what is the same, saying the biggest contribution to the norm of each force is due to the gravitational interaction we then have:

$$|\vec{F}^{(i)}| \approx \frac{GM_O m_g^{(i)}}{r^2}, \tag{3.24}$$

then the parameter α becomes

$$\alpha \approx \left[\frac{m_I^{(2)}}{m_g^{(2)}} - \frac{m_I^{(1)}}{m_g^{(1)}} \right] \frac{r |\vec{r} \times \vec{\omega}| (\vec{\omega} \cdot \vec{r})}{GM_O}, \quad (3.25)$$

$$\alpha \approx \left[\frac{m_I^{(2)}}{m_g^{(2)}} - \frac{m_I^{(1)}}{m_g^{(1)}} \right] \frac{\cos \theta \sin \theta}{300}. \quad (3.26)$$

With the result of the equation 3.26 we can see that this experiment cannot be done by someone standing at the poles nor at the equator, because there $\theta = 0$ and $\theta = \pi/2$ respectively and the product $\cos \theta \sin \theta$ vanishes, and fortunately Eötvös was working in Hungary, at a comfortable middle latitude and he found that α was zero in a part for 10^9 , this means it is zero until the ninth significant number, from there on there are errors, and mathematically from 3.26 we can see that if it is zero this means by the elementary theorem of algebra that

$$\frac{m_I^{(1)}}{m_g^{(1)}} = \frac{m_I^{(2)}}{m_g^{(2)}}. \quad (3.27)$$

3.2.3.2 Tidal forces

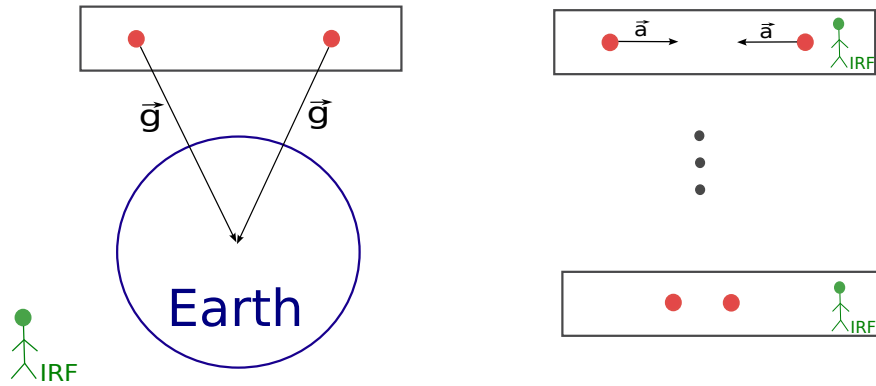


Figure 3.2: In the left there are two particles which are in a free fall to the earth and the inertial reference frame (IRF) is to the side of the earth, he observes a diagonal trajectory of the bodies. In the right we set the IRF inside the box with the two bodies, and he observes that there is a strange force which makes the bodies get together.

In the figure 3.2 we can see an scheme of the effect of the tidal forces on two particles inside a box. This box is in a free fall to the earth, an observer with an inertial reference frame next to the earth will observe a kind of diagonal trajectory for both particles towards the center of the earth but if we place our inertial reference frame inside the box we will see that both particles will join together why? In the Newtonian mechanics we would say that there is a fictitious force² that makes the

²Newton never used the term fictitious force, that term was introduced by the Mach's formalism for the mechanics.

objects attract each other but in fact, there is no imaginary force, there is a non homogeneous gravitational field, so *the tidal forces are an effect of the non homogeneity of the gravitational field, these are physical and important forces.*

3.2.3.3 Light deflection and gravitational red shift

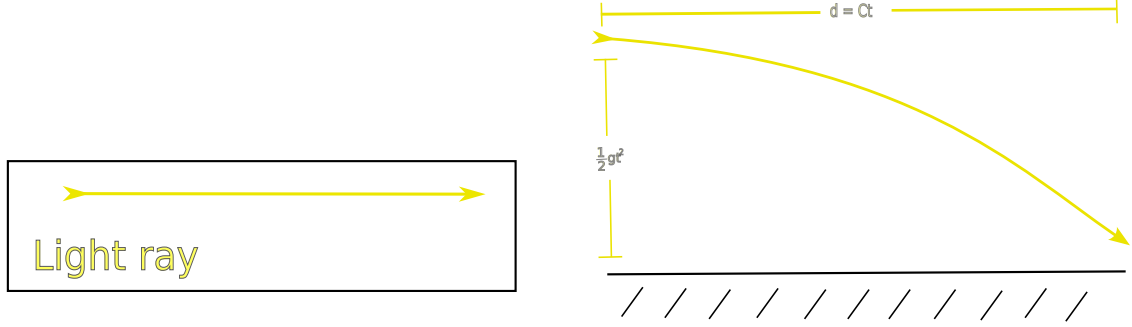


Figure 3.3: In the left there is a light ray seeing from an inertial reference frame inside the box, and in the right there is the same ray but seeing from the earth.

In the figure 3.3 we can see that even light is affected by gravity, in the Newtonian theory this was something impossible to imagine!, we need then a theory to describe this effect.

There are some other effects of the gravity on the light, like the color shift, this is a gravitational Doppler effect, so, lets suppose two observers Alice (A) and Bob (B), Alice is in a free fall at a distance h from Bob and sends a light pulse, then Bob gets it and in the moment he gets it Alice sends another beam what is the relation between the Alice and bob interval times?. To solve this we will suppose the gravity is uniform and we will set a reference frame such that $\vec{g} = \vec{a}$ so for Alice

$$Z_A = h + \frac{1}{2}gt^2 \quad (3.28)$$

$$Z_B = \frac{1}{2}gt^2. \quad (3.29)$$

Alice sends the first signal at a time t_1 then the position of the beam Z is

$$Z = Z_A(t_1) - C(t - t_1) = h + \frac{1}{2}gt_1^2 - C(t - t_1), \quad (3.30)$$

it arrives to Bob at $t = T_1$ then

$$h + \frac{1}{2}gt_1^2 - C(T_1 - t_1) = \frac{1}{2}gT_1^2. \quad (3.31)$$

Alice sends the second signal at $t = t_1 + \Delta\tau_A$ then

$$Z = Z_A(t_1 + \Delta\tau_A) - C(t - t_1 - \Delta\tau_A). \quad (3.32)$$

And it lands on Bob at $t = T_1 + \Delta\tau_B$ thus

$$h + \frac{1}{2}g(t_1 + \Delta\tau_A)^2 - C(T_1 + \Delta\tau_B - t_1 - \Delta\tau_A) = \frac{1}{2}g(T_1 + \Delta\tau_B)^2 \quad (3.33)$$

Lest expand 3.33 then

$$h + \frac{1}{2}g(t_1 + \Delta\tau_A)^2 - C(T_1 + \Delta\tau_B - t_1 - \Delta\tau_A) = \frac{1}{2}g(T_1 + \Delta\tau_B)^2, \quad (3.34)$$

$$h + \frac{1}{2}g(t_1^2 + 2t_1\Delta\tau_A + \Delta\tau_A^2) - C(T_1 + \Delta\tau_B - t_1 - \Delta\tau_A) = \frac{1}{2}g(T_1^2 + 2T_1\Delta\tau_B + \Delta\tau_B^2). \quad (3.35)$$

Using the equation 3.31 in the right hand side of the last equation and simplifying the h and $\frac{1}{2}gt_1^2$ then we have

$$\begin{aligned} gt_1\Delta\tau_A + \frac{1}{2}g\Delta\tau_A^2 - C(\Delta\tau_B - \Delta\tau_A) &= gT_1\Delta\tau_B + \frac{1}{2}g\Delta\tau_B^2, \\ \Delta\tau_A(gt_1 + C) + \frac{1}{2}g\Delta\tau_A^2 &= \Delta\tau_B(gT_1 + C) + \frac{1}{2}g\Delta\tau_B^2. \end{aligned}$$

Dividing all by C and remembering that in Earth $g\Delta\tau_i \ll C$ then we have

$$\Delta\tau_A \left(1 + \frac{gt_1}{C}\right) = \Delta\tau_B \left(1 + \frac{gT_1}{C}\right) \quad (3.36)$$

$$\Delta\tau_B = \left(1 + \frac{gt_1}{C}\right) \left(1 + \frac{gT_1}{C}\right)^{-1} \Delta\tau_A, \quad (3.37)$$

now using the geometric series $\frac{1}{1+x} = 1 - x + x^2 - x^3 + \dots$ to first order because, in the Earth $gt_i \leq C$ thus

$$\Delta\tau_B \approx \left[1 + \frac{gt_1}{C} - \frac{gT_1}{C} - g^2 \frac{t_1 T_1}{C^2}\right] \Delta\tau_A \approx \left[1 - \frac{g}{C}(T_1 - t_1)\right] \Delta\tau_A. \quad (3.38)$$

The times seems to run slower in B than in A we can conclude then that *the time runs slower where the gravitational field is stronger*. Here we can also see two things, the first one is that the compactness appears again because

$$\Delta\tau_B \approx \left[1 - \frac{gh}{C^2}\right] \Delta\tau_A = \left[1 + \frac{\phi_B - \phi_A}{C^2}\right] \Delta\tau_A, \quad (3.39)$$

then we see the compactness appears and tells us how fast will run the time of an observer. The other thing we can see is a blue shift because using the relation between time and frequency $\Delta\tau_i = \frac{\lambda_i}{C}$ we then have

$$\lambda_B \approx \left(1 - \frac{gh}{C^2}\right) \lambda_A \quad (3.40)$$

and because the correction inside the parenthesis is negative the wavelength measured by Bob is always the shorter one, so there will be a blue shift, or what is the same a Doppler effect when the ambulance (Alice with her laser) comes closer to Bob.

3.2.4 Einstein's equivalence principle and the space-time

Einstein introduced the space-time structure to his theory, this to make a coupled theory of space and time, in the Newtonian theory the space is absolute and the time is absolute but not together, this was the Einstein's contribution to physics, neither time nor space are absolute separately but joint they are, such a structure has a special geometry, called a pseudo-Riemannian geometry, this because in these geometries there are negative and positive or even null vector modulus, their physical interpretation will be discussed later. A space-time diagram is a representation of the physical events, for example for the blue shift problem the diagram is represented in the figure 3.4.

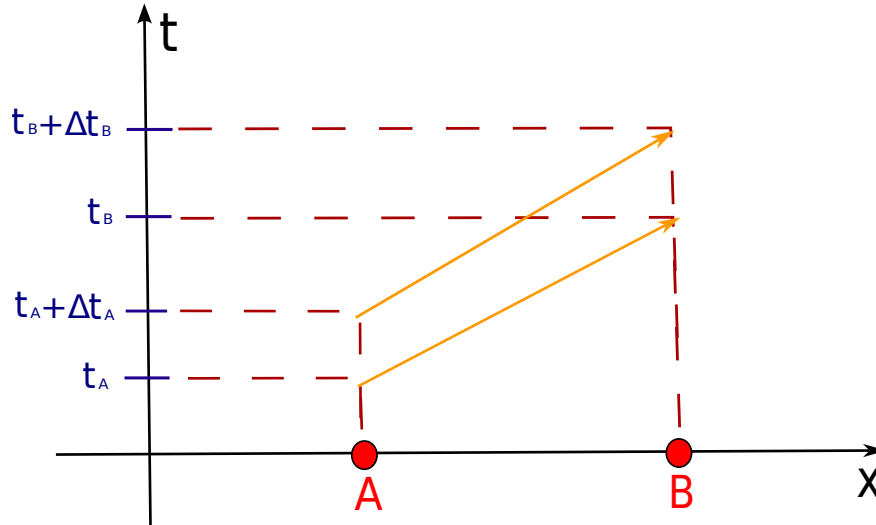


Figure 3.4: Space-time diagram representation for the Doppler effect analyzed before.

Lets study this, in a Minkowskian structure we have got the following relation

$$C^2 d\tau^2 = C^2 dt^2 - dx^2 - dy^2 - dz^2, \quad (3.41)$$

but lets see what happens if we modify it a bit like this

$$C^2 d\tau^2 = \left(1 + \frac{2\phi}{C^2}\right) C^2 dt^2 - \left(1 - \frac{2\phi}{C^2}\right) (dx^2 + dy^2 + dz^2), \quad (3.42)$$

as we can see for Alice and Bob there are no physical displacement, only temporal, so their proper times are, using the equation 3.42 are

$$\Delta\tau_A^2 = \left(1 + \frac{2\phi_A}{C^2}\right) \Delta t^2, \quad (3.43)$$

$$\Delta\tau_B^2 = \left(1 + \frac{2\phi_B}{C^2}\right) \Delta t^2. \quad (3.44)$$

Now, taking the square root of both equations and using Taylor expansion to first order we then have

$$\Delta\tau_A \approx \left(1 + \frac{\phi_A}{C^2}\right) \Delta t, \quad (3.45)$$

$$\Delta\tau_B \approx \left(1 + \frac{\phi_B}{C^2}\right) \Delta t. \quad (3.46)$$

It is easy to see that the relation between their proper times is

$$\Delta\tau_B \approx \left(1 + \frac{\phi_B}{C^2}\right) \left(1 + \frac{\phi_A}{C^2}\right)^{-1} \Delta\tau_A \quad (3.47)$$

Using now the geometrical series because in the earth the compactness is really low $\phi/C^2 \ll 1$ finally we then have

$$\Delta\tau_B \approx \left(1 + \frac{\phi_B - \phi_A}{C^2}\right) \Delta\tau_A. \quad (3.48)$$

This is a really strong result because changing the flat geometric structure of the Minkowskian space-time we recover the result we had obtained before for the gravitational blue shift, and this is a direct proof that we need a more geometrical and more general theory of relativity.

3.3 First steps on general relativity

From now on we will move to a more physical territory using all the abstract tools developed by the mathematicians, first we are going to learn how to crawl, then to walk and at the end how to run, because the general relativity is in fact a theory of *the geometry of space and time* the heart of a movement, from a physical point of view, is the notion of position and even more important, the notion of distance, this in mathematics, in a topological space can be defined using a topology on the topological space, here in the same way we're going to define a metric on the manifold, what we will call a metric tensor.

3.3.1 Metric tensor

Lets see how to make the notion of distance over M , for example, lets look at $\mathbb{R}^3$. Let $\vec{X}(t)$ a curve on $\mathbb{R}^3$ and an open interval $a < t < b$ so the arc length L is given by

$$L = \int_a^b ds = \int_a^b \left[\frac{d\vec{X}}{dt} \cdot \frac{d\vec{X}}{dt} \right]^{1/2} dt, \quad (3.49)$$

where $\vec{X}$ is the tangent vector to the curve, we see here rises the notion of inner product of these tangent vectors does it mean that always an inner product give us a notion of distance? The fast answer is yes, but the long answer is no, the notion of inner product is more related to the notion of a topology of the vector space, in our case of the tangent space to a point p on the manifold M .

The notion of distance is a kind of combinations between the notion of topology or inner product with the physical structure of the problem, there are many maps or inner products that satisfies the axioms of a topology or the inner product is the physical meaning what rises the notion of distance, therefore to find such a distance lets define the metric tensor $\mathbb{G}$

$$\mathbb{G} : T_p(M) \times T_p(M) \rightarrow \mathbb{R}, \quad (3.50)$$

and in our notation the application of the metric tensor $\mathbb{G}$ to two vectors is

$$\mathbb{G}(\vec{A}, \vec{B}) = \vec{A} \cdot \vec{B}. \quad (3.51)$$

Definition 17 A metric tensor in $p \in M$ is a tensor $\mathbb{G}$ of type $\binom{0}{2}$, known also as bilinear form, such that satisfies the following properties:

1. $\mathbb{G}(\vec{X}, \vec{Y}) = \mathbb{G}(\vec{Y}, \vec{X}) \forall \vec{X}, \vec{Y} \in T_p(M)$.
2. $\mathbb{G}(\vec{X}, \vec{Y}) = 0 \forall \vec{Y} \in T_p \Leftrightarrow \vec{X} = \vec{0}$ this means the metric tensor is not degenerated.

In differential geometry we're always interested in a metric such that it is Riemannian this means *every entry of the signature is positive*, so that the metric tensor is positive definite and $\mathbb{G}(\vec{X}, \vec{X}) > 0$ for every non zero vector. In general relativity we are interested instead in Lorentzian metrics, those whose signature has the form $- + + +$, and here the metric is no longer positive definite, which is exactly what allows a vector to have zero or negative norm.

Definition 18 *A pseudo-Riemannian manifold is a pair $(M, \mathbb{G})$ where M is a differentiable manifold and $\mathbb{G}$ is a metric tensor field on it. When the metric is positive definite the pair is called a Riemannian manifold, and when its signature is Lorentzian it is called a Lorentzian manifold.*

In general relativity such a pair is known as space-time.

In this way we can calculate a length curve L as

$$L = \int_a^b \left[\mathbb{G}(\vec{X}, \vec{X}) \right]_{\lambda(t)}^{1/2} dt, \quad (3.52)$$

and given a coordinate chart we can write the metric tensor as

$$\mathbb{G} = g_{\alpha\beta} dx^\alpha \otimes dx^\beta. \quad (3.53)$$

Definition 19 *The metric $\mathbb{G}$ is not degenerated, thus this must be invertible. The inverse metric is a symmetric tensor field of type $\binom{2}{0}$ denoted by $\mathbb{G}^{-1}$ such that*

$$g^{ac} g_{cb} = \delta_b^a. \quad (3.54)$$

Definition 20 *The Lorentzian signature is defined such that for any point p in the manifold we can choose an orthonormal basis $\{\mathbf{e}_\alpha\}$ such that*

$$\mathbb{G}(\mathbf{e}_\alpha, \mathbf{e}_\beta) = g_{\alpha\beta} \equiv \text{diag}(-1, 1, 1, 1). \quad (3.55)$$

This basis is not unique and we must set a relation between basis

$$\begin{aligned} \mathbf{e}_{\bar{\alpha}} &= \Lambda_{\bar{\alpha}}^\beta \mathbf{e}_\beta \rightarrow g_{\bar{\alpha}\bar{\beta}} = \mathbb{G}(\mathbf{e}_{\bar{\alpha}}, \mathbf{e}_{\bar{\beta}}) = \mathbb{G}(\Lambda_{\bar{\alpha}}^\mu \mathbf{e}_\mu, \Lambda_{\bar{\beta}}^\nu \mathbf{e}_\nu) \\ &= \Lambda_{\bar{\alpha}}^\mu \Lambda_{\bar{\beta}}^\nu \mathbb{G}(\mathbf{e}_\mu, \mathbf{e}_\nu) = \Lambda_{\bar{\alpha}}^\mu \Lambda_{\bar{\beta}}^\nu g_{\mu\nu}. \end{aligned}$$

All the basis are related by Lorentz transformations.

3.3.2 Proper time and an introduction to paths

Due to the Lorentzian nature of the space-time manifold $(M, \mathbb{G})$ the vectors in this space have a particular behavior.

Definition 21 On a Lorentzian manifold $(M, \mathbb{G})$ a non zero vector $\vec{X} \in T_p(M)$ is

- A *time-like* vector $\mathbb{G}(\vec{X}, \vec{X}) < 0$.
- A *null or light-like* vector $\mathbb{G}(\vec{X}, \vec{X}) = 0$.
- A *space-like* vector $\mathbb{G}(\vec{X}, \vec{X}) > 0$.

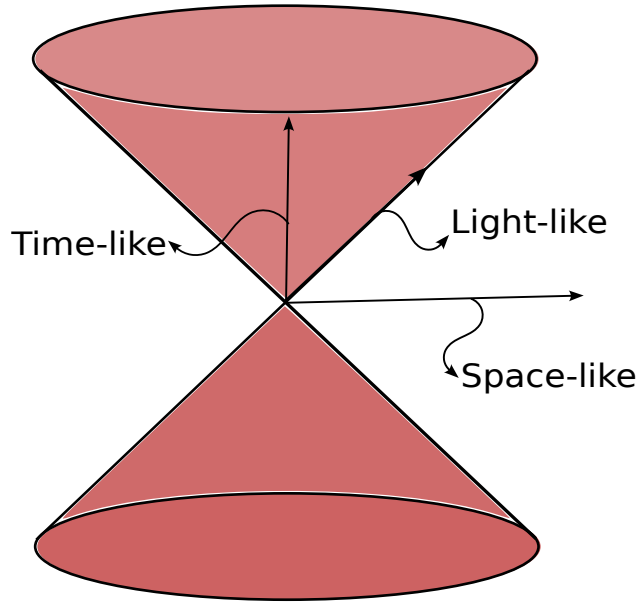


Figure 3.5: The light cone for the space-time structure, in the x -axis there is $\mathbb{R}^3$ and in the y axis there is the coordinate time t , a space-like vector is a vector which lives outwards the cone, a light-like lives on the border of the cone and a time-like lives inside the cone.

In the figure 3.5 there is a graphical representations of these three kinds of vectors, if we adopt a mostly positive signature, and from the postulates of the special relativity we can see the vectors with physical meaning are the time and light like vectors, because we can reach and not even get over the light speed so everything will be inside the cone and the objects that travel at the speed of light will be on the cone border, a space-like vector cannot be the four velocity of any observer, although it is perfectly meaningful as a separation between events.

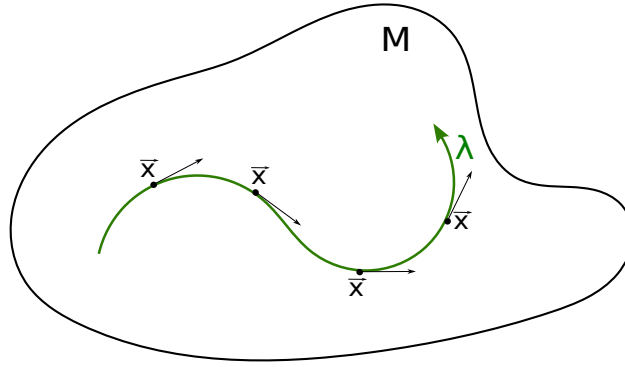


Figure 3.6: Graphical representation of an integral time-like curve on a manifold M .

We will see we can associate an observer with a time-like vector, therefore we'll focus on the properties of these vector, moreover on the characteristics of their tangent space $T_p(M)$, first we must see that if $\{\mathbf{e}_\mu\}$ is basis of orthonormal vectors in p , then the metric tensor has components

$$\mathbb{G}(\mathbf{e}_\alpha, \mathbf{e}_\beta) = \eta_{\alpha\beta} \quad (3.56)$$

such that $T_p(M)$ has exactly the same structure that the Minkowski space-time.

The time-like trajectory in a Lorentizan manifold $(M, \mathbb{G})$ is given by an integral curve such that $\vec{X}$ is always time-like or in a more formal way $\mathbb{G}(\vec{X}, \vec{X}) < 0$ see figure 3.6.

Lets see the behaviour of this curve. Let $\lambda(v)$ a smooth time-like curve such that $\lambda(0) = p$ and we can find an expression for a tangent vector $\vec{X}^a$, which is time-like, therefore we can define a proper time

$$\frac{d\tau}{dv} = \sqrt{-(g_{ab}X^aX^b)|_{\lambda(v)}} \quad \& \quad \tau(0) = 0. \quad (3.57)$$

We must remember $ds^2 = d\vec{X} \cdot d\vec{X} = g_{\alpha\beta}dx^\alpha dx^\beta$ this is given a coordinate basis in a coordinate chart $\phi = \{x^0, x^1, x^2, x^3\} \in \mathbb{R}^4$ thus

$$\frac{ds^2}{dv^2} = g_{\alpha\beta} \frac{dx^\alpha}{dv} \frac{dx^\beta}{dv}, \quad (3.58)$$

using a Minkowski space-time we have

$$d\tau^2 = -g_{\mu\nu}dx^\mu dx^\nu = -ds^2, \quad (3.59)$$

this allows us to see that $\vec{X} = X^\alpha \frac{\partial}{\partial x^\alpha}$ therefore $X^\alpha = \frac{dx^\alpha}{dv}$ we can then integrate and find an expression for the proper time

$$\tau = \int_0^{v_q} \left[- \left(g_{\mu\nu} \frac{dx^\mu}{dv} \frac{dx^\nu}{dv} \right) \right]_{\lambda(v)}^{1/2} dv. \quad (3.60)$$

We will see later that the time-like vectors can be parameterized from a proper time like 3.60, for the null-like ones we gotta chose an special curve, this is because the equation 3.60 ins related to geodesics. We're going to suppose $v = \tau$, the parameter is the proper time, then developing the equation 3.60

$$\tau = \int_0^\tau \left[- \left(g_{\mu\nu} \frac{dx^\mu}{d\tau} \frac{dx^\nu}{d\tau} \right) \right]^{1/2} d\tau,$$

$$\tau = \int_0^\tau [-U^\mu U_\mu]^{1/2} d\tau.$$

For the expression 3.60 become the proper time, we gotta impose that the tangent quadri-vector has norm -1 for any coordinate chart.

In the study of mechanics there is an important question, what is the path a test particle will follow in certain conditions? And why, from an infinity family of trajectories, did the particle choose its trajectory? To answer this a mathematical theory was developed, what we in physics known as the variational principles, let M a manifold and $\lambda(\tau)$ a time-like integral curve on M and $\lambda(\tau')$ alternative integral curves, see figure 3.7, thus the proper time is given by

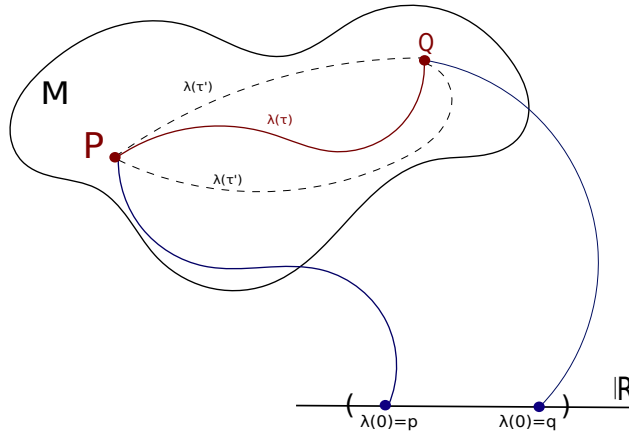


Figure 3.7: The real time-like trajectory a particle will follow is $\lambda(\tau)$ and the other two will be little different trajectories but with initial an final points fixed.

$$\tau [\lambda(v)] = \int_0^1 G(x(\lambda(v)), \dot{x}(\lambda(v))) dv, \quad (3.61)$$

$$G = [-g_{\mu\nu}(x(\lambda(v))) \dot{x}^\mu(\lambda(v)) \dot{x}^\nu(\lambda(v))]^{1/2}, \quad (3.62)$$

Using the equations 3.61 and 3.62 we want to formulate the following problem, how must the integrand G behave along the curve in order that the proper time be stationary, which is to say a maximum, a minimum or a saddle. Let us find the time-like curve $\lambda(v)$ which extremises the integral 3.61.

Let us denote the true curve as $x(\lambda)$ and a neighbouring wrong one as

$$X(\lambda) = x(\lambda) + \eta(\lambda), \quad (3.63)$$

where $\eta(\lambda)$ is the difference between the real and the wrong path, and because the initial and end point are fixed, thus

$$\eta(\lambda(0)) = \eta(\lambda(1)) = 0. \quad (3.64)$$

The integral $\tau[\lambda]$ taken along the wrong curve $X(\lambda)$ must be smaller than the one taken along the right curve $x(\lambda)$, no matter how close the former is to the latter, because for time-like curves in a Lorentzian space-time the geodesic is the path of *maximum* proper time and not of minimum length. To express this requirement, we will introduce a parameter α and redefine $X(\lambda)$ to be

$$X(\lambda) = x(\lambda) + \alpha\eta(\lambda), \quad (3.65)$$

The integral taken along the curve $X(\lambda)$ now depends on the parameter α , so let us call it $\tau(\alpha)$. The right curve $x(\lambda)$ is obtained from 3.65 by setting $\alpha = 0$. Thus the requirement that τ be extremal for the right curve $x(\lambda)$ implies that $\tau(\alpha)$ is stationary at $\alpha = 0$. With this result, we have converted our problem to the traditional problem from elementary calculus of making sure that an ordinary function [namely $\tau(\alpha)$] is stationary at the specified point $\alpha = 0$. To ensure this, we must just check that the derivative $\frac{d\tau}{d\alpha}$ is zero when $\alpha = 0$.

The integral $\tau(\alpha)$ is

$$\tau(\alpha) = \int_0^1 G(x(\lambda), \dot{x}(\lambda), \lambda) dv = \int_0^1 G(x + \alpha\eta, \dot{x} + \alpha\dot{\eta}) dv. \quad (3.66)$$

To differentiate 3.66 with respect to α we note that α appears in the integrand G , so we need to evaluate $\frac{\partial G}{\partial \alpha}$. Since α appears in two of the arguments of G , this gives two terms, namely (using the chain rule)

$$\frac{\partial G(x + \alpha\eta, \dot{x} + \alpha\dot{\eta}, \lambda)}{\partial \alpha} = \eta^\mu \frac{\partial G}{\partial x^\mu} + \dot{\eta}^\mu \frac{\partial G}{\partial \dot{x}^\mu}, \quad (3.67)$$

and for $\frac{d\tau}{d\alpha}$ which has to be zero

$$\frac{d\tau}{d\alpha} = \int_0^1 \frac{\partial G}{\partial \alpha} dv = \int_0^1 \left(\eta \frac{\partial G}{\partial x^\mu} + \dot{\eta} \frac{\partial G}{\partial \dot{x}^\mu} \right) dv = 0. \quad (3.68)$$

We need to rewrite the second term on the right using integration by parts

$$\int_0^1 \dot{\eta}(\lambda) \frac{\partial G}{\partial \dot{x}^\mu} = \left[\eta(\lambda) \frac{\partial G}{\partial \dot{x}^\mu} \right]_0^1 - \int_0^1 \eta(\lambda) \frac{d}{d\lambda} \left(\frac{\partial G}{\partial \dot{x}^\mu} \right) dv, \quad (3.69)$$

Because η vanishes at the initial and at the end point, the boundary term goes to zero, thus

$$\int_0^1 \dot{\eta}(\lambda) \frac{\partial G}{\partial \dot{x}^\mu} dv = - \int_0^1 \eta(\lambda) \frac{d}{d\lambda} \left(\frac{\partial G}{\partial \dot{x}^\mu} \right) dv. \quad (3.70)$$

Inserting this equation into 3.68 we find that

$$\int_0^1 \eta(\lambda) \left(\frac{d}{dv} \left(\frac{\partial G}{\partial \dot{x}^\mu} \right) - \frac{\partial G}{\partial x^\mu} \right) dv = 0, \quad (3.71)$$

thus we have arrived into the Euler- Lagrange equation for G :

$$\frac{d}{dv} \left(\frac{\partial G}{\partial \dot{x}^\mu} \right) - \frac{\partial G}{\partial x^\mu} = 0. \quad (3.72)$$

Computing the partial derivatives we will have

$$\frac{\partial G}{\partial \dot{x}^\mu} = -\frac{1}{G} g_{\mu\nu} \dot{x}^\nu, \quad (3.73)$$

$$\frac{\partial G}{\partial x^\mu} = -\frac{1}{2G} \partial_\mu g_{\nu\rho} \dot{x}^\nu \dot{x}^\rho. \quad (3.74)$$

Now, before solving the Euler-Lagrange equation for G lets give a deeper look into the relation between the curve, the proper time and G

$$G = \frac{d\tau}{d\lambda} \rightarrow \frac{d}{d\lambda} = \frac{d\tau}{d\lambda} \frac{d}{d\tau} = G \frac{d}{d\tau}. \quad (3.75)$$

Using this equation, the Euler lagrange equation thus we will have the following development. Multiplying the equation 3.72 by G we will have

$$G \frac{d}{d\tau} \left(\frac{\partial G}{\partial \dot{x}^\mu} \right) - G \frac{\partial G}{\partial x^\mu} = 0,$$

using the chain rule, therefore

$$\begin{aligned} \frac{d}{d\tau} \left[\frac{\partial}{\partial \dot{x}^\mu} \left(\frac{1}{2} G^2 \right) \right] - \frac{\partial}{\partial x^\mu} \left(\frac{1}{2} G^2 \right) &= 0, \\ \frac{1}{2} G^2 &= -\frac{1}{2} g_{\alpha\beta} \dot{x}^\alpha \dot{x}^\beta \equiv \mathcal{L}. \end{aligned}$$

$\mathcal{L}$ is in fact the Lagrangian of the system, thus the equation 3.72 becomes

$$\frac{d}{d\tau} \left(\frac{\partial}{\partial \dot{x}^\mu} \mathcal{L} \right) - \frac{\partial}{\partial x^\mu} \mathcal{L} = 0. \quad (3.76)$$

Now, we gotta calculate the partial derivatives

$$\frac{\partial}{\partial x^\mu} \mathcal{L} = \frac{1}{2} \frac{\partial}{\partial x^\mu} (g_{\alpha\beta} \dot{x}^\alpha \dot{x}^\beta) = \frac{1}{2} (\partial_\mu g_{\alpha\beta} \dot{x}^\alpha \dot{x}^\beta), \quad (3.77)$$

$$\frac{\partial}{\partial \dot{x}^\mu} \mathcal{L} = \frac{1}{2} \frac{\partial}{\partial \dot{x}^\mu} (g_{\alpha\beta} \dot{x}^\alpha \dot{x}^\beta) = \frac{1}{2} (g_{\alpha\beta} \delta_\mu^\alpha \dot{x}^\beta + g_{\alpha\beta} \dot{x}^\alpha \delta_\mu^\beta) = \frac{1}{2} (g_{\mu\beta} \dot{x}^\beta + g_{\alpha\mu} \dot{x}^\alpha). \quad (3.78)$$

Using both equation in the Euler-Lagrange equation then

$$\begin{aligned} \frac{d}{d\tau} \left[\frac{1}{2} (g_{\mu\beta} \dot{x}^\beta + g_{\alpha\mu} \dot{x}^\alpha) \right] &= \frac{1}{2} (g_{\mu\beta} \ddot{x}^\beta + \partial_l g_{\mu\beta} \dot{x}^\beta \dot{x}^\alpha + g_{\alpha\mu} \ddot{x}^\alpha + \partial_l g_{\alpha\mu} \dot{x}^\alpha \dot{x}^l), \\ \frac{1}{2} (g_{\mu\beta} \ddot{x}^\beta + \partial_l g_{\mu\beta} \dot{x}^\beta \dot{x}^\alpha + g_{\alpha\mu} \ddot{x}^\alpha + \partial_l g_{\alpha\mu} \dot{x}^\alpha \dot{x}^l) &- \frac{1}{2} (\partial_\mu g_{\alpha\beta} \dot{x}^\alpha \dot{x}^\beta), \end{aligned}$$

relabelling the dummy indices, calling β what was α in the first two terms, α what was l in the third and l what was β in the fourth, we arrive at

$$g_{\alpha\mu} \ddot{x}^\alpha + \frac{1}{2} (\partial_\alpha g_{\mu\beta} + \partial_\beta g_{\alpha\mu} - \partial_\mu g_{\alpha\beta}) \dot{x}^\alpha \dot{x}^\beta = 0. \quad (3.79)$$

Multiplying the equation 3.79 by $g^{\mu\lambda}$ we will have

$$\delta_\alpha^\lambda \ddot{x}^\alpha + \frac{1}{2} g^{\mu\lambda} (\partial_\alpha g_{\mu\beta} + \partial_\beta g_{\alpha\mu} - \partial_\mu g_{\alpha\beta}) \dot{x}^\alpha \dot{x}^\beta = 0, \quad (3.80)$$

$$\ddot{x}^\lambda + \frac{1}{2} g^{\mu\lambda} (\partial_\alpha g_{\mu\beta} + \partial_\beta g_{\alpha\mu} - \partial_\mu g_{\alpha\beta}) \dot{x}^\alpha \dot{x}^\beta = 0. \quad (3.81)$$

$$\frac{d^2 x^\lambda}{d\tau^2} + \Gamma_{\alpha\beta}^\lambda \frac{dx^\alpha}{d\tau} \frac{dx^\beta}{d\tau} = 0, \quad (3.82)$$

$$\Gamma_{\alpha\beta}^\lambda = \frac{1}{2} g^{\lambda\mu} (\partial_\alpha g_{\beta\mu} + \partial_\beta g_{\alpha\mu} - \partial_\mu g_{\alpha\beta}). \quad (3.83)$$

The equation 3.82 it is not more than the differential equation for the geodesics in the space-time, we shall prove this shortly, but for now we must make some definitions. $\Gamma_{\alpha\beta}^\mu$ are known as the Christoffel symbols, they need a kind of connection, that is because we have derivatives along the curve, but a curve at any point can have a different tangent space, so each tangent vector is different and we need something to connect those spaces, here that connection is the Levi-Civita connection, which one we will develop later. A particular case is when the Christoffel symbols is zero, we will obtain that the quadri-acceleration is zero, this is the domain of the special relativity so we can see physically that *if every Christoffel symbol vanishes in some chart then the four acceleration vanishes too and those coordinates are inertial. The converse, however, must be stated with care, because non vanishing Christoffel symbols do **not** by themselves mean that the space-time is curved: the flat plane written in polar coordinates already has non vanishing symbols, exactly as we found in chapter 2 when we discovered that the orthonormal polar basis is not a coordinate basis. Curvature is measured by the Riemann tensor and never by the connection alone, and building that tensor is the business of the section which follows.*

3.3.3 Covariant derivative

In the Euler-Lagrange equation for the metric determinant G we used derivatives, but here the derivatives are kind of special, why? To answer this, first let us examine a vector field V^a and see whether its gradient is or is not a tensor

$$T_\beta^\alpha = \partial_\beta V^\alpha. \quad (3.84)$$

To see how does T_β^α transforms let supposes two coordinate charts $\phi = \{x^0, x^1, x^2, x^3\}$ and $\bar{\phi} = \{\bar{x}^0, \bar{x}^1, \bar{x}^2, \bar{x}^3\}$, thus

$$\begin{aligned} T_{\bar{\beta}}^{\bar{\alpha}} &= \partial_{\bar{\beta}} V^{\bar{\alpha}} = \partial_{\bar{\beta}} \left(\frac{\partial x^{\bar{\alpha}}}{\partial x^\mu} V^\mu \right) = \frac{\partial x^\nu}{\partial x^{\bar{\beta}}} \frac{\partial}{\partial x^\nu} \left(\frac{\partial x^{\bar{\alpha}}}{\partial x^\mu} V^\mu \right) = \frac{\partial x^{\bar{\alpha}}}{\partial x^\mu} \frac{\partial x^\nu}{\partial x^{\bar{\beta}}} \left(\frac{\partial}{\partial x^\nu} V^\mu \right) + \left(\frac{\partial x^\nu}{\partial x^{\bar{\beta}}} V^\mu \right) \frac{\partial}{\partial x^\nu} \frac{\partial x^{\bar{\alpha}}}{\partial x^\mu}, \\ T_{\bar{\beta}}^{\bar{\alpha}} &\neq \frac{\partial x^{\bar{\alpha}}}{\partial x^\mu} \frac{\partial x^\nu}{\partial x^{\bar{\beta}}} T_\nu^\mu. \end{aligned}$$

Thus T_β^α as we had defined does not transforms as a tensor, therefore we must look carefully on how do we take derivatives. The problem lies in the fact that T_β^α as we had defined is only the derivatives of the components, but how about the derivatives of the basis vectors? They are also changing along the manifold, so their derivatives have to be taken into account, to solve this we must introduce the notion of the *covariant derivative*.

Definition 22 A covariant derivative $\vec{\nabla}$ on a manifold M is a map from two vector fields X, Y to a vector field with the following properties.

Let X, Y, Z three smooth vector fields and f, g two smooth functions, the covariant derivative satisfies the following properties:

1. $\vec{\nabla}_{fX+gY}Z = f\vec{\nabla}_XZ + g\vec{\nabla}_YZ.$
2. $\vec{\nabla}_X(Y + Z) = \vec{\nabla}_XY + \vec{\nabla}_XZ.$
3. $\vec{\nabla}_X(fY) = f\vec{\nabla}_XY + (\vec{\nabla}_Xf)Y.$

From these properties we can see that

$$\vec{\nabla}Y : X \rightarrow \vec{\nabla}_XY \quad (3.85)$$

is a linear map from $T_p(M)$ to itself, and

$$\vec{\nabla}f : X \rightarrow \vec{\nabla}_Xf = X(f) \quad (3.86)$$

which is the definition of gradient. Lets see an example, let $X = X^\mu \mathbf{e}_\mu$ and $Y = Y^\mu \mathbf{e}_\mu$, thus lets calculate the covariant derivative

$$\begin{aligned} \vec{\nabla}_XY &= \vec{\nabla}_X(y^\mu \mathbf{e}_\mu) = X(y^\mu) \mathbf{e}_\mu + y^\mu \vec{\nabla}_X \mathbf{e}_\mu \\ &= x^\nu \mathbf{e}_\nu(y^\mu) \mathbf{e}_\mu + y^\mu \vec{\nabla}_{x^\nu \mathbf{e}_\nu} \mathbf{e}_\mu = x^\nu \mathbf{e}_\nu(y^\mu) \mathbf{e}_\mu + y^\mu x^\nu \vec{\nabla}_{\mathbf{e}_\nu} \mathbf{e}_\mu, \\ \vec{\nabla}_XY &= X^\nu \left[\mathbf{e}_\nu(Y^\mu) \mathbf{e}_\mu + Y^\mu \vec{\nabla}_{\mathbf{e}_\nu} \mathbf{e}_\mu \right] = X^\nu \left[\mathbf{e}_\nu(Y^\mu) + \Gamma_{\nu\rho}^\mu Y^\rho \right] \mathbf{e}_\mu \end{aligned} \quad (3.87)$$

where

$$\Gamma_{\nu\mu}^\rho \mathbf{e}_\rho = \vec{\nabla}_{\mathbf{e}_\nu} \mathbf{e}_\mu. \quad (3.88)$$

From the equation 3.87 we see then the covariant derivative is a map from a tangent space to itself, and whose components are given by

$$\left[\vec{\nabla}_XY \right]^\mu = X^\nu \left[\mathbf{e}_\nu(Y^\mu) + \Gamma_{\nu\rho}^\mu Y^\rho \right], \quad (3.89)$$

and if we choose a coordinate chart, the components of the covariant derivative, denoted by $Y_{;\nu}^\mu$, are

$$Y^\mu_{;\nu} = \partial_\nu Y^\mu + \Gamma^\mu_{\nu\rho} Y^\rho. \quad (3.90)$$

Here we see the importance of the derivatives of the basis vectors and their direct relation with the Christoffel symbols, therefore we gotta do a definition

Definition 23 In a basis $\{\mathbf{e}_\mu\}$ the components of the connections are given by $\Gamma^\mu_{\rho\nu}\mathbf{e}_\mu$ called the Christoffel symbols and are defined by

$$\vec{\nabla}_{\mathbf{e}_\rho} \mathbf{e}_\nu = \Gamma^\mu_{\rho\nu} \mathbf{e}_\mu. \quad (3.91)$$

Now, lets generalize the covariant derivative to tensor fields. If T is a tensor field $\binom{r}{s}$ then $\vec{\nabla}T$ is a tensor field of kind $\binom{r}{s+1}$. To show this, lets start with the one-forms:

$$\begin{aligned} \left(\vec{\nabla}_X \eta\right)(Y) &= \vec{\nabla}_X (\eta(Y)) - \eta\left(\vec{\nabla}_X Y\right) = \vec{\nabla}_X (\eta_\mu Y^\mu) - \eta_\mu \left(\vec{\nabla}_X Y\right)^\mu, \\ &= X(\eta_\mu) Y^\mu + \eta_\mu X(Y^\mu) - \eta_\mu (X^\nu \mathbf{e}_\nu(Y^\mu) + \Gamma^\mu_{\rho\nu} Y^\rho X^\nu), \\ \left(\vec{\nabla}_X \eta\right)(Y) &= [X(\eta_\mu) - \Gamma^\rho_{\mu\nu} X^\nu \eta_\rho] Y^\mu. \end{aligned} \quad (3.92)$$

Therefore $\vec{\nabla}_X \eta$ is a co-vector field, and the application $\vec{\nabla} \eta$ is a tensor field of rank $\binom{0}{2}$ such that

$$\left(\vec{\nabla}_X \eta\right)_\mu = X(\eta_\mu) - \Gamma^\rho_{\mu\nu} X^\nu \eta_\rho = X^\nu \mathbf{e}_\nu(\eta_\mu) - \Gamma^\rho_{\mu\nu} X^\nu \eta_\rho, \quad (3.93)$$

and the covariant derivative of the one-form, denoted by $\eta_{\mu;\nu}$ is given by

$$\eta_{\mu;\nu} = \mathbf{e}_\nu(\eta_\mu) - \Gamma^\rho_{\mu\nu} \eta_\rho. \quad (3.94)$$

Now, lets generalize all this development for a tensor field T of rank $\binom{r}{s}$, so lets take the covariant derivative $\vec{\nabla}T$

$$\begin{aligned} \vec{\nabla}_\rho T^{\mu_1 \dots \mu_r}_{\nu_1 \dots \nu_s} &= \partial_\rho T^{\mu_1 \dots \mu_r}_{\nu_1 \dots \nu_s} + \Gamma^{\mu_1}_{\sigma\rho} T^{\sigma \mu_2 \dots \mu_r}_{\nu_1 \dots \nu_s} + \dots + \Gamma^{\mu_r}_{\sigma\rho} T^{\mu_1 \dots \sigma}_{\nu_1 \dots \nu_s} \\ &\quad - \Gamma^\sigma_{\nu_1\rho} T^{\mu_1 \dots \mu_r}_{\sigma \nu_2 \dots \nu_s} - \dots - \Gamma^\sigma_{\nu_s\rho} T^{\mu_1 \dots \mu_r}_{\nu_1 \dots \sigma}, \end{aligned}$$

where the positive signs represent the Christoffel symbols for the covariant derivatives of the vectors that compose the tensor and the negative ones are for the derivatives of the one-form that compose the tensor, and we must note that *the covariant derivative is taking a vector and carry it*

along another one.

The second Newton's law, the uncoupled Maxwell laws, the Navier-Stokes equation, almost all the equations in physics are written in terms of second derivatives, therefore we gotta study the properties of second covariant derivatives, the usual partial derivative has the following property

$$\partial_\nu \partial_\mu f = \partial_\mu \partial_\nu f, \quad (3.95)$$

and let f a smooth functions, lets calculate its covariant derivatives

$$\vec{\nabla}_\nu \vec{\nabla}_\mu f = \partial_\nu \partial_\mu f - \Gamma_{\mu\nu}^\rho \partial_\rho f, \quad (3.96)$$

$$\vec{\nabla}_\mu \vec{\nabla}_\nu f = \partial_\mu \partial_\nu f - \Gamma_{\nu\mu}^\rho \partial_\rho f. \quad (3.97)$$

From these equation we can calculate the anti-symmetric part of these second covariant derivatives given by

$$\vec{\nabla}_{[\nu} \vec{\nabla}_{\mu]} f = -\Gamma_{[\mu\nu]}^\rho \partial_\rho f. \quad (3.98)$$

Thus if we want the second derivative to commute the Christoffel symbol must be symmetric in the lower indices, but in general for a vector or a tensor this is not generally true.

Definition 24 A connection $\vec{\nabla}$ is free of torsion, if $\vec{\nabla}_A \vec{\nabla}_B f = \vec{\nabla}_B \vec{\nabla}_A f$ for any smooth function f which is equivalent to $\Gamma_{[\mu\nu]}^\rho = 0$.

To have a first step on the understating of the torsion, for example think on a particle in a circular motion, we know there is a curvature factor, but the particle keeps in the circular motion, if there is any force which changes the angular momentum therefore a torque appears, such a system has torsion.

Theorem 6 For a torsion free connection, if X and Y are two vector fields, then

$$[X, Y] = \vec{\nabla}_X Y - \vec{\nabla}_Y X. \quad (3.99)$$

Let $\phi = \{x^0, x^1, x^2, x^3\}$ a coordinate chart, thus

$$\begin{aligned} X^\nu \vec{\nabla}_\nu Y^\mu - Y^\nu \vec{\nabla}_\nu X^\mu &= X^\nu \partial_\nu Y^\mu + \Gamma_{\sigma\nu}^\mu X^\sigma Y^\nu - Y^\nu \partial_\nu X^\mu - \Gamma_{\nu\sigma}^\mu Y^\sigma X^\nu \\ &= (X^\nu \partial_\nu Y^\mu - Y^\nu \partial_\nu X^\mu) + \Gamma_{\sigma\nu}^\mu X^\sigma Y^\nu - \Gamma_{\nu\sigma}^\mu Y^\sigma X^\nu \\ &= [X, Y]^\mu + 2\Gamma_{[\sigma\nu]}^\mu X^\sigma Y^\nu, \end{aligned}$$

now, if there is no torsion, therefore the Christoffel symbol commutes in the lower indices, thus

$$X^\nu \vec{\nabla}_\nu Y^\mu - Y^\nu \vec{\nabla}_\nu X^\mu = [X, Y]^\mu. \quad (3.100)$$

We can also define a torsion tensor as follows

$$T(X, Y) = \vec{\nabla}_X Y - \vec{\nabla}_Y X - [X, Y]. \quad (3.101)$$

Even if the theory is free of torsion, the second derivatives don't commute for a tensor field. The torsion is analogous to the rotation plane of the particle, *we can have curvature but not torsion*.

3.3.4 The Lie derivative and the covariant derivative

We promised in chapter 2 that once both derivatives were on the table we would settle how they are related, and this is the moment to keep that promise. The question deserves care, because the answer is not the one most readers guess the first time, and because the guess that fails is repeated often enough that it has acquired an air of respectability.

Let us begin with what the two operators agree on. Acting on a scalar field there is nothing for either of them to correct, since a function carries no indices and therefore no basis vectors that could rotate or stretch from one point to the next, and both collapse onto the ordinary directional derivative,

$$\mathcal{L}_X f = X(f) = X^\mu \partial_\mu f = \vec{\nabla}_X f. \quad (3.102)$$

Equation 3.102 is the small grain of truth out of which the confusion grows. There is a second and equally honest coincidence. Taking for X one of the basis vectors of a coordinate chart, $X = \partial_\sigma$, whose components $X^\mu = \delta^\mu_\sigma$ are the same numbers at every point of the chart, the component formula for the Lie derivative of a vector field gives

$$\begin{aligned} (\mathcal{L}_{\partial_\sigma} Y)^\nu &= X^\mu \partial_\mu Y^\nu - Y^\mu \partial_\mu X^\nu = \delta^\mu_\sigma \partial_\mu Y^\nu - Y^\mu \partial_\mu \delta^\nu_\sigma \\ &= \partial_\sigma Y^\nu, \end{aligned}$$

where the term in red is the derivative of a constant and dies at once. The partial derivative is therefore the Lie derivative taken along a coordinate direction, which is a rather satisfying way of understanding why ∂_μ feels so natural inside a chart and becomes meaningless the moment we leave it.

Now for the real question, and the answer runs in the direction opposite to the expected one: it is the Lie derivative that can be written using a covariant derivative, and not the covariant derivative that can be written using a Lie derivative. We already met the statement in equation 3.100, and it

is worth seeing exactly why it works. Take a torsion free connection, which is the only kind we shall ever use, and compute the antisymmetric combination

$$\begin{aligned} X^\mu \vec{\nabla}_\mu Y^\nu - Y^\mu \vec{\nabla}_\mu X^\nu &= X^\mu (\partial_\mu Y^\nu + \Gamma_{\mu\rho}^\nu Y^\rho) - Y^\mu (\partial_\mu X^\nu + \Gamma_{\mu\rho}^\nu X^\rho) \\ &= X^\mu \partial_\mu Y^\nu - Y^\mu \partial_\mu X^\nu + \Gamma_{\mu\rho}^\nu X^\mu Y^\rho - \Gamma_{\mu\rho}^\nu Y^\mu X^\rho. \end{aligned}$$

The two terms in red carry the whole argument. Both μ and ρ are summed over in the last one, so we are free to exchange their names, and after doing so it reads $\Gamma_{\rho\mu}^\nu X^\mu Y^\rho$, which lets the pair collect into

$$(\Gamma_{\mu\rho}^\nu - \Gamma_{\rho\mu}^\nu) X^\mu Y^\rho = 2 \Gamma_{[\mu\rho]}^\nu X^\mu Y^\rho = 0,$$

vanishing identically because a torsion free connection is symmetric in its two lower indices. Every trace of the connection has cancelled, and what survives is

$$(\mathcal{L}_X Y)^\nu = [X, Y]^\nu = X^\mu \vec{\nabla}_\mu Y^\nu - Y^\mu \vec{\nabla}_\mu X^\nu. \quad (3.103)$$

The same computation performed on a tensor field of arbitrary valence, and the reader should carry it out once for a $\binom{1}{1}$ tensor in order to see the pattern with their own hands, produces

$$\mathcal{L}_X T_\nu^\mu = X^\rho \vec{\nabla}_\rho T_\nu^\mu - T_\nu^\rho \vec{\nabla}_\rho X^\mu + T_\rho^\mu \vec{\nabla}_\nu X^\rho, \quad (3.104)$$

where once again all the Christoffel symbols disappear. Equations 3.103 and 3.104 are genuinely useful in practice, because they say that we may compute a Lie derivative with whatever torsion free connection happens to be convenient and the result will not depend on which one we chose. That independence is exactly what we should demand of an object which never needed a connection in the first place.

What is false, and here is the statement we came to correct, is that the covariant derivative is a particular case of the Lie derivative. Two independent arguments rule it out, and both of them are short. The first is about how much structure each operator requires. A manifold admits infinitely many connections, all of them equally legitimate, so writing down a covariant derivative means making a genuine choice and feeding the manifold information it did not previously possess. The Lie derivative is fixed by the smooth structure alone and asks for no such choice. No amount of algebra performed on Lie derivatives can produce information that was never supplied to them.

The second argument is sharper and can be checked in a single line. The covariant derivative is linear over functions in the direction along which we differentiate,

$$\vec{\nabla}_{fX} Y = f \vec{\nabla}_X Y, \quad (3.105)$$

which is nothing but the first defining property of a connection, whereas the Lie derivative is not,

$$\mathcal{L}_f X Y = [fX, Y] = f[X, Y] - Y(f)X = f\mathcal{L}_X Y - Y(f)X.$$

The extra term written in blue refuses to go away, and it is the algebraic fingerprint of a geometrical fact. Equation 3.105 says that $\vec{\nabla}_X Y$ evaluated at a point depends on the field X only through the single arrow sitting at that point, which is why it makes perfect sense to speak of the covariant derivative of a field along a direction, and why parallel transport along a curve is well defined. The Lie derivative enjoys no such property. The value of $\mathcal{L}_X Y$ at a point depends on how X itself is varying in a neighbourhood, and therefore on the entire family of integral curves surrounding the point rather than on the arrow standing at it. Two operators which depend on their direction argument in different ways cannot be special cases of one another, and with that the question is closed.

The distinction is far easier to hold on to as a picture than as an inequality between two formulas, and figure 3.8 is that picture. To transport a vector covariantly we need a curve to travel along and a connection to tell us what remaining parallel means, and nothing more, so the neighbourhood of the curve plays no role whatsoever. To drag a vector in the sense of Lie we need an entire congruence of curves, the flow of X , and the question we are asking is how the vector joining two neighbouring curves gets stretched and turned as the flow carries it forward. Remove the neighbouring curves and the question cannot even be posed.

The two derivatives also answer two different physical questions, and general relativity needs both of them. The covariant derivative asks how a quantity changes with respect to the notion of parallelism carried by the gravitational field itself, which is why it appears in the geodesic equation and in the field equations, and why every law of physics is written with it. The Lie derivative asks how a quantity changes when the whole space-time is dragged along a flow, which is a question about diffeomorphisms rather than about gravity, and it is therefore the natural instrument for discussing symmetries. That is precisely the use we shall make of it in the next chapter, when we ask what it means for a metric to be left untouched by a flow and discover that the answer hands us the conserved energy and angular momentum of an orbiting particle.

3.3.5 Levi-Civita connection

Theorem 7 *Let M a manifold with a metric $\mathbb{G}$ so that $(M, \mathbb{G})$ is a pseudo-Riemannian manifold. Then there exists one and only one torsion free connection $\vec{\nabla}$ for which the metric is covariantly constant, $\vec{\nabla}\mathbb{G} = 0$, and it is known as the Levi-Civita connection, or the metric connection.*

Demonstration: Let X, Y, Z three vector fields, thus using the Leibniz rule and assuming the metric is covariantly constant, then

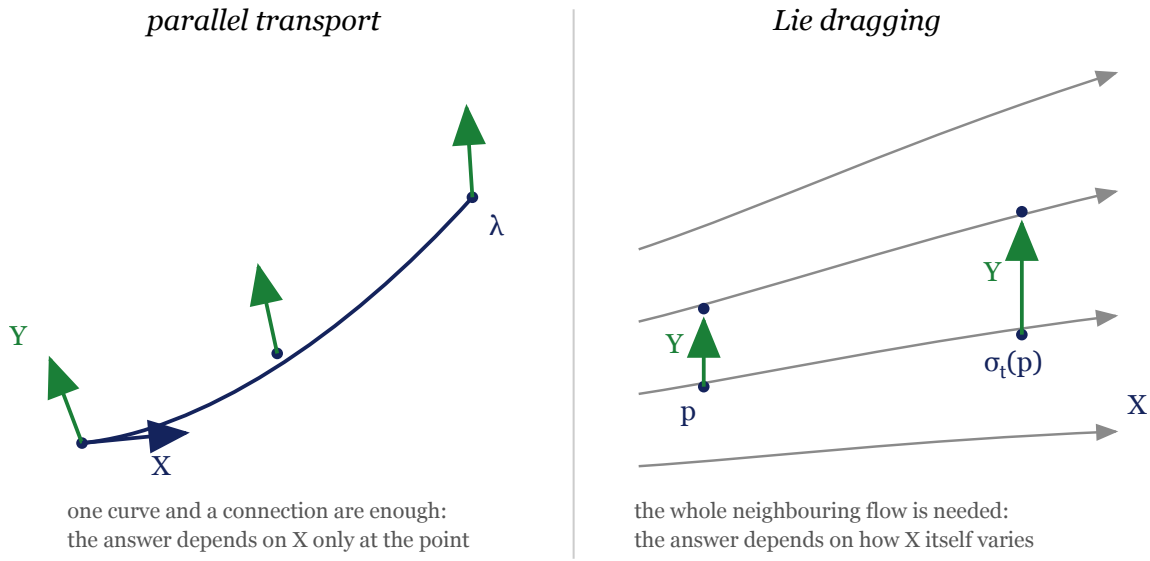


Figure 3.8: The two derivatives compared. On the left a vector is carried along a single curve by a connection, which is the only information the covariant derivative requires. On the right the vector is dragged by the flow of a vector field, and the Lie derivative measures the failure of the vector joining two neighbouring integral curves to be reproduced by the flow.

$$X (\mathbb{G}(Y, Z)) = \vec{\nabla}_X [\mathbb{G}(Y, Z)] = \mathbb{G} (\vec{\nabla}_X Y, Z) + \mathbb{G} (Y, \vec{\nabla}_X Z), \quad (3.106)$$

$$Y (\mathbb{G}(Z, X)) = \mathbb{G} (\vec{\nabla}_Y Z, X) + \mathbb{G} (Z, \vec{\nabla}_Y X), \quad (3.107)$$

$$Z (\mathbb{G}(X, Y)) = \mathbb{G} (\vec{\nabla}_Z X, Y) + \mathbb{G} (X, \vec{\nabla}_Z Y), \quad (3.108)$$

$$H \equiv X (\mathbb{G}(Y, Z)) + Y (\mathbb{G}(Z, X)) - Z (\mathbb{G}(X, Y)) \quad (3.109)$$

and if we assume the theory is free of torsion, then

$$H = 2\mathbb{G} (\vec{\nabla}_X Y, Z) - \mathbb{G} ([X, Y], Z) - \mathbb{G} ([Z, X], Y) + \mathbb{G} ([Y, Z], X), \quad (3.110)$$

thus

$$\mathbb{G} (\vec{\nabla}_X Y, Z) = \frac{1}{2} [X (\mathbb{G}(Y, Z)) + Y (\mathbb{G}(Z, X)) - Z (\mathbb{G}(X, Y)) \quad (3.111)$$

$$+ \mathbb{G} ([X, Y], Z) + \mathbb{G} ([Z, X], Y) - \mathbb{G} ([Y, Z], X)], \quad (3.112)$$

$$\mathbb{G}(\vec{\nabla}_{fX}Y, Z) = f\mathbb{G}(\vec{\nabla}_XY, Z) = \mathbb{G}(f\vec{\nabla}_XY, Z), \quad (3.113)$$

To finish the proof it only remains to check that the connection so defined really behaves as a connection when we rescale its first argument by a function, so lets do the following change of variable $X \rightarrow Xf$

$$\begin{aligned} \mathbb{G}(\vec{\nabla}_{Xf}Y, Z) &= \frac{1}{2} [fX(\mathbb{G}(Y, Z)) + Y(\mathbb{G}(Z, Xf)) - Z(\mathbb{G}(Xf, Y))] \\ &\quad + \mathbb{G}([Xf, Y], Z) + \mathbb{G}([Z, Xf], Y) - \mathbb{G}([Y, Z], Xf) \\ &= \frac{1}{2} [fX(\mathbb{G}(Y, Z)) + fY(\mathbb{G}(Z, X)) - fZ(\mathbb{G}(X, Y))] \\ &\quad + f\mathbb{G}([X, Y], Z) + f\mathbb{G}([Z, X], Y) - f\mathbb{G}([Y, Z], X) = \mathbb{G}(f\vec{\nabla}_XY, Z), \end{aligned}$$

$$\mathbb{G}(\vec{\nabla}_{fX}Y - f\vec{\nabla}_XY, Z) = 0. \quad (3.114)$$

Lets calculate the components of the Levi-Civita connection respect to a coordinate basis

$$[\mathbf{e}_\mu, \mathbf{e}_\nu] = \Gamma_{\mu\nu}^\sigma \mathbf{e}_\sigma - \Gamma_{\nu\mu}^\sigma \mathbf{e}_\sigma = 0. \quad (3.115)$$

The equation 3.115 is valid when the theory is free of torsion, thus, lets calculate the following application

$$\mathbb{G}(\vec{\nabla}_{\mathbf{e}_\rho} \mathbf{e}_\nu, \mathbf{e}_\sigma) = \frac{1}{2} [\mathbf{e}_\rho(g_{\nu\sigma}) + \mathbf{e}_\nu(g_{\sigma\rho}) - \mathbf{e}_\sigma(g_{\rho\nu})] = \frac{1}{2} [\partial_\rho g_{\nu\sigma} + \partial_\nu g_{\sigma\rho} - \partial_\sigma g_{\rho\nu}],$$

$$\mathbb{G}(\Gamma_{\rho\nu}^\tau \mathbf{e}_\tau, \mathbf{e}_\sigma) = \Gamma_{\rho\nu}^\tau g_{\tau\sigma}, \quad (3.116)$$

$$\Gamma_{\nu\rho}^\mu = \frac{1}{2} g^{\mu\sigma} (\partial_\rho g_{\sigma\nu} + \partial_\nu g_{\sigma\rho} - \partial_\sigma g_{\nu\rho}). \quad (3.117)$$

3.3.6 Geodesics

When there is only gravitational interaction the trajectories a test particle will follow are given by the geodesics of the space-time but, first of all, what is a geodesic? A *geodesic* is the straightest possible curve on a manifold, the one that parallel transports its own tangent vector. In a Riemannian space that curve is also the shortest path between two points, but in the Lorentzian space-time of general relativity a time-like geodesic is instead the curve of **maximum** proper time between two events,

which is the geometrical content of the twin paradox. In this order of ideas lets recover the equation we obtained the expression 3.82

$$\frac{d^2 x^\mu}{d\tau^2} + \Gamma_{\nu\rho}^\mu (X(\tau)) \frac{dx^\nu}{d\tau} \frac{dx^\rho}{d\tau} = 0, \quad (3.118)$$

where $\frac{dx^\mu}{d\tau} = X^\mu$ are the components of X . If we want to talk about geodesics and tangent vectors we must have something to relate different tangent vectors from one point to another on the manifold, we must make an isomorphism to be able to connect the tangent spaces, therefore lets work on the equation 3.82,

$$\frac{d^2 x^\mu}{d\tau^2} = \frac{d}{d\tau} X^\mu (X(\tau)) = \frac{dx^\nu}{d\tau} \frac{\partial X^\mu}{\partial x^\nu} = X^\nu \partial_\nu X^\mu, \quad (3.119)$$

$$X^\nu (\partial_\nu X^\mu + \Gamma_{\nu\rho}^\mu X^\rho) = 0, \quad (3.120)$$

$$X^\nu \vec{\nabla}_\nu X^\mu = 0. \quad (3.121)$$

Therefore to find the geodesic and a relation between the tangent spaces, we must take X^μ and project it along the connections over itself.

Definition 25 *Let M a manifold with connection $\vec{\nabla}$, an affinely parameterized geodesic is an integral curve of a vector field X that satisfies:*

$$\vec{\nabla}_X X = 0. \quad (3.122)$$

In geometry, an affine transformation or affine map (also called affinity) between two affine spaces (in particular, two vector spaces) consists of a linear transformation followed by a translation:

$$X \rightarrow \mathbb{A}X + b, \quad (3.123)$$

In the finite-dimensional case, every affine transformation can be represented by a matrix $\mathbb{A}$ and a vector b of the vector space. Geometrically, an affine transformation in a Euclidean space is a transformation that preserves:

- The collinearity (and coplanarity) relationships between points, that is, points that fell on the same line (or on the same plane) before the transformation, are preserved after an affine transformation.
- The ratios between distances along a line, that is, for three different aligned points P_1, P_2, P_3 the reasons $|P_2\bar{P}_1| / |P_3\bar{P}_2|$ before and after the transformation are the same.

An affine transformation $f : \mathbb{A} \rightarrow \mathbb{A}'$ between two affine spaces $(\mathbb{A}, E, \delta)$ and $(\mathbb{A}', E', \delta')$ is an application between two points that acts linearly on the vectors that join them. Formally, f is an affine transformation if there exists a linear map g such that $\forall p, q \in \mathbb{A}$

$$f(p)\vec{f}(q) = g(\vec{pq}). \quad (3.124)$$

Before going on, let's remember the usefulness of this kind of affine transformations.

3.3.6.1 Affine transformations in Galilean relativity

In the classical mechanics there is a really important group that rules the inertial frames and the mechanics, the well known as the Galilean group.

Definition 26 *The **Galilean group** is a group $G = \langle \mathbb{T}, \circ \rangle$ composed by the Galilean transformations under the compositions bilinear operation.*

The Galilean transformations are the following three

1. The displacement of the origin of the reference frame

$$T_d(\vec{r}, t) \rightarrow (\vec{r} + \vec{r}', t + t'). \quad (3.125)$$

2. The rotations of the coordinate axis of the reference frame

$$T_r(\vec{r}, t) \rightarrow (\mathbb{R}\vec{r}, t) : \mathbb{R} \in SO(3). \quad (3.126)$$

3. Motion at uniform velocity $\vec{V}$

$$T_M(\vec{r}, t) \rightarrow (\vec{r} + \vec{V}t, t). \quad (3.127)$$

It is easy to prove that the position of a particle can be determined using these three transformations, in the following form

$$(\vec{r}', t') = (T_d \circ T_r) \vec{r} + \vec{V}t, \quad (3.128)$$

we see it is naturally an affine transformation all the mechanics we studied in the school or in the first year of our degrees is written in terms of affine transformations between inertial reference frames which move one each other at constant velocity.

3.3.6.2 Affine transformations in special relativity

In the same way we're gonna deduce the affine transform for Minkowskian space-time, let $\vec{x}'$ a vector which obeys an affine transformation as:

$$\vec{x}' = \mathcal{L}(\vec{x}) = \Lambda \vec{x} + \vec{a} \quad : \quad \vec{x}, \vec{a} \in \mathbb{R}^4, \quad \Lambda \in G_L,$$

Where G_L is the Lorentz's group, but before doing that, let's discover the linear transform case:

$$\vec{x}' = \Lambda \vec{x} \tag{3.129}$$

The transform Λ must give invariant the light cone so let $\vec{v}, \vec{w} \in \mathbb{R}^4$:

$$\langle \Lambda \vec{v}, \Lambda \vec{w} \rangle = \langle \vec{v}, \vec{w} \rangle \rightarrow (\Lambda \vec{v})^T \eta (\Lambda \vec{w}) = \vec{v}^T \eta \vec{w}$$

$$\vec{v}^T \Lambda^T \eta \Lambda \vec{w} = \vec{v}^T \eta \vec{w}$$

$$\eta = \Lambda^T \eta \Lambda \tag{3.130}$$

The equation 3.130 is a similarity transform, is easily to see that Λ only could have two values for its determinant $|\Lambda| = \pm 1$.

Demonstration:

$$|\eta| = |\Lambda^T \eta \Lambda| = |\Lambda^T| |\eta| |\Lambda|, \tag{3.131}$$

Since $\eta = \text{diag}(-1, 1, 1, 1)$ we have $|\eta| = -1$, and using the properties $|A| = |A^T|$ we can rewrite the equation 3.131 as:

$$1 = |\Lambda|^2 \rightarrow |\Lambda| = \pm 1.$$

This is only possible in three cases:

1. Temporal or parity inversions:

$$\Lambda = \text{diag}[\mp 1, \pm 1, \pm 1, \pm 1]$$

Respectively

2. Unitary transformations:

$$\Lambda = \begin{bmatrix} 1 & \vec{O}^T \\ \vec{O} & R \end{bmatrix} : R \in SO(3)$$

3. Speed boosts, for example a x boost:

$$\Lambda = \begin{bmatrix} \gamma & -v_x\gamma & 0 & 0 \\ -v_x\gamma & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} : \gamma = \frac{1}{\sqrt{1-v_x^2}} \quad C = 1 \quad (3.132)$$

If we are out from the natural coordinates γ changes to $\gamma = \frac{1}{\sqrt{1-\frac{v^2}{c^2}}}$

As the Galilean group the Lorentz group has 6 parameters, 3 rotations and 3 boosts, but in general:

$$\mathcal{L}(\vec{x}) = \Lambda\vec{x} + \vec{a} : \vec{x}, \vec{a} \in \mathbb{R}^4$$

That general form is more general than the last one, we obtain it from the Lorentz's group G_L and 4 translations, we call it the *Poincaré's group* $G_{\mathcal{P}}$.

Let $\mathcal{O} \wedge \bar{\mathcal{O}}$ inertial observers such that there is a relative movement between them, thus: $\bar{\mathcal{O}}$ moves with speed $v < 1$ relative to $\mathcal{O}$, with $v > 0$ along the x direction, thus the coordinates transformed are:

$$\bar{t} = At + Bx$$

$$\bar{x} = Ct + Dx$$

$$\bar{y} = y$$

$$\bar{z} = z$$

Now, because the light cone is invariant we've got the following equation:

$$\begin{aligned} -t^2 + x^2 + y^2 + z^2 &= -\bar{t}^2 + \bar{x}^2 + \bar{y}^2 + \bar{z}^2 \\ -t^2 + x^2 &= -(A^2 - C^2)t^2 + (D^2 - B^2)x^2 + 2(CD - AB)tx \end{aligned}$$

$$A^2 - C^2 = 1$$

$$D^2 - B^2 = 1$$

$$AB = CD$$

This system is underdetermined, since we have only three equations for four unknowns! But there're some special function that satisfies that equations, look at this equation: $\cosh^2(\phi) - \sinh^2(\phi) = 1$ THAT'S IT! There is our solution, thus:

$$\begin{aligned} A &= D = \cosh(\phi) \\ C &= B = -\sinh(\phi) \end{aligned}$$

Thus our Lorentz's transformation becomes:

$$\begin{bmatrix} \bar{t} \\ \bar{x} \end{bmatrix} = \begin{bmatrix} \cosh(\phi) & -\sinh(\phi) \\ -\sinh(\phi) & \cosh(\phi) \end{bmatrix} \begin{bmatrix} t \\ x \end{bmatrix} \quad (3.133)$$

That is a beautiful solution! Looks like a rotation! But it is not a rotation, we don't even know what in the hell is ϕ so lets find out what is ϕ .

Let $\bar{x} = 0$ thus $0 = -t \sinh(\phi) + vt \cosh(\phi)$ solving for v we've got:

$$v = \tanh(\phi) \rightarrow \tanh^2(\phi) + \operatorname{sech}^2(\phi) = 1 \rightarrow \operatorname{sech}(\phi) = \sqrt{1 - v^2}$$

$$\gamma = \cosh(\phi) = \frac{1}{\sqrt{1 - v^2}} \quad (3.134)$$

$$\sinh(\phi) = \frac{v}{\sqrt{1 - v^2}} = \gamma v \quad (3.135)$$

With all of this, we can get the final form of our Lorentz's tranformation:

$$\begin{aligned} \bar{t} &= \gamma(t - vx) \\ \bar{x} &= \gamma(x - vt) \\ \bar{y} &= y \\ \bar{z} &= z \end{aligned}$$

If we are not in natural coordinates, the transformation look like this:

$$\begin{aligned} \bar{t} &= \gamma\left(t - \frac{v}{c^2}x\right) \\ \bar{x} &= \gamma(x - vt) \\ \bar{y} &= y \\ \bar{z} &= z \end{aligned}$$

That is something really beautiful, only with the postulates easily we can get nature of the mechanics for high speeds, and easily see that the nature again follows a group's structure! Having all on count we can get the Lorentz's transform:

$$[\Lambda] = \begin{bmatrix} \gamma & -\gamma v_x & -\gamma v_y & -\gamma v_z \\ -\gamma v_x & 1 + (\gamma - 1)\frac{v_x^2}{v^2} & (\gamma - 1)\frac{v_x v_y}{v^2} & (\gamma - 1)\frac{v_x v_z}{v^2} \\ -\gamma v_y & (\gamma - 1)\frac{v_y v_x}{v^2} & 1 + (\gamma - 1)\frac{v_y^2}{v^2} & (\gamma - 1)\frac{v_y v_z}{v^2} \\ -\gamma v_z & (\gamma - 1)\frac{v_z v_x}{v^2} & (\gamma - 1)\frac{v_z v_y}{v^2} & 1 + (\gamma - 1)\frac{v_z^2}{v^2} \end{bmatrix} \quad (3.136)$$

Lets turn back on the geodesics in manifolds. Let t the affine parameter for $\vec{\nabla}_X X = 0$ and suppose another parameter u such that $t = t(u)$ and the following relatio is fulfilled

$$\frac{dt}{du} > 0, \quad (3.137)$$

this condition is someway like the convex property on the Legendre's transformation, the reason why the changing rate of the parameter t respect to u must be greater than zero is to guarantee that for two different values of u they wont have the same image, or the tame, the relation $t = t(u)$ is injective and well behaved.

Lets then find out the geodesics from this other parameterization and what relation must be between these two pictures to they represent the same geodesic see 3.9.

$$Y = hX, \quad h = \frac{dt}{du}, \quad (3.138)$$

$$\vec{\nabla}_Y Y = \vec{\nabla}_{hX}(hX) = h\vec{\nabla}_X(hX) = h \left[X(h)X + h\vec{\nabla}_X X \right] = h^2\vec{\nabla}_X X + hX(h)X, \quad (3.139)$$

$$\vec{\nabla}_Y Y = X(h)hX = fY, \quad f = X(h). \quad (3.140)$$

Then we can see it is not parameterized affinely, the affine parameters are linear relations with the other parameters. Both are the same geodesic but the second one has not affine parameterization, but lets see how must be h to have such a parameterization

$$X(h) = 0 \equiv X^\alpha \partial_\alpha h = 0, \quad (3.141)$$

therefore h is covariantly constant and $\frac{dt}{du} = h$ or what is the same thing

$$u = at + b, \quad a > 0. \quad (3.142)$$

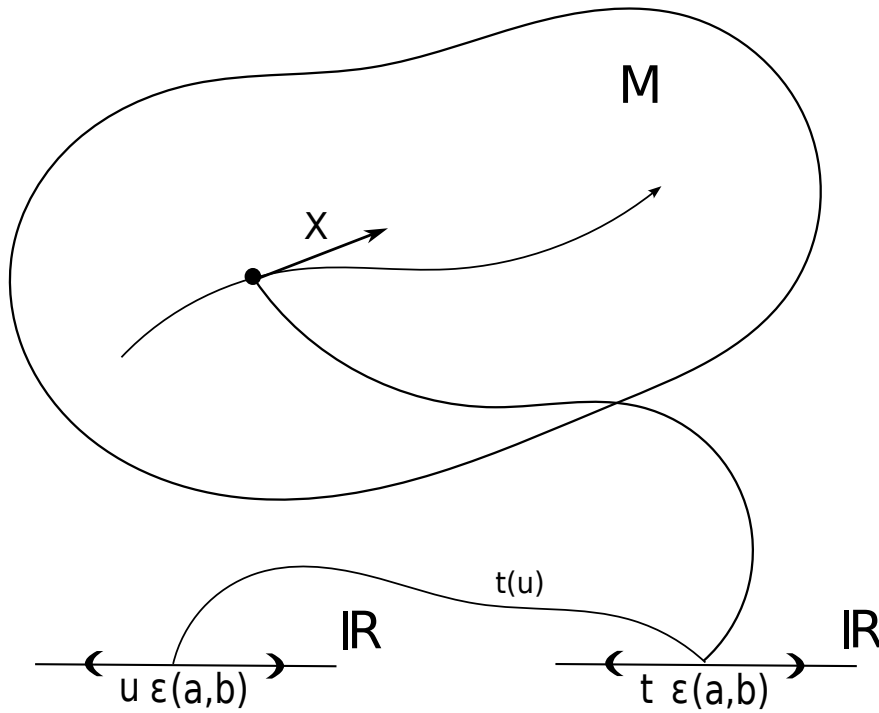


Figure 3.9: Affine parameterization of the geodesics, here we illustrate the path (geodesic) a free particle will follow which is parameterized with a parameter t but this is some application of the other parameter u .

When this is fulfilled u is also an affine parameter, thus its geodesics equation is

$$\vec{\nabla}_Y Y = 0. \quad (3.143)$$

In the general relativity the proper time τ is an affine parameters and it works to describe the time-like geodesics, *the free particles move on geodesics* this having on count the Levi-Civita connection.

3.4 Riemann's tensor

Here we're going to talk about the heart of the general relativity, the heart of the differential geometry (from a physical point of view hehe), the Riemann's tensor, inside this beautiful quantity there is the information of the curvature of our space-time the curvature of the geodesics? the curvature of the quantum vacuum? Yes, all this informations can be found inside this tensor, thus lets have fun on this topic.

3.4.1 Curvature of the Levi-Civita connection

Until here we haven't solved the problem of transporting a vector from a tangent space to another vector in another tangent space, here there is the solution of that, what is called the *parallel transport*.

Let X a tangent vector to a curve, a tensor field T is transported parallelly along the curve if and only if

$$\vec{\nabla}_X T = 0. \quad (3.144)$$

This means the tensor field does not change along the vector, therefore we had already solved the problem of parallel transport implicitly! $\vec{\nabla}_X X = 0$ this a curve whose tangent vector is parallelly transported along the same curve, but but, we need it to be unique, lets see if this is or not unique. Let T be a tensor of rank $\binom{1}{1}$ we're going to define a coordinate chart such that

$$\phi = (x^1, \dots, x^n) \quad X^\mu = \frac{dx^\mu}{d\lambda} \quad (3.145)$$

where λ is the curve parameter;

$$\left(\vec{\nabla}_X T\right)_\nu^\mu = (X^\sigma \nabla_\sigma T_\nu^\mu) = X^\sigma \partial_\sigma T_\nu^\mu + \Gamma_{\rho\sigma}^\mu T_\nu^\rho X^\sigma - \Gamma_{\nu\sigma}^\rho T_\rho^\mu X^\sigma \quad (3.146)$$

$$\left(\vec{\nabla}_X T\right)_\nu^\mu = \frac{dx^\sigma}{d\lambda} \partial_\sigma T_\nu^\mu + \Gamma_{\rho\sigma}^\mu T_\nu^\rho X^\sigma - \Gamma_{\nu\sigma}^\rho T_\rho^\mu X^\sigma = 0. \quad (3.147)$$

We have then a first order equation of the form

$$\frac{dT_\nu^\mu}{d\lambda} + \dots = 0, \quad (3.148)$$

therefore there is a unique solution once we define the initial condition, then the tensor fields in q and p are isomorphic.

Definition 27 The Riemann curvature tensor of a connection $\vec{\nabla}$ is defined by

$$R_{bcd}^a Z^b X^c Y^d = (R(X, Y) Z)^a, \quad (3.149)$$

where X, Y, Z are vector fields and $R(X, Y) Z$ is also a vector field defined by

$$R(X, Y) Z = \vec{\nabla}_X \vec{\nabla}_Y Z - \vec{\nabla}_Y \vec{\nabla}_X Z - \vec{\nabla}_{[X, Y]} Z. \quad (3.150)$$

The Riemann tensor has the following properties

$$1. R(X, Y)Z = -R(Y, X)Z.$$

2.

$$R(fX, Y)Z = \vec{\nabla}_{fX} \vec{\nabla}_Y Z - \vec{\nabla}_Y \vec{\nabla}_{fX} Z - \vec{\nabla}_{[fX, Y]} Z \quad (3.151)$$

$$= f \vec{\nabla}_X \vec{\nabla}_Y Z - \vec{\nabla}_Y (f \vec{\nabla}_X Z) - \vec{\nabla}_{f[X, Y] - Y(f)X} Z \quad (3.152)$$

$$= f \vec{\nabla}_X \vec{\nabla}_Y Z - f \vec{\nabla}_Y \vec{\nabla}_X Z - Y(f) \vec{\nabla}_X Z - f \vec{\nabla}_{[X, Y]} Z + Y(f) \vec{\nabla}_X Z = f R(X, Y)Z. \quad (3.153)$$

This means the Riemann's curvature tensors is multilinear in his two arguments, the demonstration for the second argument is entirely analogous, therefore

$$R(X, fY)Z = fR(X, Y)Z. \quad (3.154)$$

3.

$$\begin{aligned} R(X, Y)fZ &= \vec{\nabla}_X \vec{\nabla}_Y (fZ) - \vec{\nabla}_Y \vec{\nabla}_X (fZ) - \vec{\nabla}_{[X, Y]} fZ \\ &= f \vec{\nabla}_X \vec{\nabla}_Y Z - f \vec{\nabla}_Y \vec{\nabla}_X Z + \vec{\nabla}_X Z \vec{\nabla}_Y f + \vec{\nabla}_Y Z \vec{\nabla}_X f - \vec{\nabla}_{[X, Y]} fZ \\ &= f \left(\vec{\nabla}_X \vec{\nabla}_Y Z - \vec{\nabla}_Y \vec{\nabla}_X Z - \vec{\nabla}_{[X, Y]} Z \right), \end{aligned}$$

$$R(X, Y)fZ = fR(X, Y)Z. \quad (3.155)$$

In this way, the Riemann's tensor fulfills all those properties and correctly it defines a tensor, lets see how looks the Riemann's tensor given a coordinate chart and a basis $\{\mathbf{e}_\mu = \frac{\partial}{\partial x^\mu}\}$ and assuming a free of torsion theory

$$[\mathbf{e}_\mu, \mathbf{e}_\nu] = 0, \quad (3.156)$$

$$R(\mathbf{e}_\rho, \mathbf{e}_\sigma)\mathbf{e}_\mu = \vec{\nabla}_{\mathbf{e}_\rho} \vec{\nabla}_{\mathbf{e}_\sigma} \mathbf{e}_\mu - \vec{\nabla}_{\mathbf{e}_\sigma} \vec{\nabla}_{\mathbf{e}_\rho} \mathbf{e}_\mu, \quad (3.157)$$

$$\begin{aligned} R(\mathbf{e}_\rho, \mathbf{e}_\sigma)\mathbf{e}_\mu &= \vec{\nabla}_{\mathbf{e}_\rho} (\Gamma_{\mu\sigma}^\tau \mathbf{e}_\tau) - \vec{\nabla}_{\mathbf{e}_\sigma} (\Gamma_{\rho\mu}^\tau \mathbf{e}_\tau) \\ &= \vec{\nabla}_{\mathbf{e}_\rho} \Gamma_{\mu\sigma}^\tau \mathbf{e}_\tau + \Gamma_{\mu\sigma}^\tau \vec{\nabla}_{\mathbf{e}_\rho} \mathbf{e}_\tau - \vec{\nabla}_{\mathbf{e}_\sigma} \Gamma_{\rho\mu}^\tau \mathbf{e}_\tau - \Gamma_{\rho\mu}^\tau \vec{\nabla}_{\mathbf{e}_\sigma} \mathbf{e}_\tau, \end{aligned}$$

$$\begin{aligned} R(\mathbf{e}_\rho, \mathbf{e}_\sigma)\mathbf{e}_\mu &= \partial_\rho \Gamma_{\mu\sigma}^\tau \mathbf{e}_\tau + \Gamma_{\mu\sigma}^\tau \Gamma_{\tau\rho}^\nu \mathbf{e}_\nu - \partial_\sigma \Gamma_{\rho\mu}^\tau \mathbf{e}_\tau - \Gamma_{\rho\mu}^\tau \Gamma_{\sigma\tau}^\nu \mathbf{e}_\nu \\ &= \{ \partial_\rho \Gamma_{\mu\sigma}^\tau + \Gamma_{\mu\sigma}^\tau \Gamma_{\tau\rho}^\nu - \partial_\sigma \Gamma_{\rho\mu}^\tau - \Gamma_{\rho\mu}^\tau \Gamma_{\sigma\tau}^\nu \} \mathbf{e}_\nu, \end{aligned}$$

$$R^\nu_{\mu\rho\sigma} = \partial_\rho \Gamma^\nu_{\mu\sigma} + \Gamma^\tau_{\mu\sigma} \Gamma^\nu_{\tau\rho} - \partial_\sigma \Gamma^\nu_{\mu\rho} - \Gamma^\tau_{\mu\rho} \Gamma^\nu_{\sigma\tau}. \quad (3.158)$$

Definition 28 The Ricci's curvature tensor, is a tensor of rank $\binom{0}{2}$ is defined by

$$R_{ab} = R^c_{acb}. \quad (3.159)$$

With this tensor we can construct the Einstein's field equations, under the assumption of a free of torsion theory, and we can define the *Ricci's identity* as

$$R^a_{bcd} Z^b = \vec{\nabla}_c \vec{\nabla}_d Z^a - \vec{\nabla}_d \vec{\nabla}_c Z^a, \quad (3.160)$$

lets prove it

$$\begin{aligned} [R(X, Y)Z]^a &= X^c \vec{\nabla}_c (Y^d \vec{\nabla}_d Z^a) - Y^d \vec{\nabla}_d (X^c \vec{\nabla}_c Z^a) - [X, Y]^c \vec{\nabla}_c Z^a \\ &= X^c Y^d \vec{\nabla}_c \vec{\nabla}_d Z^a + X^c (\vec{\nabla}_c Y^d) \vec{\nabla}_d Z^a - Y^d X^c \vec{\nabla}_d \vec{\nabla}_c Z^a \\ &\quad - Y^d (\vec{\nabla}_d X^c) \vec{\nabla}_c Z^a - [X, Y]^c \vec{\nabla}_c Z^a, \end{aligned}$$

$$[R(X, Y)Z]^a = X^c Y^d [\vec{\nabla}_c \vec{\nabla}_d Z^a - \vec{\nabla}_d \vec{\nabla}_c Z^a] + X^c (\vec{\nabla}_c Y^d) \vec{\nabla}_d Z^a - Y^d (\vec{\nabla}_d X^c) \vec{\nabla}_c Z^a - [X, Y]^c \vec{\nabla}_c Z^a. \quad (3.161)$$

and for a free of torsion theory we can develop the commutator

$$[X, Y]^c = [\vec{\nabla}_X Y - \vec{\nabla}_Y X]^c = X^e \vec{\nabla}_e Y^c - Y^e \vec{\nabla}_e X^c, \quad (3.162)$$

$$[X, Y]^c \vec{\nabla}_c Z^a = X^e (\vec{\nabla}_e Y^c) (\vec{\nabla}_c Z^a) - Y^e (\vec{\nabla}_e X^c) (\vec{\nabla}_c Z^a), \quad (3.163)$$

$$X^c (\vec{\nabla}_c Y^d) (\vec{\nabla}_d Z^a) - Y^d (\vec{\nabla}_d X^c) (\vec{\nabla}_c Z^a) - [X, Y]^c \vec{\nabla}_c Z^a = 0, \quad (3.164)$$

$$[R(X, Y)Z]^a = (\vec{\nabla}_c \vec{\nabla}_d Z^a - \vec{\nabla}_d \vec{\nabla}_c Z^a) X^c Y^d = R^a_{bcd} Z^b X^c Y^d. \quad (3.165)$$

By comparison it only remains that

$$\vec{\nabla}_c \vec{\nabla}_d Z^a - \vec{\nabla}_d \vec{\nabla}_c Z^a = R^a_{bcd} Z^b. \quad (3.166)$$

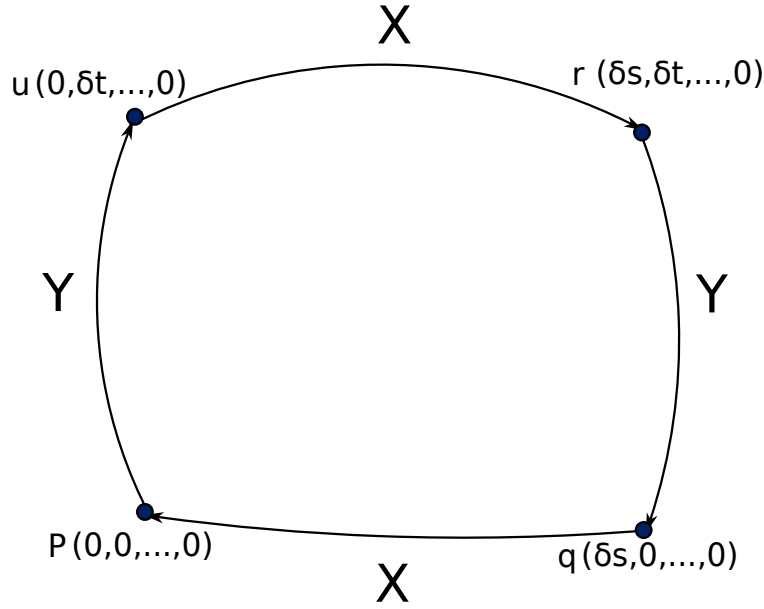


Figure 3.10: Connection between q and p with a curve along a variation of s with X as tangent vector, in the same way q and r can be connected by a curve with Y as tangent vector, p and u can be connected with a curve with Y as tangent vector and same with u and r but with X as tangent vector.

Now, let's turn back to the relation between the Riemann's tensor and the dependence of the path of the parallel transport. Let X and Y vector fields which are linearly independent everywhere such that $[X, Y] = 0$. Let $(s, t, \dots)$ a coordinate chart such that

$$X = \frac{\partial}{\partial s} \text{ \& } Y = \frac{\partial}{\partial t}. \quad (3.167)$$

Let $p \in M$ and given a coordinate chart p is represented as $(0, \dots, 0)$. Let q, r, u points with coordinates $(\delta s, 0, 0, \dots)$, $(\delta s, \delta t, 0, \dots)$, $(0, \delta t, 0, \dots)$, respectively, where $\delta s, \delta t$ are little displacement of that coordinate, we can connect the four points as is illustrated in the figure 3.10.

Now, let $Z_p \in T_p(M)$. Let's transport parallelly Z_p along pqr to obtain a vector $Z_r \in T_r(M)$, the same thing but to transport Z_p along pur to obtain the vector $Z'_r \in T_r(M)$ and let's calculate the difference $Z'_r - Z_r$ for a free of torsion connection. it is convenient to introduce a new coordinate chart known as *normal coordinates* in p , $(\Gamma^\mu_{\nu\rho}(p) = 0)$. The indices μ, ν will refer to this chart. s and t will be used as parameters along the curves whose tangent vectors are X and Y , respectively.

pq is an integral curve whose tangent vector is X and whose parameter is s . Along of pq , Z is transported parallelly: $\vec{\nabla}_X Z = 0$, therefore

$$\frac{dZ^\mu}{ds} = -\Gamma_{\nu\rho}^\mu Z^\nu X^\rho, \quad (3.168)$$

and in this way

$$\frac{d^2 Z^\mu}{ds^2} = -(\Gamma_{\nu\rho}^\mu Z^\nu X^\rho)_{,\sigma} X^\sigma. \quad (3.169)$$

Now using Taylor's expansion theorem we get:

$$Z_q^\mu = Z_p^\mu + \left(\frac{dZ^\mu}{ds}\right)_p \delta s + \frac{1}{2} \left(\frac{d^2 Z^\mu}{ds^2}\right)_p \delta s^2 + \mathcal{O}(\delta s^3). \quad (3.170)$$

We must note that:

$$\frac{d^2 Z^\mu}{ds^2} = -\Gamma_{\nu\rho,\sigma}^\mu Z^\nu X^\rho X^\sigma - \Gamma_{\nu\rho}^\mu Z_{,\sigma}^\nu X^\rho X^\sigma - \Gamma_{\nu\rho}^\mu Z^\nu X_{,\sigma}^\rho X^\sigma, \quad (3.171)$$

$$Z_q^\mu = Z_p^\mu - \frac{1}{2} (\Gamma_{\nu\rho,\sigma}^\mu Z^\nu X^\rho X^\sigma)_p \delta s^2 + \mathcal{O}(\delta s^3), \quad (3.172)$$

where we had used $\Gamma_{\nu\rho}^\mu(p) = 0$ in normal coordinates at p . Now consider the parallel transport along qr :

$$\frac{dZ^\mu}{dt} = -\Gamma_{\nu\rho}^\mu Z^\nu Y^\rho, \quad (3.173)$$

$$\frac{d^2 Z^\mu}{dt^2} = -(\Gamma_{\nu\rho}^\mu Z^\nu Y^\rho)_{,\sigma} Y^\sigma, \quad (3.174)$$

$$Z_r^\mu = Z_q^\mu - (\Gamma_{\nu\rho}^\mu Z^\nu Y^\rho)_q \delta t - \frac{1}{2} [(\Gamma_{\nu\rho}^\mu Z^\nu Y^\rho)_{,\sigma} Y^\sigma]_q \delta t^2 + \mathcal{O}(\delta t^3). \quad (3.175)$$

$$(\Gamma_{\nu\rho}^\mu Z^\nu Y^\rho)_q = (\Gamma_{\nu\rho}^\mu Z^\nu Y^\rho)_p + \left[\frac{d}{ds} (\Gamma_{\nu\rho}^\mu Z^\nu Y^\rho)\right]_p \delta s + \mathcal{O}(\delta s^2) \quad (3.176)$$

$$= [(\Gamma_{\nu\rho}^\mu Z^\nu Y^\rho)_{,\sigma} X^\sigma]_p \delta s + \mathcal{O}(\delta s^2) \quad (3.177)$$

$$= (\Gamma_{\nu\rho,\sigma}^\mu Z^\nu Y^\rho X^\sigma)_p \delta s + \mathcal{O}(\delta s^2) \quad (3.178)$$

$$[(\Gamma_{\nu\rho}^\mu Z^\nu Y^\rho)_{,\sigma}]_q = [(\Gamma_{\nu\rho}^\mu Z^\nu Y^\rho)_{,\sigma} Y^\sigma]_p + \mathcal{O}(\delta s) = (\Gamma_{\nu\rho,\sigma}^\mu Z^\nu Y^\rho Y^\sigma)_p + \mathcal{O}(\delta s), \quad (3.179)$$

$$Z_r^\mu = Z_q^\mu - \left[(\Gamma_{\nu\rho,\sigma}^\mu Z^\nu Y^\rho X^\sigma)_p \delta s + \mathcal{O}(\delta s^2) \right] \delta t - \frac{1}{2} \left[(\Gamma_{\nu\rho}^\mu Z^\nu Y^\rho X^\sigma)_p \delta s + \mathcal{O}(\delta s^2) \right] \delta t^2 + \mathcal{O}(\delta t^2), \quad (3.180)$$

$$Z_r^\mu = Z_p^\mu - \frac{1}{2} (\Gamma_{\nu\rho,\sigma}^\mu)_p \left[Z^\nu (X^\rho X^\sigma \delta s^2 + Y^\rho Y^\sigma \delta t^2 + 2Y^\rho X^\sigma \delta s \delta t) \right]_p + \mathcal{O}(\delta^3). \quad (3.181)$$

Now, consider the parallel transport along *pur*. The result can be obtained from the last expression interchanging *X* with *Y* and *s* with *t*

$$Z_r^\mu = Z_p^\mu - \frac{1}{2} (\Gamma_{\nu\rho,\sigma}^\mu)_p \left[Z^\nu (Y^\rho Y^\sigma \delta t^2 + X^\rho X^\sigma \delta s^2 + 2X^\rho Y^\sigma \delta t \delta s) \right]_p + \mathcal{O}(\delta^3) \quad (3.182)$$

in this way the difference between the two parallel transport is

$$\Delta Z_r^\mu = Z_r^\mu - Z_r^\mu \quad (3.183)$$

$$= [\Gamma_{\nu\rho,\sigma}^\mu Z^\nu (Y^\rho X^\sigma - X^\rho Y^\sigma)]_p \delta t \delta s + \mathcal{O}(\delta^3) \quad (3.184)$$

$$= [(\Gamma_{\nu\sigma,\rho}^\mu - \Gamma_{\nu\rho,\sigma}^\mu) Z^\nu X^\rho Y^\sigma]_p \delta s \delta t \mathcal{O}(\delta^3) \quad (3.185)$$

$$= [(R_{\nu\rho\sigma}^\mu - \Gamma_{\nu\sigma}^\tau \Gamma_{\tau\rho}^\mu + \Gamma_{\nu\rho}^\tau \Gamma_{\tau\sigma}^\mu) Z^\nu X^\rho Y^\sigma]_p \delta t \delta s + \mathcal{O}(\delta^3), \quad (3.186)$$

$$\Delta Z_r^\mu = (R_{\nu\rho\sigma}^\mu Z^\nu X^\rho Y^\sigma)_p \delta s \delta t + \mathcal{O}(\delta^3), \quad (3.187)$$

$$\Delta Z_r^\mu = (R_{\nu\rho\sigma}^\mu Z^\nu X^\rho Y^\sigma)_r \delta s \delta t + \mathcal{O}(\delta^3). \quad (3.188)$$

In the last equality is used that the quantities in *p* and *r* are different by $\mathcal{O}(\delta)$. this result is derived in a coordinate basis defined using normal coordinates in *p* but now both side take on count tensors in *r* therefore our equations is independent of the basis, thus

$$\left(R_{bcd}^a Z^b X^c Y^d \right)_r = \lim_{\delta \rightarrow 0} \frac{\Delta Z_r^\mu}{\delta s \delta t}. \quad (3.189)$$

The Riemann's tensor measures the dependence of the path of the parallel transport. The last result can be reinterpreted describing the effect of the parallel transport of a vector Z_r^a along the closed curve *pqrup*, see figure 3.10, to obtain the vector $Z_r'^a$. In this way ΔZ_r^a measures the change in Z_r^a when it is transported parallelly around a closed curve.

Lets see the symmetries of the Riemann's tensor

1. $R^a_{bcd} = -R^a_{bdc}$ it is anti symmetric.
2. If $\vec{\nabla}$ is a free of torsion connections then

$$R^a_{[bcd]} = \frac{1}{3!} (R^a_{bcd} + R^a_{cdb} + R^a_{dbc} - R^a_{cbd} - R^a_{bdc} - R^a_{dcb}) = 0. \quad (3.190)$$

3. If $\vec{\nabla}$ is a free of torsion connection then

$$R^a_{b[cd;e]} = 0. \quad (3.191)$$

Thanks to the fact the metric tensor is covariantly constant, then it will allow us to raise and lower indices and using these we can see the following properties

1. $R_{abcd} = R_{cdab}$.
2. $R_{ab} = R_{ba}$.

Definition 29 *The Ricci's scalar is*

$$R = g^{ab} R_{ab}. \quad (3.192)$$

Definition 30 *The Einstein's tensor, is a symmetric tensor of rank $\binom{0}{2}$ and is defined as*

$$G_{ab} = R_{ab} - \frac{1}{2} g_{ab} R. \quad (3.193)$$

Proposition: the covariant derivative of the Einstein's tensor is zero

$$\vec{\nabla}_a G^{ab} = 0. \quad (3.194)$$

Demonstration: it is worth being careful here, because this result does not follow from the Leibniz rule alone, it needs the second Bianchi identity, and any argument that avoids the Bianchi identity is quietly assuming what it wants to prove. Let us take the derivative of the Einstein tensor with its indices raised, using that the metric is covariantly constant, $\vec{\nabla}_a g^{bc} = 0$, so that the metric may be pulled in and out of any covariant derivative at will,

$$\vec{\nabla}_a G^{ab} = \vec{\nabla}_a \left[R^{ab} - \frac{1}{2} g^{ab} R \right] = \vec{\nabla}_a R^{ab} - \frac{1}{2} g^{ab} \vec{\nabla}_a R.$$

The first term is exactly what the twice contracted Bianchi identity, which we shall derive in the following subsection, gives us,

$$\vec{\nabla}_a R^{ab} = \frac{1}{2} \vec{\nabla}^b R = \frac{1}{2} g^{ab} \vec{\nabla}_a R, \quad (3.195)$$

and substituting equation 3.195 into the previous line the two contributions cancel exactly,

$$\vec{\nabla}_a G^{ab} = \frac{1}{2} g^{ab} \vec{\nabla}_a R - \frac{1}{2} g^{ab} \vec{\nabla}_a R = 0.$$

This identity is not a curiosity, it is the reason the Einstein tensor and not the Ricci tensor sits on the left hand side of the field equations, because the energy momentum tensor on the right hand side is itself conserved, and a conserved source demands a geometrical object whose divergence vanishes automatically, for every metric and without imposing any extra condition.

3.4.1.1 Einstein's equations

These equations holds under some postulates, the known as the general relativity postulates which are

1. The space-time is a pseudo-Riemannian manifold equipped with a Levi-Civita connection $(M, \mathbb{G}, \vec{\nabla})$.
2. The free particles follows geodesics.
3. The energy-momentum tensor T_{ab} is conserved: $\vec{\nabla}_a T^{ab} = 0$.
4. The Einstein's tensor satisfies that

$$G_{ab} = R_{ab} - \frac{1}{2} g_{ab} R = \frac{8\pi G}{C^4} T_{ab}. \quad (3.196)$$

If there is no matter at all, so that $T_{ab} = 0$, then the Ricci scalar and the Ricci tensor both vanish, and it is worth seeing why

$$\begin{aligned} g^{ab} \left(R_{ab} - \frac{1}{2} g_{ab} R \right) &= \alpha g^{ab} T_{ab}, \\ R - 2R &= \alpha T \rightarrow T = -\frac{1}{\alpha} R, \\ R_{ab} &= \alpha T_{ab} + \frac{1}{2} g_{ab} R = \alpha \left(T_{ab} - \frac{1}{2} g_{ab} T \right), \end{aligned} \quad (3.197)$$

we can see easily that if there is no matter the Ricci's tensor is zero.

The Einstein's equations are ten second order, non linear and coupled equations of the metric tensor, but this form is not unique. Lets think about a symmetric tensor H_{ab} , which is covariantly constant, the space-time must be a pseudo-Riemannian manifold of four dimensions, with a Levi-Civita connections, thus by the **Lovelock theorem**

$$H_{ab} = \alpha G_{ab} + \beta g_{ab}, \quad (3.198)$$

$$G_{\alpha\beta} + \Lambda g_{\alpha\beta} = \frac{8\pi G}{C^4} T_{\alpha\beta}. \quad (3.199)$$

Λ is known as the cosmological constant. The history here is worth getting right, because it is usually told backwards: Einstein *introduced* the term himself in 1917, and he did it in order to force his equations to describe a static universe, which is what everybody believed the universe to be at the time. He then abandoned it once Hubble showed in 1929 that the universe is expanding (Hubble, 1929). The accelerated expansion, which brought the constant back into fashion, was only discovered in 1998, more than forty years after Einstein's death (Riess, Adam G. et al., 1998; Perlmutter, Saul et al., 1999), and today Λ is studied in connection with dark energy.

3.4.2 Properties related to the Riemann's tensor

In this part we're going to study tensor equations related to the Riemann's tensor and some really important identities as the Jacobi and Bianchi's identities, which are really important in the development of general relativity and will guide us to the Einstein's tensor. Let $\{\mathbf{e}_\alpha\} = \left\{\frac{\partial}{\partial x^\alpha}\right\}$ a coordinate basis, then in this basis the components of the Riemann's tensor become

$$R^\alpha_{\beta\mu\nu} = \partial_\mu \Gamma^\alpha_{\beta\nu} - \partial_\nu \Gamma^\alpha_{\beta\mu} + \Gamma^\alpha_{\sigma\mu} \Gamma^\sigma_{\beta\nu} - \Gamma^\alpha_{\sigma\nu} \Gamma^\sigma_{\beta\mu}. \quad (3.200)$$

We're going to consider the components of the Riemann's tensor in a locally inertial reference frame known as *normal coordinates*, these are special such that

$$\Gamma^\alpha_{\sigma\mu}(p) = 0, \quad \partial_\mu \Gamma^\alpha_{\beta\nu}(p) \neq 0, \quad (3.201)$$

from these point of view the Riemann's tensor takes the following form:

$$R^\alpha_{\beta\mu\nu} = \partial_\mu \Gamma^\alpha_{\beta\nu} - \partial_\nu \Gamma^\alpha_{\beta\mu}, \quad (3.202)$$

and we gotta calculate the Christoffel's symbol, thus

$$\begin{aligned}
\Gamma_{\mu\nu}^{\alpha} &= \frac{1}{2} g^{\alpha\beta} (\partial_{\nu} g_{\beta\mu} + \partial_{\mu} g_{\beta\nu} - \partial_{\beta} g_{\mu\nu}) . \\
\partial_{\sigma} \Gamma_{\mu\nu}^{\alpha} &= \frac{1}{2} g^{\alpha\beta} (\partial_{\sigma} \partial_{\nu} g_{\beta\mu} + \partial_{\sigma} \partial_{\mu} g_{\beta\nu} - \partial_{\sigma} \partial_{\beta} g_{\mu\nu}) \\
&\quad + \frac{1}{2} (\partial_{\nu} g_{\beta\mu} + \partial_{\mu} g_{\beta\nu} - \partial_{\beta} g_{\mu\nu}) \partial_{\sigma} g^{\alpha\beta} \\
\partial_{\sigma} \Gamma_{\mu\nu}^{\alpha} &= \frac{1}{2} g^{\alpha\beta} (\partial_{\sigma} \partial_{\nu} g_{\beta\mu} + \partial_{\sigma} \partial_{\mu} g_{\beta\nu} - \partial_{\sigma} \partial_{\beta} g_{\mu\nu}) + \frac{1}{2} [\beta, \mu\nu] \partial_{\sigma} g^{\alpha\beta}, \tag{3.203}
\end{aligned}$$

$$\partial_{\sigma} \Gamma_{\mu\nu}^{\alpha} = \frac{1}{2} g^{\alpha\beta} (\partial_{\sigma} \partial_{\nu} g_{\beta\mu} + \partial_{\sigma} \partial_{\mu} g_{\beta\nu} - \partial_{\sigma} \partial_{\beta} g_{\mu\nu}) . \tag{3.204}$$

we're going to use the equation 3.204 on the Riemann's tensor on the point $p \in M$ in the following way

$$\begin{aligned}
R_{\beta\mu\nu}^{\alpha}|_p &= \frac{1}{2} g^{\alpha\sigma} (\partial_{\mu} \partial_{\nu} g_{\sigma\beta} + \partial_{\mu} \partial_{\beta} g_{\sigma\nu} - \partial_{\mu} \partial_{\sigma} g_{\beta\nu})|_p - \frac{1}{2} g^{\alpha\sigma} (\partial_{\nu} \partial_{\mu} g_{\sigma\beta} + \partial_{\nu} \partial_{\beta} g_{\sigma\mu} - \partial_{\mu} \partial_{\sigma} g_{\mu\beta})|_p, \\
R_{\beta\mu\nu}^{\alpha}|_p &= \frac{1}{2} g^{\alpha\sigma} (\partial_{\beta} \partial_{\mu} g_{\sigma\nu} - \partial_{\nu} \partial_{\beta} g_{\sigma\mu} + \partial_{\nu} \partial_{\sigma} g_{\mu\beta} - \partial_{\mu} \partial_{\sigma} g_{\beta\nu})|_p, \tag{3.205}
\end{aligned}$$

$$R_{\alpha\beta\mu\nu}|_p = g_{\alpha\lambda} R_{\beta\mu\nu}^{\lambda} = \frac{1}{2} (\partial_{\beta} \partial_{\mu} g_{\alpha\nu} - \partial_{\nu} \partial_{\beta} g_{\alpha\mu} + \partial_{\nu} \partial_{\alpha} g_{\mu\beta} - \partial_{\mu} \partial_{\alpha} g_{\beta\nu})|_p \tag{3.206}$$

From here on we can see the following relations, which are tensor equations and do not depend on the coordinate chart

$$R_{\alpha\beta\mu\nu} = -R_{\beta\alpha\mu\nu}, \tag{3.207}$$

$$R_{\alpha\beta\mu\nu} = -R_{\alpha\beta\nu\mu}, \tag{3.208}$$

$$R_{\alpha\beta\mu\nu} = R_{\mu\nu\alpha\beta}, \tag{3.209}$$

$$R_{\alpha\beta\mu\nu} + R_{\alpha\nu\beta\mu} + R_{\alpha\mu\nu\beta} = R_{\alpha[\beta\mu\nu]} = 0. \tag{3.210}$$

These relations and symmetries on the Riemann's tensor reduce the $4^4 = 256$ components of the Riemann tensor in four dimensions to only 20 independent ones. There are more relations beyond theses, the known as the *Bianchi identities* which will be worth to define and arrive the Einstein's tensor. In normal coordinates we have got

$$\vec{\nabla}_{\lambda} R_{\alpha\beta\mu\nu} = \partial_{\lambda} R_{\alpha\beta\mu\nu} = \frac{1}{2} (\partial_{\lambda} \partial_{\beta} \partial_{\mu} g_{\alpha\nu} - \partial_{\lambda} \partial_{\nu} \partial_{\beta} g_{\alpha\mu} + \partial_{\lambda} \partial_{\nu} \partial_{\alpha} g_{\mu\beta} - \partial_{\lambda} \partial_{\mu} \partial_{\alpha} g_{\beta\nu})|_p, \tag{3.211}$$

We can find an analogous to the Jacobi's identity with covariant derivatives

$$\vec{\nabla}_\lambda R_{\alpha\beta\mu\nu} + \vec{\nabla}_\nu R_{\alpha\beta\lambda\mu} + \vec{\nabla}_\mu R_{\alpha\beta\nu\lambda} = 0, \quad (3.212)$$

$$R_{\alpha\beta[\mu\nu;\lambda]} = \vec{\nabla}_{[\lambda} R_{\mu\nu]\alpha\beta} = 0. \quad (3.213)$$

All this geometrical development was done by Riemann with the intention of generalizing the Gauss work on curvature. Lets find out the contracted Bianchi's identities, therefore

$$\left[\vec{\nabla}_e R_{abcd} + \vec{\nabla}_d R_{abec} + \vec{\nabla}_c R_{abde} = 0 \right] g^{ac}, \quad (3.214)$$

$$\vec{\nabla}_e R_{bd} + \vec{\nabla}_d (g^{ac} R_{abec}) + g^{ac} \vec{\nabla}_c R_{abde} = 0, \quad (3.215)$$

$$\vec{\nabla}_e R_{bd} - \vec{\nabla}_d R_{be} + \vec{\nabla}^a R_{abde} = 0, \quad (3.216)$$

$$\vec{\nabla}_e R_{bd} - \vec{\nabla}_d R_{be} + \vec{\nabla}^a R_{abde} = 0. \quad (3.217)$$

From the Bianchi's identity for the Riemann's tensor we've found the respective identity for the Ricci's tensor also known as the *contracted Bianchi identity* given in the equations 3.217 but we can contract it one more time and obtain the Bianchi identity for the Ricci's scalar which is related with the Einstein's tensor and also is a clue to the Einstein's equations, so contracting the equation 3.217 we'll then have

$$g^{bd} \left[\vec{\nabla}_e R_{bd} - \vec{\nabla}_d R_{be} + \vec{\nabla}^a R_{abde} = 0 \right], \quad (3.218)$$

$$\vec{\nabla}_e R - \vec{\nabla}^b R_{be} - \vec{\nabla}^a R_{ae} = 0, \quad (3.219)$$

$$\vec{\nabla}^a R_{ae} = \frac{1}{2} \vec{\nabla}_e R. \quad (3.220)$$

The equation 3.220 is known as the Bianchi's identity twice contracted and playing a bit with this equations we will find something special, lets find out it

$$\begin{aligned} \vec{\nabla}^a R_{ae} - \frac{1}{2} \vec{\nabla}_e R &= 0, \\ \vec{\nabla}^a R_{ae} - \frac{1}{2} g_{ae} \vec{\nabla}^a R &= 0, \\ \vec{\nabla}^a \left(R_{ae} - \frac{1}{2} g_{ae} R \right) &= 0, \end{aligned}$$

$$\vec{\nabla}^a G_{ab} = 0 \ ; \ G_{ab} = R_{ab} - \frac{1}{2} g_{ab} R, \quad (3.221)$$

There appears, quite naturally, the Einstein tensor! Now, in the curved spaces the geodesics aren't always parallel there are some separation between one or another geodesic but how can we measure or have an idea of how far or deviated is a geodesic from other.

3.4.3 Geodesic deviation

Definition 31 *Let M a manifold with a Levi-Civita connection $\vec{\nabla}$. A monoparametric family of geodesics is a map*

$$\gamma : I \times I' \rightarrow M, \quad (3.222)$$

where I, I' are two open intervals on the $\mathbb{R}$ such that

1. *For a S constant, $\gamma(s, t) = \gamma_s(t)$ is a geodesic with affine parameters t , therefore S would be a parameter that tags the geodesic.*
2. *The collection of these curves define a 2D smooth surface and is a subset of the manifold $\Sigma \subset M$ as is represented in the figure 3.11*

To quantify the deviation between two adjacent geodesics, lets first, suppose a coordinate chart $x^\mu(s, t) \in M$, therefore

$$T^\mu = \frac{\partial x^\mu}{\partial t}, \quad S^\mu = \frac{\partial x^\mu}{\partial s}. \quad (3.223)$$

The first vectors T^μ are called the tangent vectors to the geodesics, and the other ones S^μ are named the deviation vectors. We're going to express the vectors of the chart in terms of the vectors of deviation.

$$x^\mu(s + \delta s, t) = x^\mu(s, t) + \left(\frac{\partial x^\mu}{\partial s} \right) \delta s + \mathcal{O}(\delta s^2). \quad (3.224)$$

We're going to analyze the vector S and its changing along the geodesic T . To do it, we can define a relative speed between geodesics

$$V^a = \left(\vec{\nabla}_T S \right)^a = T^b \vec{\nabla}_b S^a, \quad (3.225)$$

if we can talk about the relative speed between geodesics, we can also talk about the acceleration relative between geodesics, then

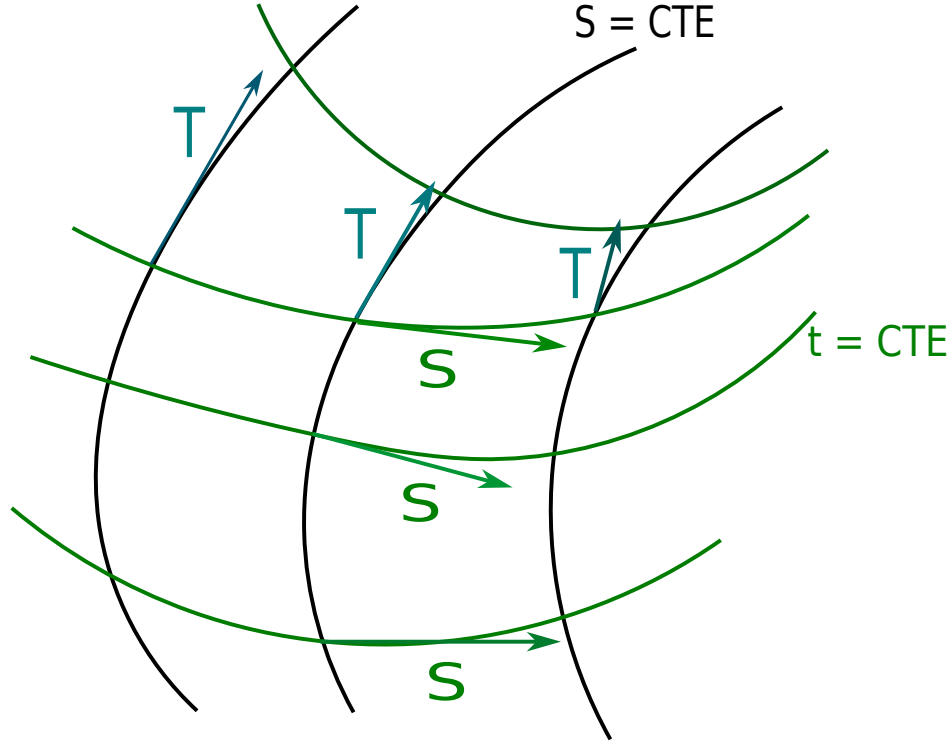


Figure 3.11: Representation of all the possible subsets of S constant and t constant, this is analogous to what is done in the formulation 3 + 1 where we separate the space-time and folds as a Cauchy space.

$$a^a = \left(\vec{\nabla}_T V \right)^a = T^b \vec{\nabla}_b V^a = T^c \vec{\nabla}_c \left(T^b \vec{\nabla}_b S^a \right) = \left(\vec{\nabla}_T \vec{\nabla}_T S \right)^a, \quad (3.226)$$

we can introduce the coordinate basis on Σ as $S = \partial_s$ and $T = \partial_t$ and because we're having a theory free of torsion so from the commutator we have

$$[T, S](f) = T[S(f)] - S[T(f)] = \partial_t[S(f)] - \partial_s[T(f)] = 0, \quad (3.227)$$

under these conditions and without torsion then

$$\vec{\nabla}_T S = \vec{\nabla}_S T \rightarrow T^b \nabla_b S^a = S^b \nabla_b T^a, \quad (3.228)$$

$$\vec{\nabla}_T \vec{\nabla}_T S = R(T, S)T. \quad (3.229)$$

Using 3.229 into the acceleration relative into geodesics, then we have

$$a^a = [\vec{\nabla}_T \vec{\nabla}_T S]^a = (R(T, S)T)^a = R^a_{bcd} T^b T^c S^d, \quad (3.230)$$

$$a^a = R^a_{bcd} T^b T^c S^d. \quad (3.231)$$

The equation 3.231 implies something really, really amazing and important ***The acceleration is curvature*** but we gotta read this equations correctly, as cause and effect, the presence of curvature produces an effect we call acceleration, we can write the equation 3.231 in other way, as

$$a^a = \frac{D^2}{Dt^2} S^a = R^a_{bcd} T^b T^c S^d. \quad (3.232)$$

3.5 Gravitation and the Einstein's field equation

We have found a relation between the curvature tensor and the relative acceleration between geodesics, and knowing a particle test has a geodesics as a path, then different curvatures produces different accelerations, and from the experiments we've talk about in the first part acceleration is indistinguishable from gravity.

Well then, how are going to relate the acceleration to curvature? To do this, we're going to solve the Einsteins field equations in the Newtonian limit to interpret this physically and see whether Newton's theory was flat or curved, and whether it was a geometrical theory from the very beginning. First we're going to set some properties of the manifolds to set the space-time and use it to develop this theory.

1. The set of all the events in a $4D$ manifold with pseudo-Riemannian metric and a Levi-Civita connections is a differential manifold called *space-time* denoted by $(M, \mathbb{G}, \vec{\nabla})$.
2. The Manifold is Lorentzian.
3. The weak equivalence principle is fulfilled.
4. The Einstein's equivalence principle is fulfilled.

3.5.1 Newtonian limit

We gotta then talk about particles whose speed is a lot less than the speed of light C and the gravitational field is stationary and weak, using geometrical coordinates $C = 1$ then

$$\frac{d\vec{X}}{d\tau} \ll \frac{dt}{d\tau}, \quad (3.233)$$

$$\frac{d\vec{X}}{d\tau} = \frac{d\vec{X}}{dt} \frac{dt}{d\tau} \ll \frac{dt}{d\tau} \rightarrow \left| \frac{d\vec{X}}{dt} \right| \ll 1. \quad (3.234)$$

Having on count all of this, we're going to see the geodesic equation

$$\frac{d^2 x^\mu}{d\tau^2} + \Gamma_{\alpha\beta}^\mu \frac{dx^\alpha}{d\tau} \frac{dx^\beta}{d\tau} = 0, \quad (3.235)$$

because we've low speed all the components related to spatial ones when multiplied are really low valued so those terms are going to be ignored, thus

$$\frac{d^2 x^\mu}{d\tau^2} + \Gamma_{00}^\mu \frac{dx^0}{d\tau} \frac{dx^0}{d\tau} = 0, \quad (3.236)$$

$$\Gamma_{00}^\mu = \frac{1}{2} g^{\mu\sigma} (\partial_0 g_{\sigma 0} + \partial_0 g_{0\sigma} - \partial_\sigma g_{00}) \quad (3.237)$$

and because the gravitational field is stationary, the zeroth derivatives are then zero, thus

$$\Gamma_{00}^\mu = -\frac{1}{2} g^{\mu\sigma} \partial_\sigma g_{00}. \quad (3.238)$$

Then the geodesic equation becomes

$$\frac{d^2 x^\mu}{d\tau^2} - \frac{1}{2} g^{\mu\sigma} \partial_\sigma g_{00} \left(\frac{dt}{d\tau} \right)^2 = 0. \quad (3.239)$$

If the gravitational field is weak, then what we have physically is a flat Minkowskian space-time which in a local region has a perturbation or tensor, then the global metric can be set as

$$g_{\alpha\beta} = \eta_{\alpha\beta} + h_{\alpha\beta}; \quad |h_{\alpha\beta}| \ll 1 \quad (3.240)$$

with this we can calculate the Christoffel symbol:

$$\Gamma_{00}^\mu = -\frac{1}{2} (\eta^{\mu\sigma} - h^{\mu\sigma}) \partial_\sigma (\eta_{00} + h_{00}) = -\frac{1}{2} (\eta^{\mu\sigma} - h^{\mu\sigma}) \partial_\sigma h_{00} \approx -\frac{1}{2} \eta^{\mu\sigma} \partial_\sigma h_{00}. \quad (3.241)$$

$$\frac{d^2 x^\mu}{d\tau^2} = \frac{1}{2} \eta^{\mu\sigma} \partial_\sigma h_{00} \left(\frac{dt}{d\tau} \right)^2. \quad (3.242)$$

Because the Minkowski metric is a diagonal metric then we can split it into two equations

1. $\mu = \sigma = 0$ we find that *in the Newton's theory t is an affine parameter* and t and τ are related affinely

$$\frac{d^2 t}{d\tau^2} = 0. \quad (3.243)$$

2. $\mu = \sigma = i$

$$\frac{d^2 x^i}{dt^2} = \frac{1}{2} \partial_i h_{00} \quad (3.244)$$

In the right hand side of the last equation we see a gradient, this is looking like a Newton's second law, lets find out what is h_{00} , therefore

$$\frac{d^2 \vec{X}}{dt^2} = -\vec{\nabla} \left(-\frac{GM}{r} \right), \quad (3.245)$$

$$h_{00} = \frac{2GM}{r} + \text{constant} = -2\phi + \text{constant}. \quad (3.246)$$

And far from the source it must be asymptotically flat, that is Minkowskian, then the constant we add to h_{00} is then zero, thus the metric becomes

$$g_{00} = -(1 + 2\phi), \quad (3.247)$$

$$g_{ii} = (1 - 2\phi). \quad (3.248)$$

Thanks to these we can introduce the *covariance principle* which sets that if we develop a theory in a locally flat neighbourhood we can generalize it changing the usual derivatives by covariant derivatives.

3.5.2 Einstein's field equations

Using the covariance principle we can generalize the following electrodynamic equations

$$\partial_\alpha T^{\alpha\beta} = 0 \rightarrow \vec{\nabla}_\alpha T^{\alpha\beta} = 0, \quad (3.249)$$

$$\partial_\beta F^{\alpha\beta} = 4\pi J^\alpha \rightarrow \vec{\nabla}_\beta F^{\alpha\beta} = 4\pi J^\alpha, \quad (3.250)$$

$$\partial_\beta^* F^{\alpha\beta} = 0 \rightarrow \vec{\nabla}_\beta^* F^{\alpha\beta} = 0. \quad (3.251)$$

We see from one side the matter and the other the geometry, the presence of matter implies a change in geometry? We will turn back to this question later. Now, we're going to start from the Einstein's equations in the Newtonian limit

$$\nabla^2 \phi = 4\pi G \rho, \quad \nabla^2 = \delta^{ij} \partial_i \partial_j, \quad (3.252)$$

and introducing the energy-momentum tensor which contains the information of the energy density, momentum flux and strains then the first component of that tensor is the mass density

(remembering from the mass-energy equivalence) is $T^{00} = \rho$, and in some point in the Newtonian limit then

$$\nabla^2 h_{00} = -8\pi G T_{00}. \quad (3.253)$$

But we've using usual derivatives and we must work with tensor equations, then to solve this problem with the Laplacian we are going to introduce the d'Alembertian $\square = \nabla^a \nabla_a$ and because we've got a second order equation we gotta use a second order geometry equation, therefore the Riemann's tensor must be introduced but this has a rank $\binom{1}{3}$ then the Ricci's tensor is a better option. Einstein's proposed the Ricci's tensor is directly proportional to the energy-momentum tensor $R_{ab} = k T_{ab}$ but from the conservation law we have got

$$\vec{\nabla}^a T_{ab} = 0 \rightarrow \vec{\nabla}^a R_{ab} = 0, \quad (3.254)$$

but

$$\vec{\nabla}^a R_{ab} = \frac{1}{2} \vec{\nabla}_b R, \quad (3.255)$$

$$\vec{\nabla}^a R_{ab} = \frac{1}{2} \vec{\nabla}_b R = \frac{k}{2} \vec{\nabla}_b T = \frac{k}{2} \partial_b T = 0. \quad (3.256)$$

This implies then T is a constant along the space-time, but this is not always true, thus this assumption between the Ricci's tensor and the energy-momentum tensor is discarded, then we're going to propose the Einstein's tensor (which is covariantly constant), then

$$\vec{\nabla}^a \left(R_{ab} - \frac{1}{2} g_{ab} R \right) = 0 \rightarrow R_{ab} - \frac{1}{2} g_{ab} R = k T_{ab}, \quad (3.257)$$

$$g^{ab} \left(R_{ab} - \frac{1}{2} g_{ab} R \right) = k g^{ab} T_{ab} \rightarrow R = -k T, \quad (3.258)$$

then finally the relation between the curvature and the matter is

$$R_{ab} = k \left(T_{ab} - \frac{1}{2} g_{ab} T \right) \quad (3.259)$$

we're going to see what happens in the Newtonian limit, then the greatest interaction is due to the component T_{00} or what is the same, due to the mass density.

$$\begin{aligned}
R_{00} &= k \left(T_{00} - \frac{1}{2} g_{00} T \right), \\
T &= g^{ab} T_{ab} = g^{00} T_{00} = - (1 + h_{00}) \rho = -T_{00}, \\
R_{00} &= k \left(T_{00} - \frac{1}{2} g_{00} T_{00} \right) = k \left(T_{00} + \frac{1}{2} (h_{00} - 1) T_{00} \right) \approx \frac{1}{2} k T_{00}.
\end{aligned}$$

Now, we are going to calculate the Ricci's tensor

$$\begin{aligned}
R_{\mu\nu} &= \partial_\lambda \Gamma_{\mu\nu}^\lambda - \partial_\nu \Gamma_{\mu\lambda}^\lambda + \Gamma_{\mu\nu}^\rho \Gamma_{\lambda\rho}^\lambda - \Gamma_{\mu\lambda}^\rho \Gamma_{\nu\rho}^\lambda, \\
R_{00} &= \partial_\lambda \Gamma_{00}^\lambda - \partial_0 \Gamma_{0\lambda}^\lambda + \Gamma_{00}^\rho \Gamma_{\lambda\rho}^\lambda - \Gamma_{0\lambda}^\rho \Gamma_{0\rho}^\lambda,
\end{aligned}$$

$$R_{00} = \partial_\lambda \Gamma_{00}^\lambda, \quad (3.260)$$

$$R_{00} = R^i_{0i0} = \partial_i \Gamma_{00}^i = \partial_i \left[-\frac{1}{2} g^{i\sigma} \partial_\sigma g_{00} \right] = \partial_i \left[-\frac{1}{2} g^{ij} \partial_j g_{00} \right] \quad (3.261)$$

$$= -\frac{1}{2} \partial_i [(\eta^{ij} - h^{ij}) \partial_j (\eta_{00} + h_{00})] \quad (3.262)$$

$$= -\frac{1}{2} \eta^{ij} \partial_i \partial_j h_{00} = -\frac{1}{2} \nabla^2 h_{00}, \quad (3.263)$$

$$\nabla^2 h_{00} = -k T_{00}. \quad (3.264)$$

We are now one single step away from the constant k , which is after all the reason we took the Newtonian limit in the first place, and the step consists in nothing more than recognising the equation we have just written. Two results obtained earlier are all we need: the perturbation of the temporal component of the metric is fixed by the Newtonian potential, $h_{00} = -2\phi$, which is what allowed us to reproduce the Newtonian equation of motion, and the temporal component of the energy momentum tensor is, to leading order in the slow motion approximation, the mass density itself, $T_{00} = \rho$. Substituting both into equation 3.264 we get

$$\begin{aligned}
\nabla^2 (-2\phi) &= -k \rho, \\
-2\nabla^2 \phi &= -k \rho, \\
\nabla^2 \phi &= \frac{k}{2} \rho,
\end{aligned}$$

and this is to be compared, term by term, with the Poisson equation of Newtonian gravity which we wrote down at the very beginning of this chapter,

$$\nabla^2 \phi = 4\pi G \rho. \quad (3.265)$$

The two equations describe the same physics in the same regime, so their coefficients must agree, and therefore

$$\frac{k}{2} = 4\pi G \longrightarrow k = 8\pi G, \quad (3.266)$$

which in the geometrised units we have been using, where $G = 1$, is simply $k = 8\pi$. Restoring the factors of C that we had quietly set to one is a matter of remembering that the energy density carried by T_{00} is ρC^2 rather than ρ , and that the metric perturbation is $h_{00} = -2\phi/C^2$, so that the same comparison gives

$$k = \frac{8\pi G}{C^4}, \quad (3.267)$$

and the field equations take the form in which they are usually quoted,

$$R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R = \frac{8\pi G}{C^4}T_{\mu\nu}. \quad (3.268)$$

It is worth pausing on the number sitting in front of the energy momentum tensor, because it explains something a student often finds puzzling, namely why gravity, which governs the motion of galaxies, is so absurdly hard to detect in a laboratory. Putting in the values of the two constants we find $8\pi G/C^4 \approx 2.1 \times 10^{-43}$ in units of $\text{s}^2 \text{kg}^{-1} \text{m}^{-1}$, which is to say that **to produce one unit of space-time curvature one needs of the order of 10^{43} units of energy density**, and this is precisely why the curvature of space-time only becomes noticeable when the source is something of the size of a planet, and why it takes an object as violent as a pair of merging black holes to shake the geometry hard enough for us to hear it a billion years later.

3.5.2.1 Einstein-Hilbert action

Can we deduce the Einstein equations from a variational principle, in the way we deduce the equations of motion of any other field theory? This is the road Hilbert took in 1915, in parallel with Einstein and by a completely different route, as we discussed in chapter 1, and it leads to the object we now call the Einstein-Hilbert action (Hilbert, 1915).

Lets define an action and a Lagrangian density as

$$S_H[g] = \int_M \mathcal{L}_H \sqrt{-g} d^4x, \quad \mathcal{L}_H = \frac{1}{16\pi} R, \quad (3.269)$$

$$S_H[g] = \frac{1}{16\pi} \int_M R \sqrt{-g} d^4x = \int L dt. \quad (3.270)$$

In fact, for a general system, we can split the action into a geometrical part and a material part as

$$S = S_H + S_m. \quad (3.271)$$

Now, lets develop more this

$$S_H[g] = \frac{1}{16\pi} \int_M g^{ab} R_{ab} \sqrt{-g} d^4x, \quad (3.272)$$

now we are going to variate the action thus

$$\delta S_H[g] = \frac{1}{16\pi} \int_M [\sqrt{-g} g^{\mu\nu} \delta R_{\mu\nu} + \sqrt{-g} R_{\mu\nu} \delta g^{\mu\nu} + R \delta \sqrt{-g}] d^4x \quad (3.273)$$

applying a variation of the gravitational field $g_{\mu\nu} \rightarrow g_{\mu\nu} + \delta g_{\mu\nu}$ we can expand each term of the integrand, thus first lets solve for $\sqrt{-g} g^{\mu\nu} \delta R_{\mu\nu}$

$$\begin{aligned} \sqrt{-g} g^{\mu\nu} \delta R_{\mu\nu} &= \sqrt{-g} g^{\mu\nu} \delta [\partial_\alpha \Gamma_{\mu\nu}^\alpha - \partial_\nu \Gamma_{\mu\alpha}^\alpha + \Gamma_{\mu\nu}^\tau \Gamma_{\tau\alpha}^\alpha - \Gamma_{\mu\alpha}^\tau \Gamma_{\nu\tau}^\alpha] \\ &= \sqrt{-g} g^{\mu\nu} [\partial_\alpha \delta \Gamma_{\mu\nu}^\alpha - \partial_\nu \delta \Gamma_{\mu\alpha}^\alpha + \delta \Gamma_{\mu\nu}^\tau \Gamma_{\tau\alpha}^\alpha + \Gamma_{\mu\nu}^\tau \delta \Gamma_{\tau\alpha}^\alpha - \delta \Gamma_{\mu\alpha}^\tau \Gamma_{\nu\tau}^\alpha - \Gamma_{\mu\alpha}^\tau \delta \Gamma_{\nu\tau}^\alpha], \end{aligned}$$

remembering the definition of the Christoffel symbols

$$\Gamma_{\nu\rho}^\mu = \frac{1}{2} g^{\mu\sigma} (\partial_\rho g_{\sigma\nu} + \partial_\nu g_{\sigma\rho} - \partial_\sigma g_{\nu\rho}), \quad (3.274)$$

$$\begin{aligned} \delta \Gamma_{\nu\rho}^\mu &= \frac{1}{2} \delta g^{\mu\sigma} (\partial_\rho g_{\sigma\nu} + \partial_\nu g_{\sigma\rho} - \partial_\sigma g_{\nu\rho}) + \frac{1}{2} g^{\mu\sigma} (\partial_\rho \delta g_{\sigma\nu} + \partial_\nu \delta g_{\sigma\rho} - \partial_\sigma \delta g_{\nu\rho}) \\ &= \frac{1}{2} \delta g^{\mu\sigma} (g_{\sigma\nu,\rho} + g_{\sigma\rho,\nu} - g_{\nu\rho,\sigma}) + \frac{1}{2} g^{\mu\sigma} [(\delta g_{\sigma\nu})_{,\rho} + (\delta g_{\sigma\rho})_{,\nu} - (\delta g_{\nu\rho})_{,\sigma}]. \end{aligned}$$

We are going to focus on the term $\delta g^{\mu\sigma}$ then

$$\begin{aligned} \delta g^{\mu\sigma} &= \delta (g^{\mu k} g_k^\sigma) = \delta (g^{\mu k} g^{l\sigma} g_{kl}) \\ &= \delta g^{\mu k} g^{l\sigma} g_{kl} + g^{\mu k} \delta g^{l\sigma} g_{kl} + g^{\mu k} g^{l\sigma} \delta g_{kl} \\ &= \delta g^{\mu k} \delta_k^\sigma + \delta g^{l\sigma} \delta_l^\mu + g^{\mu k} g^{l\sigma} \delta g_{kl} = -g^{\mu k} g^{l\sigma} \delta g_{kl}. \end{aligned}$$

Using this in the variation of the Christoffel symbol, then

$$\begin{aligned}\delta\Gamma_{\nu\rho}^{\mu} &= -\frac{1}{2}g^{\mu k}g^{l\sigma}\delta g_{kl}(g_{\sigma\nu,\rho}+g_{\sigma\rho,\nu}-g_{\nu\rho,\sigma})+\frac{1}{2}g^{\mu\sigma}[(\delta g_{\sigma\nu})_{,\rho}+(\delta g_{\sigma\rho})_{,\nu}-(\delta g_{\nu\rho})_{,\sigma}] \\ &= -g^{\mu k}\delta g_{kl}\Gamma_{\nu\rho}^l+\frac{1}{2}g^{\mu\sigma}[(\delta g_{\sigma\nu})_{,\rho}+(\delta g_{\sigma\rho})_{,\nu}-(\delta g_{\nu\rho})_{,\sigma}],\end{aligned}$$

$$\begin{aligned}(\delta g_{\alpha\beta})_{;\gamma} &= (\delta g_{\alpha\beta})_{,\gamma}-\Gamma_{\alpha\gamma}^l\delta g_{l\beta}-\Gamma_{\beta\gamma}^l\delta g_{l\alpha}, \\ (\delta g_{\sigma\nu})_{,\rho}+(\delta g_{\sigma\rho})_{,\nu}-(\delta g_{\nu\rho})_{,\sigma} \\ &= (\delta g_{\sigma\nu})_{;\rho}+(\delta g_{\sigma\rho})_{;\nu}-(\delta g_{\nu\rho})_{;\sigma} \\ &\quad +\Gamma_{\sigma\rho}^l\delta g_{l\nu}+\Gamma_{\nu\rho}^l\delta g_{l\sigma}+\Gamma_{\sigma\nu}^l\delta g_{l\rho}+\Gamma_{\rho\nu}^l\delta g_{l\sigma}-\Gamma_{\nu\sigma}^l\delta g_{l\rho}-\Gamma_{\rho\sigma}^l\delta g_{l\nu}.\end{aligned}$$

Because we have got a connection free of torsion is fulfilled the Christoffel symbols commute in their lower indices $\Gamma_{\beta\gamma}^{\alpha}=\Gamma_{\gamma\beta}^{\alpha}$ then

$$(\delta g_{\sigma\nu})_{,\rho}+(\delta g_{\sigma\rho})_{,\nu}-(\delta g_{\nu\rho})_{,\sigma}=(\delta g_{\sigma\nu})_{;\rho}+(\delta g_{\sigma\rho})_{;\nu}-(\delta g_{\nu\rho})_{;\sigma}+2\Gamma_{\nu\rho}^l\delta g_{l\sigma}, \quad (3.275)$$

$$\begin{aligned}\delta\Gamma_{\nu\rho}^{\mu} &= -g^{\mu k}\delta g_{kl}\Gamma_{\nu\rho}^l+\frac{1}{2}g^{\mu\sigma}\left[(\delta g_{\sigma\nu})_{;\rho}+(\delta g_{\sigma\rho})_{;\nu}-(\delta g_{\nu\rho})_{;\sigma}+2\Gamma_{\nu\rho}^l\delta g_{l\sigma}\right] \\ &= \frac{1}{2}g^{\mu\sigma}[(\delta g_{\sigma\nu})_{;\rho}+(\delta g_{\sigma\rho})_{;\nu}-(\delta g_{\nu\rho})_{;\sigma}] \\ \delta\Gamma_{\nu\rho}^{\mu} &= \frac{1}{2}g^{\mu\sigma}[(\delta g_{\sigma\nu})_{;\rho}+(\delta g_{\sigma\rho})_{;\nu}-(\delta g_{\nu\rho})_{;\sigma}],\end{aligned} \quad (3.276)$$

now, lest calculate the following covariant derivatives

$$(\delta\Gamma_{\nu\rho}^{\mu})_{;\alpha}=(\delta\Gamma_{\nu\rho}^{\mu})_{,\alpha}+\Gamma_{\sigma\alpha}^{\mu}(\delta\Gamma_{\nu\rho}^{\sigma})-\Gamma_{\nu\alpha}^{\sigma}(\delta\Gamma_{\sigma\rho}^{\mu})-\Gamma_{\rho\alpha}^{\sigma}(\delta\Gamma_{\sigma\nu}^{\mu}) \quad (3.277)$$

$$(\delta\Gamma_{\nu\alpha}^{\mu})_{;\rho}=(\delta\Gamma_{\nu\alpha}^{\mu})_{,\rho}+\Gamma_{\sigma\rho}^{\mu}(\delta\Gamma_{\nu\alpha}^{\sigma})-\Gamma_{\nu\rho}^{\sigma}(\delta\Gamma_{\sigma\alpha}^{\mu})-\Gamma_{\alpha\rho}^{\sigma}(\delta\Gamma_{\sigma\nu}^{\mu}) \quad (3.278)$$

we are going to, in both equation a contraction of μ with α then

$$(\delta\Gamma_{\nu\rho}^{\mu})_{;\mu}=(\delta\Gamma_{\nu\rho}^{\mu})_{,\mu}+\Gamma_{\sigma\mu}^{\mu}(\delta\Gamma_{\nu\rho}^{\sigma})-\Gamma_{\nu\mu}^{\sigma}(\delta\Gamma_{\sigma\rho}^{\mu})-\Gamma_{\rho\mu}^{\sigma}(\delta\Gamma_{\sigma\nu}^{\mu}), \quad (3.279)$$

$$(\delta\Gamma_{\nu\mu}^{\mu})_{;\rho}=(\delta\Gamma_{\nu\mu}^{\mu})_{,\rho}+\Gamma_{\sigma\rho}^{\mu}(\delta\Gamma_{\nu\mu}^{\sigma})-\Gamma_{\nu\rho}^{\sigma}(\delta\Gamma_{\sigma\mu}^{\mu})-\Gamma_{\mu\rho}^{\sigma}(\delta\Gamma_{\sigma\nu}^{\mu}). \quad (3.280)$$

Subtracting both equations then

$$(\delta\Gamma_{\nu\rho}^{\mu})_{;\mu} - (\delta\Gamma_{\nu\mu}^{\mu})_{;\rho} = (\delta\Gamma_{\nu\rho}^{\mu})_{,\mu} - (\delta\Gamma_{\nu\mu}^{\mu})_{,\rho} + \Gamma_{\sigma\mu}^{\mu}(\delta\Gamma_{\nu\rho}^{\sigma}) + \Gamma_{\nu\rho}^{\sigma}(\delta\Gamma_{\sigma\mu}^{\mu}) - \Gamma_{\nu\mu}^{\sigma}(\delta\Gamma_{\sigma\rho}^{\mu}) - \Gamma_{\sigma\rho}^{\mu}(\delta\Gamma_{\nu\mu}^{\sigma}) \quad (3.281)$$

Relabelling $\mu \rightarrow \alpha$, $\nu \rightarrow \mu$ and $\rho \rightarrow \nu$ we obtain

$$\delta R_{\mu\nu} = (\delta\Gamma_{\mu\nu}^{\alpha})_{;\alpha} - (\delta\Gamma_{\mu\alpha}^{\alpha})_{;\nu}, \quad (3.282)$$

thus

$$\sqrt{-g}g^{\mu\nu}\delta R_{\mu\nu} = \sqrt{-g}[(\delta\Gamma_{\mu\nu}^{\alpha})_{;\alpha} - (\delta\Gamma_{\mu\alpha}^{\alpha})_{;\nu}]. \quad (3.283)$$

Now, for the second term in the integrand

$$\frac{1}{\sqrt{-g}}\delta\sqrt{-g} = \frac{1}{2}\delta\ln(-g) = \frac{1}{2}\frac{\delta(-g)}{-g},$$

$$\delta(g^{\mu\nu}g_{\mu\nu}) = 0 = g^{\mu\nu}\delta g_{\mu\nu} + g_{\mu\nu}\delta g^{\mu\nu}, \quad (3.284)$$

$$\frac{1}{\sqrt{-g}}\delta\sqrt{-g} = \frac{1}{2}g^{\mu\nu}\delta g_{\mu\nu} = -\frac{1}{2}g_{\mu\nu}\delta g^{\mu\nu}. \quad (3.285)$$

Here g denotes the determinant of the metric, and we have used Jacobi's formula $\delta g = g g^{\mu\nu}\delta g_{\mu\nu}$, which is the statement that the derivative of a determinant is the determinant times the trace of the inverse matrix times the variation. Lets define the following quantity

$$W^{\lambda} = g^{\mu\nu}\delta\Gamma_{\mu\nu}^{\lambda} - g^{\mu\lambda}\delta\Gamma_{\mu\nu}^{\nu} \quad (3.286)$$

we're going to interpret this quantity later, from now it will be a useful variable change to work on the algebra, if we contract over λ its covariant derivative we get

$$W^{\lambda}_{;\lambda} = g^{\mu\nu}\delta R_{\mu\nu}, \quad (3.287)$$

using these equation on the variation of the Einstein-Hilbert action we got

$$\delta S_H[g] = \frac{1}{16\pi} \int_M \left[R_{ab}\delta g^{ab} - \frac{R}{2}g_{ab}\delta g^{ab} + \vec{\nabla}_a W^a \right] \sqrt{-g}d^4x, \quad (3.288)$$

$$\delta S_H[g] = \frac{1}{16\pi} \int \left(R_{ab} - \frac{1}{2} R g_{ab} \right) \sqrt{-g} d^4x + \frac{1}{16\pi} \vec{\nabla}_a W^a \sqrt{-g} d^4x. \quad (3.289)$$

We gotta focus on the second term, to do it, first look at the figure 3.12 where the integrand is being integrated on M but the second term is in fact a divergence over a $4D$ volume thus in analogy to the $3D$ Gauss's theorem of divergence, this becomes the following integral

$$\int_M \vec{\nabla}_a W^a \sqrt{-g} d^4x = \oint_{\partial M} W^a dS_a. \quad (3.290)$$

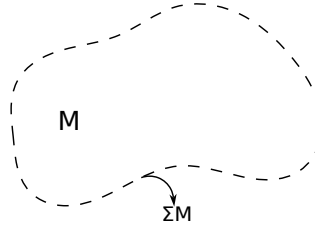


Figure 3.12: Representation of the region (manifold M) of integration and its surrounding surface ΣM .

The integral 3.290 is completely analogous to that in electrodynamics where we integrate the divergence of the Poynting vector, so we can extend the integration surface ΣM to infinity or what is the same *our manifold is asymptotically flat* and the second integral goes to zero, in analogy to electrodynamics we can interpret the vector W^λ as a generalised four-momentum or heat or losses vector. The variation of the Einstein-Hilbert action then becomes

$$\delta S_H[g] = \frac{1}{16\pi} \int_M \left(R_{ab} - \frac{1}{2} R g_{ab} \right) \delta g^{ab} \sqrt{-g} d^4x, \quad (3.291)$$

$$\frac{1}{\sqrt{-g}} \frac{\delta S_H[g]}{\delta g^{ab}} = \frac{G_{ab}}{16\pi}. \quad (3.292)$$

If we extremalize it we find immediately $G_{ab} = 0$ which are the Einstein's equations in the vacuum.

3.5.2.2 Energy-momentum tensor

Here we are going to build up the Energy-momentum tensor the heart of the Einstein's equations and most of the electrodynamics too, to do it we are going to use the Einstein-Hilbert action and a matter action as follows

$$S = S_H[g] + S_{matter}, \quad (3.293)$$

then lets see how changes the variation of the action

$$\frac{1}{\sqrt{-g}} \frac{\delta S}{\delta g^{ab}} = \frac{1}{\sqrt{-g}} \left[\frac{\delta S_H}{\delta g^{ab}} + \frac{\delta S_m}{\delta g^{ab}} \right], \quad (3.294)$$

$$\frac{1}{\sqrt{-g}} \frac{\delta S}{\delta g^{ab}} = \frac{1}{\sqrt{-g}} \left(\frac{G_{ab}}{16\pi} + \frac{\delta S_m}{\delta g^{ab}} \right) = 0, \quad (3.295)$$

$$\frac{1}{\sqrt{-g}} \frac{\delta S_m}{\delta g^{ab}} = -\frac{T_{ab}}{2} \rightarrow T_{ab} = -\frac{2}{\sqrt{-g}} \frac{\delta S_m}{\delta g^{ab}}, \quad (3.296)$$

$$\delta S_m = -\frac{1}{2} \int_M \sqrt{-g} T_{ab} \delta g^{ab} d^4x, \quad (3.297)$$

applying variational principles we then arrive to

$$R_{ab} - \frac{1}{2} g_{ab} R = 8\pi T_{ab}. \quad (3.298)$$

The definition we have just obtained, $T_{ab} = -\frac{2}{\sqrt{-g}} \frac{\delta S_m}{\delta g^{ab}}$, is beautifully compact and completely useless the first time one tries to solve a problem with it, because it tells us what the energy momentum tensor *is* without telling us what it *looks like* for any actual piece of matter. It is worth spending a page repairing that, since the right hand side of Einstein's equations is where all the physics that is not geometry has to live, and a student who cannot write down $T_{\mu\nu}$ for a cloud of dust or for a star is not yet in a position to solve anything.

The interpretation of the components follows from the conservation law itself. In a local inertial frame the statement $\partial_\mu T^{\mu\nu} = 0$ splits, exactly as in continuum mechanics, into a conservation of energy and a conservation of momentum, and reading off which quantity flows where we find that T^{00} is the energy density measured by that observer, T^{0i} is both the density of the i -th component of momentum and the flux of energy across the surface normal to x^i , the equality of those two readings being a purely relativistic statement about the equivalence of mass and energy, and T^{ij} is the flux of the i -th component of momentum across the surface normal to x^j , which is what an engineer calls the stress tensor, with the diagonal entries measuring pressure and the off diagonal ones measuring shear. The symmetry $T^{\mu\nu} = T^{\nu\mu}$, which comes for free from the variational definition because g^{ab} is symmetric, then encodes the statement that the density of momentum and the flux of energy are the same object seen twice.

The simplest matter of all is *dust*, by which we mean a swarm of particles that do not interact and therefore exert no pressure on one another, each of them carried along the same four velocity field u^μ with a rest mass density ρ measured in the local rest frame. There is only one tensor one can build out of these ingredients, and it is

$$T^{\mu\nu} = \rho u^\mu u^\nu. \quad (3.299)$$

That this is the right object is checked by evaluating it in the rest frame of the dust, where $u^\mu = (1, 0, 0, 0)$ and every component vanishes except $T^{00} = \rho$, which is exactly what we want of a fluid that carries energy and nothing else. What is remarkable about equation 3.299 is that its conservation already contains the equation of motion of the particles, and the calculation is short enough to do here in full. Applying the Leibniz rule,

$$\vec{\nabla}_\mu T^{\mu\nu} = \vec{\nabla}_\mu (\rho u^\mu u^\nu) = u^\nu \vec{\nabla}_\mu (\rho u^\mu) + \rho u^\mu \vec{\nabla}_\mu u^\nu = 0,$$

and we now extract two independent statements from this single equation by projecting it first along the four velocity and then orthogonally to it. Contracting with u_ν and using the normalisation $u^\nu u_\nu = -1$, together with the identity

$$u_\nu \vec{\nabla}_\mu u^\nu = \frac{1}{2} \vec{\nabla}_\mu (u^\nu u_\nu) = \frac{1}{2} \vec{\nabla}_\mu (-1) = 0, \quad (3.300)$$

which simply says that the four acceleration is always orthogonal to the four velocity, we obtain

$$\begin{aligned} u_\nu \vec{\nabla}_\mu T^{\mu\nu} &= u_\nu u^\nu \vec{\nabla}_\mu (\rho u^\mu) + \rho u^\mu u_\nu \vec{\nabla}_\mu u^\nu \\ &= (-1) \vec{\nabla}_\mu (\rho u^\mu) + 0 = 0, \end{aligned}$$

so that

$$\vec{\nabla}_\mu (\rho u^\mu) = 0, \quad (3.301)$$

which is the relativistic continuity equation and expresses the conservation of the number of dust particles. Putting equation 3.301 back into the original conservation law kills the first term outright and leaves

$$u^\nu \vec{\nabla}_\mu (\rho u^\mu) + \rho u^\mu \vec{\nabla}_\mu u^\nu = \rho u^\mu \vec{\nabla}_\mu u^\nu = 0,$$

and since ρ is not zero wherever there is any dust at all we conclude that

$$u^\mu \vec{\nabla}_\mu u^\nu = 0, \quad (3.302)$$

which the reader will recognise immediately as $\vec{\nabla}_u u = 0$, the geodesic equation. This deserves to be read slowly, because it says that **the motion of free matter along geodesics is not an extra postulate of general relativity at all, it is a theorem forced on us by the conservation of the energy momentum tensor**, and therefore ultimately by the Bianchi identity, which is in turn

a statement about geometry. The second postulate we listed when we set up the theory was never independent of the fourth.

A real fluid does push on itself, and the next case up is the *perfect fluid*, defined as one that has no viscosity and no heat conduction, so that in its own rest frame it is characterised completely by an energy density ρ and an isotropic pressure p . Its energy momentum tensor is

$$T^{\mu\nu} = (\rho + p) u^\mu u^\nu + p g^{\mu\nu}, \quad (3.303)$$

and once again the way to convince oneself is to go to the rest frame, where $u^\mu = (1, 0, 0, 0)$ and $g^{\mu\nu} = \eta^{\mu\nu}$, so that the temporal component becomes

$$T^{00} = (\rho + p) (1)(1) + p \eta^{00} = \rho + p + p(-1) = \rho + p - p = \rho,$$

while the spatial ones give $T^{ij} = 0 + p \delta^{ij}$, which is an isotropic pressure and nothing else, and all the mixed components vanish. The trace follows just as directly, remembering that $g_{\mu\nu} g^{\mu\nu} = 4$ in four dimensions,

$$\begin{aligned} T = g_{\mu\nu} T^{\mu\nu} &= (\rho + p) g_{\mu\nu} u^\mu u^\nu + p g_{\mu\nu} g^{\mu\nu} \\ &= (\rho + p) (-1) + p (4) = -\rho - p + 4p = 3p - \rho, \end{aligned}$$

a result we shall need the moment we want to write the field equations in their trace reversed form. Conserving equation 3.303 and projecting exactly as we did for dust, along u and orthogonally to it, gives this time two genuinely different equations, since the pressure now does work,

$$u^\mu \vec{\nabla}_\mu \rho + (\rho + p) \vec{\nabla}_\mu u^\mu = 0, \quad (3.304)$$

$$(\rho + p) u^\mu \vec{\nabla}_\mu u^\nu = -(g^{\mu\nu} + u^\mu u^\nu) \vec{\nabla}_\mu p, \quad (3.305)$$

of which the first is the relativistic statement of energy conservation and the second is the relativistic Euler equation, with the combination $\rho + p$ playing the role of the inertia of the fluid, so that a hot gas is harder to accelerate than a cold one of the same mass, and with the projector $g^{\mu\nu} + u^\mu u^\nu$ making sure that only the part of the pressure gradient orthogonal to the motion can push the fluid about. In the limit of small velocities and pressures negligible against the energy density these two equations collapse onto the continuity equation and onto $\rho d\vec{v}/dt = -\vec{\nabla} p - \rho \vec{\nabla} \phi$, which is the Euler equation every student meets in a first course on fluids.

The last case worth writing down explicitly is the electromagnetic field, which is not matter at all but certainly carries energy and therefore certainly gravitates. Built out of the Faraday tensor it reads

$$T^{\mu\nu} = \frac{1}{4\pi} \left(F^{\mu\alpha} F^{\nu}_{\alpha} - \frac{1}{4} g^{\mu\nu} F_{\alpha\beta} F^{\alpha\beta} \right), \quad (3.306)$$

whose temporal component in a local inertial frame is the familiar energy density $T^{00} = (E^2 + B^2)/8\pi$ of the electromagnetic field. What makes equation 3.306 special is its trace,

$$\begin{aligned} T = g_{\mu\nu} T^{\mu\nu} &= \frac{1}{4\pi} \left(F^{\mu\alpha} F_{\mu\alpha} - \frac{1}{4} g_{\mu\nu} g^{\mu\nu} F_{\alpha\beta} F^{\alpha\beta} \right) \\ &= \frac{1}{4\pi} \left(F_{\alpha\beta} F^{\alpha\beta} - \frac{4}{4} F_{\alpha\beta} F^{\alpha\beta} \right) = 0, \end{aligned}$$

which vanishes identically, and a glance at the trace reversed field equations then tells us that a space-time containing nothing but electromagnetic radiation has $R = 0$ even though it is certainly curved. The deep reason is that Maxwell's theory has no length scale of its own, and the tracelessness of its energy momentum tensor is the shadow that conformal invariance casts on the gravitational field equations.

Thus we've arrived to the Einstein's field equations from variational principles, now we're going to solve this equations but only under the assumption that *the gravitational field is weak but not stationary*. To do it we have to suppose the metric of our manifold is flat with a certain perturbation in a region, so the global metric will be

$$g_{\alpha\beta} = \eta_{\alpha\beta} + h_{\alpha\beta} \quad |h_{\alpha\beta}| \ll 1. \quad (3.307)$$

Before going further we must prove the tensor 3.307 is a metric, lets then prove it. The contravariant components of the metric are given by

$$g^{\alpha\beta} = \eta^{\alpha\beta} - h^{\alpha\beta} \quad (3.308)$$

and given a coordinate chart $\phi = \{x^i\}$ $i = 0, 1, 2, 3$ and another coordinate chart $\bar{\phi} = \{\bar{x}^i\}$; $i = 0, 1, 2, 3$ thus the transformation rule for the metric is

$$g_{\bar{\alpha}\bar{\beta}} = \Lambda^{\mu}_{\bar{\alpha}} \Lambda^{\nu}_{\bar{\beta}} g_{\mu\nu} \quad (3.309)$$

where $\Lambda^{\alpha}_{\bar{\beta}}$ is a Lorentz Boost then what we see is that the transformation rule is a similarity or translation transformation then

$$g_{\bar{\alpha}\bar{\beta}} = \Lambda^{\mu}_{\bar{\alpha}} \Lambda^{\nu}_{\bar{\beta}} \eta_{\mu\nu} + \Lambda^{\mu}_{\bar{\alpha}} \Lambda^{\nu}_{\bar{\beta}} h_{\mu\nu} = \eta_{\bar{\alpha}\bar{\beta}} + h_{\bar{\alpha}\bar{\beta}}. \quad (3.310)$$

Then $g_{\alpha\beta}$ is effectively a tensor and having a theory invariant under Lorentz transformations implies having at least a special relativistic theory. In the Einstein's equations there is the Ricci's

number and tensor, but to calculate them we first have to calculate the Riemann's tensor and the Christoffel symbols, therefore

$$\begin{aligned}
 \Gamma_{\nu\rho}^{\mu} &= \frac{1}{2}g^{\mu\sigma}(g_{\sigma\nu,\rho} + g_{\rho\sigma,\nu} - g_{\nu\rho,\sigma}) \\
 &= \frac{1}{2}(\eta^{\mu\sigma} - h^{\mu\sigma})[h_{\sigma\nu,\rho} + h_{\rho\sigma,\nu} - h_{\nu\rho,\sigma}] \\
 &= \frac{1}{2}\eta^{\mu\sigma}(h_{\sigma\nu,\rho} + h_{\rho\sigma,\nu} - h_{\nu\rho,\sigma}), \\
 R_{\nu\rho\sigma}^{\alpha} &= \Gamma_{\nu\sigma,\rho}^{\alpha} - \Gamma_{\nu\rho,\sigma}^{\alpha} + \mathcal{O}(\Gamma^2),
 \end{aligned} \tag{3.311}$$

$$\begin{aligned}
 \eta_{\mu\alpha}R_{\nu\rho\sigma}^{\alpha} &= R_{\mu\nu\rho\sigma} = \frac{\eta_{\mu\alpha}}{2}\left[\eta^{\alpha\beta}(h_{\beta\nu,\rho} + h_{\beta\rho,\nu} - h_{\rho\nu,\beta})\right] - \frac{\eta^{\alpha\beta}}{2}\left[\eta^{\alpha\beta}(h_{\beta\nu,\sigma} + h_{\sigma\beta,\nu} - h_{\sigma\nu,\beta})_{,\rho}\right. \\
 &\quad \left.- \eta^{\alpha\beta}(h_{\beta\nu,\rho} + h_{\rho\beta,\nu} - h_{\rho\nu,\beta})_{,\sigma}\right],
 \end{aligned}$$

$$R_{\mu\nu\rho\sigma} = \frac{1}{2}\delta_{\mu}^{\beta}(h_{\beta\sigma,\nu\rho} + h_{\rho\nu,\beta\sigma} - h_{\sigma\nu,\beta\rho} - h_{\rho\beta,\nu\sigma}), \tag{3.312}$$

$$R_{\mu\nu\rho\sigma} = \frac{1}{2}(h_{\mu\sigma,\nu\rho} + h_{\rho\nu,\mu\sigma} - h_{\sigma\nu,\mu\rho} - h_{\rho\mu,\nu\sigma}) \tag{3.313}$$

There the Ricci's curvature tensor is given by

$$R_{\mu\nu} = \eta^{\alpha\rho}R_{\alpha\mu\rho\nu} = \frac{1}{2}[\partial_{\mu}\partial_{\rho}h_{\nu}^{\rho} + \partial^{\rho}\partial_{\nu}h_{\rho\mu} - \partial^{\rho}\partial_{\rho}h_{\nu\mu} - \partial_{\mu}\partial_{\nu}h] \tag{3.314}$$

$$R_{\mu\nu} = \frac{1}{2}(\partial_{\rho}\partial_{\mu}h_{\nu}^{\rho} + \partial_{\rho}\partial_{\nu}h_{\mu}^{\rho} - \partial_{\mu}\partial_{\nu}h - \square h_{\mu\nu}), \tag{3.315}$$

where $\square = \vec{\nabla}_a \vec{\nabla}^a$ is the d'Alembertian and in a Minkowski space it corresponds to $\square = \partial_{\alpha}\partial^{\alpha}$. Well then the Ricci's number is given by

$$R = \eta^{\mu\nu}R_{\mu\nu} = \frac{1}{2}\eta^{\mu\nu}(\partial_{\rho}\partial_{\mu}h_{\nu}^{\rho} + \partial_{\rho}\partial_{\nu}h_{\mu}^{\rho} - \partial_{\mu}\partial_{\nu}h - \square h_{\mu\nu}) = \partial_{\rho}\partial_{\mu}h^{\rho\mu} - \square h. \tag{3.316}$$

Now, introducing all of this on the Einstein's field equation we will have the following equation

$$\frac{1}{2}(\partial_{\rho}\partial_{\mu}h_{\nu}^{\rho} + \partial_{\rho}\partial_{\nu}h_{\mu}^{\rho} - \partial_{\mu}\partial_{\nu}h - \square h_{\mu\nu}) - \frac{1}{2}\eta_{\mu\nu}[\partial_{\rho}\partial_{\sigma}h^{\rho\sigma} - \square h] = 8\pi T_{\mu\nu}. \tag{3.317}$$

Before going further there is the need of introducing a special kind of tensor, called the *trace reversed tensor*, which is extremely useful in general relativity and of which, as we are about to see, the Einstein tensor is itself an example. The trace reversed version of the perturbation tensor is given by

$$\bar{h}_{\mu\nu} = h_{\mu\nu} - \frac{1}{2}\eta_{\mu\nu}h, \quad (3.318)$$

it is easy to see that if we trace reverse the Ricci curvature tensor we get the Einstein's tensor like this:

$$\bar{R}_{\alpha\beta} = R_{\alpha\beta} - \frac{1}{2}g_{\alpha\beta}R. \quad (3.319)$$

We can then rewrite the perturbation tensor in term of its inverted trace tensor and also its trace as follows

$$h_{\mu\nu} = \bar{h}_{\mu\nu} - \frac{1}{2}\eta_{\mu\nu}\bar{h}, \quad (3.320)$$

$$\eta^{\mu\nu}\bar{h}_{\mu\nu} = h - 2h = -h = \bar{h}. \quad (3.321)$$

Then the Einstein's field equations in terms of the inverted trace tensor becomes

$$\begin{aligned} & \partial_\rho \partial_\mu \left(\bar{h}^\rho_\nu - \frac{1}{2}\delta^\rho_\nu \bar{h} \right) + \partial_\rho \partial_\nu \left(\bar{h}^\rho_\mu - \frac{1}{2}\delta^\rho_\mu \bar{h} \right) + \partial_\mu \partial_\nu \bar{h} \\ & - \square \left(\bar{h}_{\mu\nu} - \frac{1}{2}\eta_{\mu\nu}\bar{h} \right) - \eta_{\mu\nu} \partial_\rho \partial_\sigma \left(\bar{h}^{\rho\sigma} - \frac{1}{2}\eta^{\rho\sigma}\bar{h} \right) - \eta_{\mu\nu} \square \bar{h} = 16\pi T_{\mu\nu} \end{aligned}$$

$$\begin{aligned}
& \partial_\rho \partial_\mu \bar{h}^\rho_\nu - \frac{1}{2} \delta^\rho_\nu \partial_\rho \partial_\mu \bar{h} + \partial_\rho \partial_\nu \bar{h}^\rho_\mu - \frac{1}{2} \delta^\rho_\mu \partial_\rho \partial_\nu \bar{h} + \partial_\mu \partial_\nu \bar{h} - \square \bar{h}_{\mu\nu} + \frac{1}{2} \eta_{\mu\nu} \square \bar{h} \\
& - \eta_{\mu\nu} \partial_\rho \partial_\sigma \bar{h}^{\rho\sigma} + \frac{1}{2} \eta_{\mu\nu} \eta^{\rho\sigma} \partial_\rho \partial_\sigma \bar{h} - \eta_{\mu\nu} \square \bar{h} = 16\pi T_{\mu\nu}, \\
& = \partial_\rho \partial_\mu \eta^{\rho\alpha} \bar{h}_{\alpha\nu} + \partial_\rho \partial_\nu \eta^{\rho\alpha} \bar{h}_{\alpha\mu} - \frac{1}{2} \delta^\rho_\nu \partial_\rho \partial_\mu \bar{h} - \frac{1}{2} \delta^\rho_\mu \partial_\rho \partial_\nu \bar{h} + \partial_\mu \partial_\nu \bar{h} \\
& - \square \bar{h}_{\mu\nu} + \frac{1}{2} \eta_{\mu\nu} \square \bar{h} + \frac{1}{2} \eta_{\mu\nu} \eta^{\rho\sigma} \partial_\rho \partial_\sigma \bar{h} - \eta_{\mu\nu} \partial_\rho \partial_\sigma \eta^{\rho\alpha} \eta^{\sigma\beta} \bar{h}_{\alpha\beta} - \eta_{\mu\nu} \square \bar{h}, \\
& = \partial_\rho \partial_\mu \eta^{\rho\alpha} \bar{h}_{\alpha\nu} + \partial_\rho \partial_\nu \eta^{\rho\alpha} \bar{h}_{\alpha\mu} - \frac{1}{2} \partial_\nu \partial_\mu \bar{h} - \frac{1}{2} \partial_\mu \partial_\nu \bar{h} + \partial_\mu \partial_\nu \bar{h} - \square \bar{h}_{\mu\nu} \\
& + \frac{1}{2} \eta_{\mu\nu} \square \bar{h} + \frac{1}{2} \eta_{\mu\nu} \partial_\rho \partial^\rho \bar{h} - \eta_{\mu\nu} \eta^{\rho\alpha} \eta^{\sigma\beta} \partial_\rho \partial_\sigma \bar{h}_{\alpha\beta} - \eta_{\mu\nu} \square \bar{h}, \\
& = \partial_\rho \partial_\mu \eta^{\rho\alpha} \bar{h}_{\alpha\nu} + \partial_\rho \partial_\nu \eta^{\rho\alpha} \bar{h}_{\alpha\mu} - \square \bar{h}_{\mu\nu} - \eta_{\mu\nu} \eta^{\rho\alpha} \eta^{\sigma\beta} \partial_\rho \partial_\sigma \bar{h}_{\alpha\beta} \\
& = \partial_\mu [\partial_\rho (\eta^{\rho\alpha} \bar{h}_{\nu\alpha})] + \partial_\nu [\partial_\rho (\eta^{\rho\alpha} \bar{h}_{\mu\alpha})] - \square \bar{h}_{\mu\nu} - \eta_{\mu\nu} \eta^{\rho\alpha} \eta^{\sigma\beta} \partial_\rho \partial_\sigma \bar{h}_{\alpha\beta} \\
& = \partial_\mu W_\nu + \partial_\nu W_\mu - \square \bar{h}_{\mu\nu} - \eta_{\mu\nu} \eta^{\rho\alpha} \eta^{\sigma\beta} \partial_\rho \partial_\sigma \bar{h}_{\alpha\beta} \\
& = \partial_\mu W_\nu + \partial_\nu W_\mu - \square \bar{h}_{\mu\nu} - \eta_{\mu\nu} \eta^{\rho\alpha} \partial_\rho [\partial_\sigma (\eta^{\sigma\beta} \bar{h}_{\alpha\beta})], \\
& - [\square \bar{h}_{\mu\nu} - \partial_\mu W_\nu - \partial_\nu W_\mu + \eta_{\mu\nu} \eta^{\alpha\beta} \partial_\alpha W_\beta] = 16\pi T_{\mu\nu} \tag{3.322}
\end{aligned}$$

Multiplying both sides by -1 and changing $\mu \rightarrow \alpha$ and $\nu \rightarrow \beta$ then we have finally

$$\square \bar{h}_{\alpha\beta} - \partial_\beta W_\alpha - \partial_\alpha W_\beta + \eta_{\alpha\beta} \eta^{\mu\nu} \partial_\nu W_\mu = -16\pi T_{\alpha\beta}; \quad W_\alpha = \partial_\mu (\eta^{\mu\nu} \bar{h}_{\alpha\nu}). \tag{3.323}$$

The equation 3.323 is the Einstein's equations and they are really similar to the Maxwell's equations, the terms carrying W_α behave like flux terms on the boundary of the manifold. Our intention now is to convert the equation 3.323 into a wave equation, but how do we do it? In exactly the same way as in electromagnetism! We work with a Gauge theory, a calibration theory, there we use the Lorenz's Gauge, here we have to find and use this Lorenz gauge and later also find out the called *transverse Gauge*.

Chapter 4

Exact Solutions to Einstein's Equations: Black Holes

*Solve the equations exactly enough, and
space-time itself draws the horizon beyond which
nothing returns.*

“The black holes of nature are the most perfect macroscopic objects there
are in the universe.”

– Subrahmanyan Chandrasekhar

4.1 The Schwarzschild solution

The previous chapter left the Einstein field equations standing as ten coupled, non linear partial differential equations for ten unknown functions of four variables, with no general method available for solving them and with a weak field programme half finished, waiting for a gauge choice we had only just begun to set up. That programme will be picked up again in due course, because it is what eventually hands us gravitational radiation, but it is not the natural place to start, and history did not start there either. Within six weeks of Einstein publishing his equations in November 1915, Karl Schwarzschild had already found an exact solution, in closed form, while serving on the Russian front, as we recounted in chapter 1 (Schwarzschild, 1916). The reason such a thing is possible so soon is that symmetry is enormously powerful: if we demand from the outset that the space-time be static and spherically symmetric, the ten unknown functions collapse to two functions of a single variable, and the field equations collapse with them into ordinary differential equations that can be integrated by hand. This chapter is devoted to solutions of exactly that kind, exact and closed form, found by imposing enough symmetry on Einstein's equations to solve them outright rather than merely to linearise them, and it begins with the solution that started it all.

4.1.1 The ansatz

We are looking for the geometry outside a spherical star, so we impose two physical requirements on the metric. The first is that the space-time be *static*, meaning that nothing depends on the time coordinate and that the geometry is unchanged if we run time backwards, which forbids any cross term of the form $dt dr$. The second is that it be *spherically symmetric*, meaning that the two dimensional surfaces of constant t and constant r are ordinary round spheres, so that the angular part of the metric can only be the metric of a sphere multiplied by a function of r , and by choosing the radial coordinate appropriately we can always arrange that function to be exactly r^2 . This is worth saying carefully, because it fixes the physical meaning of r : in the Schwarzschild solution r is not the distance from the centre, it is the *areal radius*, defined so that the sphere at radius r has area $4\pi r^2$. With these two demands the most general metric we can write, in the mostly plus signature of this book, is

$$ds^2 = -e^{2\alpha(r)} dt^2 + e^{2\beta(r)} dr^2 + r^2 (d\vartheta^2 + \sin^2 \vartheta d\varphi^2), \quad (4.1)$$

where $\alpha(r)$ and $\beta(r)$ are the two unknown functions, and where writing them inside exponentials costs us nothing and saves us a great deal of algebra later, since it guarantees automatically that g_{tt} never changes sign and that g_{rr} stays positive. Reading off the components of the metric and of its inverse from equation 4.1 we have

$$g_{tt} = -e^{2\alpha}, \quad g_{rr} = e^{2\beta}, \quad g_{\vartheta\vartheta} = r^2, \quad g_{\varphi\varphi} = r^2 \sin^2 \vartheta, \quad (4.2)$$

$$g^{tt} = -e^{-2\alpha}, \quad g^{rr} = e^{-2\beta}, \quad g^{\vartheta\vartheta} = \frac{1}{r^2}, \quad g^{\varphi\varphi} = \frac{1}{r^2 \sin^2 \vartheta}, \quad (4.3)$$

the inverse being trivial to write down because the metric is diagonal, and every other component of both vanishing identically. From now on a prime will always denote a derivative with respect to r , which is the only variable anything depends on.

4.1.2 The Christoffel symbols

Everything that follows is built on the connection, so we compute all of its components from the definition

$$\Gamma_{\mu\nu}^{\lambda} = \frac{1}{2} g^{\lambda\sigma} (\partial_{\mu} g_{\sigma\nu} + \partial_{\nu} g_{\sigma\mu} - \partial_{\sigma} g_{\mu\nu}), \quad (4.4)$$

and the work is much lighter than it looks because the metric is diagonal, so the sum over σ in equation 4.4 collapses to the single term $\sigma = \lambda$, and because the only non vanishing derivatives of the metric are

$$\partial_r g_{tt} = -2\alpha' e^{2\alpha}, \quad \partial_r g_{rr} = 2\beta' e^{2\beta}, \quad \partial_r g_{\vartheta\vartheta} = 2r, \quad (4.5)$$

$$\partial_r g_{\varphi\varphi} = 2r \sin^2 \vartheta, \quad \partial_{\vartheta} g_{\varphi\varphi} = 2r^2 \sin \vartheta \cos \vartheta. \quad (4.6)$$

We now go through the symbols one at a time. Taking $\lambda = t$, $\mu = t$, $\nu = r$, the first and third terms of equation 4.4 involve $\partial_t g_{tr}$ and $\partial_t g_{tr}$, both zero because the metric is static and because g_{tr} does not exist in the first place, so that

$$\Gamma_{tr}^t = \frac{1}{2} g^{tt} (\partial_t g_{tr} + \partial_r g_{tt} - \partial_t g_{tr}) = \frac{1}{2} (-e^{-2\alpha}) (-2\alpha' e^{2\alpha}) = \alpha',$$

where the two red terms have cancelled against each other before doing anything else and the exponentials have cancelled in the product. For $\lambda = r$, $\mu = \nu = t$ only the last term of equation 4.4 survives, and it comes with a minus sign,

$$\Gamma_{tt}^r = \frac{1}{2} g^{rr} (\partial_t g_{rt} + \partial_t g_{rt} - \partial_r g_{tt}) = \frac{1}{2} e^{-2\beta} [-(-2\alpha' e^{2\alpha})] = \alpha' e^{2\alpha-2\beta}.$$

For $\lambda = \mu = \nu = r$ all three terms are the same and two of them cancel against the third only partially, leaving

$$\Gamma_{rr}^r = \frac{1}{2} g^{rr} (\partial_r g_{rr} + \partial_r g_{rr} - \partial_r g_{rr}) = \frac{1}{2} g^{rr} \partial_r g_{rr} = \frac{1}{2} e^{-2\beta} (2\beta' e^{2\beta}) = \beta'.$$

The two angular symbols with an upper radial index follow the same pattern as Γ_{tt}^r , since again only the derivative of the angular metric with respect to r survives,

$$\begin{aligned}\Gamma_{\vartheta\vartheta}^r &= \frac{1}{2}g^{rr}(\partial_{\vartheta}g_{r\vartheta} + \partial_{\vartheta}g_{r\vartheta} - \partial_r g_{\vartheta\vartheta}) = \frac{1}{2}e^{-2\beta}(-2r) = -re^{-2\beta}, \\ \Gamma_{\varphi\varphi}^r &= \frac{1}{2}g^{rr}(\partial_{\varphi}g_{r\varphi} + \partial_{\varphi}g_{r\varphi} - \partial_r g_{\varphi\varphi}) = \frac{1}{2}e^{-2\beta}(-2r \sin^2 \vartheta) = -r \sin^2 \vartheta e^{-2\beta}.\end{aligned}$$

The symbols with an upper angular index are just as quick. For $\lambda = \vartheta$, $\mu = r$, $\nu = \vartheta$ only the derivative $\partial_r g_{\vartheta\vartheta}$ survives, and the factor r^{-2} coming from the inverse metric leaves a pure $1/r$,

$$\Gamma_{r\vartheta}^{\vartheta} = \frac{1}{2}g^{\vartheta\vartheta}\partial_r g_{\vartheta\vartheta} = \frac{1}{2}\frac{1}{r^2}(2r) = \frac{1}{r},$$

and in exactly the same way, using this time the derivative of $g_{\varphi\varphi}$ with respect to r and then with respect to ϑ ,

$$\begin{aligned}\Gamma_{r\varphi}^{\varphi} &= \frac{1}{2}g^{\varphi\varphi}\partial_r g_{\varphi\varphi} = \frac{1}{2}\frac{1}{r^2 \sin^2 \vartheta}(2r \sin^2 \vartheta) = \frac{1}{r}, \\ \Gamma_{\vartheta\varphi}^{\varphi} &= \frac{1}{2}g^{\varphi\varphi}\partial_{\vartheta} g_{\varphi\varphi} = \frac{1}{2}\frac{1}{r^2 \sin^2 \vartheta}(2r^2 \sin \vartheta \cos \vartheta) = \frac{\cos \vartheta}{\sin \vartheta} = \cot \vartheta,\end{aligned}$$

while the remaining one carries an upper ϑ and two lower φ ,

$$\Gamma_{\varphi\varphi}^{\vartheta} = \frac{1}{2}g^{\vartheta\vartheta}(-\partial_{\vartheta} g_{\varphi\varphi}) = \frac{1}{2}\frac{1}{r^2}(-2r^2 \sin \vartheta \cos \vartheta) = -\sin \vartheta \cos \vartheta.$$

Collecting everything, and remembering that the connection is symmetric in its two lower indices so that each of these also gives the symbol with those indices exchanged, the complete list of non vanishing Christoffel symbols of the metric 4.1 is

$$\begin{aligned}\Gamma_{tr}^t &= \alpha', & \Gamma_{tt}^r &= \alpha' e^{2\alpha-2\beta}, & \Gamma_{rr}^r &= \beta', \\ \Gamma_{\vartheta\vartheta}^r &= -re^{-2\beta}, & \Gamma_{\varphi\varphi}^r &= -r \sin^2 \vartheta e^{-2\beta}, & \Gamma_{r\vartheta}^{\vartheta} &= \frac{1}{r}, \\ \Gamma_{\varphi\varphi}^{\vartheta} &= -\sin \vartheta \cos \vartheta, & \Gamma_{r\varphi}^{\varphi} &= \frac{1}{r}, & \Gamma_{\vartheta\varphi}^{\varphi} &= \cot \vartheta.\end{aligned}\tag{4.7}$$

There are nine of them, and every other one vanishes. It is worth noticing in passing that several of these symbols, $\Gamma_{r\vartheta}^{\vartheta}$ and $\Gamma_{r\varphi}^{\varphi}$ and $\Gamma_{\vartheta\varphi}^{\varphi}$, would be there even in perfectly flat space written in

spherical coordinates, which is the point we insisted on earlier: a non vanishing connection is a statement about the coordinates, not about the curvature.

One combination we shall need repeatedly is the contracted symbol $\Gamma_{\lambda\mu}^{\lambda}$, obtained by summing over the repeated index. Only two of its four components are non zero,

$$\begin{aligned}\Gamma_{\lambda r}^{\lambda} &= \Gamma_{tr}^t + \Gamma_{rr}^r + \Gamma_{\vartheta r}^{\vartheta} + \Gamma_{\varphi r}^{\varphi} = \alpha' + \beta' + \frac{1}{r} + \frac{1}{r} = \alpha' + \beta' + \frac{2}{r}, \\ \Gamma_{\lambda \vartheta}^{\lambda} &= \Gamma_{t\vartheta}^t + \Gamma_{r\vartheta}^r + \Gamma_{\vartheta\vartheta}^{\vartheta} + \Gamma_{\varphi\vartheta}^{\varphi} = 0 + 0 + 0 + \cot \vartheta = \cot \vartheta,\end{aligned}\tag{4.8}$$

while $\Gamma_{\lambda t}^{\lambda}$ and $\Gamma_{\lambda\varphi}^{\lambda}$ both vanish, the first because the geometry is static and the second because it is axially symmetric.

4.1.3 The Ricci tensor

We now assemble the Ricci tensor from the connection using

$$R_{\mu\nu} = \partial_{\lambda}\Gamma_{\mu\nu}^{\lambda} - \partial_{\nu}\Gamma_{\mu\lambda}^{\lambda} + \Gamma_{\lambda\rho}^{\lambda}\Gamma_{\mu\nu}^{\rho} - \Gamma_{\nu\rho}^{\lambda}\Gamma_{\mu\lambda}^{\rho},\tag{4.9}$$

and we take the four diagonal components in turn. All the off diagonal components vanish, which one can check directly and which is in any case guaranteed by the symmetry of the problem, since there is no preferred direction available with which to build them.

Starting with the temporal component, the second term of equation 4.9 dies immediately because nothing depends on t , and in the third and fourth terms only $\rho = r$ and the pairs involving t and r can contribute, because those are the only non vanishing symbols with a t among their lower indices,

$$R_{tt} = \partial_r \Gamma_{tt}^r - \partial_t \Gamma_{t\lambda}^{\lambda} + \Gamma_{\lambda r}^{\lambda} \Gamma_{tt}^r - (\Gamma_{tr}^t \Gamma_{tt}^r + \Gamma_{tt}^r \Gamma_{tr}^t).$$

The first term requires the derivative of a product of exponentials, which we do carefully,

$$\begin{aligned}\partial_r \Gamma_{tt}^r &= \frac{d}{dr} \left(\alpha' e^{2\alpha-2\beta} \right) = \alpha'' e^{2\alpha-2\beta} + \alpha' (2\alpha' - 2\beta') e^{2\alpha-2\beta} \\ &= e^{2\alpha-2\beta} [\alpha'' + 2(\alpha')^2 - 2\alpha'\beta'],\end{aligned}$$

the third term is read straight off equation 4.8,

$$\Gamma_{\lambda r}^{\lambda} \Gamma_{tt}^r = \left(\alpha' + \beta' + \frac{2}{r} \right) \alpha' e^{2\alpha-2\beta} = e^{2\alpha-2\beta} \left[(\alpha')^2 + \alpha'\beta' + \frac{2\alpha'}{r} \right],$$

and the fourth, which appears twice because the pair of indices can be taken in either order, is

$$-(\Gamma_{tr}^t \Gamma_{tt}^r + \Gamma_{tt}^r \Gamma_{tr}^t) = -2\alpha' (\alpha' e^{2\alpha-2\beta}) = -2e^{2\alpha-2\beta} (\alpha')^2.$$

Adding the three surviving pieces and factoring out the common exponential, the terms in $(\alpha')^2$ largely destroy one another,

$$\begin{aligned} R_{tt} &= e^{2\alpha-2\beta} \left[\alpha'' + 2(\alpha')^2 - 2\alpha'\beta' + (\alpha')^2 + \alpha'\beta' + \frac{2\alpha'}{r} - 2(\alpha')^2 \right] \\ &= e^{2(\alpha-\beta)} \left[\alpha'' + (\alpha')^2 - \alpha'\beta' + \frac{2\alpha'}{r} \right], \end{aligned}$$

where of the three contributions $2(\alpha')^2 + (\alpha')^2 - 2(\alpha')^2$ only one survives, and of $-2\alpha'\beta' + \alpha'\beta'$ only one half survives with its sign.

The radial component is assembled in the same way, except that now the second term of equation 4.9 does contribute, since $\Gamma_{r\lambda}^\lambda$ depends on r ,

$$R_{rr} = \partial_r \Gamma_{rr}^r - \partial_r \Gamma_{r\lambda}^\lambda + \Gamma_{\lambda r}^\lambda \Gamma_{rr}^r - \Gamma_{r\rho}^\lambda \Gamma_{r\lambda}^\rho.$$

Term by term, the first is simply β'' , the second follows from differentiating equation 4.8,

$$-\partial_r \Gamma_{r\lambda}^\lambda = -\frac{d}{dr} \left(\alpha' + \beta' + \frac{2}{r} \right) = -\alpha'' - \beta'' + \frac{2}{r^2},$$

the third is

$$\Gamma_{\lambda r}^\lambda \Gamma_{rr}^r = \left(\alpha' + \beta' + \frac{2}{r} \right) \beta' = \alpha'\beta' + (\beta')^2 + \frac{2\beta'}{r},$$

and the fourth collects the four symbols that carry a lower r together with a matching pair of indices, namely Γ_{rt}^t , Γ_{rr}^r , $\Gamma_{r\vartheta}^\vartheta$ and $\Gamma_{r\varphi}^\varphi$,

$$-\Gamma_{r\rho}^\lambda \Gamma_{r\lambda}^\rho = - \left[(\Gamma_{rt}^t)^2 + (\Gamma_{rr}^r)^2 + (\Gamma_{r\vartheta}^\vartheta)^2 + (\Gamma_{r\varphi}^\varphi)^2 \right] = - \left[(\alpha')^2 + (\beta')^2 + \frac{2}{r^2} \right].$$

Adding the four contributions, the second derivatives of β cancel, the two terms in $(\beta')^2$ cancel, and the two terms in $2/r^2$ cancel as well,

$$\begin{aligned} R_{rr} &= \beta'' - \alpha'' - \beta'' + \frac{2}{r^2} + \alpha'\beta' + (\beta')^2 + \frac{2\beta'}{r} - (\alpha')^2 - (\beta')^2 - \frac{2}{r^2} \\ &= -\alpha'' - (\alpha')^2 + \alpha'\beta' + \frac{2\beta'}{r}. \end{aligned}$$

The polar component takes a little more care because it is the only one in which the angular symbols play a role. Writing out equation 4.9 with $\mu = \nu = \vartheta$,

$$R_{\vartheta\vartheta} = \partial_r \Gamma^r_{\vartheta\vartheta} - \partial_{\vartheta} \Gamma^{\lambda}_{\vartheta\lambda} + \Gamma^{\lambda}_{\lambda r} \Gamma^r_{\vartheta\vartheta} - \Gamma^{\lambda}_{\vartheta\rho} \Gamma^{\rho}_{\vartheta\lambda},$$

the first term is

$$\partial_r \Gamma^r_{\vartheta\vartheta} = \frac{d}{dr} (-re^{-2\beta}) = -e^{-2\beta} + 2r\beta'e^{-2\beta} = e^{-2\beta} (2r\beta' - 1),$$

the second uses $\Gamma^{\lambda}_{\vartheta\lambda} = \cot \vartheta$ from equation 4.8 and produces the only trigonometric derivative of the whole calculation,

$$-\partial_{\vartheta} \Gamma^{\lambda}_{\vartheta\lambda} = -\frac{d}{d\vartheta} \cot \vartheta = -\left(-\frac{1}{\sin^2 \vartheta}\right) = \frac{1}{\sin^2 \vartheta},$$

the third is

$$\Gamma^{\lambda}_{\lambda r} \Gamma^r_{\vartheta\vartheta} = \left(\alpha' + \beta' + \frac{2}{r}\right) (-re^{-2\beta}) = -e^{-2\beta} (r\alpha' + r\beta' + 2),$$

and the fourth collects the three surviving index pairs, the first two of which are equal and appear with opposite orderings,

$$\begin{aligned} -\Gamma^{\lambda}_{\vartheta\rho} \Gamma^{\rho}_{\vartheta\lambda} &= -\left[\Gamma^r_{\vartheta\vartheta} \Gamma^{\vartheta}_{\vartheta r} + \Gamma^{\vartheta}_{\vartheta r} \Gamma^r_{\vartheta\vartheta} + \left(\Gamma^{\varphi}_{\vartheta\varphi}\right)^2\right] \\ &= -\left[2\left(-re^{-2\beta}\right) \frac{1}{r} + \cot^2 \vartheta\right] = 2e^{-2\beta} - \cot^2 \vartheta. \end{aligned}$$

Now we add the four pieces. The trigonometric terms combine through the Pythagorean identity divided by $\sin^2 \vartheta$, that is through $\csc^2 \vartheta - \cot^2 \vartheta = 1$, which we insert deliberately at this point, while among the exponential terms the constants -2 and $+2$ cancel against each other,

$$\begin{aligned} R_{\vartheta\vartheta} &= e^{-2\beta} (2r\beta' - 1) + \frac{1}{\sin^2 \vartheta} - e^{-2\beta} (r\alpha' + r\beta' + 2) + 2e^{-2\beta} - \cot^2 \vartheta \\ &= e^{-2\beta} (2r\beta' - 1 - r\alpha' - r\beta') + 1 \\ &= e^{-2\beta} [r(\beta' - \alpha') - 1] + 1. \end{aligned}$$

The azimuthal component needs no separate calculation. Spherical symmetry guarantees that the two angular directions cannot be distinguished by the geometry, and carrying the computation through confirms it,

$$R_{\varphi\varphi} = \sin^2 \vartheta R_{\vartheta\vartheta}, \quad (4.10)$$

so it gives us no information that $R_{\vartheta\vartheta}$ has not already given.

4.1.4 Solving the vacuum equations

Outside the star there is no matter, so $T_{\mu\nu} = 0$, and we showed earlier that in that case the field equations reduce to the vanishing of the Ricci tensor alone. We therefore have to solve

$$R_{tt} = R_{rr} = R_{\vartheta\vartheta} = 0, \quad (4.11)$$

which is three equations for two unknown functions, and the fact that the system is nevertheless consistent is itself a consequence of the Bianchi identity.

The first move is the one that makes the whole problem collapse. Since $e^{2(\alpha-\beta)}$ never vanishes, the condition $R_{tt} = 0$ is equivalent to the vanishing of the bracket, and we may therefore add that bracket to R_{rr} and ask what the sum gives,

$$\begin{aligned} e^{-2(\alpha-\beta)} R_{tt} + R_{rr} &= \left[\alpha'' + (\alpha')^2 - \alpha'\beta' + \frac{2\alpha'}{r} \right] + \left[-\alpha'' - (\alpha')^2 + \alpha'\beta' + \frac{2\beta'}{r} \right] \\ &= \frac{2\alpha'}{r} + \frac{2\beta'}{r} = \frac{2}{r} (\alpha' + \beta') = 0. \end{aligned}$$

Every single term involving second derivatives or products of first derivatives has cancelled, and what is left is almost embarrassingly simple: since r is not zero anywhere in the region we care about,

$$\alpha' + \beta' = 0 \quad \longrightarrow \quad \alpha + \beta = \text{constant}. \quad (4.12)$$

That constant can be set to zero without any loss of generality, and the reason is worth stating because it is a genuinely physical argument rather than an algebraic convenience. A non zero constant would amount to multiplying g_{tt} by a fixed positive number, which is the same as rescaling the time coordinate by $t \rightarrow e^{\text{const}} t$, and we are free to choose the unit of time so that a clock infinitely far from the star, where the geometry is flat, ticks at the same rate as the coordinate t . Making that choice,

$$\beta = -\alpha. \quad (4.13)$$

With equation 4.13 in hand we go to the angular equation, which is now an equation for α alone. Substituting $\beta' = -\alpha'$ and $e^{-2\beta} = e^{2\alpha}$ into $R_{\vartheta\vartheta} = 0$,

$$\begin{aligned} e^{2\alpha} [r(-\alpha' - \alpha') - 1] + 1 &= 0, \\ e^{2\alpha} (-2r\alpha' - 1) &= -1, \\ e^{2\alpha} (2r\alpha' + 1) &= 1, \end{aligned}$$

where in the last step we multiplied both sides by -1 . Now comes the only trick in the entire derivation, and it is the observation that the left hand side is an exact derivative. Differentiating the product $re^{2\alpha}$ with the chain rule,

$$\frac{d}{dr} (re^{2\alpha}) = e^{2\alpha} + r(2\alpha'e^{2\alpha}) = e^{2\alpha} (1 + 2r\alpha'), \quad (4.14)$$

which is precisely the combination we have just been left with. Our equation therefore reads

$$\frac{d}{dr} (re^{2\alpha}) = 1, \quad (4.15)$$

and this we integrate at sight. Writing the constant of integration as K ,

$$\begin{aligned} re^{2\alpha} &= r + K, \\ e^{2\alpha} &= 1 + \frac{K}{r}. \end{aligned}$$

We have solved the field equations. Before interpreting the answer it is worth performing the consistency check we promised, namely verifying that the third equation, $R_{tt} = 0$, which we never used, is satisfied automatically. Writing $f \equiv e^{2\alpha}$, so that $2\alpha' = f'/f$ and $\alpha'' = f''/2f - (f')^2/2f^2$, and substituting $\beta = -\alpha$ into the bracket of R_{tt} ,

$$\begin{aligned}\alpha'' + (\alpha')^2 - \alpha'\beta' + \frac{2\alpha'}{r} &= \alpha'' + (\alpha')^2 + (\alpha')^2 + \frac{2\alpha'}{r} = \alpha'' + 2(\alpha')^2 + \frac{2\alpha'}{r} \\ &= \frac{f''}{2f} - \frac{(f')^2}{2f^2} + \frac{2(f')^2}{4f^2} + \frac{f'}{rf} = \frac{1}{2f} \left(f'' + \frac{2f'}{r} \right),\end{aligned}$$

where the two terms in $(f')^2$ have cancelled exactly, and inserting $f = 1 + K/r$, whose derivatives are $f' = -K/r^2$ and $f'' = 2K/r^3$,

$$f'' + \frac{2f'}{r} = \frac{2K}{r^3} + \frac{2}{r} \left(-\frac{K}{r^2} \right) = \frac{2K}{r^3} - \frac{2K}{r^3} = 0,$$

so that R_{tt} vanishes identically, as the Bianchi identity promised it would.

4.1.5 Identifying the constant and reading the solution

The constant K is not fixed by the field equations, which know nothing about what sits at the centre, and it can only be determined by matching the solution to a situation we already understand. Far from the source the field is weak, and there we derived in the Newtonian limit that the temporal component of the metric must be $g_{tt} = -(1 + 2\phi)$ with $\phi = -Gm/r$ the Newtonian potential of a body of mass m . Comparing that with what we have just found,

$$g_{tt} = -e^{2\alpha} = -\left(1 + \frac{K}{r}\right) \quad \text{and} \quad g_{tt} = -\left(1 - \frac{2Gm}{r}\right),$$

we can read the constant off immediately, $K = -2Gm$, or simply $K = -2m$ in the geometrised units in which $G = C = 1$. Putting everything together, the solution of the vacuum field equations outside a static spherically symmetric body is

$$ds^2 = -\left(1 - \frac{2Gm}{C^2 r}\right) C^2 dt^2 + \left(1 - \frac{2Gm}{C^2 r}\right)^{-1} dr^2 + r^2 (d\vartheta^2 + \sin^2 \vartheta d\varphi^2), \quad (4.16)$$

which is the Schwarzschild metric, and which in geometrised units is written in the compact form one usually meets,

$$ds^2 = -\left(1 - \frac{2m}{r}\right) dt^2 + \frac{dr^2}{1 - \frac{2m}{r}} + r^2 d\Omega^2. \quad (4.17)$$

Several things about equation 4.17 deserve comment. The first is that it depends on the source only through the single number m , so that two stars of the same mass produce exactly the same

exterior geometry no matter how differently their interiors are built, which is the first hint of the no hair behaviour that black holes display in full. The second is that as $r \rightarrow \infty$ the metric tends to $\eta_{\mu\nu}$, so the solution is asymptotically flat and describes an isolated body, as it should. The third is that we assumed staticity at the very start, and it turns out we need not have: Birkhoff proved in 1923 that *every* spherically symmetric solution of the vacuum equations is necessarily static and is necessarily the Schwarzschild metric, so that a spherically pulsating star radiates no gravitational waves at all, which is the exact gravitational analogue of the statement that a spherically symmetric distribution of charge radiates no electromagnetic waves (Wald, 1984).

The fourth point is the one that occupied physicists for half a century. The metric 4.17 misbehaves at two places, at $r = 2m$, where g_{tt} vanishes and g_{rr} blows up, and at $r = 0$, where the reverse happens, and the question is whether either of these is a genuine feature of the geometry or merely a failure of the coordinates, in the same way that the spherical chart on S^2 failed at the poles in chapter 2 although nothing whatever happens to the sphere there. The way to settle such a question is to compute a scalar built from the curvature, since a scalar cannot depend on the chart, and the simplest one available is the Kretschmann scalar, which for the Schwarzschild metric works out to

$$R_{abcd}R^{abcd} = \frac{48m^2}{r^6}. \quad (4.18)$$

Equation 4.18 is perfectly finite at $r = 2m$, which tells us that the surface $r = 2m$ is a coordinate singularity and nothing more, an artefact of having insisted on a static observer's time coordinate in a region where no observer can remain static. It is the *event horizon*, and a freely falling traveller crosses it without noticing anything locally unusual, although they can never come back. At $r = 0$ the same scalar diverges, and no change of coordinates can repair a divergent scalar, so that is a genuine curvature singularity where the theory that produced it stops making sense and where we expect a quantum theory of gravity to take over. For an ordinary star none of this matters in the slightest, because $2m$ lies far inside the surface, where the vacuum solution does not apply in the first place: for the Sun the radius $2Gm/C^2$ is about three kilometres, buried under seven hundred thousand kilometres of hydrogen.

With the exterior geometry in hand one can go on to compute everything that made general relativity famous, the precession of the perihelion of Mercury, the deflection of starlight grazing the Sun and the delay of radar signals passing near it, all of which follow from integrating the geodesic equation in the metric 4.17. Before we do any of that, however, there are two pieces of machinery which the Schwarzschild solution allows us to introduce in their natural habitat, and which we shall need constantly from here onwards. The first tells us how to translate the components of a tensor into numbers an actual observer would measure, and the second tells us how to extract conserved quantities from a space-time without integrating anything at all.

4.2 Observers and symmetries

The two sections which follow share a common motivation. So far every component we have written has been a component with respect to a coordinate basis, and coordinates are bookkeeping devices

with no physical meaning of their own. A physicist eventually has to ask what an observer sitting somewhere in the space-time actually measures, and that question requires a basis attached to the observer rather than to the chart. Separately, integrating the geodesic equation directly is painful, and we know from Lagrangian mechanics that the honest way to avoid the pain is to find conserved quantities. Both needs are answered by ideas that cost us nothing new: the first by choosing a different basis, the second by asking the Lie derivative of chapter 2 a question about the metric.

4.2.1 Tetrads and what an observer measures

Everything in this chapter has been written in a coordinate basis, the set $\{\partial_\mu\}$ attached to a chart, and it is time to say why that is not good enough. The vectors ∂_μ are in general neither unit vectors nor orthogonal to one another, because their inner products are the metric components themselves,

$$\mathbb{G}(\partial_\mu, \partial_\nu) = g_{\mu\nu}, \quad (4.19)$$

and in Schwarzschild coordinates that means ∂_t has squared length $-(1 - 2m/r)$, which is not -1 , and ∂_φ has squared length $r^2 \sin^2 \vartheta$, which grows without bound. So when we compute a component such as T^{tt} we have not computed an energy density that anybody could measure with an instrument, we have computed a number expressed in units which themselves vary from place to place.

We already know the remedy, because we built it in chapter 2 when we abandoned the requirement that a basis come from a chart and allowed ourselves an arbitrary frame $\{e_{(a)}\}$ with its dual coframe $\{e^{(a)}\}$, its dictionary $e_{(a)}^\mu$ and its structure functions. What we could not do there was single out any particular frame, because choosing one requires a criterion and the only criterion a physicist wants is a metric one. Now we have a metric, so we impose it: of all the frames available at a point we keep those which satisfy

$$\mathbb{G}(e_{(a)}, e_{(b)}) = \eta_{ab}, \quad (4.20)$$

that is, those which are orthonormal with respect to the space-time metric at every point. A frame obeying 4.20 is what in four dimensions we call a *tetrad*. Latin indices from the beginning of the alphabet will always be frame indices, raised and lowered with η_{ab} , while Greek indices remain coordinate indices, raised and lowered with $g_{\mu\nu}$. That a tetrad always exists is guaranteed by the local flatness we built into the theory in the first place, since the equivalence principle says precisely that at any point we can bring the metric to the Minkowski form, and equation 4.20 is that statement written as a choice of basis rather than as a choice of coordinates.

Writing the frame vectors in the coordinate basis, $e_{(a)} = e_{(a)}^\mu \partial_\mu$, and substituting into 4.20 with the help of 4.19, the defining condition becomes a relation between the dictionary and the metric,

$$g_{\mu\nu} e_{(a)}^\mu e_{(b)}^\nu = \eta_{ab}, \quad (4.21)$$

and contracting 4.21 with the inverse dictionary of chapter 2 turns it inside out and gives the metric back as

$$g_{\mu\nu} = \eta_{ab} e_{\mu}^{(a)} e_{\nu}^{(b)}. \quad (4.22)$$

Equation 4.22 is worth pausing over, because it says that the metric can be regarded as a derived object: give me four one form fields and I will hand you back a metric. That observation is the starting point of the tetrad formulation of general relativity, and it is what makes it possible to put spinors, which have no coordinate description at all, into a curved space-time.

The translation of tensor components between the two languages is the one we already wrote in chapter 2, but it acquires here a physical meaning it did not have there. The numbers $T_{ab} = T_{\mu\nu} e_{(a)}^{\mu} e_{(b)}^{\nu}$ are components in an orthonormal basis, which is to say components measured with respect to rods and clocks that are locally identical to the rods and clocks of special relativity, and they are therefore the numbers an observer whose worldline carries that frame would actually read off an instrument. The numbers $T_{\mu\nu}$ are not.

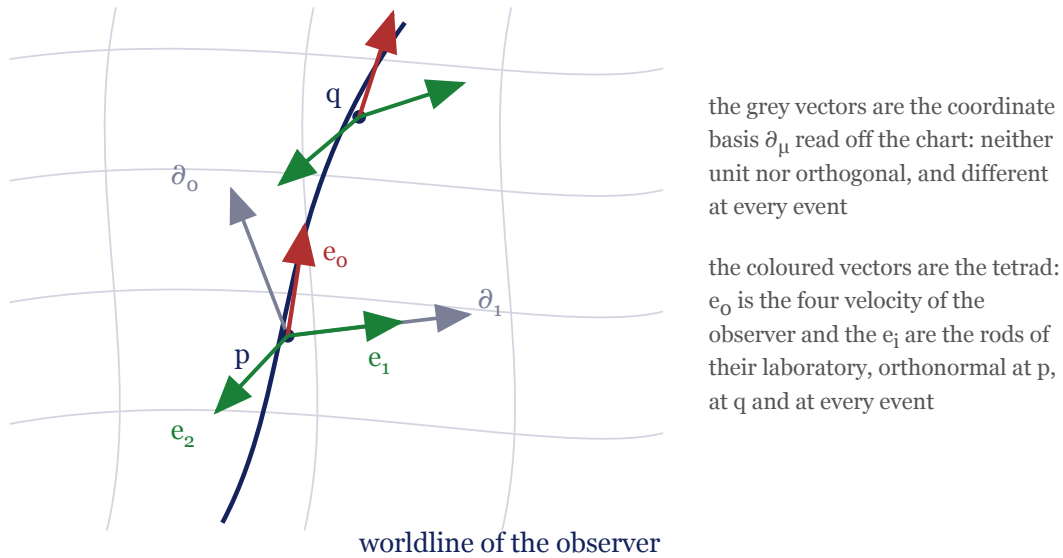


Figure 4.1: A coordinate basis and an orthonormal frame at the same event. The coordinate vectors ∂_{μ} follow the chart and are in general neither unit nor orthogonal, and their lengths change from one event to the next, while the tetrad $e_{(a)}$ is orthonormal at every event and is the basis with respect to which an observer measures. The two are related by the matrix $e_{(a)}^{\mu}$.

The freedom we have gained is not free of ambiguity, and the ambiguity is itself interesting. Equation 4.20 does not determine the tetrad uniquely, because if $\{e_{(a)}\}$ is orthonormal then so is any

$\Lambda_a^b(x) e_{(b)}$ provided the matrix Λ preserves η_{ab} , and the matrices that preserve η_{ab} are precisely the Lorentz transformations. A tetrad field therefore carries a local Lorentz symmetry, one independent Lorentz transformation at every point of the space-time, which is nothing other than the freedom of each observer to choose their own orientation and their own state of motion. General relativity written in the tetrad language looks remarkably like a gauge theory with the Lorentz group as its gauge group, and that resemblance is not an accident.

The price is the one chapter 2 warned us about, namely that the frame vectors no longer commute and their structure functions C_{ab}^c are not zero. If they were, the tetrad would be a coordinate basis, and a coordinate basis cannot be orthonormal everywhere unless the space-time is flat, so a curved geometry leaves us no choice in the matter. It bears repeating here, because the temptation is strongest at exactly this point, that those structure functions are not a measure of curvature: the plane is flat and its orthonormal polar frame already has one of them different from zero. Curvature is the Riemann tensor and nothing else.

Let us make all of this concrete with the only solution we have in hand. Consider an observer who remains at fixed r , ϑ and φ outside a Schwarzschild body, a person standing on a platform held up by rockets far above a black hole, and build the frame they carry. The timelike leg must be the normalised four velocity of such an observer, and the remaining three must be the normalised coordinate directions, which are already mutually orthogonal because the Schwarzschild metric is diagonal. Dividing each ∂_μ by the square root of the magnitude of the corresponding metric component gives

$$\begin{aligned} e_{(0)} &= \left(1 - \frac{2m}{r}\right)^{-1/2} \partial_t, & e_{(1)} &= \left(1 - \frac{2m}{r}\right)^{1/2} \partial_r, \\ e_{(2)} &= \frac{1}{r} \partial_\vartheta, & e_{(3)} &= \frac{1}{r \sin \vartheta} \partial_\varphi, \end{aligned} \quad (4.23)$$

and the reader may verify directly from 4.21 that these four satisfy $\mathbb{G}(e_{(a)}, e_{(b)}) = \eta_{ab}$, since for example

$$\mathbb{G}(e_{(0)}, e_{(0)}) = \left(1 - \frac{2m}{r}\right)^{-1} g_{tt} = \left(1 - \frac{2m}{r}\right)^{-1} \left[-\left(1 - \frac{2m}{r}\right)\right] = -1.$$

The frame 4.23 pays for itself immediately. Suppose a photon of coordinate energy E is emitted at radius r_1 and received at radius r_2 . The energy measured by a static observer is not E but the frame component

$$E_{\text{meas}} = -p_\mu e_{(0)}^\mu = \left(1 - \frac{2m}{r}\right)^{-1/2} (-p_t) = \left(1 - \frac{2m}{r}\right)^{-1/2} E,$$

because $-p_t$ is the conserved quantity, as we are about to prove in the next section, while the frame factor is what converts it into a reading. Taking the ratio at the two radii gives the gravitational redshift

$$\frac{E_2}{E_1} = \left(\frac{1 - 2m/r_1}{1 - 2m/r_2} \right)^{1/2}, \quad (4.24)$$

which reproduces the result we obtained from the equivalence principle in chapter 3, only now it has fallen out of the formalism rather than out of a thought experiment about lifts. Notice also what 4.23 says as $r \rightarrow 2m$: the normalisation factor of $e_{(0)}$ diverges, which is the frame telling us that the proper acceleration required to stay in place becomes infinite, and that no observer can remain static at the horizon. The horizon is a coordinate singularity of the metric but it is a genuine obstruction for this particular family of observers, and the tetrad is what lets us say the two things separately without confusing them.

4.2.2 Killing vectors and conserved quantities

We now come to the question that the Lie derivative was invented to answer. A symmetry of a space-time should mean that the geometry looks the same after we have moved along some flow, and we have already built the instrument that measures how a tensor changes when it is dragged along a flow, so the definition writes itself. A vector field ξ is a *Killing vector field* of the space-time $(M, \mathbb{G})$ if the metric is unchanged when dragged along its flow,

$$\mathcal{L}_\xi \mathbb{G} = 0, \quad (4.25)$$

and the flow generated by such a field is called an isometry. Equation 4.25 deserves to be read slowly. It does not say that the metric components are constant, which would be a statement about a chart and therefore meaningless. It says that if we pick up the entire geometry and slide it along the integral curves of ξ , every distance and every angle comes back exactly as it was, so that no measurement whatsoever could detect that the sliding had happened.

The definition is geometrical, but for computation we want it in components, and getting it there is a short exercise with equation 3.104. Applying the Lie derivative to the metric and using the fact that the covariant derivative of the metric vanishes for the Levi-Civita connection,

$$\begin{aligned} \mathcal{L}_\xi g_{\mu\nu} &= \xi^\rho \vec{\nabla}_\rho g_{\mu\nu} + g_{\rho\nu} \vec{\nabla}_\mu \xi^\rho + g_{\mu\rho} \vec{\nabla}_\nu \xi^\rho \\ &= \vec{\nabla}_\mu \xi_\nu + \vec{\nabla}_\nu \xi_\mu, \end{aligned}$$

where the term in red is zero by metric compatibility and where we have used the metric to lower the index inside the derivative, which is legitimate for the same reason. So the condition 4.25 is equivalent to

$$\vec{\nabla}_\mu \xi_\nu + \vec{\nabla}_\nu \xi_\mu = 2\vec{\nabla}_{(\mu} \xi_{\nu)} = 0, \quad (4.26)$$

which is the *Killing equation*. It is a system of ten coupled first order partial differential equations for four unknown functions, which is heavily overdetermined, and that is the technical reason why a generic space-time has no symmetries at all: solutions simply do not exist unless the geometry has been arranged to admit them. Minkowski space-time admits the maximum number, ten of them, corresponding to four translations, three rotations and three boosts, which is the Poincaré group reappearing as the isometry group of flat space. Schwarzschild admits four.

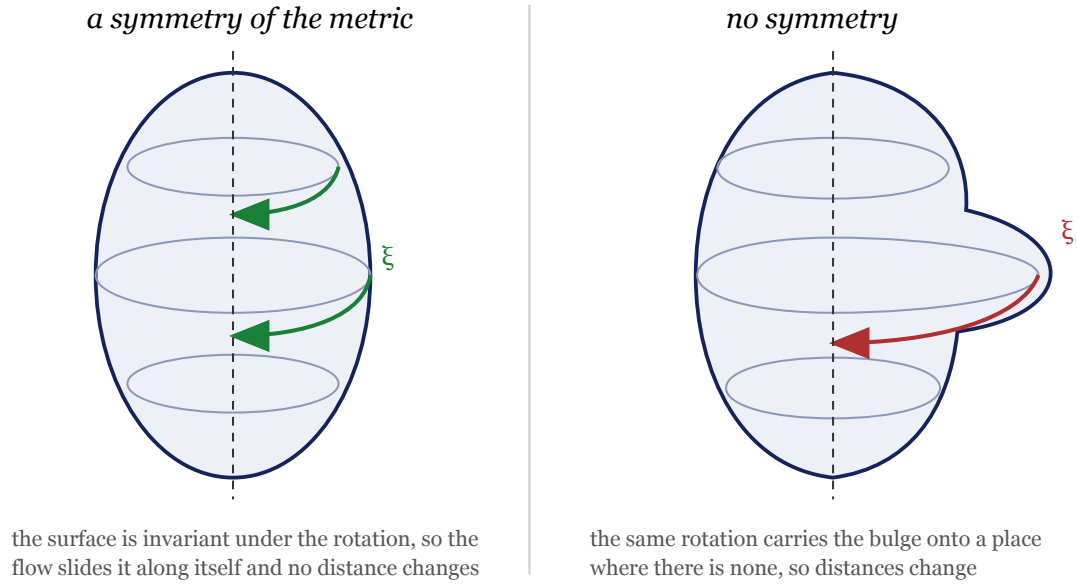


Figure 4.2: On the left the surface is invariant under rotation about its axis, so the flow generated by ξ slides the surface along itself and every distance is left exactly as it was: ξ is a Killing vector. On the right the same flow carries the bulge round onto a region where there is none, so distances are changed and ξ satisfies no Killing equation. A symmetry is not a property of the flow but of the geometry the flow is made to act on.

The reason Killing vectors matter to a physicist rather than to a geometer is the following theorem, which is the general relativistic form of Noether's theorem and which is one of the most useful results in the whole subject.

Theorem 8 *Let ξ be a Killing vector field and let γ be a geodesic with tangent u^μ . Then the quantity*

$$Q = \xi_\mu u^\mu$$

is constant along γ .

Proof. The rate of change of Q along the curve is its covariant derivative in the direction of the tangent, so

$$\frac{dQ}{d\tau} = u^\nu \vec{\nabla}_\nu (\xi_\mu u^\mu) = u^\nu u^\mu \vec{\nabla}_\nu \xi_\mu + \xi_\mu \textcolor{red}{u^\nu \vec{\nabla}_\nu u^\mu},$$

where we used the Leibniz rule. The term in red is exactly the left hand side of the geodesic equation and therefore vanishes, because γ is a geodesic. What remains is the contraction of $u^\nu u^\mu$, which is symmetric under the exchange of its two indices, with $\vec{\nabla}_\nu \xi_\mu$, and by the identity we proved in chapter 2 the contraction of a symmetric object with an antisymmetric one vanishes, so only the symmetric part of the derivative survives,

$$\frac{dQ}{d\tau} = u^\nu u^\mu \vec{\nabla}_{(\nu} \xi_{\mu)} = 0,$$

which is zero by the Killing equation 4.26. ■

That is a remarkable trade. In exchange for one symmetry of the geometry we obtain one first integral of the motion, and first integrals are precisely what turn an unmanageable system of second order equations into something we can solve. The physical interpretation of each conserved quantity is read off from the Killing vector that produced it, exactly as in Lagrangian mechanics: a timelike Killing vector gives a conserved energy, a rotational one gives a conserved angular momentum, and a translational one gives a conserved linear momentum.

There is a second version of the theorem which is just as useful and which applies to fields rather than to particles. If ξ is a Killing vector and $T^{\mu\nu}$ is the energy momentum tensor, then the current

$$J^\mu = T^{\mu\nu} \xi_\nu \tag{4.27}$$

is covariantly conserved, since

$$\vec{\nabla}_\mu J^\mu = \left(\textcolor{red}{\vec{\nabla}_\mu T^{\mu\nu}} \right) \xi_\nu + T^{\mu\nu} \vec{\nabla}_\mu \xi_\nu = T^{\mu\nu} \vec{\nabla}_{(\mu} \xi_{\nu)} = 0,$$

where the red term vanishes by the conservation law $\vec{\nabla}_\mu T^{\mu\nu} = 0$ which the Bianchi identities forced upon us, and where the second term dies because $T^{\mu\nu}$ is symmetric and the symmetric part of $\vec{\nabla}_\mu \xi_\nu$ is zero. This is worth emphasising because it rescues something we lost earlier. We remarked when we wrote down $\vec{\nabla}_\mu T^{\mu\nu} = 0$ that this is a local statement and does not by itself give a globally conserved energy, since one cannot integrate a vector equation over a curved space-time in any invariant way. Equation 4.27 is a scalar equation for a genuine current, and a current can be integrated. *Energy in general relativity is globally conserved when, and only when, the space-time*

possesses a *timelike Killing vector*, and since a generic space-time possesses none, the total energy of a generic space-time is simply not a defined quantity.

Let us apply all of this to Schwarzschild, where the payoff is immediate. The metric 4.17 has components which depend on r and ϑ but on neither t nor φ , and whenever the components in some chart are independent of a coordinate x^λ the corresponding coordinate vector is a Killing vector, because then

$$\mathcal{L}_{\partial_\lambda} g_{\mu\nu} = \partial_\lambda g_{\mu\nu} = 0$$

directly from the definition. So $\xi_{(t)} = \partial_t$ and $\xi_{(\varphi)} = \partial_\varphi$ are Killing vectors of the Schwarzschild geometry, the first expressing the fact that the geometry does not change with time and the second that it does not change under rotation about the axis. There are two more coming from the full spherical symmetry, which together with $\xi_{(\varphi)}$ close into the algebra $\mathfrak{so}(3)$ we computed on the sphere in chapter 2, but these two suffice for what follows.

Lowering the index on each with the metric, so that $\xi_{(t)\mu} = g_{\mu t}$ and $\xi_{(\varphi)\mu} = g_{\mu\varphi}$, and contracting with the four velocity $u^\mu = (\dot{t}, \dot{r}, \dot{\vartheta}, \dot{\varphi})$ of a particle in free fall, where the dot is the derivative with respect to proper time, the theorem hands us two constants of the motion,

$$E = -\xi_{(t)\mu} u^\mu = \left(1 - \frac{2m}{r}\right) \dot{t}, \quad L = \xi_{(\varphi)\mu} u^\mu = r^2 \sin^2 \vartheta \dot{\varphi}, \quad (4.28)$$

the minus sign in the first being conventional so that E comes out positive for a future directed particle. These are the energy per unit mass and the angular momentum per unit mass of the orbit, as we can check by letting $r \rightarrow \infty$, where $E \rightarrow \dot{t}$, which is the special relativistic γ and therefore the energy per unit mass of a free particle, and where L reduces to the familiar $r^2 \dot{\varphi}$ of Newtonian mechanics, which is Kepler's second law in disguise.

With 4.28 in hand the geodesic problem becomes almost easy. Restricting to the equatorial plane $\vartheta = \pi/2$, which we may always do by a rotation, and substituting both constants into the normalisation condition $g_{\mu\nu} u^\mu u^\nu = -1$, a little algebra gives

$$\frac{1}{2} \dot{r}^2 + \underbrace{\left[-\frac{m}{r} + \frac{L^2}{2r^2} - \frac{mL^2}{r^3} \right]}_{V_{\text{eff}}(r)} = \frac{E^2 - 1}{2}, \quad (4.29)$$

which is the energy equation of a single particle moving in one dimension under an effective potential, and which we have obtained without ever writing down a Christoffel symbol. Equation 4.29 is the whole of orbital dynamics around a spherical body. Its first two terms are exactly Newton's, the attractive potential and the centrifugal barrier, and the third term, the one proportional to mL^2/r^3 , is the entire contribution of general relativity. It is small at large radii, which is why Newton worked so well for two centuries, and it is attractive and grows faster than the centrifugal

barrier as r decreases, which has two consequences with names attached to them. The first is that orbits no longer close on themselves, and their slow rotation is the perihelion precession of Mercury. The second is that at small enough radius the barrier is overwhelmed altogether, so that unlike in Newtonian gravity a particle with any amount of angular momentum whatsoever can fall in. That is a genuinely new possibility, and it is the origin of the innermost stable circular orbit at $r = 6m$ which sets the inner edge of the accretion discs we observe around black holes.

So the two ideas of this section do what we advertised. The tetrad tells us what an observer measures, and the Killing vectors tell us what does not change, and between them they convert the geometry of the previous sections into predictions.

4.3 The Einstein-Maxwell equations

Every source we have fed to the field equations so far has been matter of one kind or another, dust or a perfect fluid, and in each case we wrote down an energy momentum tensor and then forgot where it came from. There is one source that does not allow us that luxury, and it is the electromagnetic field. Electromagnetism is, together with gravity, one of the only two interactions of unbounded range in classical physics, it is the only other field theory Einstein had available when he was building general relativity, and unlike a fluid it comes with field equations of its own which must be solved simultaneously with Einstein's. The result is the first genuinely coupled system in the theory, and the first one in which we can see gravitation and another field determining each other rather than one merely sitting inside the other.

The physical statement behind the mathematics is simple enough to be worth saying before any equation. The electromagnetic field carries energy and momentum, and by the equivalence principle anything carrying energy and momentum must gravitate. A region of space containing nothing but light is therefore not flat, and two beams of light attract one another, a conclusion no one would have drawn from Maxwell's theory alone and which follows here with no additional hypothesis whatever. Conversely, the geometry produced in this way tells the field how to propagate, so the two theories are locked together and neither can be solved first.

4.3.1 Maxwell's theory on a curved background

The transition from flat to curved space-time is governed by the rule we have used throughout, minimal coupling, which instructs us to replace $\eta_{\mu\nu}$ by $g_{\mu\nu}$ and every partial derivative by a covariant one. Applying it blindly to the definition of the Faraday tensor would give

$$F_{\mu\nu} = \vec{\nabla}_\mu A_\nu - \vec{\nabla}_\nu A_\mu = \partial_\mu A_\nu - \Gamma_{\mu\nu}^\lambda A_\lambda - \partial_\nu A_\mu + \Gamma_{\nu\mu}^\lambda A_\lambda,$$

and the two terms in red cancel each other exactly, because the Levi-Civita connection is symmetric in its lower indices. The Faraday tensor is therefore unchanged by the passage to curved space-time,

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu, \quad (4.30)$$

and this is the promise made in chapter 2 being kept. We noticed there that the antisymmetrised partial derivative of a one form is a tensor even though the plain partial derivative is not, and called it the first appearance of the exterior derivative. Equation 4.30 is that observation cashed in: the electromagnetic field strength needs no connection to be defined, and a physicist may go on writing it with ordinary derivatives in the most violently curved space-time without committing any error at all.

The homogeneous Maxwell equations inherit the same protection. They state that the exterior derivative of F vanishes,

$$\partial_{[\lambda} F_{\mu\nu]} = \vec{\nabla}_{[\lambda} F_{\mu\nu]} = 0, \quad (4.31)$$

which in three dimensional language is the absence of magnetic monopoles together with Faraday's law of induction, and which holds identically once 4.30 is granted, since it amounts to the statement that mixed partial derivatives commute. The inhomogeneous pair is where the geometry finally enters,

$$\vec{\nabla}_\nu F^{\mu\nu} = 4\pi J^\mu, \quad (4.32)$$

and here the order of the indices genuinely matters, because F is antisymmetric and writing the divergence on the first slot instead of the second flips the sign, $\vec{\nabla}_\mu F^{\mu\nu} = -4\pi J^\nu$. We shall keep the form 4.32 throughout. There is a practical simplification worth recording, which is that the divergence of an antisymmetric tensor never needs a Christoffel symbol either,

$$\vec{\nabla}_\nu F^{\mu\nu} = \frac{1}{\sqrt{-g}} \partial_\nu (\sqrt{-g} F^{\mu\nu}), \quad (4.33)$$

a formula we shall lean on heavily when we solve the equations, and which is again a consequence of the symmetry of the connection contracting against the antisymmetry of the field.

Charge conservation is no longer an independent assumption but a theorem. Taking the divergence of 4.32 and commuting the two covariant derivatives, the commutator produces two Riemann terms which contract down to the Ricci tensor, and since the Ricci tensor is symmetric while $F^{\mu\nu}$ is antisymmetric the identity we proved in chapter 2 kills them both,

$$4\pi \vec{\nabla}_\mu J^\mu = \vec{\nabla}_\mu \vec{\nabla}_\nu F^{\mu\nu} = \frac{1}{2} [\vec{\nabla}_\mu, \vec{\nabla}_\nu] F^{\mu\nu} = \frac{1}{2} (R_{\lambda\nu} F^{\lambda\nu} + R_{\lambda\mu} F^{\mu\lambda}) = 0,$$

so that $\vec{\nabla}_\mu J^\mu = 0$ in any space-time whatsoever. That the curvature drops out of the argument is not an accident of the calculation, it is the statement that the conservation of electric charge is a topological fact about the theory and not a dynamical one, and that no gravitational field however strong can create or destroy a single electron.

4.3.2 The coupled system from a single action

The most economical way to obtain both sets of equations at once, and the way that makes the structure of the theory clearest, is to write a single action and vary it twice. Adding the Maxwell Lagrangian to the Einstein-Hilbert action of the previous section gives

$$S[g_{\mu\nu}, A_\mu] = \int \left(\frac{R}{16\pi} - \frac{1}{16\pi} F_{\alpha\beta} F^{\alpha\beta} \right) \sqrt{-g} d^4x, \quad (4.34)$$

in which the dynamical variables are the metric and the potential, and in which the only coupling between them is hidden in the index raising and in the volume element. Everything follows from 4.34.

Varying with respect to the potential is the easier of the two, since the gravitational term does not contain A_μ at all. Using $\delta F_{\alpha\beta} = \partial_\alpha \delta A_\beta - \partial_\beta \delta A_\alpha$ and the antisymmetry of $F^{\alpha\beta}$, which lets us combine the two resulting terms into one,

$$\delta S = -\frac{1}{16\pi} \int F^{\alpha\beta} 2 \delta F_{\alpha\beta} \sqrt{-g} d^4x = -\frac{1}{4\pi} \int F^{\alpha\beta} (\partial_\alpha \delta A_\beta) \sqrt{-g} d^4x,$$

where the factor in blue comes from the two identical contributions of $F_{\alpha\beta} F^{\alpha\beta}$ and the second one from the antisymmetry, and integrating by parts while discarding the boundary term in the usual way,

$$\delta S = \frac{1}{4\pi} \int \partial_\alpha \left(\sqrt{-g} F^{\alpha\beta} \right) \delta A_\beta d^4x = \frac{1}{4\pi} \int \vec{\nabla}_\alpha F^{\alpha\beta} \delta A_\beta \sqrt{-g} d^4x,$$

where the last step used 4.33 backwards. Demanding that this vanish for every variation δA_β returns the source free Maxwell equation, and had we included a coupling term $-\int A_\mu J^\mu \sqrt{-g} d^4x$ in the action it would have returned 4.32 complete with its source.

Varying with respect to the metric is the calculation that produces the coupling. We showed when deriving the field equations that the variation of the gravitational term gives the Einstein tensor, so we need only the matter part, and for it we need two standard results. The first is the variation of the determinant,

$$\delta \sqrt{-g} = -\frac{1}{2} \sqrt{-g} g_{\mu\nu} \delta g^{\mu\nu}, \quad (4.35)$$

and the second is that $F_{\alpha\beta}$, being built from partial derivatives of A_μ alone by equation 4.30, does not depend on the metric at all, so that the whole metric dependence of the Maxwell term sits in the two inverse metrics needed to raise its indices,

$$F_{\alpha\beta}F^{\alpha\beta} = g^{\alpha\mu}g^{\beta\nu}F_{\alpha\beta}F_{\mu\nu}.$$

Varying that product, the two inverse metrics contribute one term each and the two terms are equal after relabelling, so

$$\delta \left(F_{\alpha\beta}F^{\alpha\beta} \right) = 2 F_{\mu\beta}F_{\nu}{}^{\beta} \delta g^{\mu\nu},$$

and assembling this with 4.35 the variation of the Maxwell part of 4.34 becomes

$$\begin{aligned} \delta S_M &= -\frac{1}{16\pi} \int \left[2 F_{\mu\alpha}F_{\nu}{}^{\alpha} \sqrt{-g} - \frac{1}{2} \sqrt{-g} g_{\mu\nu} F_{\alpha\beta}F^{\alpha\beta} \right] \delta g^{\mu\nu} d^4x \\ &= -\frac{1}{8\pi} \int \left[F_{\mu\alpha}F_{\nu}{}^{\alpha} - \frac{1}{4} g_{\mu\nu} F_{\alpha\beta}F^{\alpha\beta} \right] \delta g^{\mu\nu} \sqrt{-g} d^4x. \end{aligned}$$

Comparing the bracket with the general definition of the energy momentum tensor as the response of the matter action to a change of the metric, we recover precisely the tensor we wrote down without justification in chapter 3,

$$T_{\mu\nu} = \frac{1}{4\pi} \left(F_{\mu\alpha}F_{\nu}{}^{\alpha} - \frac{1}{4} g_{\mu\nu} F_{\alpha\beta}F^{\alpha\beta} \right), \quad (4.36)$$

which is a satisfying closure, since what was then an educated guess is now a consequence of the action. Putting the two variations together, the Einstein-Maxwell system reads

$$\begin{aligned} G_{\mu\nu} &= 8\pi T_{\mu\nu} = 2 \left(F_{\mu\alpha}F_{\nu}{}^{\alpha} - \frac{1}{4} g_{\mu\nu} F_{\alpha\beta}F^{\alpha\beta} \right), \\ \vec{\nabla}_{\nu} F^{\mu\nu} &= 4\pi J^{\mu}, \quad \partial_{[\lambda} F_{\mu\nu]} = 0. \end{aligned} \quad (4.37)$$

One simplification is available immediately and it is worth having in hand before we attempt a solution. We showed earlier that the tensor 4.36 is traceless in four dimensions, and a traceless source forces the Ricci scalar to vanish through the trace of the field equations, $R = -8\pi T = 0$. The Einstein-Maxwell equations therefore collapse to the simpler form

$$R_{\mu\nu} = 8\pi T_{\mu\nu}, \quad (4.38)$$

with no trace term to carry around, so that an electrovacuum space-time is not Ricci flat, as a vacuum one is, but it does have vanishing curvature scalar. The tracelessness itself reflects the fact that Maxwell's theory contains no length and no mass, and it is the classical shadow of the conformal invariance of electromagnetism.

4.3.3 Why the system is consistent, and where the Lorentz force comes from

Before solving anything we should check that the system 4.37 is not self contradictory, and the check turns out to deliver a good deal more than we asked of it. The Bianchi identity guarantees that $\vec{\nabla}^\mu G_{\mu\nu} = 0$ identically, so the field equations can only be consistent if the right hand side is also divergence free. Let us compute that divergence honestly and see what the theory demands.

Taking the divergence of 4.36 and applying the Leibniz rule to each of the two terms,

$$4\pi \vec{\nabla}_\mu T^{\mu\nu} = \underbrace{\left(\vec{\nabla}_\mu F^{\mu\alpha}\right) F^\nu{}_\alpha}_{(I)} + \underbrace{F^{\mu\alpha} \vec{\nabla}_\mu F^\nu{}_\alpha}_{(II)} - \underbrace{\frac{1}{2} F^{\alpha\beta} \vec{\nabla}^\nu F_{\alpha\beta}}_{(III)},$$

where the factor of one half in the third term appeared because the two factors of F in the trace part contribute equally. The first term is the only one that knows about the sources, and by 4.32 written with the divergence on the first index it is

$$(I) = -4\pi J^\alpha F^\nu{}_\alpha.$$

The remaining two terms contain no sources and we now show that they cancel each other identically, which is where the homogeneous Maxwell equations earn their keep. Since $F^{\mu\alpha}$ is antisymmetric in μ and α , only the antisymmetric part of what it multiplies survives the contraction, so we may write the second term as

$$(II) = \frac{1}{2} F^{\mu\alpha} \left(\vec{\nabla}_\mu F^\nu{}_\alpha - \vec{\nabla}_\alpha F^\nu{}_\mu \right),$$

and now the homogeneous equation 4.31, written out as the cyclic identity $\vec{\nabla}_\mu F_{\rho\alpha} + \vec{\nabla}_\rho F_{\alpha\mu} + \vec{\nabla}_\alpha F_{\mu\rho} = 0$, lets us rewrite the bracket. Lowering the free index for a moment and using the antisymmetry of F in the middle term of the cycle,

$$\vec{\nabla}_\mu F_{\rho\alpha} - \vec{\nabla}_\alpha F_{\rho\mu} = \vec{\nabla}_\mu F_{\rho\alpha} + \vec{\nabla}_\alpha F_{\mu\rho} = -\vec{\nabla}_\rho F_{\alpha\mu} = \vec{\nabla}_\rho F_{\mu\alpha},$$

so that raising the index ρ back up

$$(II) = \frac{1}{2} F^{\mu\alpha} \vec{\nabla}^\nu F_{\mu\alpha} = (III),$$

and the two cancel exactly. What survives is

$$\vec{\nabla}_\mu T^{\mu\nu} = -F^{\nu\alpha} J_\alpha, \quad (4.39)$$

and equation 4.39 deserves to be read carefully, because it is one of the most eloquent results in the whole theory. In a genuine vacuum, with $J^\mu = 0$, the right hand side vanishes and the system is consistent, which is what we set out to verify. But when charges are present the electromagnetic energy momentum tensor is *not* conserved on its own, and the rate at which it fails to be conserved is exactly the four force per unit volume that the field exerts on the charges. The total energy momentum of field plus matter is conserved, and what 4.39 describes is the transfer between the two.

The consequence is remarkable. The Lorentz force law, which in flat space-time is an independent postulate that has to be supplied by hand alongside Maxwell's equations, is here a theorem. If the charges are carried by a dust of charge density ϱ and four velocity u^μ , so that $J^\mu = \varrho u^\mu$, then demanding the conservation of the total energy momentum tensor and using 4.39 gives for the matter part

$$\rho u^\mu \vec{\nabla}_\mu u^\nu = \varrho F^\nu{}_\mu u^\mu, \quad (4.40)$$

which is precisely the equation of motion of a charged particle, mass times four acceleration equal to charge times the Faraday tensor contracted with the four velocity. General relativity therefore does to electromagnetism what it did to gravity itself: it demotes a force law to a consequence of a conservation law, exactly as the geodesic equation for neutral dust fell out of $\vec{\nabla}_\mu T^{\mu\nu} = 0$ in chapter 3. A charged particle in a curved space-time does not follow a geodesic, and 4.40 says by exactly how much it deviates.

4.3.4 The Reissner-Nordström solution

The obvious thing to do with a new system of equations is to solve it in the simplest case, and the simplest case is the charged analogue of Schwarzschild: the geometry outside a static, spherically symmetric body carrying a total charge q . Reissner found it in 1916 and Nordström independently in 1918, within months of Schwarzschild's own solution and while the theory was barely a year old (Reissner, 1916; Nordström, 1918).

Since the symmetries we impose are exactly those of the Schwarzschild problem, we may reuse the ansatz 4.1 without any modification, along with every Christoffel symbol and every component of the Ricci tensor we computed there. The only new ingredient is the electromagnetic field, and symmetry pins it down almost completely. A static configuration forbids any magnetic field, since a magnetic field would single out a direction and break spherical symmetry unless it were radial, and a radial magnetic field would require a monopole, which 4.31 forbids. The electric field must be radial for the same reason and can depend on r alone. The only surviving components of the Faraday tensor are therefore

$$F_{tr} = -F_{rt} = \mathcal{E}(r), \quad (4.41)$$

with everything else zero, and one unknown function $\mathcal{E}$ to determine alongside α and β .

We solve Maxwell first, because it decouples. Outside the body there are no charges, so $J^\mu = 0$, and the only non trivial component of 4.32 is the temporal one. Raising both indices with the diagonal metric,

$$F^{tr} = g^{tt}g^{rr}F_{tr} = (-e^{-2\alpha})\left(e^{-2\beta}\right)\mathcal{E} = -e^{-2(\alpha+\beta)}\mathcal{E},$$

and with the determinant of the ansatz being $\sqrt{-g} = e^{\alpha+\beta}r^2 \sin \vartheta$, formula 4.33 gives for $\mu = t$ a single surviving derivative,

$$\vec{\nabla}_\nu F^{t\nu} = \frac{1}{e^{\alpha+\beta}r^2 \sin \vartheta} \partial_r \left(e^{\alpha+\beta}r^2 \sin \vartheta \cdot \left(-e^{-2(\alpha+\beta)}\mathcal{E} \right) \right) = 0,$$

where the factors in red cancel because nothing depends on ϑ . The equation therefore states that the quantity inside the derivative is a constant, which we may name at our convenience,

$$r^2 e^{-(\alpha+\beta)} \mathcal{E} = -q,$$

and the choice of sign is made so that the potential comes out in the familiar Coulomb form, as we verify in a moment. Solving for the field,

$$\mathcal{E}(r) = -\frac{q}{r^2} e^{\alpha+\beta}. \quad (4.42)$$

We shall find shortly that $\alpha + \beta = 0$ for this solution exactly as it did for Schwarzschild, in which case $F_{tr} = -q/r^2$ and the potential $A_t = -q/r$ reproduces the Coulomb field of a point charge q , which identifies the integration constant physically and justifies the name we gave it.

Now for the geometry. Since the source is traceless we may use the reduced form 4.38 and never compute a Ricci scalar. Substituting 4.42 into 4.36, the invariant is

$$F_{\alpha\beta}F^{\alpha\beta} = 2 F_{tr}F^{tr} = 2\mathcal{E} \left(-e^{-2(\alpha+\beta)}\mathcal{E} \right) = -2e^{-2(\alpha+\beta)}\mathcal{E}^2 = -\frac{2q^2}{r^4},$$

the factor in blue counting the two orderings tr and rt , which contribute equally because both F factors change sign. For the temporal component of the source we need $F_{t\alpha}F_t{}^\alpha = F_{tr}g^{rr}F_{tr} = e^{-2\beta}\mathcal{E}^2 = q^2 e^{2\alpha}/r^4$, so that

$$T_{tt} = \frac{1}{4\pi} \left(\frac{q^2 e^{2\alpha}}{r^4} - \frac{1}{4} (-e^{2\alpha}) \left(-\frac{2q^2}{r^4} \right) \right) = \frac{1}{4\pi} \left(\frac{q^2 e^{2\alpha}}{r^4} - \frac{q^2 e^{2\alpha}}{2r^4} \right) = \frac{q^2 e^{2\alpha}}{8\pi r^4},$$

and repeating the same two steps for the radial and polar directions, where $F_{r\alpha} F_r{}^\alpha = F_{rt} g^{tt} F_{rt} = -q^2 e^{2\beta}/r^4$ and where the angular directions receive no contribution from the first term at all,

$$T_{rr} = -\frac{q^2 e^{2\beta}}{8\pi r^4}, \quad T_{\vartheta\vartheta} = \frac{q^2}{8\pi r^2}.$$

The sign pattern is the familiar one of a radial electric field, positive energy density, negative radial pressure, positive tangential pressure, and the reader may check that the trace does vanish as promised.

The field equations 4.38 now read, using the Ricci components computed for the Schwarzschild ansatz,

$$\begin{aligned} R_{tt} &= e^{2(\alpha-\beta)} \left[\alpha'' + (\alpha')^2 - \alpha' \beta' + \frac{2\alpha'}{r} \right] = \frac{q^2 e^{2\alpha}}{r^4}, \\ R_{rr} &= -\alpha'' - (\alpha')^2 + \alpha' \beta' + \frac{2\beta'}{r} = -\frac{q^2 e^{2\beta}}{r^4}, \\ R_{\vartheta\vartheta} &= e^{-2\beta} [r(\beta' - \alpha') - 1] + 1 = \frac{q^2}{r^2}. \end{aligned} \tag{4.43}$$

The first move is the one that worked before. Multiplying the first equation by $e^{-2(\alpha-\beta)}$ and adding it to the second, the entire left hand side collapses exactly as it did in the vacuum case, and the right hand sides collapse too,

$$\underbrace{\frac{2}{r} (\alpha' + \beta')}_{\text{left hand sides}} = e^{-2(\alpha-\beta)} \underbrace{\left(\frac{q^2 e^{2\alpha}}{r^4} - \frac{q^2 e^{2\beta}}{r^4} \right)}_{\text{right hand sides}} = \frac{\textcolor{red}{q^2 e^{2\beta}}}{\textcolor{red}{r^4}} - \frac{\textcolor{red}{q^2 e^{2\beta}}}{\textcolor{red}{r^4}} = 0,$$

the two terms in red being identical and cancelling. This is a small miracle and it is worth pausing on: the charge has disappeared from this particular combination, so we recover the same conclusion as in vacuum,

$$\alpha' + \beta' = 0 \implies \beta = -\alpha, \tag{4.44}$$

where the constant of integration has been absorbed into a rescaling of t , chosen so that the metric approaches $\eta_{\mu\nu}$ at infinity. The reason this happens is not a coincidence of algebra: the

combination that vanishes is $T_{tt}e^{-2\alpha} + T_{rr}e^{-2\beta}$, which is zero precisely because a radial electric field has energy density equal and opposite to its radial pressure, and that in turn is a consequence of the tracelessness of the Maxwell tensor. With 4.44 in hand the relation $F_{tr} = -q/r^2$ promised above follows at once from 4.42.

Only the angular equation is left, and with $\beta = -\alpha$ it becomes an ordinary differential equation in one unknown. Substituting $e^{-2\beta} = e^{2\alpha}$ and $\beta' = -\alpha'$,

$$e^{2\alpha}(-2r\alpha' - 1) + 1 = \frac{q^2}{r^2},$$

and the equation becomes transparent under the substitution $\mathcal{F}(r) \equiv e^{2\alpha(r)}$, whose derivative is $\mathcal{F}' = 2\alpha'e^{2\alpha}$, so that the first term is simply $-r\mathcal{F}'$,

$$-r\mathcal{F}' - \mathcal{F} + 1 = \frac{q^2}{r^2} \implies r\mathcal{F}' + \mathcal{F} = 1 - \frac{q^2}{r^2}.$$

The left hand side is an exact derivative, since $(r\mathcal{F})' = r\mathcal{F}' + \mathcal{F}$, so the equation integrates in one step,

$$(r\mathcal{F})' = 1 - \frac{q^2}{r^2} \implies r\mathcal{F} = r + \frac{q^2}{r} + K \implies \mathcal{F} = 1 + \frac{q^2}{r^2} + \frac{K}{r},$$

with K an integration constant which the field equations cannot determine, exactly as in the Schwarzschild case, and which we fix by the same argument: far from the body the charge term dies faster than the mass term, the field is weak, and the Newtonian limit demands $g_{tt} \rightarrow -(1 - 2Gm/C^2r)$, so that $K = -2m$ in geometrised units. Collecting everything,

$$ds^2 = -\left(1 - \frac{2m}{r} + \frac{q^2}{r^2}\right)dt^2 + \frac{dr^2}{1 - \frac{2m}{r} + \frac{q^2}{r^2}} + r^2d\Omega^2, \quad (4.45)$$

which is the Reissner-Nordström metric, accompanied by the electromagnetic field $F_{tr} = -q/r^2$. Setting $q = 0$ returns Schwarzschild, as it must.

The charge term enters with the opposite sign to the mass term, and that single observation contains most of the physics of 4.45. Gravitationally, charge is repulsive. The reason is not that like charges repel, since the metric depends on q^2 and is blind to the sign of the charge, but rather that the electric field surrounding the body carries positive energy which is spread throughout the exterior region, so that the effective mass contained within a radius r decreases as we descend, and less of it remains to pull.

The consequences for the causal structure are dramatic. The metric function vanishes where $r^2 - 2mr + q^2 = 0$, that is at

$$r_{\pm} = m \pm \sqrt{m^2 - q^2}, \quad (4.46)$$

so that instead of the single horizon of Schwarzschild there are now two, an outer event horizon at r_+ and an inner Cauchy horizon at r_- , provided $m > |q|$. As the charge grows the two approach each other, and at $|q| = m$, the *extremal* case, they merge into a single degenerate horizon at $r = m$. Beyond that, for $|q| > m$, equation 4.46 has no real solutions, the metric function never vanishes, and the singularity at $r = 0$ stands exposed to the outside universe with no horizon to hide it. Such a *naked singularity* would allow the breakdown of predictability to propagate into the region where we live, and the conjecture that nature refuses to produce one is Penrose's cosmic censorship hypothesis (Penrose, 1969), still unproved and one of the central open problems of the subject.

That $r = 0$ is a genuine singularity and that both horizons are not is settled as before by a curvature scalar, and computing the Kretschmann invariant for 4.45 gives

$$R_{abcd}R^{abcd} = \frac{48m^2}{r^6} - \frac{96mq^2}{r^7} + \frac{56q^4}{r^8}, \quad (4.47)$$

which reduces to the Schwarzschild value when $q = 0$, is perfectly finite at $r_{\pm}$, and diverges at the origin. There is one further difference from the uncharged case which is worth knowing: the singularity of 4.45 is timelike rather than spacelike, so that unlike in Schwarzschild an infalling observer is not obliged to hit it, and can in principle avoid it and pass through the inner horizon. What lies beyond is a different story, because the inner horizon turns out to be violently unstable to any infalling radiation, and the smooth continuation predicted by the exact solution is almost certainly destroyed in any realistic collapse.

A last word on whether any of this describes a real object. Astrophysical black holes are expected to carry essentially no charge, and the reason is the overwhelming strength of electromagnetism relative to gravity: the ratio of the electrostatic to the gravitational force between two protons is of order 10^{36} , so a body that acquired any appreciable charge would attract opposite charges from the surrounding plasma and neutralise itself almost instantly. The Reissner-Nordström solution is therefore not a model of anything we expect to observe, and its importance is entirely theoretical, as the simplest laboratory in which to study horizons, extremality and the structure of the field equations when a second field is present.

4.3.5 The importance and reach of the Einstein-Maxwell equations

It is worth closing this chapter by saying plainly why a system that describes no astrophysical object we can point to should occupy so central a place in relativity, because the reasons are several and they are not the ones a first reading might suggest.

The first is conceptual, and we have already met it. The Einstein-Maxwell system is the simplest case in which gravity is coupled to something that is not a phenomenological fluid but a field with

equations of its own, and it is therefore the place where one learns what coupling actually means. Everything that is difficult about general relativity as a physical theory, the fact that the source depends on the metric, that the metric depends on the source, and that the two must be solved together, appears here in its mildest possible form, and every lesson learnt in this setting carries over unchanged to the Einstein-Yang-Mills and Einstein-Dirac systems and to the coupling of gravity to the whole Standard Model.

The second is mathematical. The Einstein-Maxwell equations were the first setting in which the initial value problem of general relativity was brought under control, and the existence and uniqueness theorems that Choquet-Bruhat proved for them in the nineteen fifties are what made it legitimate to speak of general relativity as a predictive theory at all (Fourès-Bruhat, 1952). Every numerical simulation of a black hole merger performed today rests on the well posedness that was first established in this context. There is also a striking structural result due to Rainich, who showed as early as 1925 that a source free electromagnetic field leaves such a distinctive fingerprint on the curvature that one can recognise an electrovacuum space-time from the metric alone and reconstruct the field from it, so that in this case electromagnetism is not something added to geometry but something already encoded in it (Rainich, 1925).

The third is that the system is the natural home of the black hole uniqueness theorems. Israel proved in 1968 that the only static, asymptotically flat electrovacuum black hole is Reissner-Nordström (Israel, 1968), and when rotation is allowed the unique stationary solution is the Kerr-Newman metric found in 1965 (Newman et al., 1965), which depends on exactly three numbers, the mass, the angular momentum and the charge. This is the precise content of the statement that black holes have no hair, and it is a genuinely astonishing result: an object formed from the collapse of a star with all its chemistry, its magnetic fields and its history retains, once it settles down, exactly three of them. The extremal limit $|q| = m$ that appeared in our solution is the boundary of the allowed region of that three parameter family, and it plays a distinguished role in the laws of black hole mechanics, where it corresponds to zero temperature and is the subject of the third law (Bardeen et al., 1973). Extremal and near extremal charged black holes are also the cases in which string theory has succeeded in counting the microscopic states responsible for the entropy, which is why a solution describing nothing in the sky remains a central object of study in quantum gravity.

The fourth reason is that the Einstein-Maxwell equations were the first serious attempt at unification, and it failed in an instructive way. Weyl tried in 1918 to obtain electromagnetism by enlarging the geometry with a scale connection, and although the theory was wrong the word he coined for the idea, *gauge*, survived and now organises all of fundamental physics (Weyl, 1918). Kaluza in 1921 and Klein in 1926 showed that general relativity written in five dimensions, with the fifth curled into a small circle, produces exactly the four dimensional Einstein-Maxwell system plus a scalar field (Kaluza, 1921; Klein, 1926), which is as close to a geometrical unification of gravity and electromagnetism as anyone has come, and which is the direct ancestor of every extra dimensional theory since.

The last reason is the most practical, and it is the one that has come to dominate. Although charge is negligible in astrophysics, magnetic fields are not, and the equations governing a magnetised plasma in a strong gravitational field are the Einstein-Maxwell equations with the source supplied

by the plasma rather than left in vacuum. That system, general relativistic magnetohydrodynamics, is what is actually solved on supercomputers to model accretion discs, relativistic jets and neutron star mergers. The mechanism by which a rotating black hole can extract its own rotational energy through the magnetic field threading its horizon was identified by Blandford and Znajek in 1977 and is the leading explanation for the jets emitted by active galactic nuclei (Blandford and Znajek, 1977). Magnetars, neutron stars whose magnetic fields reach 10^{15} gauss and are strong enough for the magnetic energy density to distort the star's own geometry, are another regime in which the two theories must be solved together (Duncan and Thompson, 1992). And the images of the shadows of black holes produced in recent years are compared against simulations in which nothing but Einstein's equations, Maxwell's equations and the coupling we have derived in this section are integrated forward in time.

So the equations we have written down, which describe no real charged black hole, nevertheless underlie the initial value formulation of the theory, the classification of its black holes, the first attempts at unification, and most of what is currently computed in relativistic astrophysics. That is a considerable return on a Lagrangian with two terms in it.

4.4 The Kerr solution

4.5 The Kerr-Newman solution

We now return to the weak field programme we interrupted, because the linearised equations we derived before this detour are about to give us something that neither Schwarzschild's static solution nor the charged one we have just built can provide, namely radiation.

Chapter 5

Gravitational Waves

*A century after Einstein first wrote them down,
the ripples of space-time finally reached our
instruments.*

“Ladies and gentlemen, we have detected gravitational waves. We did it.”

– David Reitze

5.1 Lorenz Gauge and gravitational waves

Lets remember how are the conditions of the gravitational field we're working on which is a *theory of weak field* this means we have a flat (Minkowski) space-time but there is a certain perturbation on a region we represent it as $g_{\alpha\beta} = \eta_{\alpha\beta} + h_{\alpha\beta}$ $|h_{\alpha\beta}| \ll 1$. We can do these perturbations on the coordinate charts as $x'^{\alpha} = x^{\alpha} + \xi^{\alpha}$ where $\xi^{\alpha} = \xi^{\alpha}(x^{\alpha})$ are infinitesimal transformations such that

$$\left| \frac{\partial \xi^{\alpha}}{\partial x^{\beta}} \right| \approx |h_{\alpha\beta}| \rightarrow \left| \frac{\partial \xi^{\alpha}}{\partial x^{\beta}} \right| \ll 1. \quad (5.1)$$

We are looking for a gauge theory that leaves invariant certain quantity, in our case such quantity is the Riemann's tensor, to do it we gotta see how is the behavior of our space-time

$$g'_{\alpha\beta} = g_{\mu\nu} \frac{\partial x^{\mu}}{\partial x'^{\alpha}} \frac{\partial x^{\nu}}{\partial x'^{\beta}}, \quad (5.2)$$

$$\frac{\partial x^{\mu}}{\partial x'^{\alpha}} = \delta^{\mu}_{\alpha} - \frac{\partial \xi^{\mu}}{\partial x^{\nu}} \frac{\partial x^{\nu}}{\partial x'^{\alpha}} = \delta^{\mu}_{\alpha} - \frac{\partial \xi^{\mu}}{\partial x^{\nu}} \left(\delta^{\nu}_{\alpha} - \frac{\partial \xi^{\nu}}{\partial x^{\rho}} \frac{\partial x^{\rho}}{\partial x'^{\alpha}} \right), \quad (5.3)$$

$$\frac{\partial x^{\mu}}{\partial x'^{\alpha}} = \delta^{\mu}_{\alpha} - \frac{\partial \xi^{\mu}}{\partial x^{\alpha}}, \quad (5.4)$$

$$g'_{\alpha\beta} = g_{\mu\nu} \left(\delta^{\mu}_{\alpha} - \frac{\partial \xi^{\mu}}{\partial x^{\alpha}} \right) \left(\delta^{\nu}_{\beta} - \frac{\partial \xi^{\nu}}{\partial x^{\beta}} \right) = g_{\alpha\beta} - g_{\alpha\nu} \frac{\partial \xi^{\nu}}{\partial x^{\beta}} - g_{\beta\mu} \frac{\partial \xi^{\mu}}{\partial x^{\alpha}}, \quad (5.5)$$

$$g'_{\alpha\beta} = (\eta_{\alpha\beta} + h_{\alpha\beta}) - \eta_{\alpha\nu} \frac{\partial \xi^{\nu}}{\partial x^{\beta}} - \eta_{\beta\mu} \frac{\partial \xi^{\mu}}{\partial x^{\alpha}}, \quad (5.6)$$

$$g'_{\alpha\beta} = \eta_{\alpha\beta} + h'_{\alpha\beta} \rightarrow h'_{\alpha\beta} = h_{\alpha\beta} - \eta_{\alpha\nu} \frac{\partial \xi^{\nu}}{\partial x^{\beta}} - \eta_{\beta\mu} \frac{\partial \xi^{\mu}}{\partial x^{\alpha}}. \quad (5.7)$$

In analogy to see how transforms the Riemann's tensor, lets first calculate the following transform of the tensor $R'^{\alpha\beta}_{\mu\nu}$ this because it will be easier mathematically calculate in this form and in the end using the Minkowski metric tensor we can lower one index, therefore

$$R'^{\alpha\beta}_{\mu\nu} = \frac{\partial x'^{\alpha}}{\partial x^a} \frac{\partial x'^{\beta}}{\partial x^b} \frac{\partial x^c}{\partial x'^{\mu}} \frac{\partial x^d}{\partial x'^{\nu}} R^{ab}_{cd} \quad (5.8)$$

we gotta calculate the derivative of the prime coordinates respect to the non primed one , then

$$\frac{\partial x'^{\alpha}}{\partial x^{\mu}} = \delta^{\alpha}_{\mu} + \frac{\partial \xi^{\alpha}}{\partial x^{\mu}}. \quad (5.9)$$

Using the equation 5.4 and 5.9 into the equation 5.8 we then have

$$\begin{aligned}
R'^{\alpha\beta}_{\mu\nu} &= \left[\delta_a^\alpha + \frac{\partial \xi^\alpha}{\partial x^a} \right] \left[\delta_b^\beta + \frac{\partial \xi^\beta}{\partial x^b} \right] \left[\delta_\mu^c - \frac{\partial \xi^c}{\partial x^\mu} \right] \left[\delta_\nu^d - \frac{\partial \xi^d}{\partial x^\nu} \right] R^{ab}_{cd} \\
&= \left[\delta_a^\alpha \delta_b^\beta + \delta_a^\alpha \frac{\partial \xi^\beta}{\partial x^b} + \delta_b^\beta \frac{\partial \xi^\alpha}{\partial x^a} \right] \left[\delta_\mu^c \delta_\nu^d - \delta_\mu^c \frac{\partial \xi^d}{\partial x^\nu} - \delta_\nu^d \frac{\partial \xi^c}{\partial x^\mu} \right] R^{ab}_{cd} \\
&= \left[\delta_a^\alpha \delta_b^\beta \delta_\mu^c \delta_\nu^d - \delta_a^\alpha \delta_b^\beta \delta_\mu^c \frac{\partial \xi^d}{\partial x^\nu} - \delta_a^\alpha \delta_b^\beta \delta_\nu^d \frac{\partial \xi^c}{\partial x^\mu} + \delta_a^\alpha \delta_\mu^c \delta_\nu^d \frac{\partial \xi^\beta}{\partial x^b} + \delta_b^\beta \delta_\mu^c \delta_\nu^d \frac{\partial \xi^\alpha}{\partial x^a} \right] R^{ab}_{cd} \\
&= R^{\alpha\beta}_{\mu\nu} + \left\{ \delta_b^\beta \delta_\mu^c \left[\delta_\nu^d \frac{\partial \xi^\alpha}{\partial x^a} - \delta_a^\alpha \frac{\partial \xi^d}{\partial x^\nu} \right] + \delta_a^\alpha \delta_\nu^d \left[\delta_\mu^c \frac{\partial \xi^\beta}{\partial x^b} - \delta_b^\beta \frac{\partial \xi^c}{\partial x^\mu} \right] \right\} R^{ab}_{cd} \\
&= R^{\alpha\beta}_{\mu\nu} + \left\{ \delta_b^\beta \delta_\mu^c \left[\delta_\nu^d \delta_a^\alpha \frac{\partial \xi^\alpha}{\partial x^\nu} - \delta_a^\alpha \delta_\alpha^d \frac{\partial \xi^\alpha}{\partial x^\nu} \right] + \delta_a^\alpha \delta_\nu^d \left[\delta_\mu^c \delta_b^\beta \frac{\partial \xi^\beta}{\partial x^\mu} - \delta_b^\beta \delta_\beta^c \frac{\partial \xi^\beta}{\partial x^\mu} \right] \right\} R^{ab}_{cd} \\
&= R^{\alpha\beta}_{\mu\nu} + \left\{ \delta_b^\beta \delta_\mu^c \delta_a^d \left[\frac{\partial \xi^\alpha}{\partial x^\nu} - \frac{\partial \xi^\alpha}{\partial x^\nu} \right] + \delta_a^\alpha \delta_\nu^d \delta_b^c \left[\frac{\partial \xi^\beta}{\partial x^\mu} - \frac{\partial \xi^\beta}{\partial x^\mu} \right] \right\} R^{ab}_{cd},
\end{aligned}$$

$$R'^{\alpha\beta}_{\mu\nu} = R^{\alpha\beta}_{\mu\nu} \quad (5.10)$$

$$\eta_{b\beta} R'^{\alpha\beta}_{\mu\nu} = R'^{\alpha}_{b\mu\nu} = \eta_{b\beta} R^{\alpha\beta}_{\mu\nu} = R^{\alpha}_{b\mu\nu} \quad (5.11)$$

Doing the variable change $b = \beta$ we then arrive that the Riemann's tensor keep invariant under these infinitesimal transformations:

$$R'^{\alpha}_{\beta\mu\nu} = R^{\alpha}_{\beta\mu\nu}, \quad (5.12)$$

and thus

$$R'_{\alpha\beta} = R_{\alpha\beta} \quad (5.13)$$

and the Lorenz gauge, the one we already know from electromagnetism, is

$$\partial_\alpha (\eta^{\alpha\beta} A_\beta) = \partial^\beta A_\beta = 0 \quad (5.14)$$

and then by analogy with the electromagnetism $\bar{h}_{\alpha\beta}$ fulfil a Lorenz Gauge as follows

$$\frac{\partial}{\partial x^\mu} (\eta^{\mu\nu} \bar{h}_{\alpha\nu}) = 0. \quad (5.15)$$

Well then, in the Einstein's equation 3.323 there is the trace and the trace reversed tensors, so we have to calculate their transformed versions too, in this way lets do it

$$h' = h - 2\partial_\mu \xi^\mu, \quad (5.16)$$

$$\bar{h}'_{\alpha\beta} = h'_{\alpha\beta} - \frac{1}{2}\eta_{\alpha\beta}h' = h_{\alpha\beta} - \frac{\partial\xi_\alpha}{\partial x^\beta} - \frac{\partial\xi_\beta}{\partial x^\alpha} - \frac{1}{2}\eta_{\alpha\beta}\left(h - 2\frac{\partial\xi^\mu}{\partial x^\mu}\right), \quad (5.17)$$

$$\bar{h}'_{\alpha\beta} = \bar{h}_{\alpha\beta} - \frac{\partial\xi_\alpha}{\partial x^\beta} - \frac{\partial\xi_\beta}{\partial x^\alpha} + \eta_{\alpha\beta}\frac{\partial\xi^\mu}{\partial x^\mu}, \quad (5.18)$$

$$W'_\alpha = \frac{\partial}{\partial x^\mu} \left(\eta^{\mu\nu} \bar{h}'_{\alpha\nu} \right), \quad (5.19)$$

$$W'_\alpha = \partial_\mu \left[\eta^{\mu\nu} \left(\bar{h}_{\alpha\nu} - \partial_\nu \xi_\alpha - \partial_\alpha \xi_\nu + \eta_{\alpha\nu} \partial_\sigma \xi^\sigma \right) \right], \quad (5.20)$$

$$W'_\alpha = \partial_\mu \left[\eta^{\mu\nu} \bar{h}_{\alpha\nu} - \eta^{\mu\nu} \partial_\nu \xi_\alpha - \partial_\alpha \xi^\mu + \delta_\alpha^\mu \partial_\sigma \xi^\sigma \right], \quad (5.21)$$

$$W'_\alpha = W_\alpha - \eta^{\mu\nu} \frac{\partial^2 \xi_\alpha}{\partial x^\mu \partial x^\nu} \rightarrow W'_\alpha = W_\alpha - \square \xi_\alpha, \quad (5.22)$$

$$W'_\alpha = 0 \iff W_\alpha = \square \xi_\alpha. \quad (5.23)$$

where the two terms marked in red cancel against each other once the derivative is carried out, because mixed partial derivatives commute. We therefore still have the freedom to choose the four functions ξ_α , and choosing them so that $\square \xi_\alpha = W_\alpha$ makes W'_α vanish without disturbing the Riemann tensor, which leaves the field equations in the form

$$\square \bar{h}_{\alpha\beta} = -16\pi T_{\alpha\beta}, \quad (5.24)$$

and restoring the units therefore

$$\square \bar{h}_{\alpha\beta} = -\frac{16\pi G}{C^4} T_{\alpha\beta}. \quad (5.25)$$

The equation 5.25 is the wave equation for gravitational waves! Lets see how are their solutions, first we are going to analyze the vacuum case $T_{\alpha\beta} = 0$ and the simplest solution would be a traveling monochromatic plane wave as

$$\bar{h}_{\alpha\beta}(x^\mu) = A_{\alpha\beta} e^{ik_\mu x^\mu}, \quad (5.26)$$

using this solution in the equation 5.25 we find that $k_\alpha k^\alpha = 0$ this means the wave vector k^μ is a null or light-like vector, this implies that *the gravitational waves travels at the speed of light in vacuum* and from the Lorenz Gauge we find this

$$\frac{\partial}{\partial x^\mu} \left(\eta^{\mu\nu} A_{\alpha\nu} e^{ik_\sigma x^\sigma} \right) = 0, \quad (5.27)$$

$$\eta^{\mu\nu} A_{\alpha\nu} \frac{\partial}{\partial x^\mu} e^{ik_\sigma x^\sigma} = 0, \quad (5.28)$$

$$\eta^{\mu\nu} A_{\alpha\nu} e^{ik_\sigma x^\sigma} i k_\beta \frac{\partial x^\beta}{\partial x^\mu} = 0, \quad (5.29)$$

$$\eta^{\mu\nu} A_{\alpha\nu} i k_\mu e^{ik_\sigma x^\sigma} = 0. \quad (5.30)$$

The equation 5.30 means that the wave vector is orthogonal to the amplitude tensor.

5.2 Gravitational waves

Here we are going to keep developing the wave equation for the perturbation $h_{\alpha\beta}$ see how are the equations for the perturbations of the coordinates components and we will work with a new Gauge, the called *traceless transverse gauge* and we will find out the gravitational waves have got polarisation properties.

From our Lorenz gauge we had found that $W'_\alpha = 0$ but this also means that the perturbations of the coordinates satisfies a wave equation $W_\alpha = \square \xi_\alpha$ so we can suppose the most basic solution to this equation is a plane wave, therefore

$$\xi^\alpha = B^\alpha e^{ik_\sigma x^\sigma}, \quad (5.31)$$

where B^α are four infinitesimal constants, and we gotta find them to have a complete solution for the wave equation to do it lets start from the primed and bar perturbation

$$\bar{h}'_{\alpha\beta} = \bar{h}_{\alpha\beta} - \frac{\partial \xi_\alpha}{\partial x^\beta} - \frac{\partial \xi_\beta}{\partial x^\alpha} + \eta_{\alpha\beta} \frac{\partial \xi_\mu}{\partial x^\mu}, \quad (5.32)$$

$$\bar{h}'_{\alpha\beta} = A_{\alpha\beta} e^{ik_\sigma x^\sigma} - i B_\alpha k_\beta e^{ik_\sigma x^\sigma} - i B_\beta k_\alpha e^{ik_\sigma x^\sigma} + i \eta_{\alpha\beta} k_\sigma B^\sigma e^{ik_\sigma x^\sigma} = A'_{\alpha\beta} e^{ik_\sigma x^\sigma}, \quad (5.33)$$

thus

$$A'_{\alpha\beta} = A_{\alpha\beta} - i B_\alpha k_\beta - i B_\beta k_\alpha + i \eta_{\alpha\beta} k_\sigma B^\sigma. \quad (5.34)$$

5.2.1 Traceless transverse gauge

Here we are going to choose a new gauge, the *traceless transverse gauge* which will allow us to work as well with $\bar{h}_{\alpha\beta}$ as $h_{\alpha\beta}$ because, as we shall see in a moment, both satisfy the same wave equations. The traceless transverse gauge is:

$$\eta^{\mu\nu} A'_{\mu\nu} = 0, \quad (5.35)$$

$$A'_{0i} = 0. \quad (5.36)$$

Calculating the Gauge of the equation 5.34 we have

$$\eta^{\mu\nu} A'_{\mu\nu} = \eta^{\mu\nu} A_{\mu\nu} - i\eta^{\mu\nu} B_\mu k_\nu - i\eta^{\mu\nu} B_\nu k_\mu + i\eta^{\mu\nu} \eta_{\mu\nu} k_\sigma B^\sigma = 0, \quad (5.37)$$

$$A'_{0i} = A_{0i} - iB_0 k_i - iB_i k_0 + \eta_{0i} k_\sigma B^\sigma = 0, \quad (5.38)$$

$$\eta^{\mu\nu} A'_{\mu\nu} = \eta^{\mu\nu} A_{\mu\nu} + 2ik_\mu B^\mu = 0, \quad (5.39)$$

$$A_{0i} - iB_0 k_i - iB_i k_0 = 0. \quad (5.40)$$

Or, written in a more convenient way,

$$\eta^{\mu\nu} A_{\mu\nu} = -2ik_\mu B^\mu, \quad (5.41)$$

$$B_0 k_i + B_i k_0 = -iA_{0i}. \quad (5.42)$$

Now, from the orthogonality condition 5.30 we can calculate how are the components of $A_{\mu\nu}$ therefore

$$A_{0\beta} k^\beta = A_{00} k^0 + A_{0i} k^i = A_{00} k^0 = 0 \rightarrow A_{00} = 0 \rightarrow A_{0\alpha} = 0, \quad (5.43)$$

$$A_{i\beta} k^\beta = A_{i0} k^0 + A_{ij} k^j = A_{ij} k^j = 0, \quad (5.44)$$

where the first term has dropped out because the gauge condition $A_{0i} = 0$ has already killed it. It is worth reading the surviving statement carefully, because it does *not* say that the spatial amplitudes vanish, which would leave us with no wave at all: what it says is that the amplitude tensor is *transverse*, that is, orthogonal to the direction in which the wave travels, so that a gravitational

wave can only shake the two directions perpendicular to its propagation. This is the origin of the word transverse in the name of the gauge.

Before going further it will be worth to calculate the trace of the perturbation under these gauge, thus

$$\bar{h}_{0\alpha} = 0, \quad \bar{h} = \eta^{\alpha\beta} \bar{h}_{\alpha\beta} = \eta^{\alpha\beta} A_{\alpha\beta} e^{ik_\sigma x^\sigma} = 0 \quad \longrightarrow \quad h = -\bar{h} = 0, \quad (5.45)$$

$$\bar{h}_{\alpha\beta} = h_{\alpha\beta}. \quad (5.46)$$

Here we can see that using this gauge the wave equation for $h_{\alpha\beta}$ is the same that for $\bar{h}_{\alpha\beta}$ lets write the perturbation in a matrix form under the assumption that the wave travels along the z axis, so that its contravariant wave vector is $k^\alpha = [\omega/C, 0, 0, \omega/C]$ and, lowering the index with the mostly plus metric, $k_\alpha = [-\omega/C, 0, 0, \omega/C]$, thus

$$A_{\alpha\beta} k^\beta = A_{\alpha 0} k^0 + A_{\alpha i} k^i = A_{\alpha 0} \frac{\omega}{C} + A_{\alpha z} \frac{\omega}{C} = 0 \quad \longrightarrow \quad A_{\alpha z} = -A_{\alpha 0} = 0, \quad (5.47)$$

$$\eta^{\mu\nu} A_{\mu\nu} = 0 \quad \rightarrow \quad A_{xx} + A_{yy} = 0. \quad (5.48)$$

We now set $A_{xx} = a_+$, which is called the *plus mode*, and $A_{xy} = a_\times$, which is called the *cross mode*, and since the trace condition forces $A_{yy} = -A_{xx}$ while transversality has already emptied the whole z row and column, the perturbation written in the traceless transverse gauge is

$$h_{\alpha\beta}^{TT} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & h_+ & h_\times & 0 \\ 0 & h_\times & -h_+ & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad (5.49)$$

then we can define now the ***gravitational polarisation modes*** as

$$h_+ = a_+ e^{i\omega(t - \frac{z}{C})}, \quad (5.50)$$

$$h_\times = a_\times e^{i\omega(t - \frac{z}{C})}. \quad (5.51)$$

5.2.1.1 Polarisation and Stokes parameters for gravitational waves

We are going to start from the perturbation under the influence of a traceless transverse gauge 5.49 this because it has got a really familiar form so we have to ask ourselves, *what is the difference between light's polarisation and gravitational polarisation?* Well, lets figure it out.

First, we have to use the fact there is an isomorphism between the vector space $\mathbb{M}_{n \times n}$ (square $n \times n$ matrices) and $\mathbb{R}^{n^2}$, therefore from the perturbation matrix 5.49 we can define our *perturbation state vector*

$$\mathbf{h}^{TT} = \begin{bmatrix} h_+ \\ h_\times \\ h_\times \\ -h_+ \end{bmatrix}. \quad (5.52)$$

This vector will tell us the polarisation mode state where our space-time is under the perturbation 5.49 but before going further we must take on count an important fact, $a_+, a_\times$ must have their own spatial phase because not always h_+ and $h_\times$ will be on phase or will represent the same oscillation but with different amplitude, both are different, in general, polarisation modes, then if $A_+, \phi_+, A_\times, \phi_\times$ are their amplitudes and phases respectively then the polarisation mode state vector will be

$$\mathbf{h}^{TT} = \begin{bmatrix} A_+ e^{i\phi_+} \\ A_\times e^{i\phi_\times} \\ A_\times e^{i\phi_\times} \\ -A_+ e^{i\phi_+} \end{bmatrix} e^{i(\omega t - kz)}. \quad (5.53)$$

We have got only two degrees of freedom in the state vector 5.53 one for the $+$ modes and the other for the x modes, lets see how are they related geometrically. We can see easily the following two equations

$$\frac{a_+}{A_+} e^{-i\phi_\times} = e^{i(\omega t - kz)} e^{i\Delta\phi}, \quad \frac{a_\times}{A_\times} e^{-i\phi_+} = e^{i(\omega t - kz)} e^{-i\Delta\phi} \quad (5.54)$$

where $\Delta\phi = \phi_+ - \phi_\times$ having all of this on count lets calculate the following product

$$\left(\frac{a_+}{A_+} e^{-i\phi_\times} - \frac{a_\times}{A_\times} e^{-i\phi_+} \right) \left(\frac{a_+}{A_+} e^{-i\phi_\times} - \frac{a_\times}{A_\times} e^{-i\phi_+} \right)^*, \quad (5.55)$$

$$\begin{aligned}
& \left(\frac{a_+}{A_+} e^{-i\phi_-} - \frac{a_-}{A_-} e^{-i\phi_+} \right) \left(\frac{a_+}{A_+} e^{-i\phi_-} - \frac{a_-}{A_-} e^{-i\phi_+} \right)^* \\
&= \frac{a_+^2}{A_+^2} - \frac{a_+ a_-^*}{A_+ A_-} e^{i\Delta\phi} - \frac{a_- a_+^*}{A_+ A_-} e^{-i\Delta\phi} + \frac{a_-^2}{A_-^2} \\
&= \frac{a_+^2}{A_+^2} + \frac{a_-^2}{A_-^2} - 2 \frac{\text{Re}\{a_+ a_-^*\}}{A_+ A_-} \cos \Delta\phi, \\
& \left(\frac{a_+}{A_+} e^{-i\phi_-} - \frac{a_-}{A_-} e^{-i\phi_+} \right) \left(\frac{a_+}{A_+} e^{-i\phi_-} - \frac{a_-}{A_-} e^{-i\phi_+} \right)^* = \sin^2 \Delta\phi
\end{aligned} \tag{5.56}$$

or joining all

$$\frac{a_+^2}{A_+^2} + \frac{a_-^2}{A_-^2} - 2 \frac{\text{Re}\{a_+ a_-^*\}}{A_+ A_-} \cos \Delta\phi = \sin^2 \Delta\phi. \tag{5.57}$$

The polarisation modes describe an hyper-ellipse in the complex plane $\mathbb{C}^2$ in the same way as the light's polarisation! Then this implies the polarisation mode state vector 5.53 lives, in some way as a vector of this ellipse. Now, we haven't got stationary perturbations, this in the real life usually oscillates very fast and has some kind of time-dependence, under the same calculation we can arrive to an ellipse in the real space (this time taking its real part and ignoring the imaginary part)

$$\frac{a_+^2(t)}{A_+^2(t)} + \frac{a_-^2(t)}{A_-^2(t)} - 2 \frac{a_+(t) a_-^*(t)}{A_+(t) A_-(t)} \cos \Delta\phi = \sin^2 \Delta\phi. \tag{5.58}$$

To solve this problem of the time dependence we can take time averages, why? Because we in fact measure time averages and we, for simplicity, we have suppose monochromatic waves thus

$$\frac{\langle a_+^2(t) \rangle}{A_+^2} + \frac{\langle a_-^2(t) \rangle}{A_-^2} - 2 \frac{\langle a_+(t) a_-^*(t) \rangle}{A_+ A_-} \cos \Delta\phi = \sin^2 \Delta\phi \tag{5.59}$$

where

$$\langle a_i(t) a_j(t) \rangle = \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T a_i(t) a_j(t) dt \quad i, j = +, -. \tag{5.60}$$

Multiplying the equation 5.59 by $4A_+^2 A_-^2$ we can see that

$$4A_-^2 \langle a_+^2(t) \rangle + 4A_+^2 \langle a_-^2(t) \rangle - 8A_+ A_- \langle a_+(t) a_-(t) \rangle \cos \Delta\phi = (2A_+ A_- \sin \Delta\phi)^2. \tag{5.61}$$

Because in the perturbation 5.49 we have plane and harmonic waves, is easy to see the time averages are

$$\langle a_+(t)^2 \rangle = \frac{1}{2} A_+^2 \quad (5.62)$$

$$\langle a_\times(t)^2 \rangle = \frac{1}{2} A_\times^2 \quad (5.63)$$

$$\langle a_+(t)a_\times(t) \rangle = \frac{1}{2} A_+ A_\times \cos \Delta\phi, \quad (5.64)$$

substituting these equation in 5.59 yields to

$$2A_+^2 A_\times^2 + 2A_+^2 A_\times^2 - (2A_+ A_\times \cos \Delta\phi)^2 = (2A_+ A_\times \sin \Delta\phi)^2, \quad (5.65)$$

Since we want to express the result in terms of a *gravitational intensity*, we use on the left hand side of equation 5.65 the algebraic identity $4A_+^2 A_\times^2 = (A_+^2 + A_\times^2)^2 - (A_+^2 - A_\times^2)^2$, which turns the two leading terms into a difference of perfect squares and gives

$$(A_+^2 + A_\times^2)^2 - (A_+^2 - A_\times^2)^2 - (2A_+ A_\times \cos \Delta\phi)^2 = (2A_+ A_\times \sin \Delta\phi)^2, \quad (5.66)$$

or defining the following quantities

$$S_0^G = A_+^2 + A_\times^2, \quad (5.67)$$

$$S_1^G = A_+^2 - A_\times^2, \quad (5.68)$$

$$S_2^G = 2A_+ A_\times \cos \Delta\phi, \quad (5.69)$$

$$S_3^G = 2A_+ A_\times \sin \Delta\phi \quad (5.70)$$

letting us to write the equation 5.66 as

$$(S_0^G)^2 = (S_1^G)^2 + (S_2^G)^2 + (S_3^G)^2 \quad (5.71)$$

this is the same equation for the Stokes parameters for a well polarised light beam. The analogy can be pushed one step further, because exactly as in optics the three parameters S_1^G , S_2^G and S_3^G can be read as the Cartesian coordinates of a point in an abstract space, and the relation we have just derived says that for a perfectly polarised wave that point is forced to lie on a sphere of radius S_0^G . This is the *Poincaré sphere* of the gravitational field, drawn in figure 5.1, and a partially polarised signal, which is what a real detector buried in noise actually records, falls strictly inside the sphere rather than on its surface, the ratio between the two radii being what in optics is called the degree of polarisation.

Poincaré Sphere

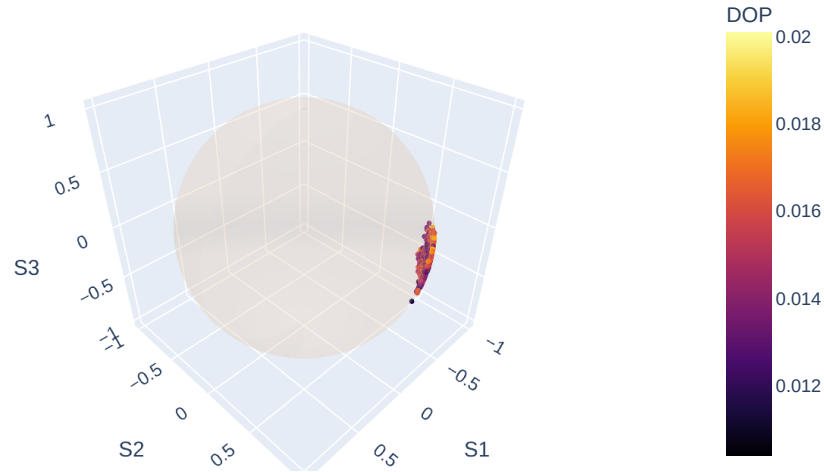


Figure 5.1: The Poincaré sphere built out of the gravitational Stokes parameters S_1^G , S_2^G and S_3^G . A perfectly polarised wave is a single point on the surface of the sphere, while the colour scale gives the degree of polarisation of the distribution of states, which for a partially polarised signal places the point strictly inside the sphere.

5.2.1.2 Helicity, and why gravitational polarisation is not optical polarisation

We opened the previous subsection by asking what the difference is between the polarisation of light and the polarisation of a gravitational wave, and having found that the two obey the very same Stokes relation the reader would be forgiven for concluding that there is no difference at all. There is one, it is fundamental, and it shows itself the moment we ask not what the polarisation states look like but how they behave when we rotate the apparatus that measures them.

Let us therefore rotate our frame by an angle ϑ about the direction in which the wave travels, which we have taken to be the z axis, so that the two transverse coordinates mix among themselves according to

$$\mathbb{R} = \begin{bmatrix} \cos \vartheta & \sin \vartheta \\ -\sin \vartheta & \cos \vartheta \end{bmatrix}. \quad (5.72)$$

The amplitude of the wave is a tensor of rank two, not a vector, so it needs one copy of the

rotation matrix for each of its two indices, which is to say $h' = \mathbb{R} h \mathbb{R}^T$. Writing out the transverse block of the perturbation 5.49 and multiplying the three matrices together,

$$h' = \begin{bmatrix} \cos \vartheta & \sin \vartheta \\ -\sin \vartheta & \cos \vartheta \end{bmatrix} \begin{bmatrix} h_+ & h_\times \\ h_\times & -h_+ \end{bmatrix} \begin{bmatrix} \cos \vartheta & -\sin \vartheta \\ \sin \vartheta & \cos \vartheta \end{bmatrix},$$

and carrying out the two products one after the other, the entry in the first row and first column comes out as

$$\begin{aligned} h'_+ &= \cos \vartheta (h_+ \cos \vartheta + h_\times \sin \vartheta) + \sin \vartheta (h_\times \cos \vartheta - h_+ \sin \vartheta) \\ &= h_+ (\cos^2 \vartheta - \sin^2 \vartheta) + h_\times (2 \sin \vartheta \cos \vartheta) = h_+ \cos 2\vartheta + h_\times \sin 2\vartheta, \end{aligned}$$

where we have used the two double angle identities marked in blue, and the entry in the first row and second column comes out, by exactly the same route, as

$$\begin{aligned} h'_\times &= \cos \vartheta (h_\times \cos \vartheta - h_+ \sin \vartheta) - \sin \vartheta (h_\times \sin \vartheta + h_+ \cos \vartheta) \\ &= h_\times (\cos^2 \vartheta - \sin^2 \vartheta) - h_+ (2 \sin \vartheta \cos \vartheta) = -h_+ \sin 2\vartheta + h_\times \cos 2\vartheta, \end{aligned}$$

while the remaining two entries add nothing new, since the matrix stays symmetric and traceless as it must. The cleanest way to read these two results is to combine them into a single complex amplitude, which gives

$$\begin{aligned} h'_+ + ih'_\times &= h_+ (\cos 2\vartheta - i \sin 2\vartheta) + h_\times (\sin 2\vartheta + i \cos 2\vartheta) \\ &= h_+ e^{-2i\vartheta} + h_\times i e^{-2i\vartheta}, \end{aligned}$$

so that finally

$$h'_+ + ih'_\times = e^{-2i\vartheta} (h_+ + ih_\times). \quad (5.73)$$

Equation 5.73 is the whole answer, and it is worth comparing it at once with the corresponding statement for light. The electric field is a vector and therefore carries a single index, so under the same rotation it picks up a single copy of the rotation matrix, and the identical manipulation gives

$$E'_x + iE'_y = e^{-i\vartheta} (E_x + iE_y). \quad (5.74)$$

The two expressions differ by nothing except a factor of two in the exponent, and that factor of two is the difference between electromagnetism and gravitation. A field whose polarisation state

acquires a phase $e^{-is\vartheta}$ under a rotation by ϑ about its direction of propagation is said to have *helicity* s , so light has helicity one and the gravitational wave has helicity two, and in the quantum language this is the statement that the photon is a particle of spin one while the graviton, should it exist, must be a particle of spin two.

Three consequences follow immediately and all three are measurable. The first is that rotating the apparatus by half a turn, $\vartheta = \pi$, leaves the gravitational amplitude exactly as it was, since $e^{-2i\pi} = 1$, whereas the electric field would have been reversed, so that **a gravitational wave repeats itself every 180 degrees while light needs a full 360**. The second is that setting $\vartheta = \pi/4$ turns h_+ into $h_\times$ and $h_\times$ into $-h_+$, so the two linear polarisation states of a gravitational wave are separated by forty five degrees and not by the ninety degrees that separate the two linear polarisations of light, which is precisely why the plus and the cross patterns look rotated into one another by half a right angle. The third concerns the Poincaré sphere of figure 5.1: since the Stokes parameters are quadratic in the amplitudes, a rotation of the amplitudes by 2ϑ carries S_1^G and S_2^G around the S_3^G axis by 4ϑ , twice as fast as the corresponding rotation in optics, so the gravitational Poincaré sphere is traversed at double the optical rate.

The same factor of two is also the reason behind the quadrupolar pattern we promised to explain. A field of helicity two cannot have a monopole or a dipole moment radiating, because those would transform with helicity zero and one respectively, and the lowest multipole that carries helicity two is the quadrupole. We shall see that statement emerge again, from an entirely different direction, when we come to the generation of waves in the next section.

5.2.2 Effects of the gravitational radiation on the matter

To study the effects of the gravitational radiation under the traceless transverse gauge we are going to suppose a particle A which interacts only gravitationally and follows a geodesic parameterized with its proper time τ this particle stays at rest at the origin of coordinates, this means its world line will be a vertical line, therefore given a coordinate chart its coordinates are

$$X_A^0(\tau) = Ct, \quad \& \quad X_A^i(\tau) = 0, \quad (5.75)$$

lets solve its geodesic equation

$$\frac{d^2 X_A^\alpha}{d\tau^2} + \Gamma_{\mu\nu}^\alpha \dot{x}_A^\mu \dot{x}_A^\nu = 0 \quad (5.76)$$

where

$$\Gamma_{\beta\gamma}^0 = -\frac{1}{2} (h_{0\gamma,\beta} + h_{0\beta,\gamma} - h_{\gamma\beta,0}), \quad (5.77)$$

$$\Gamma_{\beta\gamma}^i = \frac{1}{2} (h_{i\gamma,\beta} + h_{i\beta,\gamma} - h_{\gamma\beta,i}). \quad (5.78)$$

Using the TTG

$$\Gamma_{\beta\gamma}^0 = \frac{1}{2C} h^{TT}_{\gamma\beta,t}, \quad (5.79)$$

$$\Gamma_{0\alpha}^0 = 0, \quad (5.80)$$

$$\Gamma_{ij}^0 = \frac{1}{2C} h^{TT}_{ij,t}, \quad (5.81)$$

$$\Gamma_{00}^i = 0, \quad (5.82)$$

$$\Gamma_{0j}^i = \frac{1}{2C} h^{TT}_{ij,t}, \quad (5.83)$$

$$\Gamma_{jk}^i = \frac{1}{2} (h^{TT}_{ik,j} + h^{TT}_{ji,k} - h^{TT}_{jk,i}). \quad (5.84)$$

Putting these symbols into the geodesic equation for a particle initially at rest, every term on the left hand side vanishes, and therefore a particle that starts at rest in these coordinates stays at the same coordinate values for ever. This result is famous for being misread, so it is worth stating what it does and does not mean. It does **not** mean that the wave has no effect on the particle, it means that the traceless transverse coordinates are comoving with the free particles, so that the coordinates themselves are dragged along and the coordinate positions stop being informative. What a detector actually measures is the proper distance between two such particles, and that distance is computed with the metric, which is precisely what the wave is modulating, so the separation does oscillate even though the coordinates do not change. We shall see that the pattern of that oscillation is quadrupolar, which is the deep reason a gravitational wave stretches one transverse direction while squeezing the other.

5.3 The generation of gravitational waves

Everything we have done so far has described a wave that was already there, travelling through empty space, and we have said nothing whatever about where it came from. That is the question we take up now, and it is the question that decides whether the whole subject is observational physics or merely an elegant exercise, because a wave nobody can produce is a wave nobody can detect. The answer, which Einstein worked out in 1918 after correcting an error in his own paper of two years earlier (Einstein, 1918), is that gravitational radiation is generated by the second time derivative of the quadrupole moment of the source, and the purpose of this section is to derive that statement and then to ask how much energy such a source loses.

5.3.1 The retarded solution of the wave equation

We begin again from the field equations in the Lorenz gauge, which we obtained in equation 5.25 and which we now keep with their source term rather than setting it to zero,

$$\square \bar{h}_{\mu\nu} = -\frac{16\pi G}{C^4} T_{\mu\nu}, \quad \square = -\frac{1}{C^2} \frac{\partial^2}{\partial t^2} + \nabla^2. \quad (5.85)$$

Equation 5.85 is a set of ten uncoupled wave equations, each of them with a source, and it is exactly the equation that electromagnetism hands us for the vector potential in the same gauge, so we may borrow the solution wholesale. Writing it in the form $\square \bar{h}_{\mu\nu} = -4\pi S_{\mu\nu}$ with

$$S_{\mu\nu} = \frac{4G}{C^4} T_{\mu\nu}, \quad (5.86)$$

the retarded Green function of the d'Alembertian gives

$$\bar{h}_{\mu\nu}(t, \vec{x}) = \frac{4G}{C^4} \int \frac{T_{\mu\nu} \left(t - \frac{|\vec{x} - \vec{x}'|}{C}, \vec{x}' \right)}{|\vec{x} - \vec{x}'|} d^3x'. \quad (5.87)$$

The physical content of equation 5.87 is worth stating in words before we do anything with it, because it is the precise statement of what Newton could not have. The field at the point $\vec{x}$ and at the time t does not depend on where the matter is now, it depends on where the matter was at the earlier time $t - |\vec{x} - \vec{x}'|/C$, which is to say at the moment when a signal leaving the source would have had to set out in order to arrive here now. Gravity propagates, and it propagates at the speed of light, and the instantaneous action at a distance that so disturbed Newton in the letter we quoted in chapter 1 has finally disappeared from the theory.

We are interested in the field far from the source, in what is called the *wave zone*, so let $r = |\vec{x}|$ be the distance from the source to the observer and suppose that the source is small and confined, so that $|\vec{x}'| \ll r$ for every point of it. To leading order we may then replace $|\vec{x} - \vec{x}'|$ by r in both places where it appears, in the denominator because the correction is of relative order $|\vec{x}'|/r$, and inside the retarded time because the light travel time across the source is short compared with the time scale on which the source itself changes, an approximation which amounts to demanding that the source move slowly compared with light. With that,

$$\bar{h}_{\mu\nu}(t, \vec{x}) \approx \frac{4G}{C^4 r} \int T_{\mu\nu} \left(t - \frac{r}{C}, \vec{x}' \right) d^3x', \quad (5.88)$$

and the whole problem has been reduced to computing three volume integrals of the energy momentum tensor, one for each kind of index.

5.3.2 Why there is no monopole and no dipole radiation

Before evaluating the spatial integral, which is the one that will give us the answer, it pays to look at the other two, because what they give us is nothing, and the reason they give us nothing is one of the most instructive facts in the whole theory.

Consider first the purely temporal component. The integral appearing in equation 5.88 is

$$\int T^{00} d^3x' = C^2 \int \rho d^3x' = MC^2, \quad (5.89)$$

which is nothing other than the total energy of the source, and conservation of the energy momentum tensor tells us at once that it cannot change. Indeed, taking the temporal component of $\partial_\mu T^{\mu\nu} = 0$ and integrating over all space,

$$\frac{1}{C} \frac{d}{dt} \int T^{00} d^3x = - \int \partial_i T^{i0} d^3x = - \oint_\Sigma T^{i0} dS_i = 0,$$

where the volume integral of a divergence has been turned into a surface integral by the divergence theorem, and that surface integral vanishes because the source is confined and $T^{\mu\nu}$ is zero on a surface far enough out. The total mass is therefore a constant, and a constant produces a static field and not a wave. This is the statement that **there is no monopole gravitational radiation**, and it is the exact counterpart of the statement in electromagnetism that a spherically symmetric pulsating charge radiates nothing, which we already met from the geometrical side as Birkhoff's theorem in chapter 4.

Now the mixed component, whose integral is

$$\int T^{0i} d^3x' = C P^i, \quad (5.90)$$

the total momentum of the source, and the spatial component of the conservation law treats it exactly as before,

$$\frac{1}{C} \frac{d}{dt} \int T^{0i} d^3x = - \int \partial_j T^{ji} d^3x = - \oint_\Sigma T^{ji} dS_j = 0,$$

so the total momentum is conserved too and this component is static as well. It is worth pausing on how different this is from electromagnetism, because the contrast is the whole point. The electromagnetic dipole moment is $\vec{d} = \sum_a q_a \vec{r}_a$, and there is nothing in Maxwell's theory to stop it oscillating, since positive and negative charges can perfectly well move in opposite directions, which is exactly what an antenna is built to do and why dipole radiation dominates the electromagnetic world. The gravitational analogue of the dipole moment is $\sum_a m_a \vec{r}_a = M \vec{R}_{cm}$, and its second derivative is

$$\frac{d^2}{dt^2} (M \vec{R}_{cm}) = \frac{d\vec{P}}{dt} = 0, \quad (5.91)$$

which vanishes identically because momentum is conserved and because, crucially, there is no negative mass with which to build a gravitational antenna. **Conservation of momentum forbids**

dipole gravitational radiation, and the same argument applied to the antisymmetric part of the second moment forbids the magnetic dipole term too, since the quantity that would have to oscillate there is the total angular momentum, which is also conserved. The first multipole left standing is the quadrupole, exactly as the helicity argument of the previous section predicted, and this is why gravitational radiation is so much feebler than electromagnetic radiation and why it took a century to detect.

5.3.3 The quadrupole formula

Everything therefore hangs on the spatial integral $\int T^{ij} d^3x$, and at first sight there is no reason it should be related to anything familiar. That it is, is the content of a small and rather beautiful identity which follows from the conservation law used twice.

Start from the two components of $\partial_\mu T^{\mu\nu} = 0$, written out with $x^0 = Ct$,

$$\frac{1}{C} \frac{\partial T^{00}}{\partial t} + \partial_i T^{i0} = 0, \quad (5.92)$$

$$\frac{1}{C} \frac{\partial T^{0j}}{\partial t} + \partial_i T^{ij} = 0, \quad (5.93)$$

and differentiate the first with respect to time once more, then use the second to eliminate the mixed component,

$$\frac{1}{C^2} \frac{\partial^2 T^{00}}{\partial t^2} = -\frac{1}{C} \frac{\partial}{\partial t} \partial_i T^{i0} = -\partial_i \left(\frac{1}{C} \frac{\partial T^{0i}}{\partial t} \right) = -\partial_i (-\partial_j T^{ji}) = \partial_i \partial_j T^{ij}.$$

Multiply both sides by $x^k x^l$ and integrate over all space. On the left we obtain the second time derivative of the second moment of the energy density, and on the right we integrate by parts twice, throwing away both surface terms because the source is confined,

$$\int (\partial_i \partial_j T^{ij}) x^k x^l d^3x = \int T^{ij} \partial_i \partial_j (x^k x^l) d^3x.$$

The derivatives of the product are elementary, and this is the step that produces the factor of two,

$$\begin{aligned} \partial_j (x^k x^l) &= \delta_j^k x^l + x^k \delta_j^l, \\ \partial_i \partial_j (x^k x^l) &= \delta_j^k \delta_i^l + \delta_i^k \delta_j^l, \end{aligned}$$

so that, substituting back and letting the two Kronecker deltas do their work,

$$\int T^{ij} (\delta_j^k \delta_i^l + \delta_i^k \delta_j^l) d^3x = \int (T^{lk} + T^{kl}) d^3x = 2 \int T^{kl} d^3x,$$

where the last equality uses the symmetry of the energy momentum tensor. Putting the two sides together and introducing the second mass moment of the source,

$$I^{kl}(t) = \int \rho(t, \vec{x}) x^k x^l d^3x, \quad (5.94)$$

and remembering that $T^{00} = \rho C^2$, the two factors of C^2 cancel and we are left with the identity we were after,

$$\int T^{kl} d^3x = \frac{1}{2} \frac{d^2 I^{kl}}{dt^2}. \quad (5.95)$$

There is something almost unreasonable about equation 5.95. On the left sits an integral of stresses, of pressures and shears inside the source, quantities that depend on the detailed internal constitution of the star or the binary, and on the right sits nothing but the second derivative of how the mass is distributed in space. All the complicated internal physics has cancelled, and it has cancelled because of the conservation law alone.

Substituting equation 5.95 into the far field expression 5.88 gives the result we have been working towards,

$$\bar{h}^{ij}(t, \vec{x}) = \frac{4G}{C^4 r} \cdot \frac{1}{2} \frac{d^2 I^{ij}}{dt^2} = \frac{2G}{C^4 r} \ddot{I}^{ij} \left(t - \frac{r}{C} \right), \quad (5.96)$$

which is the *quadrupole formula*. To obtain the physical wave one still has to project onto the traceless transverse gauge, discarding the trace and the components along the direction of propagation exactly as we did in the previous section, and since the projection removes the trace anyway one may equally well write the formula in terms of the traceless quadrupole moment

$$Q^{ij} = I^{ij} - \frac{1}{3} \delta^{ij} \delta_{kl} I^{kl}, \quad (5.97)$$

which gives the same answer, so that $h_{ij}^{TT} = \frac{2G}{C^4 r} \ddot{Q}_{ij}^{TT}$.

Three readings of equation 5.96 are worth carrying away. The amplitude falls off as $1/r$ and not as $1/r^2$, which is what makes distant sources detectable at all and is the signature of a genuine radiation field rather than a near field. The amplitude is governed by $\ddot{I}$, so a source whose mass distribution moves but does not change shape, a perfectly spherical star collapsing or a rigid body rotating about its symmetry axis, radiates nothing whatever, and this is why an axisymmetric pulsar is gravitationally silent while a deformed one is not. And the prefactor G/C^4 is the same absurdly small number we met when we fixed the coupling constant in chapter 3, which is the honest reason the strains we are chasing are of order 10^{-21} .

5.3.4 The energy carried away by the waves

A wave that does no work is a mathematical curiosity, so the last question of this chapter is how much energy a gravitational wave carries, and here we run immediately into a difficulty of principle that has no counterpart in electromagnetism. The equivalence principle says that at any single event we may choose a freely falling frame in which the gravitational field vanishes, and if the field can be transformed away at a point then so can any energy density we try to attribute to it at that point. There is therefore no meaningful local energy density of the gravitational field, and any expression claiming to be one is a coordinate artefact.

What does survive is an average. If we average over a region several wavelengths across, larger than the wave but much smaller than the scale on which the background curves, the result is well defined and behaves exactly as an energy momentum tensor should. This is the construction due to Isaacson (Isaacson, 1968), and for a wave in the traceless transverse gauge it reads

$$t_{\mu\nu} = \frac{C^4}{32\pi G} \langle \partial_\mu h_{ij}^{TT} \partial_\nu h^{TTij} \rangle, \quad (5.98)$$

where the angle brackets denote precisely that average over several wavelengths. The object $t_{\mu\nu}$ is not a tensor under general coordinate changes, which is the mathematical way of saying what the equivalence principle already told us, but it is a perfectly good tensor under the Lorentz transformations of the flat background, and that is enough to compute a flux.

The energy crossing unit area per unit time in the radial direction is the component t_{0r} , and integrating it over a large sphere surrounding the source, then substituting the quadrupole formula 5.96 and performing the angular integral that the traceless transverse projection leaves behind, one arrives at the luminosity

$$L = -\frac{dE}{dt} = \frac{G}{5C^5} \langle \ddot{Q}_{ij} \ddot{Q}^{ij} \rangle, \quad (5.99)$$

which is the gravitational analogue of the Larmor formula of electrodynamics, with the third derivative of the quadrupole moment playing the part that the second derivative of the dipole moment plays there.

The constant in front of equation 5.99 deserves the same treatment we gave the coupling constant of the field equations. The combination C^5/G has the dimensions of a power and is worth about 3.6×10^{52} watts, a quantity sometimes called the Planck luminosity, and since the luminosity of any source is that enormous number multiplied by the square of a dimensionless third derivative, two things follow at once and they seem to contradict each other. On the one hand almost nothing radiates measurably, because for ordinary matter that dimensionless factor is minute. On the other hand a source that is both very compact and very asymmetric, such as two black holes in their final orbits, can drive the factor towards unity, and in the last fraction of a second before they merge such a pair momentarily outshines, in gravitational waves, the entire electromagnetic output of the observable universe. Both statements are consequences of the same formula, and the reconciliation is simply that the energy is spread over cosmological distances by the time it reaches us.

Equation 5.99 was not merely believed, it was tested, and tested long before any wave was detected directly. In 1974 Russell Hulse and Joseph Taylor discovered a pulsar in a binary orbit with another neutron star (Hulse and Taylor, 1975), a system whose orbital period is now known to extraordinary precision, and that period is shrinking year by year at exactly the rate the quadrupole formula predicts for a pair of masses losing energy to gravitational radiation. The agreement is better than a fraction of a percent, it earned the Nobel Prize of 1993, and it meant that when the detectors of chapter 1 finally recorded a passing wave in 2015 nobody seriously doubted what they were looking at.

Bibliography

- Abbott, B. P. et al. (2016). Observation of gravitational waves from a binary black hole merger. *Physical Review Letters*, 116:061102.
- Aristotle (1984). Physics. In Barnes, J., editor, *The Complete Works of Aristotle: The Revised Oxford Translation*. Princeton University Press, Princeton.
- Bardeen, J. M., Carter, B., and Hawking, S. W. (1973). The four laws of black hole mechanics. *Communications in Mathematical Physics*, 31(2):161–170.
- Blandford, R. D. and Znajek, R. L. (1977). Electromagnetic extraction of energy from kerr black holes. *Monthly Notices of the Royal Astronomical Society*, 179(3):433–456.
- Bolyai, J. (1832). Appendix: Scientiam spatii absolute veram exhibens. In *Tentamen Juventutem Studiosam in Elementa Matheseos Purae Introducendi, by Farkas Bolyai*. Typis Collegii Reformatorum, Maros-Vásárhely.
- Christoffel, E. B. (1869). Über die transformation der homogenen differentialausdrücke zweiten grades. *Journal für die reine und angewandte Mathematik*, 70:46–70.
- Clagett, M. (1959). *The Science of Mechanics in the Middle Ages*. University of Wisconsin Press, Madison.
- Corry, L., Renn, J., and Stachel, J. (1997). Belated decision in the hilbert-einstein priority dispute. *Science*, 278(5341):1270–1273.
- Duncan, R. C. and Thompson, C. (1992). Formation of very strongly magnetized neutron stars: Implications for gamma-ray bursts. *The Astrophysical Journal Letters*, 392:L9–L13.
- Dyson, F. W., Eddington, A. S., and Davidson, C. (1920). A determination of the deflection of light by the sun’s gravitational field, from observations made at the total eclipse of may 29, 1919. *Philosophical Transactions of the Royal Society A*, 220:291–333.
- Einstein, A. (1915). Die feldgleichungen der gravitation. *Sitzungsberichte der Königlich Preußischen Akademie der Wissenschaften*, pages 844–847.
- Einstein, A. (1918). Über gravitationswellen. *Sitzungsberichte der Königlich Preußischen Akademie der Wissenschaften*, pages 154–167.
- Einstein, A. (1949). Autobiographical notes. In Schilpp, P. A., editor, *Albert Einstein: Philosopher-Scientist*. Open Court, La Salle, Illinois.
- Einstein, A. and Grossmann, M. (1913). Entwurf einer verallgemeinerten relativitätstheorie und einer theorie der gravitation. *Zeitschrift für Mathematik und Physik*, 62:225–261.

- Euclid (1908). *The Thirteen Books of Euclid's Elements*. Cambridge University Press, Cambridge. Translated with introduction and commentary by Thomas L. Heath.
- Fourès-Bruhat, Y. (1952). Théorème d'existence pour certains systèmes d'équations aux dérivées partielles non linéaires. *Acta Mathematica*, 88:141–225.
- Friedmann, A. (1922). Über die krümmung des raumes. *Zeitschrift für Physik*, 10:377–386.
- Galilei, G. (1953). *Dialogue Concerning the Two Chief World Systems*. University of California Press, Berkeley. Translated by Stillman Drake.
- Galilei, G. (1974). *Two New Sciences*. University of Wisconsin Press, Madison. Translated by Stillman Drake.
- Gauss, C. F. (1828). Disquisitiones generales circa superficies curvas. *Commentationes Societatis Regiae Scientiarum Gottingensis Recentiores*, 6.
- Herodotus (2003). *The Histories*. Penguin Books, London. Translated by Aubrey de Sélincourt.
- Hilbert, D. (1915). Die grundlagen der physik (erste mitteilung). *Nachrichten von der Gesellschaft der Wissenschaften zu Göttingen, Mathematisch-Physikalische Klasse*, pages 395–407.
- Hubble, E. (1929). A relation between distance and radial velocity among extra-galactic nebulae. *Proceedings of the National Academy of Sciences*, 15(3):168–173.
- Hulse, R. A. and Taylor, J. H. (1975). Discovery of a pulsar in a binary system. *The Astrophysical Journal*, 195:L51–L53.
- Isaacson, R. A. (1968). Gravitational radiation in the limit of high frequency. ii. nonlinear terms and the effective stress tensor. *Physical Review*, 166(5):1272–1280.
- Israel, W. (1968). Event horizons in static electrovac space-times. *Communications in Mathematical Physics*, 8(3):245–260.
- Kaluza, T. (1921). Zum unitätsproblem der physik. *Sitzungsberichte der Preussischen Akademie der Wissenschaften zu Berlin*, pages 966–972.
- Klein, F. (1872). *Vergleichende Betrachtungen über neuere geometrische Forschungen*. Verlag von Andreas Deichert, Erlangen. The Erlangen Programme.
- Klein, O. (1926). Quantentheorie und fünfdimensionale relativitätstheorie. *Zeitschrift für Physik*, 37(12):895–906.
- Kline, M. (1972). *Mathematical Thought from Ancient to Modern Times*. Oxford University Press, New York.
- Lemaître, G. (1927). Un univers homogène de masse constante et de rayon croissant rendant compte de la vitesse de récession des nébuleuses extra-galactiques. *Annales de la Société Scientifique de Bruxelles*, 47:49–59.

- Levi-Civita, T. (1917). Nozione di parallelismo in una varietà qualunque e conseguente specificazione geometrica della curvatura riemanniana. *Rendiconti del Circolo Matematico di Palermo*, 42:173–204.
- Lobachevsky, N. I. (1829). On the principles of geometry. *Kazan Messenger*.
- Misner, C. W., Thorne, K. S., and Wheeler, J. A. (1973). *Gravitation*. W. H. Freeman, San Francisco.
- Nakahara, M. (2003). *Geometry, Topology and Physics*. Institute of Physics Publishing, Bristol, 2nd edition.
- Newman, E. T., Couch, E., Chinnapared, K., Exton, A., Prakash, A., and Torrence, R. (1965). Metric of a rotating, charged mass. *Journal of Mathematical Physics*, 6(6):918–919.
- Newton, I. (1958). *Isaac Newton's Papers and Letters on Natural Philosophy*. Harvard University Press, Cambridge, MA. Edited by I. Bernard Cohen.
- Newton, I. (1999). *The Principia: Mathematical Principles of Natural Philosophy*. University of California Press, Berkeley. Translated by I. Bernard Cohen and Anne Whitman.
- Nordström, G. (1918). On the energy of the gravitational field in einstein's theory. *Koninklijke Nederlandse Akademie van Wetenschappen Proceedings*, 20:1238–1245.
- Oppenheimer, J. R. and Snyder, H. (1939). On continued gravitational contraction. *Physical Review*, 56:455–459.
- Pais, A. (1982). *Subtle is the Lord: The Science and the Life of Albert Einstein*. Oxford University Press, Oxford.
- Penrose, R. (1969). Gravitational collapse: The role of general relativity. *Rivista del Nuovo Cimento*, 1:252–276.
- Perlmutter, Saul et al. (1999). Measurements of ω and λ from 42 high-redshift supernovae. *The Astrophysical Journal*, 517(2):565–586.
- Ptolemy, C. (1998). *Ptolemy's Almagest*. Princeton University Press, Princeton. Translated and annotated by G. J. Toomer.
- Rainich, G. Y. (1925). Electrodynamics in the general relativity theory. *Transactions of the American Mathematical Society*, 27(1):106–136.
- Rashed, R., editor (1996). *Encyclopedia of the History of Arabic Science*. Routledge, London.
- Reissner, H. (1916). Über die eigengravitation des elektrischen felde nach der einsteinschen theorie. *Annalen der Physik*, 355(9):106–120.
- Ricci-Curbastro, G. and Levi-Civita, T. (1900). Méthodes de calcul différentiel absolu et leurs applications. *Mathematische Annalen*, 54:125–201.

- Riemann, B. (1873). On the hypotheses which lie at the bases of geometry. *Nature*, 8:14–17, 36–37. Translated by William Kingdon Clifford.
- Riess, Adam G. et al. (1998). Observational evidence from supernovae for an accelerating universe and a cosmological constant. *The Astronomical Journal*, 116(3):1009–1038.
- Saccheri, G. (1733). *Euclides ab Omni Naevo Vindicatus*. Paolo Antonio Montano, Milan.
- Sakurai, J. J. and Napolitano, J. (2011). *Modern Quantum Mechanics*. Addison-Wesley, Boston, 2nd edition.
- Saliba, G. (1987). The height of the atmosphere according to mu’ayyad al-din al-’urdi, qutb al-din al-shirazi, and ibn al-shatir. *Annals of the New York Academy of Sciences*, 500:121–181.
- Schutz, B. F. (1980). *Geometrical Methods of Mathematical Physics*. Cambridge University Press, Cambridge.
- Schwarzschild, K. (1916). Über das gravitationsfeld eines massenpunktes nach der einsteinschen theorie. *Sitzungsberichte der Königlich Preußischen Akademie der Wissenschaften*, pages 189–196.
- Wald, R. M. (1984). *General Relativity*. University of Chicago Press, Chicago.
- Weyl, H. (1918). *Raum, Zeit, Materie*. Springer, Berlin.
- Wheeler, J. A. and Ford, K. (1998). *Geons, Black Holes, and Quantum Foam: A Life in Physics*. W. W. Norton, New York.
- Whitney, H. (1936). Differentiable manifolds. *Annals of Mathematics*, 37(3):645–680.